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Reading Notes

What Is Mathematics?

Modern mathematical reading notes: examples, solved exercises, diagrams, and proof mechanics.

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Contents

Foreword and Prefaces

Source Map

The foreword and the prefaces do not merely introduce the book as an educational text. They formulate a philosophy of mathematical culture. The book is presented as an elementary approach to ideas and methods, but the word "elementary" should not be confused with "shallow." The intended object is not a list of facts, but the living structure of mathematics.

0.1 The Status of the Book

The opening material makes clear that *What Is Mathematics?* should be read neither as a textbook of technical exercises nor as a popular exposition without proof. Its more precise status is this:

$$\boxed{\textit{What Is Mathematics?} = \text{an elementary reconstruction of mathematical thought.}}$$

Here "elementary" means that the book begins from accessible material: numbers, geometry, functions, limits, extrema, and calculus. But its actual purpose is structural. It tries to show how mathematics grows from examples into concepts, from concepts into methods, and from methods into theories.

For our reading, the object is therefore not simply to understand each statement. The object is to recover the generative mechanism behind each mathematical construction.

Definition 0.1: Generative Reading

A generative reading of a mathematical text is a reading that asks not only what is proved, but why the concept was introduced, what problem forced its appearance, which structure it stabilizes, and which further theory it makes possible.

Thus the book should be read as a sequence of conceptual transitions:

$$\text{problem} \longrightarrow \text{example} \longrightarrow \text{definition} \longrightarrow \text{theorem} \longrightarrow \text{method} \longrightarrow \text{new structure.}$$

This is the first guiding principle of these notes.

0.2 Mathematics Against Mechanical Training

The prefaces repeatedly warn against a degeneration of mathematical education into mechanical manipulation. This is not merely a pedagogical complaint. It is a mathematical warning.

A student may learn to perform operations without knowing the structure in which those operations are meaningful. In that case, the student possesses an algorithm but not a concept.

We may express this distinction as follows. Let P be a mathematical procedure and let S be the structure in which P is valid. Mechanical training teaches

$$P(x) \rightsquigarrow y.$$

Mathematical understanding asks instead:

$$\begin{aligned}
\text{Domain:} \quad & x && \in S, \\
\text{Operation:} \quad & P && : S \to S', \\
\text{Invariant:} \quad & I(x) && = I(P(x)), \\
\text{Justification:} \quad & P && \text{ is valid because of structural law } L.
\end{aligned}$$

The decisive difference is the presence of S, I, and L. Without them, computation is detached from mathematics.

Takeaway

The central enemy is not computation. The enemy is computation without structure. A mathematical operation becomes meaningful only when its domain, invariants, and governing laws are understood.

This point is essential for the later chapters. Arithmetic is not merely calculation with integers. Geometry is not merely drawing. Calculus is not merely differentiation and integration. Each of these is a structured domain with internal laws, transformations, and invariants.

0.3 What Counts as Mathematical Understanding?

The prefaces suggest a strong criterion of understanding. One understands a mathematical idea only when one can see its role inside a larger conceptual system.

For these notes, I will use the following working model. A mathematical object or theorem X is understood only after one has identified the tuple

$$\begin{aligned}
\mathcal{U}(X) = \big(&\mathrm{Origin}(X), \mathrm{Definition}(X), \mathrm{Examples}(X), \mathrm{NonExamples}(X), \\
&\mathrm{Theorems}(X), \mathrm{Proofs}(X), \mathrm{Invariants}(X), \mathrm{Extensions}(X)\big).
\end{aligned}$$

Each component has a different mathematical function.

$$\begin{aligned}
\mathrm{Origin}(X) \quad &: \quad \text{the problem or intuition that forces } X, \\
\mathrm{Definition}(X) \quad &: \quad \text{the formal stabilization of } X, \\
\mathrm{Examples}(X) \quad &: \quad \text{models where } X \text{ appears naturally}, \\
\mathrm{NonExamples}(X) \quad &: \quad \text{boundary cases that clarify meaning}, \\
\mathrm{Theorems}(X) \quad &: \quad \text{statements made possible by } X, \\
\mathrm{Proofs}(X) \quad &: \quad \text{logical mechanisms involving } X, \\
\mathrm{Invariants}(X) \quad &: \quad \text{quantities or structures preserved}, \\
\mathrm{Extensions}(X) \quad &: \quad \text{higher theories generated from } X.
\end{aligned}$$

This tuple is a reading discipline. It prevents the notes from becoming a collection of isolated remarks.

Remark

For a theoretical mathematician, a concept is not mastered when it can be repeated. It is mastered when its necessity, formal role, examples, counterexamples, and future extensions are visible.

0.4 The Meaning of "Elementary"

The word "elementary" in this book should be interpreted carefully. In mathematics, elementary material often contains the most primitive and therefore the most foundational structures.

For example, the natural numbers appear simple, but they already contain:

$$\text{successor structure}, \qquad \text{recursion}, \qquad \text{induction}, \qquad \text{order}, \qquad \text{arithmetic operations}.$$

These are not merely school-level topics. They are prototypes for much deeper mathematical mechanisms.

$$\begin{gathered}\text{successor} \rightsquigarrow \text{recursion and induction},\\ \text{arithmetic law} \rightsquigarrow \text{algebraic structure},\\ \text{geometrical construction} \rightsquigarrow \text{field extension},\\ \text{limit} \rightsquigarrow \text{analysis and topology},\\ \text{extremum} \rightsquigarrow \text{variational principles}.\end{gathered}$$

Thus the elementary is not the opposite of the profound. Rather, the elementary is the place where profound structures first appear in their simplest form.

Takeaway

The right way to read elementary mathematics is to look for the first appearance of universal mechanisms.

0.5 The Book as a Map of Mathematical Methods

The subtitle, *An Elementary Approach to Ideas and Methods*, is important. The book does not organize mathematics merely by subject matter. It organizes mathematics by methods.

A method is more than a technique. It is a repeatable mode of mathematical thought. Examples include:

$$\begin{aligned}\text{induction} &: \quad \text{proof by the successor structure of } \mathbb{N},\\ \text{axiomatization} &: \quad \text{separation of structure from representation},\\ \text{construction} &: \quad \text{production of objects under constraints},\\ \text{invariance} &: \quad \text{recognition of what survives transformation},\\ \text{limiting process} &: \quad \text{control of infinite approximation},\\ \text{optimization} &: \quad \text{study of extremal structure},\\ \text{formalization} &: \quad \text{conversion of reasoning into checkable form}.\end{aligned}$$

This suggests a second reading principle:

$$\boxed{\text{Every chapter should be read as an introduction to a method,} \quad \text{not merely to a topic.}}$$

For example, Chapter I is not merely about natural numbers. It is about calculation, representation, induction, and arithmetic structure. Chapter III is not merely about ruler-and-compass constructions. It is about the algebraic content hidden inside geometric operations.

0.6 A Mathematician's Standard for These Notes

Since these are not casual reading notes, the standard must be explicit. Each section of the book should eventually be rewritten in these notes according to the following mathematical protocol.

Key Points

For each chapter or section, the notes should extract:

(1) the central mathematical objects,
(2) the definitions and hidden assumptions,
(3) the main theorems or structural claims,
(4) the proof mechanisms,
(5) the examples and counterexamples,
(6) the conceptual transitions,
(7) the modern reinterpretation,
(8) the possible Lean/formalization viewpoint.

This protocol reflects a theoretical mathematician's way of reading. The goal is not speed. The goal is structural compression without loss of rigor.

Let T be a portion of the text. The output of our reading should be a structured object

$$\mathfrak{N}(T) = \big(\mathcal{D}, \mathcal{L}, \mathcal{P}, \mathcal{E}, \mathcal{I}, \mathcal{Q}, \mathcal{F}\big),$$

where

$$\begin{aligned}
\mathcal{D} &: \text{ definitions,} \\
\mathcal{L} &: \text{ lemmas and theorems,} \\
\mathcal{P} &: \text{ proof strategies,} \\
\mathcal{E} &: \text{ examples and counterexamples,} \\
\mathcal{I} &: \text{ invariants and structural principles,} \\
\mathcal{Q} &: \text{ questions and possible generalizations,} \\
\mathcal{F} &: \text{ formalization data.}
\end{aligned}$$

In this sense, a reading note is not a summary. It is a mathematical reconstruction.

0.7 The Modern Extension: AI and Formal Proof

The prefaces emphasize living mathematical understanding over formal drill. This warning becomes even more important in the age of artificial intelligence and formal proof assistants.

AI can process large mathematical corpora, suggest analogies, reorganize arguments, search for relevant lemmas, and help convert informal reasoning into structured proof plans. But AI does not remove the need for mathematical judgment. It changes where that judgment must be applied.

The new workflow may be written as

$$\begin{aligned}&\text{human intuition} \longrightarrow \text{conceptual decomposition} \longrightarrow \text{AI-assisted search}\\ &\qquad\longrightarrow \text{formal proof planning} \longrightarrow \text{human verification} \longrightarrow \text{machine checking.}\end{aligned}$$

Lean and similar proof assistants add another layer. They make explicit the dependency structure that is often compressed in ordinary mathematical prose.

$$\begin{aligned}&\text{informal theorem} \longrightarrow \text{formal statement} \longrightarrow \text{lemma dependency graph}\\ &\qquad\longrightarrow \text{proof term} \longrightarrow \text{kernel verification.}\end{aligned}$$

This does not contradict Courant's warning against empty formalism. On the contrary, it sharpens it. Empty formalism is symbol manipulation without mathematical necessity. Formalization, properly understood, is the precise externalization of genuine mathematical necessity.

Takeaway

The AI–Lean era does not abolish the classical spirit of mathematics. It requires us to make that spirit more explicit: definitions must be exact, dependencies must be visible, and intuition must be converted into checkable structure.

Thus these notes should read Courant not as a historical monument, but as a training ground for a new kind of mathematical practice.

0.8 Reading Principle for the Whole Book

The foreword and prefaces justify the following global rule.

Key Points

Every topic in the book should be read simultaneously at four levels:

$$\begin{aligned}\text{Level 1} &: \text{ the concrete example,}\\ \text{Level 2} &: \text{ the formal definition,}\\ \text{Level 3} &: \text{ the structural theorem,}\\ \text{Level 4} &: \text{ the modern conceptual or formalized viewpoint.}\end{aligned}$$

This four-level reading prevents two common failures. The first failure is elementary superficiality: treating the material as easy because the examples are familiar. The second failure is abstract emptiness: treating the material as formal symbols without seeing the concrete source.

The correct reading is dialectical:

$$\text{concrete intuition} \rightleftarrows \text{formal structure} \rightleftarrows \text{proof} \rightleftarrows \text{general theory.}$$

0.9 Questions Before Entering the Main Text

Questions to Resolve

1. What distinguishes a mathematical method from a computational procedure?
2. When does an elementary example contain the seed of a general theory?
3. How can one identify the hidden assumptions behind a familiar operation such as addition, multiplication, measurement, or taking a limit?
4. What is the difference between formalism as empty symbol manipulation and formalization as structural clarification?
5. In what sense can a theorem be understood before it is formalized, and in what sense is it not fully explicit until it is formalized?
6. How should AI be used in mathematical study without weakening the human responsibility for judgment, taste, and proof?
7. Can the dependency graph of a mathematical theory be regarded as a modern replacement for the classical linear textbook?
8. Does this book implicitly teach not only mathematics, but also a theory of how mathematical concepts are born?

0.10 Personal Resolution

Takeaway

I will read this book not as a collection of elementary facts, but as a compressed map of mathematical civilization. The task is to extract the structures beneath the examples, the proofs beneath the claims, the methods beneath the computations, and the future formalization beneath the classical prose.

The guiding formula for the entire project is

$$\boxed{\begin{aligned}\text{Mathematical reading} &= \text{conceptual excavation} + \text{deductive reconstruction}\\ &\quad + \text{structural generalization} + \text{formalizable precision}.\end{aligned}}$$

This is the standard by which the following notes should be written.

How to Use the Book

Source Map

The author's advice on how to use the book is more important than it may first appear. The book is systematic, but it is not meant to be read mechanically. One may move forward, return backward, postpone difficult sections, and choose a path according to one's mathematical maturity. This is already a statement about the nature of mathematics: mathematical understanding is not linear accumulation, but repeated reorganization.

0.11 Reading as Reconstruction

A serious mathematical book should not be read as a sequence of statements to be consumed. It should be read as a structure to be reconstructed.

The printed order of a book is necessarily linear, but the mathematical order is not. Definitions often become meaningful only after their consequences are seen. Examples may reveal their full force only when they later reappear in a different theory. A proof may be less important for the result it establishes than for the method it quietly introduces.

For this reason, I should not read this book with the mentality of finishing pages. I should read it with the intention of recovering the architecture of the subject. The question after each section is not simply: "What did the author say?" The better question is: "What mathematical necessity is being revealed here?"

Takeaway

The aim of reading is not to reproduce the exposition, but to reconstruct the mathematics behind the exposition.

0.12 Against Passive Linearity

The book itself permits a non-linear mode of reading. This is not merely a convenience for beginners. It is also how mathematicians actually read.

A difficult argument may be unreadable at first because one has not yet seen why it matters. After several later examples, the same argument may become transparent. Conversely, an elementary discussion near the beginning may contain the seed of a much deeper method. The reader who insists on complete understanding before moving on may confuse local difficulty with genuine obstruction.

A mature reading strategy therefore allows controlled postponement. To postpone is not to evade. It is to mark a difficulty, continue until the conceptual landscape becomes clearer, and then return with better questions.

> **Remark**
>
> In mathematics, rereading is not repetition. Rereading is often the moment at which the real structure first appears.

This is especially true for this book. Its early discussions of number, calculation, construction, and geometric intuition are not isolated elementary topics. They are first appearances of ideas that later become algebraic, geometric, topological, and analytic.

0.13 What to Look For

The most important habit is to distinguish between a topic and a method.

A chapter on natural numbers is not only about natural numbers. It is also about representation, induction, and the way arithmetic laws become structural laws. A chapter on geometrical construction is not only about ruler and compass. It is about the hidden algebraic content of geometric operations. A chapter on limits is not only about convergence. It is about the control of infinite processes and the birth of modern analysis.

Thus each chapter should be read with several questions in mind.

What is the central object? Why is it introduced? Which examples make it natural? Which operations are legitimate? Which invariants are preserved? What kind of proof is being used? What would fail if one of the assumptions were changed? Where does this idea reappear in a more advanced theory?

These questions are more important than the desire to summarize. A summary usually flattens mathematics. A good note should sharpen it.

> **Takeaway**
>
> A mathematical note should not merely record content. It should expose necessity, structure, and method.

0.14 Examples and Definitions

Courant's style often begins with concrete examples. This should not be mistaken for a concession to simplicity. In mathematics, examples are not ornaments; they are instruments of discovery.

A good example shows why a definition is needed. A bad example, or a non-example, shows where the boundary lies. A borderline example often reveals which hypothesis is doing the real work. The definition comes later, but the mathematical pressure comes first.

Therefore, when reading this book, I should not rush from examples to formal statements. I should ask what the examples are trying to force. If a definition appears, I should ask which earlier difficulty it resolves. If a theorem appears, I should ask why the definition was strong enough to make the theorem true.

> **Remark**
>
> Definitions in good mathematics are rarely arbitrary. They are compressed answers to prior difficulties.

This is one of the central lessons I want to extract from the book. Elementary mathematics becomes profound when one sees that its definitions are not conventions alone, but responses to structural necessity.

0.15 Proofs as Mathematical Behavior

A proof should be read not merely as verification, but as behavior.

The question is not only whether the conclusion follows. The question is how the proof moves. Does it use induction? Does it construct an object? Does it find an invariant? Does it pass from geometry to algebra? Does it reduce a complicated case to a simpler one? Does it introduce a new viewpoint from which the result becomes inevitable?

This distinction matters. A theorem may be useful, but a proof method is often more valuable. The same proof behavior may reappear in a different subject with different objects and different language. To read well is to notice this mobility.

Takeaway

The result of a proof belongs to one theorem. The method of a proof may belong to an entire field.

For this reason, I should record not only what is proved, but what kind of mathematical movement the proof represents.

0.16 The Role of Exercises

Problems and exercises should not be treated as appendices to the real text. They are part of the mathematical argument.

An exercise tests whether the reader has truly acquired a method. Some exercises consolidate technique. Some reveal hidden assumptions. Some extend a result beyond the text. Some are small research problems disguised as elementary questions.

The right attitude is not to solve exercises only for correctness. The right attitude is to ask what each problem is designed to test. Does it test calculation, conceptual understanding, construction, generalization, or counterexample-building?

A problem is most valuable when it forces a method to move outside the exact context in which it was introduced.

0.17 Reading Across Chapters

This book should be read with constant attention to connections between chapters.

The movement from arithmetic to algebra, from geometry to field theory, from projective geometry to invariance, from topology to qualitative structure, and from limits to calculus is not accidental. The book gradually teaches that mathematics grows by transporting ideas from one setting to another.

This is one of the deepest reasons for reading classical texts. A modern reader already knows the names of many advanced theories. But a classical text often shows the moment before those theories have become fully abstract. It shows the pressure that produced them.

For example, ruler-and-compass construction should be read with field extensions in the background. Projective geometry should be read with invariance in mind. Topology should be read as a liberation from metric geometry. Limits should be read as a disciplined response to infinity.

Key Points

The task is to read elementary material as the visible surface of deeper mathematical structures.

0.18 A Modern Layer: Formalization

Although this book belongs to the classical tradition of mathematical exposition, it can also be read from the perspective of contemporary formalization.

This does not mean forcing every page into Lean immediately. That would be a bad reading strategy. The first task is still to understand the mathematics. But once a concept is understood, it is natural to ask how it would be made completely explicit.

What are the objects? What are the assumptions? Which facts are definitions, and which are lemmas? Which steps are suppressed because they are obvious to a human reader? Which background structures are silently being used?

Formalization is useful because it pressures the reader to distinguish these things. It exposes the hidden dependency structure of ordinary mathematical prose.

Remark

Formalization is not a substitute for mathematical taste. It is a test of mathematical exactness.

In the age of AI and proof assistants, this becomes especially important. AI can help organize, search, and suggest. Lean can check formal correctness. But neither replaces the mathematician's responsibility to understand why a definition is natural, why a theorem matters, and why a proof is the right proof.

0.19 My Working Method

For this notebook, I will follow a simple discipline.

First, I will identify the main idea of each section. Then I will isolate the definitions, claims, examples, and proof methods. After that, I will write my own interpretation: what is really happening mathematically, what hidden structure the author is using, and how the idea connects to later mathematics.

I will not try to make every note equally long. Some sections deserve only a brief conceptual comment. Others require a full reconstruction. The standard is not length, but depth.

I will also avoid filling the notes with decorative formalism. Formulas should appear when they clarify structure, not when prose would be more honest. A mathematician's rigor is not measured by the density of symbols, but by the precision of thought.

Takeaway

The rule for these notes is: write only what improves understanding. Use formulas when they reveal structure; use prose when prose is sharper.

0.20 Personal Resolution

I will read this book as a mathematician, not as a spectator.

That means I will not merely accept the author's exposition. I will test it, reconstruct it, compare it with modern viewpoints, and ask what kind of mathematical instinct it is trying to cultivate. I will treat elementary chapters as opportunities to recover first principles, and difficult chapters as invitations to reorganize my understanding.

The goal is not to finish the book quickly. The goal is to let the book reshape my sense of mathematical structure.

Key Points

A good reading of this book should leave behind more than notes. It should produce a sharper mathematical instinct: a better eye for structure, a better ear for necessity, and a better discipline for proof.

Part I

Arithmetic, Algebra, and Number

CHAPTER 1

The Natural Numbers

Source Map

The chapter begins with a deliberately elementary object: the natural numbers. But the point is not elementary calculation. The point is to examine the first place where mathematics separates object from representation, operation from symbol, empirical pattern from proof, and finite construction from infinite law.

1.1 The First Abstraction

The natural numbers enter as the first stable abstraction of mathematics. A number is not the collection being counted, nor the word used to name it, nor the symbol used to write it. It is what remains invariant when the particular objects disappear.

This is already a profound step. To say that two collections have the same number is to ignore their material, color, position, and physical nature, and to retain only a structural feature: they can be matched element by element. Thus number begins not as arithmetic, but as abstraction from concrete multiplicity.

Courant's use of natural numbers is deliberately foundational. Before one speaks about fractions, negative numbers, real numbers, or complex numbers, one must understand the primitive system from which these later systems are constructed. The Kroneckerian slogan that the natural numbers are given while the rest is human construction should not be read naively. It is not a metaphysical theorem. It is a methodological statement: the natural numbers are the first arena where mathematical construction becomes disciplined.

Takeaway

The natural numbers are not merely the first objects of arithmetic. They are the first model of mathematical abstraction: a passage from concrete collections to invariant structure.

1.2 Number and Numeral

One of the most important distinctions in the chapter is the distinction between a number and its notation. The integer five is not the symbol 5, nor the Roman symbol V, nor any word used to pronounce it. The symbol is a representation; the number is the represented object.

This distinction looks trivial only because decimal notation has become psychologically invisible. In fact, it is one of the first lessons in mathematical maturity. Mathematics constantly separates an object from its coordinate system. The same point may have different coordinates; the same linear map may have different matrices; the same group may have different presentations. Here, in the most elementary setting, the same number already has many possible names.

The chapter's discussion of decimal notation and other bases should therefore be read as an early lesson in invariance. A number remains the same while its notation changes. Arithmetic laws belong to the object; carrying procedures belong to the chosen notation.

Remark

Positional notation is a coordinate system for integers. Changing the base is not changing the integer, but changing the coordinates in which the integer is written.

This point is more modern than it first appears. To write an integer in base B is to choose an expansion in powers of B. The familiar decimal system is only the special case $B = 10$. The preference for base ten is historical and biological, not mathematical. From the mathematical point of view, every base greater than one is legitimate.

1.3 Arithmetic Laws as Structure

Courant introduces the basic laws of arithmetic: commutativity and associativity of addition and multiplication, together with distributivity. The important point is not that these laws are familiar, but that they are laws. They are not properties of symbols alone; they describe the structure of a system.

In modern language, the natural numbers with addition and multiplication form a commutative semiring. Courant does not use this terminology, but the idea is already present. The laws identify which manipulations are legitimate and which are not.

This is why the chemical analogy in the text is important. If one interprets "addition" as physically adding one substance to another, commutativity may fail in practice. The formal symbol + has no meaning by itself. It acquires meaning only inside a structure.

Thus the laws of arithmetic should be read as the first appearance of the algebraic attitude. We do not merely compute with particular numbers; we study operations satisfying specified identities.

Takeaway

Arithmetic becomes mathematics when calculation is replaced by the study of operations and their laws.

This is the conceptual seed of algebra. Later, groups, rings, fields, modules, and algebras will all be built by the same gesture: isolate operations, record their laws, and study what follows from those laws alone.

1.4 The Dot Model and the Meaning of Operations

The chapter uses boxes of dots to explain addition and multiplication. This should not be dismissed as a childish illustration. It is a concrete model of the laws of arithmetic.

Addition corresponds to joining two finite collections without overlap. Multiplication corresponds to arranging objects in a rectangular array, or more abstractly, to forming ordered pairs. The visual model explains why the laws are true: the same finite collection can be decomposed or rearranged in different ways without changing its number of elements.

From a modern viewpoint, the dot model is the shadow of a deeper principle: cardinality sends disjoint union to addition and Cartesian product to multiplication. Courant does not formulate it this way, but this is the structural content of the pictures.

This is a typical feature of the book. A simple figure often contains a general mathematical mechanism. The reader's task is to see the mechanism without destroying the simplicity of the example.

Key Points

The dot model teaches that arithmetic laws are not arbitrary symbolic rules. They arise from operations on finite collections.

This also explains why the natural numbers are more than a list of marks. They are the universal book-keeping system for finite sets.

1.5 Order and Partial Subtraction

The order relation on natural numbers is introduced through addition: to say that $a < b$ means that b can be obtained from a by adding a positive remainder. This is a very natural definition. Order is not imposed externally; it is extracted from the additive structure.

Subtraction then appears as a partial inverse to addition. If $b > a$, one can write $b - a = c$, where c is the missing part needed to pass from a to b. But subtraction is not yet a total operation on the natural numbers. The expression $a - b$ may fail to denote a natural number.

This is an important moment. The failure of closure is the pressure that will later force an enlargement of the number system. Negative integers are not introduced by whim; they are introduced because subtraction wants to be a total operation.

Remark

A new number system often appears when a natural operation fails to be closed in the old one.

This is one of the major hidden themes of the book. The passage from natural numbers to integers, rationals, reals, and complex numbers is not an arbitrary accumulation of objects. It is a sequence of completions forced by operations, equations, and limiting processes.

1.6 The Role of Zero

The chapter's treatment of zero is modest but conceptually important. Zero is introduced as the number of the empty collection. This makes addition and multiplication behave uniformly: adding nothing leaves a number unchanged, while multiplying by nothing produces nothing.

Zero is a small symbol with large structural consequences. It gives addition an identity element and allows subtraction to be extended at least to the case $a - a = 0$. It also separates two different roles that are easily confused: zero is neutral for addition but absorbing for multiplication.

This double role is one of the first signs that operations must be studied in relation to their specific laws. The same object can behave differently under different operations.

1.7 Positional Notation as an Algorithmic Revolution

The discussion of positional notation is not merely historical. It explains why arithmetic became widely usable. Additive numeral systems such as Roman numerals can represent numbers, but they are badly adapted to calculation. Positional notation compresses information and makes algorithms possible.

This is the crucial distinction: a notation may be representationally adequate but computationally poor. A good notation does not merely name objects; it makes operations efficient.

Decimal notation does this by encoding an integer through coefficients of powers of ten. More generally, base B notation encodes an integer through successive remainders after division by B. This is already an algorithmic view of number. The integer is not simply written down; it is decomposed by a procedure.

Takeaway

Positional notation is not just a way of writing numbers. It is a technology for calculation.

This point has a modern continuation. Binary notation, which Courant presents as theoretically minimal but practically long, later becomes fundamental for digital computation. The mathematics of base representation thus belongs not only to arithmetic, but also to coding, algorithms, and computation.

1.8 Base Systems and Coordinate Freedom

The examples of seven-base, twelve-base, twenty-base, and binary notation have a clear conceptual function. They detach arithmetic from decimal habit. Once one sees that the same integer may be written in different bases, one understands that the decimal system is not part of the essence of number.

The calculation rules remain formally the same, but the tables change. This is exactly what happens whenever one changes coordinates. The invariant object is preserved, while the local expressions of operations are altered.

This is a good place to cultivate a mathematician's instinct: whenever a calculation seems natural, ask whether it belongs to the object or to the chosen representation. Decimal carrying is not an intrinsic law of number; it is a feature of decimal coordinates.

The same distinction will reappear throughout mathematics. Coordinates are useful, sometimes indispensable, but they must not be mistaken for the object.

1.9 Infinity and the Successor Principle

The second major movement of the chapter begins with the infinitude of the natural numbers. The natural number sequence has no last element because every natural number has a successor. This simple observation contains the entire logic of induction.

Mathematical induction is not empirical induction. It is not the observation that a statement is true for many cases, followed by the expectation that it will continue to hold. It is a proof principle based on the structure of the natural numbers.

The difference is decisive. Empirical induction is probabilistic in spirit; mathematical induction is deductive. It proves an infinite family of statements by proving a first case and a mechanism that transports truth from one case to the next.

Takeaway

Induction is the proof-theoretic expression of the successor structure of the natural numbers.

This is why induction is so fundamental. It is not merely a technique for proving formulas about sums. It is the way finite reasoning controls an infinite sequence of assertions.

1.10 Induction as Verification and Discovery

The examples of arithmetic progressions, geometric progressions, and sums of squares show the effectiveness of induction. Once the correct formula is known, induction verifies it cleanly.

But Courant also makes an important methodological point: induction often does not explain how the formula was discovered. The proof confirms the answer, but the discovery may have come from experiment, analogy, pattern recognition, or geometric insight.

This distinction is essential for serious mathematics. A proof is not always a record of discovery. It is often a reconstruction after discovery. The finished proof may hide the route by which the theorem was found.

Remark

Mathematical rigor and mathematical discovery are different acts. A proof may certify a result without revealing the intuition that produced it.

For this reason, when reading an inductive proof, one should ask two questions. First, why is the proof correct? Second, why was this formula reasonable to guess in the first place? The second question is often the more mathematical one.

1.11 The Formulas for Sums

The formulas for the sums of the first n integers, for finite geometric series, and for sums of squares are not isolated tricks. They are early examples of a general problem: how does one compress a finite but variable process into a closed expression?

The sum $1 + 2 + \cdots + n$ has a geometric and symmetric explanation: pair the terms from opposite ends. The finite geometric series has an algebraic explanation: multiply by the ratio and subtract. The sum of squares is less immediate; induction verifies it, but the formula itself demands a deeper source.

These examples show three different kinds of mathematical explanation.

First, there is symmetry. Second, there is algebraic cancellation. Third, there is inductive verification. A mature reader should distinguish these styles, because they reappear throughout mathematics in far more advanced forms.

Key Points

The chapter's elementary formulas should be read as prototypes of three methods: symmetry, cancellation, and induction.

1.12 A Modern Reading

From a contemporary standpoint, this chapter quietly contains several modern themes.

First, the natural numbers are the prototype of an inductively generated structure. In a formal system such as Lean, the natural numbers are not introduced as an informal sequence written on a page. They are generated from zero and successor, and their induction principle is part of their logical meaning.

Second, the arithmetic laws point toward algebraic structures. What Courant presents as the laws of arithmetic becomes, in modern language, the beginning of semiring theory.

Third, positional notation is an early form of encoding. The base expansion of an integer is not only a notation but a finite data structure. The algorithm of successive division by the base is already a computational procedure.

Fourth, induction shows the difference between checking many examples and proving all cases. This distinction is indispensable in the AI era. A machine or model may recognize many patterns, but mathematics requires a mechanism that turns pattern into proof.

Takeaway

The chapter is not merely about natural numbers. It is about the first appearance of structure, representation, algorithm, and proof by recursion.

1.13 Questions to Carry Forward

Questions to Resolve

1. In what sense are natural numbers primitive, and in what sense are they already constructed objects?
2. Does the dot model merely illustrate arithmetic, or does it contain a genuine structural explanation of the arithmetic laws?
3. When a new number system is introduced, which failure of closure or solvability is being repaired?
4. How much of arithmetic depends on the object itself, and how much depends on the representation chosen for it?
5. Is induction best understood as a proof technique, as a property of $\mathbb{N}$, or as the universal principle characterizing inductively generated objects?
6. When induction proves a formula, what separate argument explains how the formula was discovered?
7. How would the chapter look if rewritten entirely in the language of finite sets, cardinality, semirings, and inductive types?

1.14 Personal Working Summary

The natural numbers should be read as the first laboratory of mathematical thought. In them one already sees the distinction between object and notation, the emergence of operations and laws, the role of models, the need for closure, the power of representation, and the logical force of induction.

This chapter is therefore not preliminary in a weak sense. It is preliminary because it contains the first forms of ideas that later become central: algebraic structure, coordinate freedom, algorithmic representation, formal proof, and abstraction from concrete objects.

The correct lesson is not that natural numbers are simple. The correct lesson is that simplicity, when examined properly, is the first form of depth.

CHAPTER 2

Supplement to Chapter I: The Theory of Numbers

Source Map

This supplement is about arithmetic structure inside the integers. The notes below are organized around explicit computations: prime sieving, Euclid's proof, congruence tables, Fermat's theorem, quadratic residues, Pythagorean triples, the Euclidean algorithm, Euler's function, continued fractions, and linear Diophantine equations.

2.1 Prime Numbers by Sieve, Not by Guessing

A prime number is an integer $p > 1$ whose only positive divisors are 1 and p. The first practical way to detect primes is the sieve of Eratosthenes.

Worked Example: Sieve of Eratosthenes up to 50

Start with the integers $2, 3, \ldots, 50$. Keep 2, then cross out all multiples of 2 greater than 2. Keep 3, then cross out all multiples of 3 greater than 3. Continue with 5 and 7. Since

$$\sqrt{50} < 8,$$

it is enough to test primes up to 7.

$$\begin{array}{cccccccccc}
2 & 3 & \cancel{4} & 5 & \cancel{6} & 7 & \cancel{8} & \cancel{9} & \cancel{10} & 11 \\
\cancel{12} & 13 & \cancel{14} & \cancel{15} & \cancel{16} & 17 & \cancel{18} & 19 & \cancel{20} & \cancel{21} \\
\cancel{22} & 23 & \cancel{24} & \cancel{25} & \cancel{26} & \cancel{27} & \cancel{28} & 29 & \cancel{30} & 31 \\
\cancel{32} & \cancel{33} & \cancel{34} & \cancel{35} & \cancel{36} & 37 & \cancel{38} & \cancel{39} & \cancel{40} & 41 \\
\cancel{42} & 43 & \cancel{44} & \cancel{45} & \cancel{46} & 47 & \cancel{48} & \cancel{49} & \cancel{50} &
\end{array}$$

Therefore the primes not exceeding 50 are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.$$

The stopping rule is important. If n is composite, then $n = ab$ with $1 < a \le b < n$. Hence

$$a^2 \le ab = n,$$

so

$$a \le \sqrt{n}.$$

Thus to prove n prime, it is enough to test divisibility by primes $\le \sqrt{n}$.

Solved Exercise: Is 221 Prime?

Since

$$14^2 = 196 < 221 < 225 = 15^2,$$

we only test primes up to 14:

$$2, 3, 5, 7, 11, 13.$$

Now

$$221 \text{ is odd}, \qquad 2 + 2 + 1 = 5,$$

so it is not divisible by 2 or 3. It does not end in 0 or 5, so it is not divisible by 5. Also

$$221 = 7 \cdot 31 + 4, \qquad 221 = 11 \cdot 20 + 1.$$

Finally,

$$221 = 13 \cdot 17.$$

Therefore 221 is not prime.

Solved Exercise: Is 223 Prime?

Again $\sqrt{223} < 15$, so we test $2, 3, 5, 7, 11, 13$.

$$223 \not\equiv 0 \pmod 2, \qquad 2 + 2 + 3 = 7 \not\equiv 0 \pmod 3,$$

and 223 does not end in 0 or 5. Next,

$$223 = 7 \cdot 31 + 6, \qquad 223 = 11 \cdot 20 + 3, \qquad 223 = 13 \cdot 17 + 2.$$

No prime $\le \sqrt{223}$ divides 223. Therefore 223 is prime.

2.2 Euclid's Proof of Infinitely Many Primes

The classical proof is short, but it is worth writing it in a way that shows the actual mechanism.

Proof: There Are Infinitely Many Primes

Assume there are only finitely many primes:

$$p_1, p_2, \ldots, p_n.$$

Construct

$$N = p_1 p_2 \cdots p_n + 1.$$

For every i,

$$N \equiv 1 \pmod{p_i}.$$

Hence no p_i divides N.

But every integer $N > 1$ has at least one prime divisor. That prime divisor cannot be any of $p_1, \ldots, p_n$. This contradicts the assumption that the list contained all primes.
Therefore there are infinitely many primes.

Here is the construction numerically. If someone falsely claims that

$$2, 3, 5, 7$$

are all the primes, then

$$2 \cdot 3 \cdot 5 \cdot 7 + 1 = 211.$$

The number 211 is not divisible by 2, 3, 5, or 7. In fact 211 is prime. Even when the constructed number is not prime, the argument still works.

Example Where Euclid's Number Is Composite

Using the first six primes,

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031.$$

This is not prime:

$$30031 = 59 \cdot 509.$$

But the point remains: the prime divisors 59 and 509 are not in the original list

$$2, 3, 5, 7, 11, 13.$$

Euclid's argument does not require $p_1 \cdots p_n + 1$ itself to be prime; it only requires it to have a prime divisor outside the list.

2.3 Prime-Generating Formulas: A Trap

A polynomial can produce many primes at first and then fail. This is a useful warning against trusting finite data.

Worked Example: Euler's Polynomial Eventually Fails

Consider

$$f(n) = n^2 + n + 41.$$

For small n, this often gives primes:

$$f(0) = 41, \quad f(1) = 43, \quad f(2) = 47, \quad f(3) = 53.$$

But at $n = 40$,

$$f(40) = 40^2 + 40 + 41 = 1681 = 41^2.$$

Thus $f(n)$ cannot generate only primes.

There is a general obstruction. If $f(a) = p$ is prime, then for many integer-coefficient polynomials,

$$f(a + kp) \equiv f(a) \equiv 0 \pmod{p}.$$

Thus the same prime that appears as a value of the polynomial can later become a forced divisor of another value.

Concrete Check

For

$$f(n) = n^2 + n + 41,$$

we have $f(0) = 41$. Then

$$f(41) = 41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \cdot 43.$$

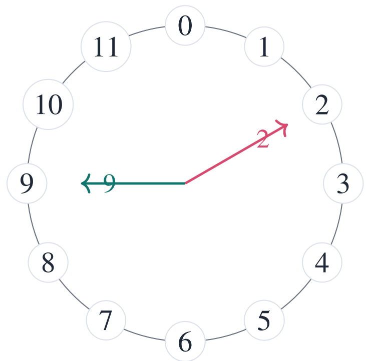
So the polynomial fails again at $n = 41$.

2.4 Congruence as Clock Arithmetic

Congruence modulo m means

$$a \equiv b \pmod{m} \iff m \mid (a - b).$$

This is arithmetic on a clock with m positions.



Modulo 12: $26 \equiv 2$, $-3 \equiv 9$.

For example,

$$26 \equiv 2 \pmod{12}$$

because

$$26 - 2 = 24 = 2 \cdot 12.$$

Also

$$-3 \equiv 9 \pmod{12}$$

because

$$-3 - 9 = -12.$$

Solved Exercise: Compute $2026 \bmod 17$

Divide:

$$17 \cdot 119 = 2023.$$

Therefore
$$2026 = 17 \cdot 119 + 3,$$
so
$$2026 \equiv 3 \pmod{17}.$$

Solved Exercise: Compute $37 \cdot 58 \bmod 11$

Reduce first:
$$37 \equiv 4 \pmod{11}, \qquad 58 \equiv 3 \pmod{11}.$$
Hence
$$37 \cdot 58 \equiv 4 \cdot 3 = 12 \equiv 1 \pmod{11}.$$
Therefore
$$37 \cdot 58 \equiv 1 \pmod{11}.$$

2.5 Solving Linear Congruences

The congruence
$$ax \equiv b \pmod{m}$$
is not always solvable. The criterion is
$$\gcd(a, m) \mid b.$$

Solved Exercise: Solve $14x \equiv 8 \pmod{30}$

First compute
$$\gcd(14, 30) = 2.$$
Since $2 \mid 8$, solutions exist. Divide the congruence by 2:
$$7x \equiv 4 \pmod{15}.$$
Now $7^{-1} \pmod{15}$ is 13, because
$$7 \cdot 13 = 91 \equiv 1 \pmod{15}.$$
Multiply both sides by 13:
$$x \equiv 13 \cdot 4 = 52 \equiv 7 \pmod{15}.$$
Thus the solutions modulo 30 are
$$x \equiv 7 \pmod{15},$$
namely
$$x \equiv 7 \quad \text{or} \quad x \equiv 22 \pmod{30}.$$
Check:
$$14 \cdot 7 = 98 \equiv 8 \pmod{30}, \qquad 14 \cdot 22 = 308 \equiv 8 \pmod{30}.$$

Solved Exercise: Show $12x \equiv 5 \pmod{18}$ Has No Solution

Here

$$\gcd(12, 18) = 6.$$

But

$$6 \nmid 5.$$

Therefore the congruence has no solution.

2.6 Fermat's Theorem with Actual Computations

Fermat's theorem says that if p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof by Permuting Nonzero Residues

Modulo a prime p, the nonzero residues are

$$1, 2, \ldots, p-1.$$

If $p \nmid a$, then multiplication by a permutes these residues:

$$a, 2a, 3a, \ldots, (p-1)a.$$

Therefore

$$a \cdot 2a \cdot 3a \cdots (p-1)a \equiv 1 \cdot 2 \cdot 3 \cdots (p-1) \pmod{p}.$$

So

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since p divides none of $1, 2, \ldots, p-1$, we may cancel $(p-1)!$ modulo p. Hence

$$a^{p-1} \equiv 1 \pmod{p}.$$

Solved Exercise: Compute $3^{100} \bmod 7$

Since 7 is prime and $7 \nmid 3$,

$$3^6 \equiv 1 \pmod{7}.$$

Now

$$100 = 6 \cdot 16 + 4.$$

Therefore

$$3^{100} \equiv 3^4 \pmod{7}.$$

But

$$3^2 = 9 \equiv 2 \pmod{7},$$

so

$$3^4 \equiv 2^2 = 4 \pmod{7}.$$

Thus

$$3^{100} \equiv 4 \pmod 7.$$

Solved Exercise: Last Digit of 7^{222}

The last digit is the residue modulo 10. The powers of 7 modulo 10 cycle:

$$7^1 \equiv 7, \quad 7^2 \equiv 49 \equiv 9, \quad 7^3 \equiv 63 \equiv 3, \quad 7^4 \equiv 21 \equiv 1 \pmod{10}.$$

The period is 4. Since

$$222 \equiv 2 \pmod 4,$$

we get

$$7^{222} \equiv 7^2 \equiv 9 \pmod{10}.$$

Therefore the last digit is 9.

2.7 Quadratic Residues: Squares Modulo a Prime

A quadratic residue modulo m is a residue class that occurs as x^2 modulo m.

Squares Modulo 7

Compute:

x	0	1	2	3	4	5	6
$x^2 \bmod 7$	0	1	4	2	2	4	1

Therefore the quadratic residues modulo 7 are

$$0, 1, 2, 4.$$

The nonzero quadratic residues are

$$1, 2, 4,$$

and the nonresidues are

$$3, 5, 6.$$

Solved Exercise: Does $x^2 \equiv 3 \pmod 7$ Have a Solution?

From the table above, the possible values of $x^2 \bmod 7$ are

$$0, 1, 2, 4.$$

Since 3 is not in this list,

$$x^2 \equiv 3 \pmod 7$$

has no solution.

Solved Exercise: Prove $x^2 + y^2 = 8z + 7$ Has No Integer Solution

Work modulo 8. Every square modulo 8 is either 0, 1, or 4:

$x \bmod 8$	0	1	2	3	4	5	6	7
$x^2 \bmod 8$	0	1	4	1	0	1	4	1

Thus

$$x^2 + y^2 \bmod 8$$

can only be one of

$$0, 1, 2, 4, 5.$$

But

$$8z + 7 \equiv 7 \pmod{8}.$$

Since 7 is impossible as a sum of two squares modulo 8, the equation

$$x^2 + y^2 = 8z + 7$$

has no integer solution.

This is the local obstruction method: reduce a global integer equation modulo a small integer and show that the required residue is impossible.

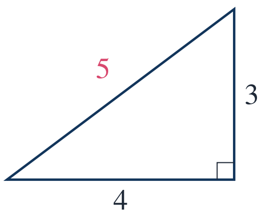
2.8 Pythagorean Triples

A Pythagorean triple is an integer solution of

$$x^2 + y^2 = z^2.$$

The basic examples are

$$3^2 + 4^2 = 5^2, \qquad 5^2 + 12^2 = 13^2, \qquad 8^2 + 15^2 = 17^2.$$



The 3-4-5 triangle

The complete primitive parametrization is

$$x = m^2 - n^2, \qquad y = 2mn, \qquad z = m^2 + n^2,$$

where

$$m > n, \qquad \gcd(m, n) = 1, \qquad m \not\equiv n \pmod{2}.$$

Check the Formula

Compute:

$$(m^2 - n^2)^2 + (2mn)^2 = m^4 - 2m^2n^2 + n^4 + 4m^2n^2.$$

Thus

$$(m^2 - n^2)^2 + (2mn)^2 = m^4 + 2m^2n^2 + n^4 = (m^2 + n^2)^2.$$

So the formula always gives a Pythagorean triple.

Solved Exercise: Generate Primitive Triples with $z < 30$

We need

$$z = m^2 + n^2 < 30, \qquad m > n, \quad \gcd(m, n) = 1, \quad m, n \text{ of opposite parity.}$$

Try $m = 2$:

$$n = 1 \quad \Longrightarrow \quad (x, y, z) = (3, 4, 5).$$

Try $m = 3$:

$$n = 2 \quad \Longrightarrow \quad (x, y, z) = (5, 12, 13).$$

The choice $n = 1$ has both m, n odd, so it gives a nonprimitive parity failure:

$$(8, 6, 10) = 2(4, 3, 5).$$

Try $m = 4$:

$$\begin{aligned} n = 1 \quad &\Longrightarrow \quad (x, y, z) = (15, 8, 17), \\ n = 3 \quad &\Longrightarrow \quad (x, y, z) = (7, 24, 25). \end{aligned}$$

The choice $n = 2$ is not coprime to $m = 4$.
Try $m = 5$:

$$n = 2 \quad \Longrightarrow \quad z = 25 + 4 = 29,$$

and

$$(x, y, z) = (21, 20, 29).$$

The choices $n = 1, 3$ have same parity as m, and $n = 4$ gives $z = 41 > 30$.
Therefore the primitive triples with $z < 30$, up to order of the two legs, are

$$(3, 4, 5), \quad (5, 12, 13), \quad (8, 15, 17), \quad (7, 24, 25), \quad (20, 21, 29).$$

2.9 The Euclidean Algorithm

The Euclidean algorithm computes greatest common divisors by repeated division with remainder.

Worked Example: Compute $\gcd(252, 198)$

Divide:

$$\begin{aligned} 252 &= 1 \cdot 198 + 54, \\ 198 &= 3 \cdot 54 + 36, \end{aligned}$$

$$54 = 1 \cdot 36 + 18,$$

$$36 = 2 \cdot 18 + 0.$$

Therefore

$$\gcd(252, 198) = 18.$$

The same computation can be reversed to write the gcd as an integer combination of the original two numbers.

Bezout Identity for 252 and 198

From the Euclidean algorithm:

$$18 = 54 - 36.$$

But

$$36 = 198 - 3 \cdot 54.$$

So

$$18 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198.$$

Also

$$54 = 252 - 198.$$

Therefore

$$18 = 4(252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198.$$

So

$$18 = 4 \cdot 252 - 5 \cdot 198.$$

Takeaway

The Euclidean algorithm does two things at once:

$$\text{it computes } \gcd(a, b) \quad \text{and} \quad \text{it gives } \gcd(a, b) = ua + vb.$$

2.10 Unique Factorization and Euclid's Lemma

The fundamental theorem of arithmetic says that every integer $n > 1$ has a prime factorization, unique up to order:

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}.$$

Worked Example: Factor 27720

Divide by small primes:

$$27720 = 2772 \cdot 10 = 2772 \cdot 2 \cdot 5.$$

Now

$$2772 = 2 \cdot 1386 = 2^2 \cdot 693.$$

Next

$$693 = 3 \cdot 231 = 3^2 \cdot 77 = 3^2 \cdot 7 \cdot 11.$$

Therefore

$$27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11.$$

The uniqueness part depends on Euclid's lemma.

Proof of Euclid's Lemma

Let p be prime. If

$$p \mid ab$$

and

$$p \nmid a,$$

then $\gcd(p, a) = 1$. By Bezout's identity, there exist integers u, v such that

$$up + va = 1.$$

Multiply by b:

$$upb + vab = b.$$

Since $p \mid upb$ and $p \mid vab$, it follows that $p \mid b$.
Therefore

$$p \mid ab \implies p \mid a \text{ or } p \mid b.$$

Solved Exercise: Use Prime Factorization to Compute gcd **and** lcm

Let

$$a = 252 = 2^2 \cdot 3^2 \cdot 7, \qquad b = 198 = 2 \cdot 3^2 \cdot 11.$$

Then

$$\gcd(a, b) = 2^1 \cdot 3^2 = 18.$$

The least common multiple uses the maximum exponents:

$$\operatorname{lcm}(a, b) = 2^2 \cdot 3^2 \cdot 7 \cdot 11.$$

So

$$\operatorname{lcm}(252, 198) = 4 \cdot 9 \cdot 7 \cdot 11 = 2772.$$

Check:

$$ab = \gcd(a, b)\operatorname{lcm}(a, b).$$

Indeed,

$$252 \cdot 198 = 18 \cdot 2772.$$

2.11 Euler's Function

Euler's function $\varphi(n)$ counts the integers

$$1 \leq k \leq n$$

that are relatively prime to n.

For a prime power,

$$\varphi(p^a) = p^a - p^{a-1} = p^a\left(1 - \frac{1}{p}\right).$$

If

$$n = \prod_i p_i^{a_i},$$

then

$$\varphi(n) = n \prod_{p_i \mid n}\left(1 - \frac{1}{p_i}\right).$$

Worked Example: Compute $\varphi(360)$

Factor:

$$360 = 36 \cdot 10 = 2^3 \cdot 3^2 \cdot 5.$$

Therefore

$$\varphi(360) = 360\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right).$$

Thus

$$\varphi(360) = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5}.$$

Compute:

$$360 \cdot \frac{1}{2} = 180, \qquad 180 \cdot \frac{2}{3} = 120, \qquad 120 \cdot \frac{4}{5} = 96.$$

So

$$\varphi(360) = 96.$$

Euler's theorem says that if $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Solved Exercise: Compute $7^{222} \bmod 40$

First,

$$\gcd(7, 40) = 1.$$

Also,

$$\varphi(40) = \varphi(2^3 \cdot 5) = 40\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 40 \cdot \frac{1}{2} \cdot \frac{4}{5} = 16.$$

So

$$7^{16} \equiv 1 \pmod{40}.$$

Now

$$222 = 16 \cdot 13 + 14.$$

Hence

$$7^{222} \equiv 7^{14} \pmod{40}.$$

Compute powers:

$$7^2 = 49 \equiv 9 \pmod{40},$$
$$7^4 \equiv 9^2 = 81 \equiv 1 \pmod{40}.$$

Thus

$$7^{14} = 7^{12}7^2 = (7^4)^3 7^2 \equiv 1^3 \cdot 9 = 9 \pmod{40}.$$

Therefore

$$7^{222} \equiv 9 \pmod{40}.$$

2.12 Chinese Remainder Computation

Although the Chinese remainder theorem is often treated later, it is a natural application of congruences.

Solved Exercise: Solve a System of Congruences

Solve

$$x \equiv 2 \pmod{3}, \qquad x \equiv 3 \pmod{5}, \qquad x \equiv 2 \pmod{7}.$$

First use the first two congruences. Numbers congruent to 3 modulo 5 are

$$3, 8, 13, 18, 23, 28, \ldots$$

Among these,

$$8 \equiv 2 \pmod{3}.$$

Therefore

$$x \equiv 8 \pmod{15}.$$

Now impose

$$x \equiv 2 \pmod{7}.$$

Write

$$x = 8 + 15k.$$

Modulo 7,

$$8 + 15k \equiv 1 + k \pmod{7}.$$

We need

$$1 + k \equiv 2 \pmod{7},$$

so

$$k \equiv 1 \pmod{7}.$$

Take $k = 1$. Then

$$x = 8 + 15 = 23.$$

The moduli $3, 5, 7$ are pairwise coprime, so the solution is unique modulo

$$3 \cdot 5 \cdot 7 = 105.$$

Thus

$$x \equiv 23 \pmod{105}.$$

2.13 Quadratic Residues and Fermat's Theorem

Quadratic residues reveal a finer structure than ordinary congruence.

Squares Modulo 11

Compute:

x	0	1	2	3	4	5	6	7	8	9	10
x^2 mod 11	0	1	4	9	5	3	3	5	9	4	1

Therefore the nonzero quadratic residues modulo 11 are

$$1, 3, 4, 5, 9.$$

The nonresidues are

$$2, 6, 7, 8, 10.$$

Solved Exercise: Is 5 a Quadratic Residue Modulo 11?

Yes. From the table,

$$4^2 = 16 \equiv 5 \pmod{11}.$$

Also

$$7^2 = 49 \equiv 5 \pmod{11}.$$

Thus the solutions to

$$x^2 \equiv 5 \pmod{11}$$

are

$$x \equiv 4,\ 7 \pmod{11}.$$

Solved Exercise: Show -1 Is Not a Square Modulo 7

Modulo 7,

$$-1 \equiv 6.$$

The quadratic residues modulo 7 are

$$0, 1, 2, 4.$$

Since

$$6 \notin \{0, 1, 2, 4\},$$

the congruence

$$x^2 \equiv -1 \pmod{7}$$

has no solution.

A useful consequence: if an integer equation implies

$$x^2 \equiv -1 \pmod{7},$$

then the original equation cannot have integer solutions.

2.14 A Diophantine Impossibility by Congruence

A Diophantine equation asks for integer solutions. Congruences often prove that no solution exists.

Solved Exercise: Prove $x^2 = 3y^2 + 2$ Has No Integer Solution

Work modulo 3. The equation gives

$$x^2 \equiv 2 \pmod 3.$$

But the squares modulo 3 are only

$$0^2 \equiv 0, \qquad 1^2 \equiv 1, \qquad 2^2 \equiv 1.$$

So a square modulo 3 is either 0 or 1, never 2.
Therefore

$$x^2 \equiv 2 \pmod 3$$

is impossible, and the equation

$$x^2 = 3y^2 + 2$$

has no integer solution.

Solved Exercise: Prove $x^2 + y^2 = 4z + 3$ Has No Integer Solution

Work modulo 4. Every square is 0 or 1 modulo 4:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 0, \quad 3^2 \equiv 1 \pmod 4.$$

Hence

$$x^2 + y^2 \equiv 0, 1, \text{ or } 2 \pmod 4.$$

But

$$4z + 3 \equiv 3 \pmod 4.$$

Contradiction. Therefore the equation has no integer solution.

2.15 Linear Diophantine Equations

The equation

$$ax + by = c$$

has integer solutions if and only if

$$\gcd(a, b) \mid c.$$

Solved Exercise: Solve $35x + 22y = 1$

Use the Euclidean algorithm:

$$\begin{aligned} 35 &= 1 \cdot 22 + 13, \\ 22 &= 1 \cdot 13 + 9, \\ 13 &= 1 \cdot 9 + 4, \\ 9 &= 2 \cdot 4 + 1. \end{aligned}$$

Now reverse:

$$1 = 9 - 2 \cdot 4.$$

Since

$$4 = 13 - 9,$$

we get

$$1 = 9 - 2(13 - 9) = 3 \cdot 9 - 2 \cdot 13.$$

Since

$$9 = 22 - 13,$$

$$1 = 3(22 - 13) - 2 \cdot 13 = 3 \cdot 22 - 5 \cdot 13.$$

Since

$$13 = 35 - 22,$$

$$1 = 3 \cdot 22 - 5(35 - 22) = 8 \cdot 22 - 5 \cdot 35.$$

Thus

$$35(-5) + 22(8) = 1.$$

One solution is

$$x = -5, \qquad y = 8.$$

The general solution is

$$x = -5 + 22t, \qquad y = 8 - 35t, \qquad t \in \mathbb{Z}.$$

Solved Exercise: Solve $35x + 22y = 14$

Since

$$35(-5) + 22(8) = 1,$$

multiply by 14:

$$35(-70) + 22(112) = 14.$$

One solution is

$$x = -70, \qquad y = 112.$$

The general solution is

$$x = -70 + 22t, \qquad y = 112 - 35t, \qquad t \in \mathbb{Z}.$$

For a smaller solution, choose $t = 3$:

$$x = -70 + 66 = -4, \qquad y = 112 - 105 = 7.$$

Check:

$$35(-4) + 22(7) = -140 + 154 = 14.$$

Thus

$$x = -4, \qquad y = 7$$

is a simple solution.

2.16 The Euclidean Algorithm and Continued Fractions

The Euclidean algorithm also produces continued fractions.

Worked Example: Continued Fraction of $415/93$

Divide:

$$415 = 4 \cdot 93 + 43,$$

so

$$\frac{415}{93} = 4 + \frac{43}{93} = 4 + \frac{1}{93/43}.$$

Next,

$$93 = 2 \cdot 43 + 7,$$

so

$$\frac{93}{43} = 2 + \frac{7}{43} = 2 + \frac{1}{43/7}.$$

Next,

$$43 = 6 \cdot 7 + 1,$$

so

$$\frac{43}{7} = 6 + \frac{1}{7}.$$

Finally,

$$\frac{7}{1} = 7.$$

Therefore

$$\frac{415}{93} = [4; 2, 6, 7].$$

In full form:

$$\frac{415}{93} = 4 + \cfrac{1}{2 + \cfrac{1}{6 + \cfrac{1}{7}}}.$$

Convergents of $[4; 2, 6, 7]$

The successive convergents are

$$4, \qquad 4 + \frac{1}{2} = \frac{9}{2},$$

$$4 + \cfrac{1}{2 + \cfrac{1}{6}} = 4 + \frac{1}{13/6} = 4 + \frac{6}{13} = \frac{58}{13},$$

and

$$[4; 2, 6, 7] = \frac{415}{93}.$$

Numerically,

$$4 = 4.0000, \quad \frac{9}{2} = 4.5000, \quad \frac{58}{13} = 4.4615\ldots, \quad \frac{415}{93} = 4.4623\ldots.$$

Continued fractions are not decoration: they give strong rational approximations.

2.17 The Continued Fraction of $\sqrt{2}$

The number $\sqrt{2}$ satisfies

$$\sqrt{2} = 1 + (\sqrt{2} - 1).$$

Now

$$\frac{1}{\sqrt{2}-1} = \frac{\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)} = \sqrt{2}+1.$$

Therefore

$$\sqrt{2} = 1 + \frac{1}{1+\sqrt{2}} = 1 + \frac{1}{2 + (\sqrt{2}-1)}.$$

Repeating gives

$$\sqrt{2} = [1; 2, 2, 2, \ldots].$$

Convergents of $\sqrt{2}$

The first convergents are

$$1, \qquad 1 + \frac{1}{2} = \frac{3}{2},$$

$$1 + \cfrac{1}{2 + \cfrac{1}{2}} = 1 + \frac{1}{5/2} = \frac{7}{5},$$

$$1 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2}}} = \frac{17}{12}.$$

Thus

$$1, \quad \frac{3}{2}, \quad \frac{7}{5}, \quad \frac{17}{12}, \quad \frac{41}{29}, \ldots$$

are rational approximations to $\sqrt{2}$.
Check the Pell-type errors:

$$1^2 - 2 \cdot 1^2 = -1,$$

$$3^2 - 2 \cdot 2^2 = 9 - 8 = 1,$$

$$7^2 - 2 \cdot 5^2 = 49 - 50 = -1,$$

$$17^2 - 2 \cdot 12^2 = 289 - 288 = 1.$$

The convergents alternately solve

$$x^2 - 2y^2 = \pm 1.$$

2.18 Fermat's Last Theorem as a Contrast

For $n = 2$, the equation

$$x^2 + y^2 = z^2$$

has infinitely many integer solutions, and we can parametrize all primitive ones:

$$x = m^2 - n^2, \qquad y = 2mn, \qquad z = m^2 + n^2.$$

For $n = 3$, the equation

$$x^3 + y^3 = z^3$$

has no positive integer solutions. Courant–Robbins presents this as part of the classical number-theoretic landscape; the full theorem for all $n > 2$ is Fermat's Last Theorem.

The useful point for these notes is not historical, but structural:

$$n = 2 \quad \text{admits a parametrization, while} \quad n > 2 \quad \text{does not.}$$

Concrete Contrast

For $n = 2$:

$$m = 2, \quad n = 1$$

gives

$$x = 2^2 - 1^2 = 3, \qquad y = 2 \cdot 2 \cdot 1 = 4, \qquad z = 2^2 + 1^2 = 5.$$

So

$$3^2 + 4^2 = 5^2.$$

If one tries the same small numbers for cubes:

$$3^3 + 4^3 = 27 + 64 = 91,$$

but

$$4^3 = 64 < 91 < 125 = 5^3.$$

There is no cube z^3 equal to 91. This numerical check is not a proof of Fermat's theorem; it only shows how quickly the quadratic parametrization breaks when the exponent changes.

2.19 A Small Prime Distribution Experiment

Prime distribution is not random, but a short table shows why it can look irregular.

Prime Gaps up to 50

The primes up to 50 are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.$$

The gaps between consecutive primes are

$$1, 2, 2, 4, 2, 4, 2, 4, 6, 2, 6, 4, 2, 4.$$

prime pair	(2, 3)	(3, 5)	(5, 7)	(7, 11)	(11, 13)	(13, 17)	(17, 19)
gap	1	2	2	4	2	4	2

prime pair	(19, 23)	(23, 29)	(29, 31)	(31, 37)	(37, 41)	(41, 43)	(43, 47)
gap	4	6	2	6	4	2	4

Twin primes in this range are pairs with gap 2:

$$(3,5), (5,7), (11,13), (17,19), (29,31), (41,43).$$

This kind of table is useful, but limited. It suggests patterns; it does not prove them. For example, the twin prime conjecture asks whether infinitely many prime pairs $p, p+2$ exist. The table gives evidence, not a proof.

2.20 Summary by Computable Tests

Key Points

1. To test whether n is prime, divide only by primes $\leq \sqrt{n}$.
2. To prove there are infinitely many primes, take any finite prime list and form $p_1 p_2 \cdots p_n + 1$.
3. To compute large powers modulo m, reduce exponents using Fermat's theorem or Euler's theorem when possible.
4. To prove an integer equation has no solution, reduce it modulo a small modulus and list possible residues.
5. To solve $ax + by = c$, first compute $\gcd(a,b)$. The equation is solvable exactly when $\gcd(a,b) \mid c$.
6. To find rational approximations, use the Euclidean algorithm as a continued-fraction machine.
7. To generate primitive Pythagorean triples, use

$$x = m^2 - n^2, \qquad y = 2mn, \qquad z = m^2 + n^2$$

with $m > n$, $\gcd(m,n) = 1$, and opposite parity.

2.21 Exercises Solved in This Note

Checklist of Concrete Results

This chapter note explicitly computed or proved:

1. primes up to 50 by sieve;
2. compositeness of 221;
3. primality of 223;
4. Euclid's proof of infinitely many primes;
5. failure of $n^2 + n + 41$ as an all-prime formula;
6. $2026 \bmod 17 = 3$;
7. $37 \cdot 58 \equiv 1 \pmod{11}$;

8. all solutions of $14x \equiv 8 \pmod{30}$;
9. non-solvability of $12x \equiv 5 \pmod{18}$;
10. $3^{100} \equiv 4 \pmod{7}$;
11. last digit of 7^{222};
12. squares modulo 7, 8, and 11;
13. impossibility of $x^2 + y^2 = 8z + 7$;
14. primitive Pythagorean triples with $z < 30$;
15. $\gcd(252, 198) = 18$ and Bezout identity

$$18 = 4 \cdot 252 - 5 \cdot 198;$$

16. prime factorization of 27720;
17. $\varphi(360) = 96$;
18. $7^{222} \equiv 9 \pmod{40}$;
19. Chinese remainder solution

$$x \equiv 23 \pmod{105};$$

20. all solutions to $35x + 22y = 1$ and $35x + 22y = 14$;
21. continued fraction

$$\frac{415}{93} = [4; 2, 6, 7];$$

22. continued fraction of $\sqrt{2}$:

$$\sqrt{2} = [1; 2, 2, 2, \ldots].$$

CHAPTER 3

The Number System of Mathematics

Source Map

This chapter is read as a sequence of forced enlargements of the number system:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

Each enlargement is justified by a concrete failure of the previous system: subtraction, division, measurement, limits, or algebraic solvability. The notes below therefore emphasize worked examples, solved exercises, diagrams, and explicit constructions rather than general commentary.

3.1 A Map of the Enlargements

The number systems may be organized by the equations they make solvable.

system	new operation made possible	typical equation
$\mathbb{N}$	counting and addition	$x + 3 = 7$
$\mathbb{Z}$	unrestricted subtraction	$x + 7 = 3$
$\mathbb{Q}$	division by nonzero integers	$5x = 2$
$\mathbb{R}$	limits and geometric lengths	$x^2 = 2$
$\mathbb{C}$	roots of all polynomial equations	$x^2 + 1 = 0$

For example, the equation

$$x + 7 = 3$$

has no solution in $\mathbb{N}$, but has the solution $x = -4$ in $\mathbb{Z}$. The equation

$$5x = 2$$

has no solution in $\mathbb{Z}$, but has the solution $x = 2/5$ in $\mathbb{Q}$. The equation

$$x^2 = 2$$

has no solution in $\mathbb{Q}$, but has the solution $x = \sqrt{2}$ in $\mathbb{R}$. Finally,

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$, but has the two solutions $x = \pm i$ in $\mathbb{C}$.

Takeaway

A number system is not enlarged for decoration. It is enlarged when a natural operation or equation cannot be handled inside the old system.

3.2 Rational Numbers as Measurement

A rational number is a quotient

$$\frac{a}{b}, \qquad a \in \mathbb{Z}, \quad b \in \mathbb{Z} \setminus \{0\}.$$

Its basic meaning is not only "part of a whole." It is also a measurement ratio. If a segment A is measured by a unit segment u, then saying

$$A = \frac{a}{b}u$$

means that $bA = au$.

Worked Example: A Common Unit of Measurement

Suppose two rods have lengths 35 cm and 84 cm. Their greatest common unit is

$$\gcd(35, 84) = 7.$$

Thus

$$35 = 5 \cdot 7, \qquad 84 = 12 \cdot 7.$$

The ratio of the two rods is therefore

$$\frac{35}{84} = \frac{5}{12}.$$

This is exactly the rational-number idea: two magnitudes are commensurable when they can both be measured by the same unit.

The arithmetic of rational numbers is controlled by the need to compare different units. For example,

$$\frac{2}{3} + \frac{5}{12} = \frac{8}{12} + \frac{5}{12} = \frac{13}{12}.$$

The common denominator 12 is a common measuring unit.

3.3 Density of the Rational Numbers

The rational numbers are not separated like the integers. Between any two different rational numbers there is another rational number.

Solved Exercise: There Is a Rational Number Between Any Two Rationals

Let $a, b \in \mathbb{Q}$ and $a < b$. Define

$$c = \frac{a+b}{2}.$$

Since $\mathbb{Q}$ is closed under addition and division by 2, we have $c \in \mathbb{Q}$. Also,

$$a < \frac{a+b}{2} < b.$$

Therefore there is always a rational number strictly between a and b.

This proves more than existence of one intermediate rational. Repeating the argument gives infinitely many rationals between a and b. For example, between $1/3$ and $1/2$,

$$\frac{1}{3} < \frac{1/3 + 1/2}{2} = \frac{5}{12} < \frac{1}{2}.$$

Then between $1/3$ and $5/12$,

$$\frac{1}{3} < \frac{1/3 + 5/12}{2} = \frac{3/12 + 5/12}{2} = \frac{1}{3} < \frac{3}{8} < \frac{5}{12}.$$

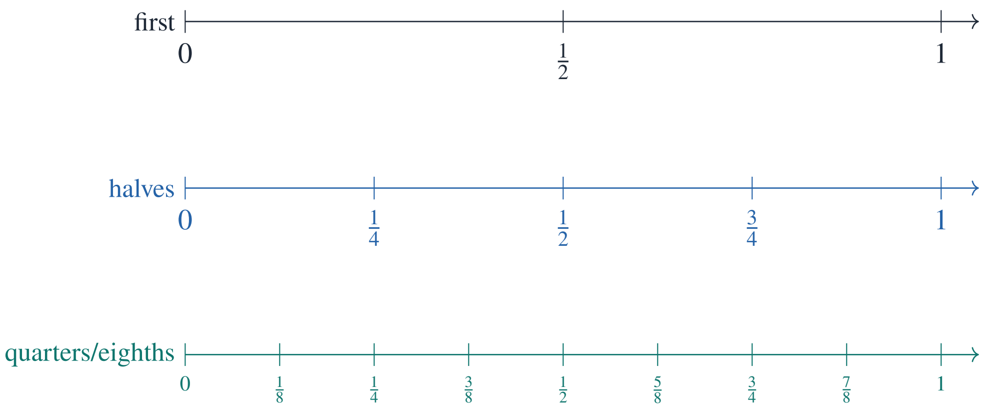
Thus we obtain

$$\frac{1}{3} < \frac{3}{8} < \frac{5}{12} < \frac{1}{2}.$$

> **Remark**
>
> The integers are discrete; the rationals are dense. This is the first major change in the geometry of the number line.

3.4 A Picture of Density

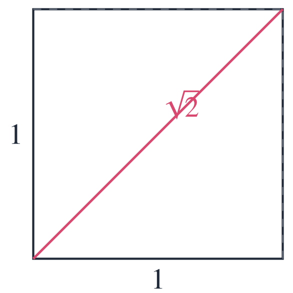
The following diagram shows that after inserting midpoints repeatedly between 0 and 1, the number line becomes increasingly crowded.



This picture should not be mistaken for the whole rational line. It only shows dyadic rationals, numbers of the form $m/2^n$. The full rational line also contains thirds, fifths, sevenths, and all other denominators.

3.5 Incommensurable Segments and $\sqrt{2}$

The diagonal of a unit square gives the first unavoidable irrational number.



By the Pythagorean theorem, the diagonal length d satisfies

$$d^2 = 1^2 + 1^2 = 2.$$

Therefore $d = \sqrt{2}$.

Solved Exercise: $\sqrt{2}$ Is Irrational

Assume, for contradiction, that

$$\sqrt{2} = \frac{p}{q}$$

where $p, q \in \mathbb{Z}$, $q \neq 0$, and the fraction is in lowest terms. Then

$$2 = \frac{p^2}{q^2},$$

so

$$p^2 = 2q^2.$$

Thus p^2 is even, hence p is even. Write $p = 2r$. Then

$$(2r)^2 = 2q^2,$$

so

$$4r^2 = 2q^2, \qquad q^2 = 2r^2.$$

Hence q^2 is even, so q is even. Therefore both p and q are even, contradicting the assumption that p/q was in lowest terms.
Thus

$$\sqrt{2} \notin \mathbb{Q}.$$

This proof shows a real failure of $\mathbb{Q}$. The rational numbers are dense, but they are not complete enough to contain every geometrically natural length.

3.6 Approximating $\sqrt{2}$ by Rationals

Although $\sqrt{2}$ is not rational, it can be trapped by rational intervals.

interval	lower square	upper square
$[1.4, 1.5]$	$1.4^2 = 1.96 < 2$	$1.5^2 = 2.25 > 2$
$[1.41, 1.42]$	$1.41^2 = 1.9881 < 2$	$1.42^2 = 2.0164 > 2$
$[1.414, 1.415]$	$1.414^2 = 1.999396 < 2$	$1.415^2 = 2.002225 > 2$
$[1.4142, 1.4143]$	$1.4142^2 = 1.99996164 < 2$	$1.4143^2 = 2.00024449 > 2$

Therefore

$$1.4142 < \sqrt{2} < 1.4143.$$

Worked Computation: Newton's Method for $\sqrt{2}$

To solve $x^2 = 2$, use

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

Starting from $x_0 = 1$,

$$x_1 = \frac{1}{2}(1+2) = 1.5.$$

Then

$$x_2 = \frac{1}{2}\left(1.5 + \frac{2}{1.5}\right) = \frac{1}{2}\left(\frac{3}{2} + \frac{4}{3}\right) = \frac{17}{12} = 1.416666\ldots.$$

Next,

$$x_3 = \frac{1}{2}\left(\frac{17}{12} + \frac{24}{17}\right) = \frac{1}{2} \cdot \frac{577}{204} = \frac{577}{408} = 1.414215686\ldots.$$

This is already correct to four decimal places.

This computation illustrates why the real numbers are needed: rational approximations can converge toward a value that is not itself rational.

3.7 Infinite Decimals

A terminating decimal is rational. For example,

$$0.375 = \frac{375}{1000} = \frac{3}{8}.$$

A repeating decimal is also rational.

Solved Exercise: Convert $0.\overline{142857}$ to a Fraction

Let

$$x = 0.\overline{142857}.$$

Then

$$10^6 x = 142857.\overline{142857}.$$

Subtracting,

$$10^6 x - x = 142857.$$

Hence

$$999999x = 142857,$$

so

$$x = \frac{142857}{999999}.$$

Since

$$142857 \cdot 7 = 999999,$$

we get

$$x = \frac{1}{7}.$$

Therefore

$$\frac{1}{7} = 0.\overline{142857}.$$

The same calculation explains why

$$0.\overline{9} = 1.$$

Solved Exercise: $0.999\ldots = 1$

Let
$$x = 0.\overline{9}.$$
Then
$$10x = 9.\overline{9}.$$
Subtract:
$$10x - x = 9.\overline{9} - 0.\overline{9} = 9.$$
Therefore
$$9x = 9,$$
so
$$x = 1.$$
Thus
$$0.999\ldots = 1.$$

Takeaway

A decimal expansion is not merely a notation. It is a limiting process. The symbol $0.999\ldots$ denotes the limit of
$$0.9, \quad 0.99, \quad 0.999, \quad \ldots$$
and this limit is exactly 1.

3.8 Rational Decimals Are Eventually Periodic

A real number is rational if and only if its decimal expansion is eventually periodic.

Proof Sketch with an Example

Consider division of 1 by 7. The possible remainders are
$$0, 1, 2, 3, 4, 5, 6.$$
If the remainder becomes 0, the decimal terminates. If not, then some nonzero remainder must repeat, since there are only six nonzero remainders. Once a remainder repeats, the entire division process repeats.
For $1/7$, the remainders cycle:
$$1,\ 3,\ 2,\ 6,\ 4,\ 5,\ 1, \ldots$$
and the digits cycle:
$$142857142857\ldots.$$
Thus
$$\frac{1}{7} = 0.\overline{142857}.$$

Conversely, if a decimal is eventually periodic, multiplying by a suitable power of 10 shifts the repeating block into alignment, and subtraction turns it into a fraction. For example,

$$y = 0.23\overline{45}$$

gives

$$100y = 23.\overline{45}, \qquad 10000y = 2345.\overline{45}.$$

Subtract:

$$9900y = 2322,$$

so

$$y = \frac{2322}{9900} = \frac{129}{550}.$$

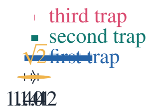
3.9 Completeness of the Real Line

The rational numbers are dense, but they have holes. The real numbers fill those holes.

One concrete way to see this is by nested intervals. For $\sqrt{2}$, the intervals

$$[1, 2] \supset [1.4, 1.5] \supset [1.41, 1.42] \supset [1.414, 1.415] \supset \cdots$$

trap a unique real number.



The real number $\sqrt{2}$ is not any finite decimal approximation. It is the limit determined by all compatible approximations.

Remark

The passage $\mathbb{Q} \to \mathbb{R}$ is not about adding a few famous numbers such as $\sqrt{2}$ and π. It is about completing the rational line so that Cauchy sequences and nested intervals have limits.

3.10 Dense Irrationals

The irrational numbers are also dense in the real line.

Solved Exercise: There Is an Irrational Number Between Any Two Reals

Let $a, b \in \mathbb{R}$ with $a < b$. Since the rationals are dense, there exists $r \in \mathbb{Q}$ such that

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

Then

$$a < r + \sqrt{2} < b.$$

Now $r + \sqrt{2}$ is irrational. Indeed, if $r + \sqrt{2} \in \mathbb{Q}$, then

$$\sqrt{2} = (r + \sqrt{2}) - r$$

would be rational, contradicting $\sqrt{2} \notin \mathbb{Q}$.
Therefore every open interval contains an irrational number.

This gives a useful picture of the real line:

between any two rationals there are irrationals,

and

between any two irrationals there are rationals.

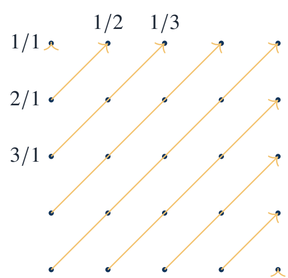
The rational and irrational numbers are densely interlaced.

3.11 Countability of the Rational Numbers

Although $\mathbb{Q}$ is dense, it is still countable. One can enumerate positive rationals by arranging them in a grid:

	$q = 1$	$q = 2$	$q = 3$	$q = 4$	$q = 5$
$p = 1$	1	$1/2$	$1/3$	$1/4$	$1/5$
$p = 2$	2	$2/2$	$2/3$	$2/4$	$2/5$
$p = 3$	3	$3/2$	$3/3$	$3/4$	$3/5$
$p = 4$	4	$4/2$	$4/3$	$4/4$	$4/5$
$p = 5$	5	$5/2$	$5/3$	$5/4$	$5/5$

Repeated fractions such as $2/2$, $3/3$, and $4/2$ may be skipped. The remaining reduced fractions can be listed by moving along diagonals $p + q = 2, 3, 4, \ldots$.



diagonal enumeration of positive rationals

Solved Exercise: $\mathbb{Q}$ Is Countable

Every rational number can be written as p/q, where $p \in \mathbb{Z}$, $q \in \mathbb{N}$, and $\gcd(|p|, q) = 1$. For each integer $N \geq 1$, there are only finitely many pairs satisfying

$$|p| + q = N.$$

List all reduced pairs with $|p| + q = 1$, then all with $|p| + q = 2$, then $|p| + q = 3$, and so on. This gives a sequence containing every rational number exactly once.
Hence $\mathbb{Q}$ is countable.

3.12 Uncountability of the Real Numbers

The real numbers are not countable. Cantor's diagonal argument proves this directly.

Solved Exercise: Cantor Diagonal Argument

Assume, for contradiction, that all real numbers in $(0, 1)$ can be listed:

$$\begin{array}{rcl} x_1 & = & 0.a_{11}a_{12}a_{13}a_{14}\dots \\ x_2 & = & 0.a_{21}a_{22}a_{23}a_{24}\dots \\ x_3 & = & 0.a_{31}a_{32}a_{33}a_{34}\dots \\ x_4 & = & 0.a_{41}a_{42}a_{43}a_{44}\dots \\ \vdots & & \vdots \end{array}$$

Construct a new decimal

$$y = 0.b_1b_2b_3b_4\dots$$

by choosing

$$b_n = \begin{cases} 1, & a_{nn} \neq 1, \\ 2, & a_{nn} = 1. \end{cases}$$

Then y differs from x_n in the n-th decimal place. Therefore $y \neq x_n$ for every n. But $y \in (0, 1)$, contradicting the assumption that the list contained all real numbers in $(0, 1)$.
Thus $\mathbb{R}$ is uncountable.

This gives a sharp result:

$$\mathbb{Q} \text{ is dense and countable,} \qquad \mathbb{R} \text{ is dense and uncountable.}$$

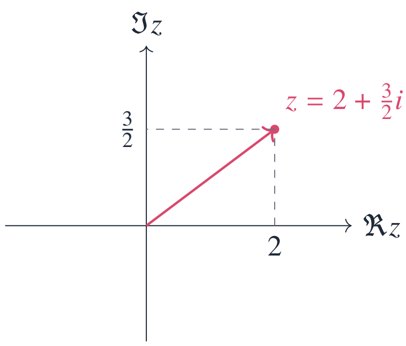
Density and cardinality are different properties.

3.13 Complex Numbers as Points and Operations

The complex numbers are defined by

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R},\ i^2 = -1\}.$$

As a set, $\mathbb{C}$ is the plane $\mathbb{R}^2$. The number $a + bi$ corresponds to the point (a, b).



Addition is vector addition:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Multiplication is determined by $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Worked Example: Complex Multiplication

Compute

$$(2+3i)(1-4i).$$

Expanding,

$$(2+3i)(1-4i) = 2-8i+3i-12i^2.$$

Since $i^2 = -1$,

$$2-8i+3i-12i^2 = 2-5i+12 = 14-5i.$$

3.14 Multiplication as Rotation and Scaling

If

$$z = r(\cos\theta + i\sin\theta), \qquad w = s(\cos\phi + i\sin\phi),$$

then

$$zw = rs(\cos(\theta+\phi) + i\sin(\theta+\phi)).$$

Thus complex multiplication multiplies lengths and adds angles.

Worked Example: $(1+i)^6$

Write

$$1+i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right).$$

Then

$$(1+i)^6 = (\sqrt{2})^6\left(\cos\frac{6\pi}{4} + i\sin\frac{6\pi}{4}\right).$$

Since

$$(\sqrt{2})^6 = 8, \qquad \frac{6\pi}{4} = \frac{3\pi}{2},$$

we get

$$(1+i)^6 = 8\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 8(0-i) = -8i.$$

Takeaway

In $\mathbb{R}$, multiplication mainly changes size. In $\mathbb{C}$, multiplication changes size and direction.

3.15 Roots of Unity

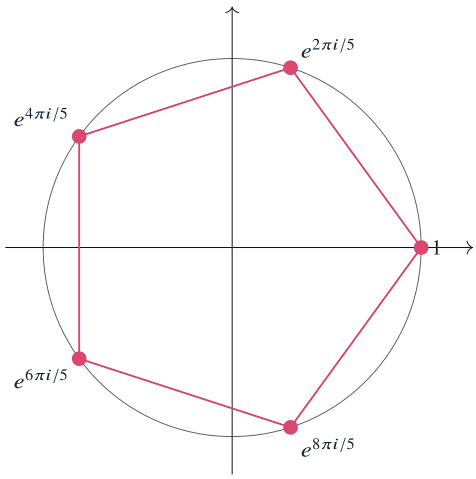
The equation

$$z^5 = 1$$

has five complex solutions:

$$z_k = e^{2\pi ik/5} = \cos\frac{2\pi k}{5} + i\sin\frac{2\pi k}{5}, \qquad k = 0,1,2,3,4.$$

They are equally spaced points on the unit circle.



Solved Exercise: Sum of the Fifth Roots of Unity

Let

$$\omega = e^{2\pi i/5}.$$

Then the five roots are

$$1, \omega, \omega^2, \omega^3, \omega^4.$$

Since $\omega^5 = 1$ and $\omega \neq 1$,

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 = \frac{\omega^5 - 1}{\omega - 1} = 0.$$

Geometrically, the five vectors form a regular pentagon centered at the origin, so their vector sum is 0.

3.16 The Fundamental Theorem of Algebra

The complex numbers are algebraically closed: every nonconstant polynomial with complex coefficients has a complex root. Equivalently, every degree n polynomial factors into n linear factors over $\mathbb{C}$, counting multiplicity.

For example,

$$x^2 + 1$$

does not factor over $\mathbb{R}$, but over $\mathbb{C}$,

$$x^2 + 1 = (x - i)(x + i).$$

Another example:

$$x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x - i)(x + i).$$

Worked Example: Solve $z^4 = 16$

Write

$$16 = 16e^{2\pi i k}, \qquad k \in \mathbb{Z}.$$

Taking fourth roots gives

$$z = 2e^{2\pi ik/4}.$$

For $k = 0, 1, 2, 3$, the distinct solutions are

$$2, \quad 2i, \quad -2, \quad -2i.$$

Therefore

$$z^4 - 16 = (z-2)(z-2i)(z+2)(z+2i).$$

3.17 Algebraic Numbers

A complex number α is algebraic if it is a root of a nonzero polynomial with integer coefficients:

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0, \qquad a_j \in \mathbb{Z}.$$

Examples:

$$\frac{2}{3}$$

is algebraic because it solves

$$3x - 2 = 0.$$

Also,

$$\sqrt{2}$$

is algebraic because it solves

$$x^2 - 2 = 0.$$

The number i is algebraic because it solves

$$x^2 + 1 = 0.$$

Solved Exercise: $\sqrt{2} + \sqrt{3}$ Is Algebraic

Let

$$x = \sqrt{2} + \sqrt{3}.$$

Then

$$x^2 = 2 + 3 + 2\sqrt{6} = 5 + 2\sqrt{6}.$$

Hence

$$x^2 - 5 = 2\sqrt{6}.$$

Squaring again,

$$(x^2 - 5)^2 = 24.$$

Therefore

$$x^4 - 10x^2 + 25 = 24,$$

so

$$x^4 - 10x^2 + 1 = 0.$$

Thus $\sqrt{2} + \sqrt{3}$ is a root of the integer polynomial

$$X^4 - 10X^2 + 1.$$

Therefore $\sqrt{2} + \sqrt{3}$ is algebraic.

This example shows that algebraic numbers are not limited to simple radicals. They form a large field, closed under addition, subtraction, multiplication, and division by nonzero elements.

3.18 Transcendental Numbers Exist

A complex number is transcendental if it is not algebraic.

The existence of transcendental numbers follows from a counting argument.

Solved Exercise: There Are Transcendental Real Numbers

There are countably many integer polynomials, because a polynomial is determined by a finite list of integers:

$$(a_0, a_1, \ldots, a_n).$$

Each nonzero polynomial has only finitely many roots. Therefore the set of all algebraic numbers is a countable union of finite sets, hence countable.
But $\mathbb{R}$ is uncountable by Cantor's diagonal argument. Therefore the set

$$\mathbb{R} \setminus \{\text{algebraic real numbers}\}$$

is nonempty; in fact, it is uncountable.
Thus transcendental real numbers exist.

This argument proves existence without constructing a specific example. Famous specific transcendental numbers include e and π, but proving their transcendence requires deeper methods.

3.19 A Concrete Classification Table

The following table records typical numbers and the smallest system among $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ that naturally contains them.

number	smallest standard system here	reason
7	$\mathbb{N}$	counting number
-4	$\mathbb{Z}$	needed for subtraction
$\frac{5}{12}$	$\mathbb{Q}$	ratio of integers
$\sqrt{2}$	$\mathbb{R}$	irrational length
π	$\mathbb{R}$	real but not rational
i	$\mathbb{C}$	solution of $x^2 + 1 = 0$
$1 + i$	$\mathbb{C}$	nonzero real and imaginary parts
$e^{2\pi i/5}$	$\mathbb{C}$	fifth root of unity

3.20 Three Exercises That Capture the Chapter

Exercise 1: Find a Rational and an Irrational Between 2 and 2.1

A rational number is

$$2.05 = \frac{205}{100}.$$

An irrational number can be obtained by taking a small irrational perturbation:

$$2 + \frac{\sqrt{2}}{100}.$$

Since

$$0 < \frac{\sqrt{2}}{100} < \frac{2}{100} = 0.02,$$

we have

$$2 < 2 + \frac{\sqrt{2}}{100} < 2.02 < 2.1.$$

The number is irrational because if $2 + \sqrt{2}/100$ were rational, then

$$\sqrt{2} = 100\left(2 + \frac{\sqrt{2}}{100} - 2\right)$$

would be rational, contradiction.

Exercise 2: Decide Whether $0.12\overline{3}$ Is Rational

Let

$$x = 0.123333\ldots.$$

Then

$$100x = 12.3333\ldots, \qquad 1000x = 123.3333\ldots.$$

Subtract:

$$1000x - 100x = 123.3333\ldots - 12.3333\ldots = 111.$$

Thus

$$900x = 111,$$

so

$$x = \frac{111}{900} = \frac{37}{300}.$$

Therefore

$$0.12\overline{3} = \frac{37}{300}.$$

Exercise 3: Factor $x^4 + 4$ Over $\mathbb{R}$ and Over $\mathbb{C}$

First write

$$x^4 + 4 = x^4 + 4x^2 + 4 - 4x^2 = (x^2 + 2)^2 - (2x)^2.$$

Therefore
$$x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2).$$
Over $\mathbb{R}$, both quadratics are irreducible because their discriminants are
$$(-2)^2 - 8 = -4 < 0, \qquad 2^2 - 8 = -4 < 0.$$
Over $\mathbb{C}$,
$$x^2 - 2x + 2 = 0$$
has roots
$$x = \frac{2 \pm \sqrt{-4}}{2} = 1 \pm i,$$
and
$$x^2 + 2x + 2 = 0$$
has roots
$$x = \frac{-2 \pm \sqrt{-4}}{2} = -1 \pm i.$$
Thus
$$x^4 + 4 = (x - (1 + i))(x - (1 - i))(x - (-1 + i))(x - (-1 - i)).$$
Equivalently,
$$x^4 + 4 = (x - 1 - i)(x - 1 + i)(x + 1 - i)(x + 1 + i).$$

3.21 Working Summary

The chapter can be compressed into five concrete lessons.

Key Points

1. $\mathbb{Z}$ repairs the failure of subtraction in $\mathbb{N}$.
2. $\mathbb{Q}$ repairs the failure of division in $\mathbb{Z}$.
3. $\mathbb{R}$ repairs the failure of $\mathbb{Q}$ to contain limits and geometric lengths such as $\sqrt{2}$.
4. $\mathbb{C}$ repairs the failure of $\mathbb{R}$ to solve equations such as $x^2 + 1 = 0$.
5. Algebraic numbers solve polynomial equations with integer coefficients; transcendental numbers exist because $\mathbb{R}$ is too large to be exhausted by countably many polynomial roots.

The most important computational thread is this:
$$\text{division} \longrightarrow \text{decimals} \longrightarrow \text{limits} \longrightarrow \text{real numbers}.$$
The most important algebraic thread is this:
$$\text{equations} \longrightarrow \text{field extensions} \longrightarrow \text{complex roots} \longrightarrow \text{algebraic numbers}.$$
Together they explain why the standard number systems are not arbitrary collections, but successive answers to concrete mathematical failures.

CHAPTER 4

Supplement to Chapter II: The Algebra of Sets

Source Map

This supplement develops the algebra of sets and then applies it to two concrete domains: mathematical logic and probability. The main point is that the operations

$$\cup, \qquad \cap, \qquad {}^c, \qquad \setminus$$

behave like an algebra. Instead of treating sets as passive collections, this chapter treats them as objects on which one can calculate.

4.1 Basic Set Operations Through a Fixed Example

Let the universal set be

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

Define

$$A = \{1, 2, 3, 4, 5, 6\}, \qquad B = \{2, 4, 6, 8, 10\}, \qquad C = \{1, 3, 5, 7, 9\}.$$

Then A is the set of integers from 1 to 6, B is the set of even elements of U, and C is the set of odd elements of U.

The union is

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}.$$

The intersection is

$$A \cap B = \{2, 4, 6\}.$$

The difference is

$$A \setminus B = \{1, 3, 5\}.$$

The complement of A in U is

$$A^c = U \setminus A = \{7, 8, 9, 10\}.$$

Worked Computation: Symmetric Difference

The symmetric difference is

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

In this example,

$$A \setminus B = \{1, 3, 5\},$$

and

$$B \setminus A = \{8, 10\}.$$

Therefore

$$A \triangle B = \{1, 3, 5, 8, 10\}.$$

Equivalently,

$$A \triangle B = (A \cup B) \setminus (A \cap B).$$

Indeed,

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\},$$

and

$$A \cap B = \{2, 4, 6\}.$$

Removing the intersection gives

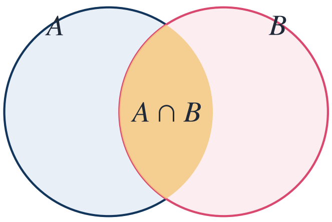
$$\{1, 3, 5, 8, 10\}.$$

Takeaway

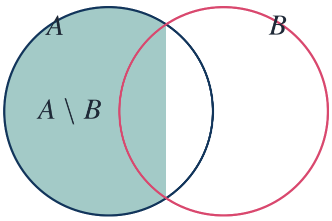
The operations $\cup$, $\cap$, complement, and set difference are not notation only. They are calculational operations, just as $+$, $\cdot$, and $-$ are operations on numbers.

4.2 Venn Diagrams as Calculations

The Venn diagram below shows $A \cap B$, the part common to both sets.



The next diagram shows $A \setminus B$, the part of A outside B.



The most important use of such diagrams is not artistic. A diagram gives a quick check of a set identity before giving the formal proof by membership.

4.3 Proving Set Identities by Element Chasing

To prove an identity of sets, one usually proves double inclusion. That means one proves

$$X \subseteq Y \qquad \text{and} \qquad Y \subseteq X.$$

Equivalently, one takes an arbitrary element x and checks that

$$x \in X \iff x \in Y.$$

Solved Exercise: Distributive Law

Prove

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Let x be arbitrary. Then

$$\begin{aligned} x \in A \cap (B \cup C) &\iff x \in A \text{ and } x \in B \cup C \\ &\iff x \in A \text{ and } (x \in B \text{ or } x \in C) \\ &\iff (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C) \\ &\iff x \in A \cap B \text{ or } x \in A \cap C \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

Therefore

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Solved Exercise: De Morgan's Law

Prove

$$(A \cup B)^c = A^c \cap B^c.$$

For any $x \in U$,

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff \text{not}(x \in A \text{ or } x \in B) \\ &\iff x \notin A \text{ and } x \notin B \\ &\iff x \in A^c \text{ and } x \in B^c \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

Thus

$$(A \cup B)^c = A^c \cap B^c.$$

The second De Morgan law is

$$(A \cap B)^c = A^c \cup B^c.$$

It is proved in exactly the same way, replacing "and" by "or" at the critical step.

4.4 A Truth-Table Proof of a Set Identity

Set algebra can be checked by membership tables. Let 1 mean "belongs" and 0 mean "does not belong." For the identity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

we compute:

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

The two relevant columns are equal in every row. Therefore the set identity is valid.

Remark

Element chasing and truth tables are the same method in two languages. Element chasing uses membership statements; the truth table encodes membership statements as 0 and 1.

4.5 The Boolean Algebra of Subsets

For a fixed universe U, the collection

$$\mathcal{P}(U) = \{A : A \subseteq U\}$$

is closed under union, intersection, and complement. This is the Boolean algebra of subsets of U.

The algebraic laws include

$$\begin{aligned} A \cup B &= B \cup A, & A \cap B &= B \cap A, \\ A \cup (B \cup C) &= (A \cup B) \cup C, & A \cap (B \cap C) &= (A \cap B) \cap C, \\ A \cup (A \cap B) &= A, & A \cap (A \cup B) &= A, \end{aligned}$$

and

$$A \cup A^c = U, \qquad A \cap A^c = \varnothing.$$

Solved Exercise: Absorption Law

Prove

$$A \cup (A \cap B) = A.$$

First, since $A \subseteq A$, and $A \cap B \subseteq A$, every element of $A \cup (A \cap B)$ belongs to A. Thus

$$A \cup (A \cap B) \subseteq A.$$

Conversely,

$$A \subseteq A \cup (A \cap B)$$

because a union contains each of its parts. Therefore

$$A \cup (A \cap B) = A.$$

Concrete Check with Finite Sets

Let

$$A = \{1, 2, 3, 4, 5, 6\}, \qquad B = \{2, 4, 6, 8, 10\}.$$

Then

$$A \cap B = \{2, 4, 6\}.$$

Therefore

$$A \cup (A \cap B) = \{1, 2, 3, 4, 5, 6\} \cup \{2, 4, 6\} = \{1, 2, 3, 4, 5, 6\} = A.$$

4.6 Power Sets and Binary Choice

The power set $\mathcal{P}(S)$ is the set of all subsets of S.

Worked Example: Power Set of a Three-Element Set

Let

$$S = \{a, b, c\}.$$

Then

$$\mathcal{P}(S) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

Thus

$$|\mathcal{P}(S)| = 8 = 2^3.$$

The reason is that every element has two choices: included or not included.

subset	a	b	c	binary code
$\varnothing$	0	0	0	000
$\{c\}$	0	0	1	001
$\{b\}$	0	1	0	010
$\{b, c\}$	0	1	1	011
$\{a\}$	1	0	0	100
$\{a, c\}$	1	0	1	101
$\{a, b\}$	1	1	0	110
$\{a, b, c\}$	1	1	1	111

Solved Exercise: Prove $|\mathcal{P}(S)| = 2^n$ for $|S| = n$

Suppose

$$S = \{s_1, s_2, \ldots, s_n\}.$$

To choose a subset $T \subseteq S$, for each s_i one must choose one of two possibilities:

$$s_i \in T \qquad \text{or} \qquad s_i \notin T.$$

These n binary choices are independent. Hence the number of subsets is

$$2 \cdot 2 \cdots 2 = 2^n.$$

Therefore

$$|\mathcal{P}(S)| = 2^n.$$

4.7 Cartesian Products and Relations

The Cartesian product of two sets is

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

For example, if

$$A = \{1, 2, 3\}, \qquad B = \{x, y\},$$

then

$$A \times B = \{(1, x), (1, y), (2, x), (2, y), (3, x), (3, y)\}.$$

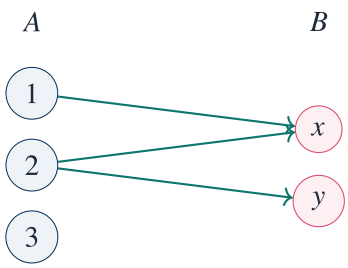
Thus

$$|A \times B| = |A||B| = 3 \cdot 2 = 6.$$

A relation from A to B is any subset of $A \times B$. For example,

$$R = \{(1, x), (2, x), (2, y)\}$$

is a relation from A to B.



Solved Exercise: How Many Relations Are There from A to B?

A relation from A to B is a subset of $A \times B$. If

$$|A| = m, \qquad |B| = n,$$

then

$$|A \times B| = mn.$$

The number of subsets of $A \times B$ is therefore

$$2^{mn}.$$

Hence the number of relations from A to B is

$$2^{mn}.$$

For $A = \{1, 2, 3\}$ and $B = \{x, y\}$, this gives

$$2^{3 \cdot 2} = 2^6 = 64$$

relations.

4.8 Functions as Special Relations

A function $f : A \to B$ is a relation with two extra properties:

$$\text{for each } a \in A, \text{ there exists exactly one } b \in B \text{ such that } (a, b) \in f.$$

Thus every element of the domain must have one and only one output.

Worked Example: Relations That Are and Are Not Functions

Let

$$A = \{1, 2, 3\}, \qquad B = \{x, y\}.$$

The relation

$$R_1 = \{(1, x), (2, x), (3, y)\}$$

is a function, because each element of A appears exactly once as a first coordinate.
The relation

$$R_2 = \{(1, x), (2, x)\}$$

is not a function $A \to B$, because $3 \in A$ has no output.
The relation

$$R_3 = \{(1, x), (1, y), (2, x), (3, y)\}$$

is not a function, because $1 \in A$ has two outputs.

Solved Exercise: How Many Functions $A \to B$ Are There?

If

$$|A| = m, \qquad |B| = n,$$

then each element of A has n possible images. Since there are m elements of A, the total number of functions is

$$n^m.$$

For

$$A = \{1, 2, 3\}, \qquad B = \{x, y\},$$

there are

$$2^3 = 8$$

functions $A \to B$.

4.9 Images and Preimages

Let $f : X \to Y$. If $A \subseteq X$, the image of A is

$$f(A) = \{f(x) : x \in A\}.$$

If $B \subseteq Y$, the preimage of B is

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

Take

$$X = \{1, 2, 3, 4\}, \qquad Y = \{a, b, c\},$$

and define $f : X \to Y$ by

$$f(1) = a, \qquad f(2) = b, \qquad f(3) = b, \qquad f(4) = c.$$

Worked Computation: Images and Preimages

Let

$$A = \{1, 2, 4\} \subseteq X.$$

Then

$$f(A) = \{f(1), f(2), f(4)\} = \{a, b, c\}.$$

Let

$$B = \{b, c\} \subseteq Y.$$

Then

$$f^{-1}(B) = \{2, 3, 4\},$$

because

$$f(2) = b, \quad f(3) = b, \quad f(4) = c.$$

Preimages behave better than images with respect to intersections.

Solved Exercise: Preimages Preserve Intersections

Prove

$$f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2).$$

For any $x \in X$,

$$\begin{aligned} x \in f^{-1}(B_1 \cap B_2) &\iff f(x) \in B_1 \cap B_2 \\ &\iff f(x) \in B_1 \text{ and } f(x) \in B_2 \\ &\iff x \in f^{-1}(B_1) \text{ and } x \in f^{-1}(B_2) \\ &\iff x \in f^{-1}(B_1) \cap f^{-1}(B_2). \end{aligned}$$

Therefore

$$f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2).$$

Images do not always preserve intersections.

Counterexample: Image of an Intersection

Let

$$X = \{1, 2\}, \qquad Y = \{a\},$$

and define

$$f(1) = a, \qquad f(2) = a.$$

Let

$$A = \{1\}, \qquad B = \{2\}.$$

Then

$$A \cap B = \varnothing,$$

so

$$f(A \cap B) = f(\varnothing) = \varnothing.$$

But

$$f(A) = \{a\}, \qquad f(B) = \{a\},$$

hence

$$f(A) \cap f(B) = \{a\}.$$

Therefore

$$f(A \cap B) \neq f(A) \cap f(B).$$

4.10 Equivalence Relations and Partitions

An equivalence relation on a set X is a relation $\sim$ satisfying:

$$x \sim x \quad \text{for all } x \in X,$$
$$x \sim y \implies y \sim x,$$
$$x \sim y \text{ and } y \sim z \implies x \sim z.$$

The standard example is congruence modulo n:

$$a \sim b \qquad \iff \qquad a \equiv b \pmod{n}.$$

Worked Example: Equivalence Classes Modulo 3

On $\mathbb{Z}$, define

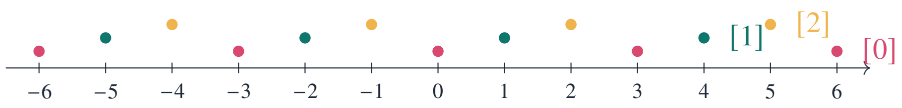
$$a \sim b \qquad \iff \qquad 3 \mid (a - b).$$

The equivalence classes are

$$[0] = \{\dots, -6, -3, 0, 3, 6, 9, \dots\},$$
$$[1] = \{\dots, -5, -2, 1, 4, 7, 10, \dots\},$$
$$[2] = \{\dots, -4, -1, 2, 5, 8, 11, \dots\}.$$

These three classes are disjoint and cover all of $\mathbb{Z}$. Thus they form a partition:

$$\mathbb{Z} = [0] \sqcup [1] \sqcup [2].$$



Takeaway

Equivalence relations are machines for replacing equality by a coarser notion of sameness. The output is a partition of the original set.

4.11 Logic as Algebra of Sets

Suppose U is a universe of objects. A predicate $P(x)$ determines a set

$$P^* = \{x \in U : P(x) \text{ is true}\}.$$

Similarly,

$$Q^* = \{x \in U : Q(x) \text{ is true}\}.$$

Then logical operations correspond to set operations:

logic	set operation
$P(x)$ and $Q(x)$	$P^* \cap Q^*$
$P(x)$ or $Q(x)$	$P^* \cup Q^*$
not $P(x)$	$(P^*)^c$
$P(x) \Rightarrow Q(x)$	$P^* \subseteq Q^*$

Worked Example: Predicates as Sets

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

Define

$$P(x) : \quad x \text{ is even},$$
$$Q(x) : \quad x \leq 6.$$

Then

$$P^* = \{2, 4, 6, 8, 10\},$$

and

$$Q^* = \{1, 2, 3, 4, 5, 6\}.$$

The statement

$$P(x) \text{ and } Q(x)$$

corresponds to

$$P^* \cap Q^* = \{2, 4, 6\}.$$

The statement

$$P(x) \text{ or } Q(x)$$

corresponds to

$$P^* \cup Q^* = \{1, 2, 3, 4, 5, 6, 8, 10\}.$$

The statement

$$P(x) \Rightarrow Q(x)$$

is false on this universe because

$$8 \in P^* \quad \text{but} \quad 8 \notin Q^*.$$

Equivalently,

$$P^* \nsubseteq Q^*.$$

4.12 Truth Tables and Implication

The implication

$$P \Rightarrow Q$$

is false only when P is true and Q is false.

P	Q	$P \Rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1

This truth table explains why

$$P \Rightarrow Q$$

is logically equivalent to

$$\neg P \vee Q.$$

Truth-Table Verification

P	Q	$\neg P$	$\neg P \vee Q$	$P \Rightarrow Q$
0	0	1	1	1
0	1	1	1	1
1	0	0	0	0
1	1	0	1	1

The last two columns agree in every row. Therefore

$$P \Rightarrow Q \iff \neg P \vee Q.$$

In set language,

$$P^* \subseteq Q^*$$

is equivalent to

$$(P^*)^c \cup Q^* = U.$$

That is exactly the set-theoretic form of

$$\neg P \vee Q.$$

4.13 Quantifiers and Their Negations

Quantifiers become much clearer when translated into sets.

The statement

$$\forall x \in U,\ P(x)$$

means

$$P^* = U.$$

The statement

$$\exists x \in U,\ P(x)$$

means

$$P^* \neq \varnothing.$$

The negation rules are

$$\neg(\forall x,\ P(x)) \Longleftrightarrow \exists x,\ \neg P(x),$$

and

$$\neg(\exists x,\ P(x)) \Longleftrightarrow \forall x,\ \neg P(x).$$

Solved Exercise: Negate a Quantified Statement

Negate the statement:

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ |a_n - L| < \epsilon.$$

Step by step, the negation is

$$\exists \epsilon > 0$$

such that

$$\forall N \in \mathbb{N},$$

there exists

$$n \geq N$$

such that

$$|a_n - L| \geq \epsilon.$$

Thus the negation is

$$\exists \epsilon > 0,\ \forall N \in \mathbb{N},\ \exists n \geq N \text{ such that } |a_n - L| \geq \epsilon.$$

This example is important because it is the exact logical form behind the definition of failure of convergence.

4.14 The Order of Quantifiers Matters

The statements

$$\forall x\ \exists y\ P(x, y)$$

and

$$\exists y\ \forall x\ P(x, y)$$

are usually not equivalent.

Worked Example: Quantifier Order

Let the universe be $\mathbb{Z}$, and let

$$P(x, y): \quad y > x.$$

The statement

$$\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z} \text{ such that } y > x$$

is true. Given x, choose

$$y = x + 1.$$

But the statement

$$\exists y \in \mathbb{Z},\ \forall x \in \mathbb{Z},\ y > x$$

is false. No fixed integer y is larger than every integer x, because choosing

$$x = y + 1$$

gives

$$x > y.$$

Remark

The first statement allows y to depend on x. The second demands one single y that works for all x. This dependency distinction is often the source of mistakes in analysis and probability.

4.15 Inclusion–Exclusion for Two and Three Sets

For finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The subtraction is needed because elements of $A \cap B$ have been counted twice.

Worked Example: Two Sets

In a class of 40 students, 25 study algebra, 18 study analysis, and 10 study both. How many study algebra or analysis?
Let

$$A = \{\text{students studying algebra}\},$$

$$B = \{\text{students studying analysis}\}.$$

Then

$$|A| = 25, \qquad |B| = 18, \qquad |A \cap B| = 10.$$

Thus

$$|A \cup B| = 25 + 18 - 10 = 33.$$

So 33 students study at least one of the two subjects.

For three sets,

$$\begin{aligned} |A \cup B \cup C| = |A| &+ |B| + |C| \\ &- |A \cap B| - |A \cap C| - |B \cap C| \\ &+ |A \cap B \cap C|. \end{aligned}$$

Solved Exercise: Three-Set Inclusion–Exclusion

Suppose among 100 students:

$$|A| = 52, \quad |B| = 41, \quad |C| = 35,$$

$$|A \cap B| = 20, \quad |A \cap C| = 18, \quad |B \cap C| = 15,$$

and

$$|A \cap B \cap C| = 8.$$

Then

$$|A \cup B \cup C| = 52 + 41 + 35 - 20 - 18 - 15 + 8$$
$$= 128 - 53 + 8$$
$$= 83.$$

Thus 83 students belong to at least one of the three sets.

4.16 Probability as Set Algebra

A probability space consists of a sample space Ω, a collection of events, and a probability function $\mathbb{P}$. In elementary finite cases, events are subsets of Ω.

The set operations become probability operations:

$$A \cup B = \text{“}A \text{ or } B \text{ occurs”},$$

$$A \cap B = \text{“}A \text{ and } B \text{ occur”},$$

$$A^c = \text{“}A \text{ does not occur”}.$$

Worked Example: Rolling One Die

Let

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Define

$$A = \{\text{even outcome}\} = \{2, 4, 6\},$$

$$B = \{\text{outcome at least } 4\} = \{4, 5, 6\}.$$

Then

$$A \cup B = \{2, 4, 5, 6\},$$

$$A \cap B = \{4, 6\},$$

and

$$A^c = \{1, 3, 5\}.$$

Since the die is fair,

$$\mathbb{P}(A) = \frac{3}{6} = \frac{1}{2},$$

$$\mathbb{P}(B) = \frac{3}{6} = \frac{1}{2},$$

$$\mathbb{P}(A \cap B) = \frac{2}{6} = \frac{1}{3}.$$

Therefore

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{2}{3}.$$

4.17 Disjoint Events and Independence Are Different

Two events are disjoint if

$$A \cap B = \varnothing.$$

Two events are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

These are different concepts.

Counterexample: Disjoint Does Not Mean Independent

Roll one fair die. Let

$$A = \{1, 2, 3\}, \qquad B = \{4, 5, 6\}.$$

Then

$$A \cap B = \varnothing,$$

so A and B are disjoint.
But

$$\mathbb{P}(A) = \frac{1}{2}, \qquad \mathbb{P}(B) = \frac{1}{2},$$

so

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{4}.$$

Meanwhile

$$\mathbb{P}(A \cap B) = 0.$$

Thus

$$\mathbb{P}(A \cap B) \neq \mathbb{P}(A)\mathbb{P}(B).$$

The events are not independent.

Example: Independent but Not Disjoint

Toss two fair coins:

$$\Omega = \{HH, HT, TH, TT\}.$$

Let

$$A = \{\text{first toss is } H\} = \{HH, HT\},$$

$$B = \{\text{second toss is } H\} = \{HH, TH\}.$$

Then

$$A \cap B = \{HH\}.$$

So the events are not disjoint.
But

$$\mathbb{P}(A) = \frac{1}{2}, \qquad \mathbb{P}(B) = \frac{1}{2},$$

and

$$\mathbb{P}(A \cap B) = \frac{1}{4}.$$

Therefore

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

So A and B are independent.

Takeaway

Disjointness is a set-theoretic exclusion condition. Independence is a multiplicative probability condition. Except for zero-probability cases, nonempty disjoint events cannot be independent.

4.18 Conditional Probability

Conditional probability is defined by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) > 0.$$

It means: restrict the universe to B, then measure how much of B lies inside A.

Worked Example: Conditional Probability with a Die

Roll one fair die. Let

$$A = \{\text{even}\} = \{2, 4, 6\},$$

and

$$B = \{\text{at least } 4\} = \{4, 5, 6\}.$$

Then

$$A \cap B = \{4, 6\}.$$

So

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{2/6}{3/6} = \frac{2}{3}.$$

This is also clear directly: if the outcome is known to be in $\{4, 5, 6\}$, then the favorable outcomes are 4 and 6, two out of three.

Solved Exercise: Conditional Probability from a Table

Suppose a group of 100 students is classified as follows:

	takes algebra	does not take algebra	total
takes analysis	18	12	30
does not take analysis	22	48	70
total	40	60	100

Let

$$A = \{\text{takes algebra}\}, \qquad B = \{\text{takes analysis}\}.$$

Then

$$\mathbb{P}(A) = \frac{40}{100}, \qquad \mathbb{P}(B) = \frac{30}{100}, \qquad \mathbb{P}(A \cap B) = \frac{18}{100}.$$

Thus

$$\mathbb{P}(A \mid B) = \frac{18/100}{30/100} = \frac{18}{30} = \frac{3}{5}.$$

So among students taking analysis, the probability of also taking algebra is

$$\frac{3}{5}.$$

4.19 Bayes' Formula as Set Algebra

Bayes' formula follows directly from the definition of conditional probability:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

When $A_1, \ldots, A_n$ partition the sample space, then

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

Worked Example: Medical Test Base Rate

Suppose a disease has prevalence

$$\mathbb{P}(D) = 0.01.$$

A test has sensitivity

$$\mathbb{P}(+ \mid D) = 0.99,$$

and false positive rate

$$\mathbb{P}(+ \mid D^c) = 0.05.$$

Find

$$\mathbb{P}(D \mid +).$$

First compute the total probability of a positive test:

$$\begin{aligned}\mathbb{P}(+) &= \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c) \\ &= 0.99 \cdot 0.01 + 0.05 \cdot 0.99 \\ &= 0.0099 + 0.0495 \\ &= 0.0594.\end{aligned}$$

Then

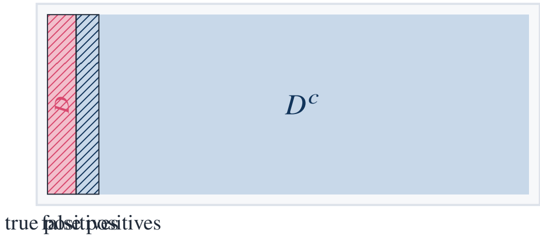
$$\begin{aligned}\mathbb{P}(D \mid +) &= \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)} \\ &= \frac{0.99 \cdot 0.01}{0.0594} \\ &= \frac{0.0099}{0.0594} \\ &= \frac{1}{6}.\end{aligned}$$

So even after a positive test, the probability of disease is only

$$\frac{1}{6} \approx 16.7\%.$$

The calculation is not paradoxical. The false positive rate acts on the large set D^c, while sensitivity acts on the small set D. Set size matters.

positive tests come from both D and D^c



4.20 Counting with Sets: Exactly One, At Least One, None

Set algebra is useful for counting questions where the wording matters.

Solved Exercise: Exactly One of Two Events

Let

$$|A| = 30, \qquad |B| = 22, \qquad |A \cap B| = 12.$$

The number of elements in exactly one of A, B is

$$|A \triangle B|.$$

Compute:

$$|A \triangle B| = |A \setminus B| + |B \setminus A|.$$

Now

$$|A \setminus B| = |A| - |A \cap B| = 30 - 12 = 18,$$

and

$$|B \setminus A| = |B| - |A \cap B| = 22 - 12 = 10.$$

Thus

$$|A \triangle B| = 18 + 10 = 28.$$

Equivalently,

$$|A \triangle B| = |A| + |B| - 2|A \cap B| = 30 + 22 - 24 = 28.$$

Solved Exercise: None of Three Sets

Suppose

$$|U| = 100,$$

and

$$|A \cup B \cup C| = 83.$$

Then the number of elements in none of A, B, C is

$$|(A \cup B \cup C)^c| = |U| - |A \cup B \cup C| = 100 - 83 = 17.$$

4.21 A Probability Problem Solved by Inclusion–Exclusion

Solved Exercise: Drawing a Card

A card is drawn from a standard 52-card deck. Find the probability that the card is either a heart or a face card.
Let

$$H = \{\text{hearts}\}, \qquad F = \{\text{face cards}\}.$$

There are

$$|H| = 13$$

hearts and

$$|F| = 12$$

face cards. The face cards that are hearts are

$$J\heartsuit,\ Q\heartsuit,\ K\heartsuit,$$

so

$$|H \cap F| = 3.$$

Therefore

$$|H \cup F| = |H| + |F| - |H \cap F| = 13 + 12 - 3 = 22.$$

Hence

$$\mathbb{P}(H \cup F) = \frac{22}{52} = \frac{11}{26}.$$

This example shows why one cannot simply add probabilities without correcting for overlap:

$$\frac{13}{52} + \frac{12}{52}$$

counts the three heart face cards twice.

4.22 A More Serious Probability Example: At Least One Match

This is the classical matching problem in a small finite case.

Solved Exercise: Three Letters and Three Envelopes

Suppose three letters are randomly placed into three addressed envelopes. What is the probability that at least one letter goes into the correct envelope?
Let

$$A_i = \{\text{letter } i \text{ goes into its correct envelope}\}.$$

There are

$$3! = 6$$

possible assignments.
For each i,

$$|A_i| = 2!,$$

because once letter i is fixed, the other two letters can be permuted freely. Thus

$$|A_i| = 2.$$

For each pair $i < j$,

$$|A_i \cap A_j| = 1!,$$

so

$$|A_i \cap A_j| = 1.$$

Finally,

$$|A_1 \cap A_2 \cap A_3| = 1.$$

By inclusion–exclusion,

$$\begin{aligned} |A_1 \cup A_2 \cup A_3| &= (|A_1| + |A_2| + |A_3|) - (|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|) \\ &\quad + |A_1 \cap A_2 \cap A_3| \\ &= (2 + 2 + 2) - (1 + 1 + 1) + 1 \\ &= 4. \end{aligned}$$

Therefore the probability of at least one correct match is

$$\frac{4}{6} = \frac{2}{3}.$$

For n letters, the same set method gives the derangement formula:

$$!n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right).$$

Thus set algebra scales from a small counting exercise to a general formula.

4.23 Set Partitions and the Law of Total Probability

A collection

$$A_1, A_2, \ldots, A_n$$

is a partition of Ω if

$$A_i \cap A_j = \varnothing \quad (i \neq j),$$

and

$$A_1 \cup \cdots \cup A_n = \Omega.$$

For any event B,

$$B = (B \cap A_1) \sqcup (B \cap A_2) \sqcup \cdots \sqcup (B \cap A_n).$$

Therefore

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \cap A_i) = \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

Worked Example: Total Probability

A factory has three machines:

$$\mathbb{P}(M_1) = 0.5, \qquad \mathbb{P}(M_2) = 0.3, \qquad \mathbb{P}(M_3) = 0.2.$$

Their defect rates are

$$\mathbb{P}(D \mid M_1) = 0.01, \qquad \mathbb{P}(D \mid M_2) = 0.03, \qquad \mathbb{P}(D \mid M_3) = 0.04.$$

Then

$$\begin{aligned} \mathbb{P}(D) &= 0.01 \cdot 0.5 + 0.03 \cdot 0.3 + 0.04 \cdot 0.2 \\ &= 0.005 + 0.009 + 0.008 \\ &= 0.022. \end{aligned}$$

So the overall defect probability is

$$2.2\%.$$

4.24 A Bayes Exercise with the Same Factory

Solved Exercise: Posterior Probability of the Source Machine

Using the factory data above, suppose an item is defective. What is the probability it came from machine M_3?
Bayes' formula gives

$$\mathbb{P}(M_3 \mid D) = \frac{\mathbb{P}(D \mid M_3)\mathbb{P}(M_3)}{\mathbb{P}(D)}.$$

We already computed

$$\mathbb{P}(D) = 0.022.$$

Also,

$$\mathbb{P}(D \mid M_3)\mathbb{P}(M_3) = 0.04 \cdot 0.2 = 0.008.$$

Therefore

$$\mathbb{P}(M_3 \mid D) = \frac{0.008}{0.022} = \frac{8}{22} = \frac{4}{11}.$$

Thus, given that an item is defective, the probability that it came from machine M_3 is

$$\frac{4}{11} \approx 36.36\%.$$

Notice the shift:

$$\mathbb{P}(M_3) = 20\%,$$

but

$$\mathbb{P}(M_3 \mid D) \approx 36.36\%.$$

The posterior probability is higher because M_3 has the largest defect rate.

4.25 Sets, Logic, and Probability in One Example

Consider a population of 1000 applicants to a program. Define

$$A = \{\text{applicants with strong algebra background}\},$$

$$B = \{\text{applicants with strong analysis background}\},$$

$$C = \{\text{applicants with research experience}\}.$$

Suppose

$$|A| = 420, \quad |B| = 380, \quad |C| = 260,$$

$$|A \cap B| = 210, \quad |A \cap C| = 150, \quad |B \cap C| = 130,$$

and

$$|A \cap B \cap C| = 90.$$

Solved Exercise: Count Applicants with at Least One Strength

By inclusion–exclusion,

$$\begin{aligned} |A \cup B \cup C| &= 420 + 380 + 260 - 210 - 150 - 130 + 90 \\ &= 1060 - 490 + 90 \\ &= 660. \end{aligned}$$

Thus 660 applicants have at least one of the three listed strengths.

Solved Exercise: Count Applicants with All Three Strengths but Avoid Double Counting

The applicants with all three strengths are exactly

$$A \cap B \cap C.$$

So the number is

$$|A \cap B \cap C| = 90.$$

The probability that a uniformly chosen applicant has all three strengths is

$$\frac{90}{1000} = 0.09.$$

Solved Exercise: Conditional Probability

What is the probability that an applicant has research experience, given that the applicant has both algebra and analysis background?

We need

$$\mathbb{P}(C \mid A \cap B) = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(A \cap B)}.$$

Using counts,

$$\mathbb{P}(C \mid A \cap B) = \frac{|A \cap B \cap C|}{|A \cap B|} = \frac{90}{210} = \frac{3}{7}.$$

This example uses all three parts of the supplement:

$$\text{sets: } A \cup B \cup C,$$
$$\text{logic: “algebra and analysis imply stronger evidence”},$$
$$\text{probability: } \mathbb{P}(C \mid A \cap B) = \frac{3}{7}.$$

4.26 A Small Boolean Algebra Table

For two sets A, B, every point of U lies in exactly one of four regions:

$$A \cap B, \qquad A \cap B^c, \qquad A^c \cap B, \qquad A^c \cap B^c.$$

region	membership pattern	interpretation
$A \cap B$	$(1, 1)$	in both
$A \cap B^c$	$(1, 0)$	in A only
$A^c \cap B$	$(0, 1)$	in B only
$A^c \cap B^c$	$(0, 0)$	in neither

This partition is useful because every expression in A and B is a union of some of these four atoms.

Worked Example: Express $A \triangle B$ by Atoms

The symmetric difference contains points that lie in exactly one of A, B. Therefore

$$A \triangle B = (A \cap B^c) \cup (A^c \cap B).$$

The two pieces are disjoint, so

$$|A \triangle B| = |A \cap B^c| + |A^c \cap B|.$$

4.27 One Nontrivial Identity: Difference Distributes over Union

Solved Exercise

Prove

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

Use complements:

$$A \setminus (B \cup C) = A \cap (B \cup C)^c.$$

By De Morgan’s law,

$$(B \cup C)^c = B^c \cap C^c.$$

Therefore

$$A \setminus (B \cup C) = A \cap B^c \cap C^c.$$

On the other hand,

$$(A \setminus B) \cap (A \setminus C) = (A \cap B^c) \cap (A \cap C^c).$$

Associating and using $A \cap A = A$, this becomes

$$A \cap B^c \cap C^c.$$

Hence

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

4.28 Exercises Solved in This Note

Checklist of Concrete Results

This chapter note explicitly computed or proved:

1. union, intersection, complement, difference, and symmetric difference for concrete finite sets;
2. the distributive law
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C);$$
3. De Morgan's law
$$(A \cup B)^c = A^c \cap B^c;$$
4. the absorption law
$$A \cup (A \cap B) = A;$$
5. the formula
$$|\mathcal{P}(S)| = 2^{|S|};$$
6. the number of relations from A to B:
$$2^{|A||B|};$$
7. the number of functions from A to B:
$$|B|^{|A|};$$
8. preservation of intersections by preimages;
9. a counterexample showing images do not preserve intersections;
10. equivalence classes modulo 3;
11. implication as
$$P \Rightarrow Q \iff \neg P \vee Q;$$
12. negation of a quantified convergence statement;
13. the fact that quantifier order matters;
14. two-set and three-set inclusion–exclusion;
15. conditional probability from a finite table;

16. Bayes' formula with a numerical base-rate example;
17. the difference between disjointness and independence;
18. a derangement computation for three letters;
19. the law of total probability and a posterior probability calculation.

4.29 Working Summary

The algebra of sets gives a common language for three types of calculation.

Key Points

1. In pure set theory, it computes regions:
$$A \cup B, \quad A \cap B, \quad A^c, \quad A \setminus B.$$
2. In logic, it computes truth:
$$\text{and} \leftrightarrow \cap, \qquad \text{or} \leftrightarrow \cup, \qquad \text{not} \leftrightarrow {}^c.$$
3. In probability, it computes events:
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

The practical lesson is that one should not memorize set formulas as isolated rules. Each formula can be checked by one of three concrete methods:

element chasing, truth tables, finite counting examples.

These three methods are mathematically equivalent but useful in different situations.

Part II

Geometry and Spatial Structure

CHAPTER 5

Geometrical Constructions and Number Fields

Source Map

Courant and Robbins do not present straightedge-and-compass construction as a collection of isolated Euclidean tricks. The chapter follows a clear path: first, basic constructions show what the classical tools can actually do; second, coordinates translate these operations into algebra; third, constructible lengths are recognized as elements of number fields; finally, the classical impossibility problems are explained by field-degree obstructions. The guiding idea is:

$$\text{geometry of construction} \longrightarrow \text{algebra of operations} \longrightarrow \text{number fields} \longrightarrow \text{impossibility theorems.}$$

5.1 The Chapter's Actual Direction

The chapter begins with familiar Euclidean constructions, but its real purpose is not to teach drawing. Its purpose is to reveal a hidden algebra behind drawing.

With an unmarked straightedge, one may draw the line through two constructed points. With a compass, one may draw a circle whose center is a constructed point and whose radius is the distance between two constructed points. Therefore all new points arise from three kinds of intersections:

geometric intersection	algebraic equations	new operation
line with line	two linear equations	field operations
line with circle	linear + quadratic	one square root
circle with circle	two quadratics	one square root after subtraction

This table is the structural core of the chapter. It explains why classical construction problems are not merely problems of cleverness. They are problems about which numbers can be reached by repeated arithmetic operations and square roots.

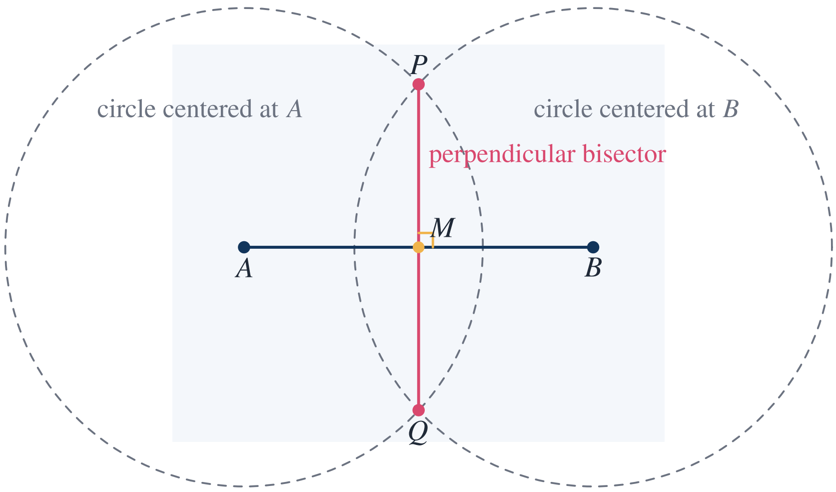
Takeaway

The straightedge and compass are quadratic tools. They can generate field operations and square roots, but they cannot in general generate cube roots or solve irreducible cubic equations.

5.2 The First Operation: Bisecting a Segment

The construction of the perpendicular bisector is the first example where a geometric procedure has a precise algebraic meaning.

Given a segment AB, draw two circles of the same radius, one centered at A, the other centered at B. Their two intersection points determine the perpendicular bisector of AB.



Proof that the Construction Works

Let P and Q be the two intersection points of the equal-radius circles. Since P lies on both circles,

$$PA = PB.$$

Since Q lies on both circles,

$$QA = QB.$$

Therefore both P and Q are equidistant from A and B. The locus of points equidistant from A and B is the perpendicular bisector of $\overline{AB}$. Hence the line PQ is the perpendicular bisector.
In coordinates, take

$$A = (-1, 0), \qquad B = (1, 0).$$

A point $X = (x, y)$ is equidistant from A and B exactly when

$$(x + 1)^2 + y^2 = (x - 1)^2 + y^2.$$

Expanding gives

$$x^2 + 2x + 1 + y^2 = x^2 - 2x + 1 + y^2,$$

so

$$4x = 0, \qquad x = 0.$$

Thus the locus is the vertical line $x = 0$, which is perpendicular to $\overline{AB}$ and passes through its midpoint.

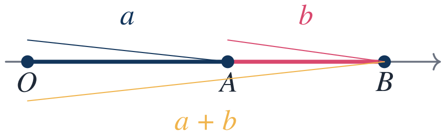
This is typical of the chapter's method: a classical construction is first understood geometrically, then translated into equations.

5.3 Constructing Arithmetic on a Line

Once a unit length is fixed, the real line becomes a geometric model of arithmetic. The point 0 is chosen, a point 1 is marked, and every constructible length can be placed on the same line.

5.3.1 Addition

If a and b are constructible lengths, then $a + b$ is constructible by laying the two segments consecutively on the same ray.



5.3.2 Subtraction

If $a > b > 0$, then $a - b$ is constructed by placing b inside a segment of length a.

$$OB = a, \qquad OA = b \quad \Longrightarrow \quad AB = a - b.$$

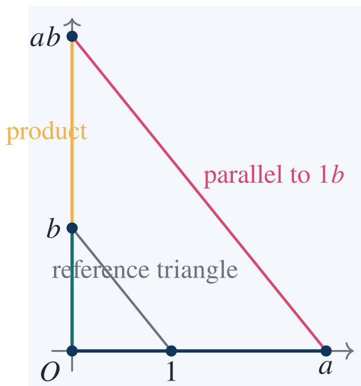
Takeaway

The first algebraic closure property of constructible lengths is closure under addition and subtraction.

5.4 Multiplication by Similar Triangles

Courant and Robbins then shift from simple laying-off of lengths to the stronger fact that multiplication and division can also be constructed. This is where similar triangles become algebra.

Assume a unit segment has been fixed. To construct ab, place 1 and a on one ray and b on another ray. Draw the line joining 1 to b, then draw through a a parallel line. The intercept on the second ray has length ab.



Proof of the Product Construction

The two triangles

$$\triangle OAP \qquad \text{and} \qquad \triangle OUB$$

are similar because the line through A is drawn parallel to UB. Hence

$$\frac{OP}{OB} = \frac{OA}{OU}.$$

Substitute

$$OB = b, \qquad OA = a, \qquad OU = 1.$$

Then

$$\frac{OP}{b} = a,$$

so

$$OP = ab.$$

Thus multiplication is constructible.

5.5 Division by Similar Triangles

The division construction is the same proportional idea reversed. To construct

$$\frac{a}{b},$$

place b and a on one ray, place 1 on another ray, and use a parallel line to transfer the ratio.

Construction and Proof of a/b

Place

$$OB = b, \qquad OP = a$$

on one ray and

$$OU = 1$$

on another ray. Draw the line BU. Through P, draw a line parallel to BU, meeting the second ray at X.
By similarity,

$$\frac{OX}{OU} = \frac{OP}{OB}.$$

Since $OU = 1$, $OP = a$, and $OB = b$,

$$OX = \frac{a}{b}.$$

Thus division by a nonzero constructible length is constructible.

Key Points

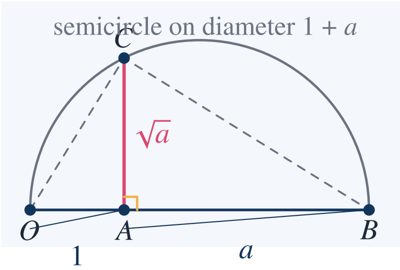
At this stage the constructible lengths are already closed under the field operations:

$$a, b \text{ constructible} \quad \Longrightarrow \quad a + b,\ a - b,\ ab,\ \frac{a}{b}\ (b \neq 0) \text{ constructible.}$$

This is the first reason the chapter's title contains "number fields."

5.6 Constructing Square Roots

The next essential step is square-root extraction. If $a > 0$ is constructible, then $\sqrt{a}$ is constructible by a semicircle construction.



Proof of the Square-Root Construction

Construct a segment OB of length $1 + a$, and choose A on it so that

$$OA = 1, \qquad AB = a.$$

Draw the semicircle with diameter OB. At A, erect a perpendicular meeting the semicircle at C. Since C lies on the semicircle with diameter OB, Thales' theorem gives

$$\angle OCB = 90^\circ.$$

In the right triangle OCB, the altitude to the hypotenuse satisfies

$$AC^2 = OA \cdot AB.$$

Substituting $OA = 1$ and $AB = a$, we obtain

$$AC^2 = a.$$

Therefore

$$AC = \sqrt{a}.$$

Concrete Example: Constructing $\sqrt{3}$

Take $a = 3$. Construct $OB = 4$, choose A with $OA = 1$, $AB = 3$, and erect the perpendicular to the semicircle. Then

$$AC^2 = 1 \cdot 3 = 3,$$

so

$$AC = \sqrt{3}.$$

This is not an approximation; it is an exact Euclidean construction of the length $\sqrt{3}$.

5.7 The Algebraic Meaning of the Constructions

Now the chapter's viewpoint changes. The constructions above show that constructible lengths are closed under

$$+, \quad -, \quad \times, \quad \div, \quad \sqrt{\ }.$$

This naturally suggests the language of fields.

A set $K \subseteq \mathbb{R}$ is a field if

$$a, b \in K$$

implies

$$a+b, \quad a-b, \quad ab, \quad \frac{a}{b}\ (b \neq 0)$$

also lie in K. The rational numbers $\mathbb{Q}$ are the basic example.

Takeaway

The constructible real numbers form a field containing $\mathbb{Q}$. Moreover, if $a > 0$ is constructible, then $\sqrt{a}$ is constructible.

This is the chapter's central conceptual move: Euclidean construction becomes a problem about field extensions.

5.8 Intersections Explain Why Only Square Roots Appear

Choose coordinates so that the initial points are

$$(0,0) \qquad \text{and} \qquad (1,0).$$

Suppose all currently constructed points have coordinates in a field K. We ask what kind of coordinates can appear after one more construction step.

A line through two known points has equation

$$ax + by = c, \qquad a, b, c \in K.$$

A circle with known center and radius has equation

$$(x-h)^2 + (y-k)^2 = r^2, \qquad h, k, r \in K.$$

Line–Line Intersection

Solving

$$a_1x + b_1y = c_1, \qquad a_2x + b_2y = c_2$$

uses only addition, subtraction, multiplication, and division. Hence the intersection point still has coordinates in K, provided the two lines are not parallel.

Line–Circle Intersection

If the line is

$$y = mx + d$$

and the circle is

$$(x - h)^2 + (y - k)^2 = r^2,$$

then substitution gives

$$(x - h)^2 + (mx + d - k)^2 = r^2.$$

This is a quadratic equation in x:

$$Ax^2 + Bx + C = 0, \qquad A, B, C \in K.$$

Therefore

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}.$$

The new coordinate lies in

$$K\left(\sqrt{B^2 - 4AC}\right).$$

Thus one construction step introduces at most one square root.

Circle–Circle Intersection

Two circles have equations

$$(x - h_1)^2 + (y - k_1)^2 = r_1^2,$$
$$(x - h_2)^2 + (y - k_2)^2 = r_2^2.$$

Subtracting them cancels x^2 and y^2, leaving a linear equation:

$$\alpha x + \beta y = \gamma.$$

Thus the two-circle intersection reduces to a line–circle intersection. Once again, the new coordinates require at most a square root over K.

5.9 Quadratic Field Towers

Repeated construction therefore gives a tower of fields

$$\mathbb{Q} = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_m,$$

where each step has degree at most 2:

$$[K_{j+1} : K_j] \leq 2.$$

Thus

$$[K_m : \mathbb{Q}] = 2^r$$

for some integer $r \leq m$, after removing redundant steps.

Key Points

If an algebraic number α is constructible, then

$$[\mathbb{Q}(\alpha) : \mathbb{Q}]$$

must divide a power of 2. Therefore an algebraic number of irreducible degree 3, 5, 6, 7, or any degree with an odd prime factor cannot be constructed by straightedge and compass.

This is where the chapter becomes powerful. It changes ancient geometry into an algebraic obstruction.

5.10 A Constructible Number: The Golden Ratio

The regular pentagon and the golden ratio are natural examples of constructibility. The golden ratio is

$$\phi = \frac{1+\sqrt{5}}{2}.$$

Since $\sqrt{5}$ is constructible, ϕ is constructible.

Algebraic Equation of the Golden Ratio

Let

$$\phi = \frac{1+\sqrt{5}}{2}.$$

Then

$$2\phi - 1 = \sqrt{5}.$$

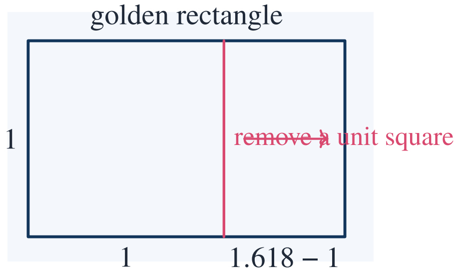
Squaring gives

$$4\phi^2 - 4\phi + 1 = 5.$$

Hence

$$\phi^2 - \phi - 1 = 0.$$

Thus ϕ is a quadratic algebraic number.



The equation

$$\phi^2 = \phi + 1$$

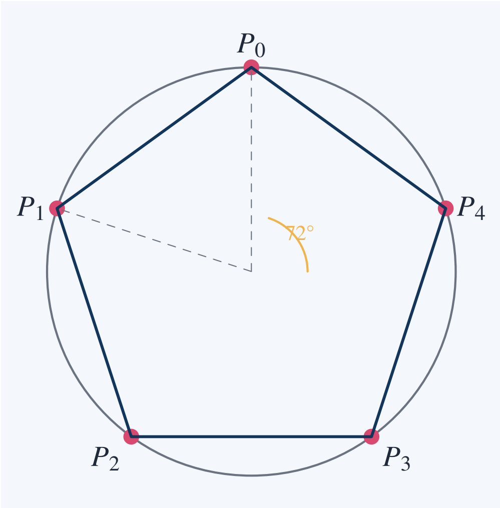
expresses the self-similarity of the golden rectangle: removing a square from a $\phi : 1$ rectangle leaves a smaller rectangle with the same ratio.

5.11 Regular Pentagon: A Successful Construction

The regular pentagon is constructible because its vertices require

$$\cos\frac{2\pi}{5},$$

and this number lies in a quadratic field.



regular pentagon on a constructible circle

Computation of $\cos(72^\circ)$

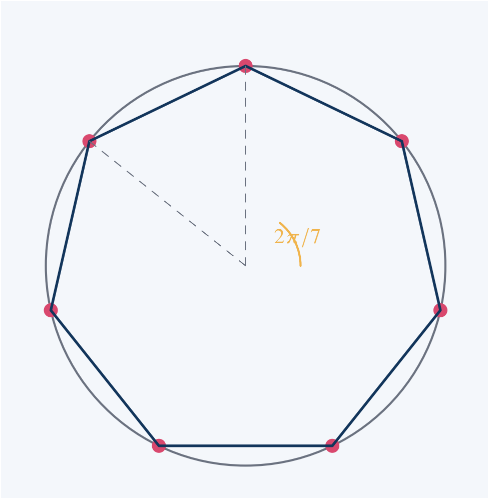
Let
$$\zeta = e^{2\pi i/5}.$$
Since $\zeta^5 = 1$ and $\zeta \neq 1$,
$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0.$$
Divide by ζ^2:
$$\zeta^{-2} + \zeta^{-1} + 1 + \zeta + \zeta^2 = 0.$$
Let
$$t = \zeta + \zeta^{-1} = 2\cos\frac{2\pi}{5}.$$
Then
$$\zeta^2 + \zeta^{-2} = t^2 - 2.$$
Hence
$$(t^2 - 2) + t + 1 = 0,$$
so
$$t^2 + t - 1 = 0.$$
Therefore
$$t = \frac{-1 + \sqrt{5}}{2},$$
and
$$\cos\frac{2\pi}{5} = \frac{t}{2} = \frac{\sqrt{5} - 1}{4}.$$
This is constructible because it belongs to $\mathbb{Q}(\sqrt{5})$.

This calculation is typical of the chapter: a polygon construction becomes a field calculation involving roots of unity.

5.12 Regular Heptagon: A Failed Construction

The regular heptagon is not constructible. The obstruction is already visible in the algebra of

$$2\cos\frac{2\pi}{7}.$$



regular heptagon: visually simple, algebraically cubic

Deriving the Cubic for the Heptagon

Let

$$\zeta = e^{2\pi i/7}.$$

Since

$$1+\zeta+\zeta^2+\zeta^3+\zeta^4+\zeta^5+\zeta^6 = 0,$$

divide by ζ^3:

$$\zeta^{-3}+\zeta^{-2}+\zeta^{-1}+1+\zeta+\zeta^2+\zeta^3 = 0.$$

Put

$$t = \zeta+\zeta^{-1} = 2\cos\frac{2\pi}{7}.$$

Then

$$\zeta^2+\zeta^{-2} = t^2-2,$$

and

$$\zeta^3+\zeta^{-3} = t^3-3t.$$

Substitute into the previous equation:

$$(t^3-3t)+(t^2-2)+t+1 = 0.$$

Thus

$$t^3+t^2-2t-1 = 0.$$

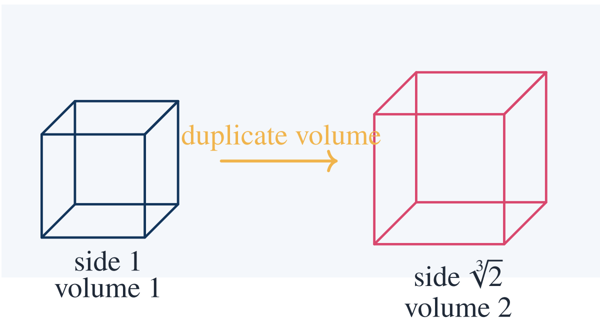
Irreducibility and Nonconstructibility

Consider
$$f(t) = t^3 + t^2 - 2t - 1.$$
The rational root test says any rational root must be
$$\pm 1.$$
But
$$f(1) = 1 + 1 - 2 - 1 = -1,$$
and
$$f(-1) = -1 + 1 + 2 - 1 = 1.$$
So f has no rational root. Since f is cubic, it is irreducible over $\mathbb{Q}$. Therefore
$$[\mathbb{Q}(t) : \mathbb{Q}] = 3.$$
But a constructible algebraic number must have degree dividing a power of 2. Since $3 \nmid 2^r$ for every r, t is not constructible.
Therefore the regular heptagon cannot be constructed by straightedge and compass.

This example shows the chapter's central method at full strength: the diagram is easy to imagine, but the number field forbids the construction.

5.13 Duplication of the Cube

The classical problem of duplicating the cube asks for a cube whose volume is twice that of the unit cube. If the original cube has side 1, the desired side length x satisfies
$$x^3 = 2.$$
Hence
$$x = \sqrt[3]{2}.$$



Nonconstructibility of $\sqrt[3]{2}$

The number $\sqrt[3]{2}$ is a root of
$$x^3 - 2 = 0.$$

By the rational root test, the possible rational roots are

$$\pm 1, \quad \pm 2.$$

None of them is a root:

$$1^3 - 2 = -1, \qquad (-1)^3 - 2 = -3,$$
$$2^3 - 2 = 6, \qquad (-2)^3 - 2 = -10.$$

Thus $x^3 - 2$ has no rational root. Since it is cubic, it is irreducible over $\mathbb{Q}$. Therefore

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3.$$

A constructible algebraic number must have degree dividing a power of 2. Since 3 does not divide any 2^r, the length $\sqrt[3]{2}$ is not constructible.
Therefore the cube cannot be duplicated by straightedge and compass.

5.14 Trisection of an Angle

Some angles can be trisected. For example, a right angle can be trisected because $30°$ is constructible. The classical problem asks whether every angle can be trisected. The answer is no.

A standard obstruction comes from the angle $60°$. If it could be trisected, then $20°$ would be constructible, and so

$$\cos 20°$$

would be constructible.

Use the triple-angle identity:

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta.$$

For $\theta = 20°$,

$$\cos 60° = 4\cos^3 20° - 3\cos 20°.$$

Since

$$\cos 60° = \frac{1}{2},$$

let

$$x = 2\cos 20°.$$

Then the equation becomes

$$x^3 - 3x - 1 = 0.$$

Irreducibility of the Trisection Cubic

Consider

$$f(x) = x^3 - 3x - 1.$$

The rational root test gives possible rational roots

$$\pm 1.$$

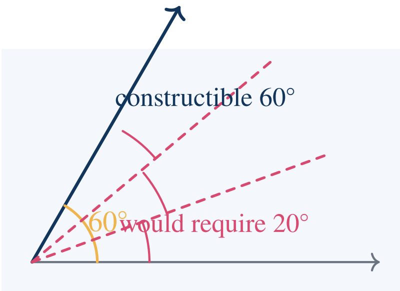
But

$$f(1) = 1 - 3 - 1 = -3,$$

and

$$f(-1) = -1 + 3 - 1 = 1.$$

So f has no rational root. Since f is cubic, it is irreducible over $\mathbb{Q}$. Therefore a root has degree 3 over $\mathbb{Q}$.
Thus $2\cos 20^\circ$ is not constructible, and 20° is not constructible from 60° by straightedge and compass.



The diagram is simple; the obstruction is algebraic. The desired trisection would force a cubic number of degree 3, but straightedge and compass cannot produce such a number.

5.15 Regular Polygons and Fermat Primes

The regular n-gon problem asks for which n a regular n-gon can be constructed. The answer is one of the most striking results connecting Euclidean geometry with arithmetic.

A regular n-gon is constructible if and only if

$$n = 2^k p_1 p_2 \cdots p_r,$$

where the p_i are distinct Fermat primes:

$$p_i = 2^{2^{m_i}} + 1.$$

The known Fermat primes begin

$$3, \quad 5, \quad 17, \quad 257, \quad 65537.$$

Concrete Examples

The regular triangle is constructible:

$$3 = 2^{2^0} + 1.$$

The square is constructible:

$$4 = 2^2.$$

The regular pentagon is constructible:

$$5 = 2^{2^1} + 1.$$

The regular 15-gon is constructible:

$$15 = 3 \cdot 5,$$

a product of distinct Fermat primes.
The regular 17-gon is constructible:

$$17 = 2^{2^2} + 1.$$

The regular 7-gon is not constructible, because 7 is not a Fermat prime and its associated cosine has cubic degree.

5.16 Squaring the Circle

The problem of squaring the circle asks for a square with the same area as a given circle. If the circle has radius 1, its area is

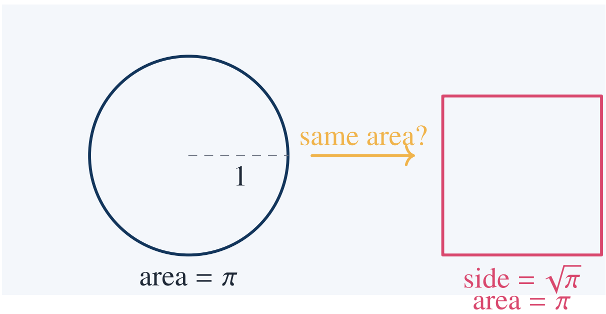
$$\pi.$$

A square of equal area must have side length s satisfying

$$s^2 = \pi,$$

so

$$s = \sqrt{\pi}.$$



The impossibility of this construction is deeper than the previous cubic examples. It depends on the fact that π is transcendental, so it is not algebraic over $\mathbb{Q}$. Every constructible number is algebraic. Therefore

$$\sqrt{\pi}$$

cannot be constructible, because if $\sqrt{\pi}$ were algebraic, then

$$\pi = (\sqrt{\pi})^2$$

would also be algebraic, contradicting the transcendence of π.

Takeaway

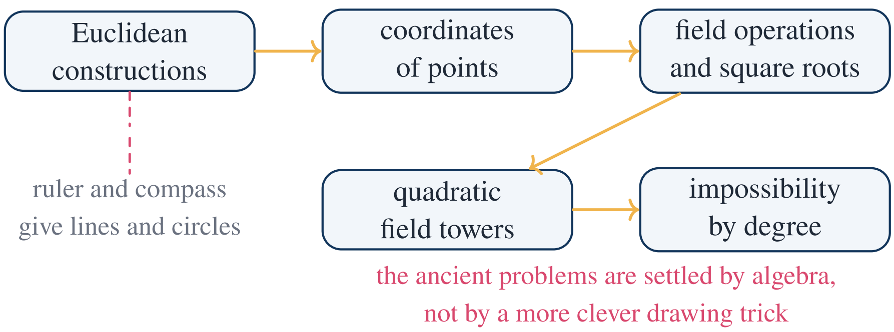
Duplication of the cube and angle trisection fail because of cubic algebraic degree. Squaring the circle fails for a stronger reason: it would require a transcendental length.

5.17 A Worked Classification Table

The following table summarizes what the algebraic criterion predicts.

object	required number	degree/type	constructible?
segment sum	$a+b$	field operation	yes
product	ab	field operation	yes
square root	$\sqrt{a}$	quadratic	yes
golden ratio	$\frac{1+\sqrt{5}}{2}$	2	yes
regular pentagon	$\cos(2\pi/5)$	2	yes
cube duplication	$\sqrt[3]{2}$	3	no
trisect 60°	$2\cos 20^\circ$	3	no
regular heptagon	$2\cos(2\pi/7)$	3	no
squaring circle	$\sqrt{\pi}$	transcendental	no

5.18 The Chapter's Method in One Diagram



This is the way the chapter advances. It begins with constructions one can perform by hand, but the endpoint is a classification principle: if a proposed construction would require a number outside all quadratic field towers, then no straightedge-and-compass procedure can produce it.

5.19 Exercises Solved in This Note

Concrete Results Established Above

This note explicitly constructed or proved:

1. the perpendicular bisector of a segment;
2. the algebraic equation of the perpendicular bisector in coordinates;
3. construction of $a+b$ and $a-b$;
4. construction of ab by similar triangles;
5. construction of a/b by similar triangles;
6. construction of $\sqrt{a}$ by a semicircle;
7. closure of constructible lengths under field operations and square roots;
8. the fact that each construction step introduces at most a quadratic extension;
9. the quadratic field-tower obstruction to constructibility;
10. constructibility of the golden ratio;

11. constructibility of the regular pentagon through

$$\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4};$$

12. nonconstructibility of the regular heptagon through the irreducible cubic

$$t^3 + t^2 - 2t - 1 = 0;$$

13. impossibility of duplicating the cube through

$$x^3 - 2 = 0;$$

14. impossibility of trisecting 60° through

$$x^3 - 3x - 1 = 0;$$

15. impossibility of squaring the circle using the transcendence of π.

5.20 Working Summary

The chapter is not merely about drawing figures. It is about discovering that Euclidean construction has an algebraic boundary.

Key Points

straightedge and compass $\Rightarrow$ linear and quadratic equations $\Rightarrow$ quadratic field towers $\Rightarrow$ degree obs

The most important lesson is that impossibility can be proved positively. To show that the cube cannot be duplicated, the angle cannot always be trisected, or the regular heptagon cannot be constructed, one does not need to test every possible drawing. One proves that every possible drawing is trapped inside a quadratic algebraic world, and then shows that the desired object lies outside that world.

CHAPTER 6

Projective Geometry, Axiomatics, and Non-Euclidean Geometries

Source Map

This chapter changes the meaning of geometry. In Euclidean geometry, the central quantities are length, angle, congruence, and area. In projective geometry, these metric notions are deliberately discarded, and the surviving notions are incidence, alignment, concurrence, and cross ratio. Courant and Robbins use this shift to prepare the axiomatic point of view: a geometry is not necessarily the geometry of physical space, but a system of undefined objects and relations satisfying specified axioms. Non-Euclidean geometries then appear not as contradictions of Euclid, but as consistent alternatives to Euclid's parallel postulate.

6.1 The Route of the Chapter

The chapter proceeds through a precise chain of ideas:

$$\text{perspective drawing} \longrightarrow \text{points at infinity} \longrightarrow \text{projective transformations} \longrightarrow \text{axioms and models} \longrightarrow \text{non-Euclidean geo}$$

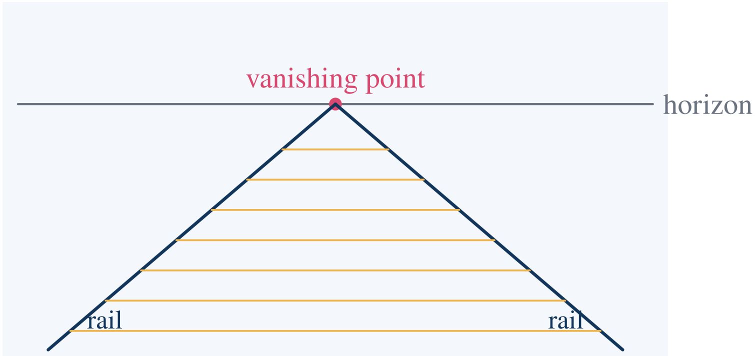
The first move is visual: in perspective, parallel railway tracks appear to meet at a vanishing point. The second move is mathematical: instead of saying that parallel lines "do not meet," projective geometry says that they meet on the line at infinity. The third move is algebraic: homogeneous coordinates make this statement exact. The fourth move is logical: once geometry is formulated by incidence axioms, alternative geometries become possible.

Takeaway

The chapter's main idea is not that Euclidean geometry is false. It is that Euclidean geometry is only one possible geometry. Projective and non-Euclidean geometries arise by changing which properties are treated as fundamental.

6.2 Perspective: Parallel Lines Meet in a Vanishing Point

In ordinary Euclidean geometry, two distinct parallel lines do not meet. In perspective drawing, however, they appear to meet at a vanishing point on the horizon. Projective geometry makes this visual fact into an exact mathematical principle.



Parallel tracks in space become converging lines in a projection.

The mathematical lesson is that projection does not preserve length or angle. The sleepers in the drawing above do not have equal visible length, although they may represent equal physical lengths. What is preserved is incidence: points on a line still project to points on a line.

Key Points

Projective geometry keeps:

$$\text{collinearity}, \quad \text{concurrence}, \quad \text{incidence}, \quad \text{cross ratio}.$$

It discards:

$$\text{length}, \quad \text{angle}, \quad \text{parallelism as a separate relation}, \quad \text{area}.$$

6.3 Affine Plane versus Projective Plane

The affine plane $\mathbb{A}^2$ is the ordinary coordinate plane with points

$$(x, y).$$

The projective plane $\mathbb{P}^2$ adds points at infinity. The clean way to do this is to use homogeneous coordinates:

$$[X : Y : Z],$$

where

$$[X : Y : Z] = [\lambda X : \lambda Y : \lambda Z] \qquad \lambda \neq 0.$$

Ordinary affine points are embedded as

$$(x, y) \longmapsto [x : y : 1].$$

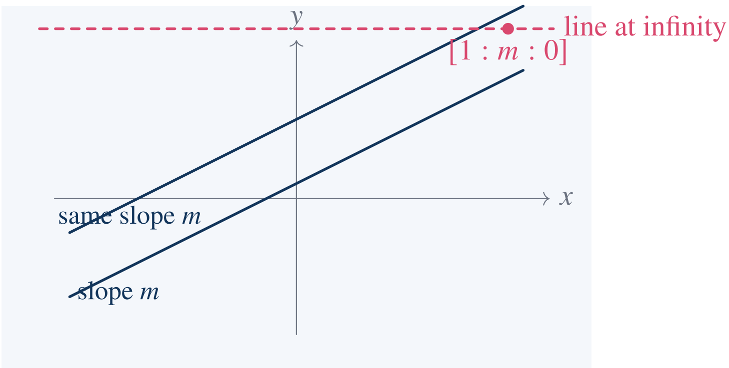
Points with

$$Z = 0$$

are points at infinity. The line

$$Z = 0$$

is called the line at infinity.



Two parallel affine lines with the same slope meet at the same point at infinity. This is why projective geometry removes the special exception for parallel lines.

Solved Exercise: Intersection of Parallel Lines in Projective Coordinates

Consider the two affine lines

$$y = 2x + 1, \qquad y = 2x - 3.$$

In homogeneous coordinates $[X : Y : Z]$, these become

$$Y = 2X + Z, \qquad Y = 2X - 3Z.$$

Equivalently,

$$-2X + Y - Z = 0,$$
$$-2X + Y + 3Z = 0.$$

Subtract the two equations:

$$4Z = 0.$$

Thus

$$Z = 0.$$

Substitute into

$$-2X + Y - Z = 0$$

to get

$$Y = 2X.$$

Therefore the intersection point is

$$[X : Y : Z] = [1 : 2 : 0].$$

This is the point at infinity corresponding to slope 2.

6.4 Lines and Incidence in Homogeneous Coordinates

In $\mathbb{P}^2$, a line has equation

$$aX + bY + cZ = 0.$$

A point $[X : Y : Z]$ lies on this line exactly when

$$aX + bY + cZ = 0.$$

Thus incidence becomes a dot product:

$$(a, b, c) \cdot (X, Y, Z) = 0.$$

Worked Example: Line Through Two Projective Points

Find the line through

$$P = [1 : 2 : 1], \qquad Q = [3 : 1 : 1].$$

A line

$$aX + bY + cZ = 0$$

through P and Q must satisfy

$$a + 2b + c = 0,$$

$$3a + b + c = 0.$$

Subtract:

$$-2a + b = 0,$$

so

$$b = 2a.$$

Substitute into the first equation:

$$a + 4a + c = 0,$$

so

$$c = -5a.$$

Taking $a = 1$, the line is

$$X + 2Y - 5Z = 0.$$

Check:

$$1 + 2 \cdot 2 - 5 = 0,$$

$$3 + 2 \cdot 1 - 5 = 0.$$

Worked Example: Intersection of Two Projective Lines

Find the intersection of

$$L_1 : X + Y - Z = 0, \qquad L_2 : 2X - Y + Z = 0.$$

From L_1,

$$Z = X + Y.$$

Substitute into L_2:

$$2X - Y + (X + Y) = 0,$$

so

$$3X = 0.$$

Hence $X = 0$. Then $Z = Y$. Taking $Y = 1$, the intersection point is

$$[0 : 1 : 1].$$

This coordinate method shows why projective geometry is algebraically clean: points and lines are treated symmetrically.

6.5 Duality: Points and Lines Exchange Roles

Projective geometry has a duality principle: many theorems remain true when one exchanges the words "point" and "line," and exchanges "collinear" with "concurrent."

For example:

statement	dual statement
two points determine a line	two lines determine a point
three points are collinear	three lines are concurrent
a point lies on a line	a line passes through a point

This is not just wordplay. In homogeneous coordinates, a point is represented by a triple

$$[X : Y : Z],$$

and a line is represented by a triple

$$[a : b : c],$$

with incidence

$$aX + bY + cZ = 0.$$

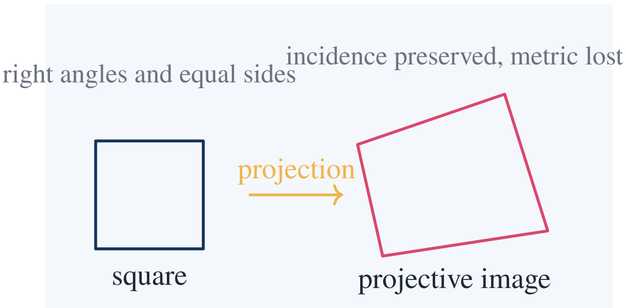
The formal symmetry makes duality natural.

Takeaway

Duality is one of the reasons projective geometry is more symmetric than Euclidean geometry. Euclidean geometry distinguishes points and lines by metric intuition; projective geometry treats incidence as primary.

6.6 Projection Does Not Preserve Distance

A central lesson of the chapter is that different geometries study different invariants. Projection can turn a square into a quadrilateral that is not a square. The straight lines remain straight, but equal lengths and right angles disappear.



Concrete Point

In the square, the diagonals bisect each other and the sides have equal length. After a projective transformation, the image remains a quadrilateral whose diagonals meet, but the side lengths and angles are generally changed.
Thus the property

"the four vertices form a quadrilateral"

is projective, while the property

"the four vertices form a square"

is not projective.

6.7 Cross Ratio: A Metric-Free Quantity

Although projective transformations do not preserve ordinary distance, they do preserve the cross ratio of four collinear points. For points on the affine line with coordinates a, b, c, d, define

$$(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}.$$

Worked Example: A Cross Ratio

Compute

$$(0, 1; 3, \infty).$$

Using the limiting form as $d \to \infty$,

$$(0, 1; 3, \infty) = \lim_{d\to\infty} \frac{(3-0)(d-1)}{(3-1)(d-0)}.$$

This gives

$$(0, 1; 3, \infty) = \frac{3}{2} \lim_{d\to\infty} \frac{d-1}{d} = \frac{3}{2}.$$

Verification Under a Projective Map

Let

$$T(x) = \frac{2x+1}{x+1}.$$

Then

$$T(0) = 1, \qquad T(1) = \frac{3}{2}, \qquad T(3) = \frac{7}{4}, \qquad T(\infty) = 2.$$

Compute

$$(1, \tfrac{3}{2}; \tfrac{7}{4}, 2) = \frac{(\frac{7}{4}-1)(2-\frac{3}{2})}{(\frac{7}{4}-\frac{3}{2})(2-1)}.$$

Now

$$\frac{7}{4} - 1 = \frac{3}{4}, \qquad 2 - \frac{3}{2} = \frac{1}{2},$$

$$\frac{7}{4} - \frac{3}{2} = \frac{1}{4}, \qquad 2 - 1 = 1.$$

Therefore

$$(1, \tfrac{3}{2}; \tfrac{7}{4}, 2) = \frac{(\frac{3}{4})(\frac{1}{2})}{(\frac{1}{4})(1)} = \frac{3/8}{1/4} = \frac{3}{2}.$$

This equals

$$(0, 1; 3, \infty).$$

Thus this example shows the invariance of cross ratio under a projective transformation.

Takeaway

Cross ratio plays the role of a projective numerical invariant. It survives projection even though lengths and angles do not.

6.8 Desargues' Theorem

Desargues' theorem is one of the central results of projective geometry. It states that if two triangles are in perspective from a point, then the three intersection points of corresponding sides are collinear.

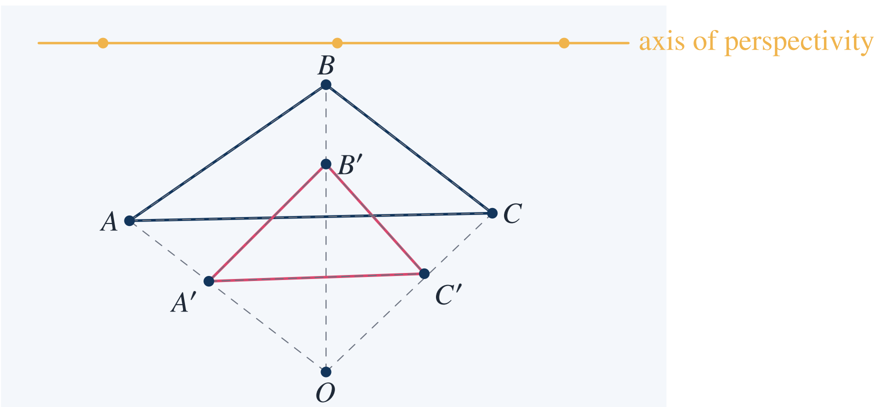
In concrete terms, if the lines

$$AA', \quad BB', \quad CC'$$

meet at one point, then the three points

$$AB \cap A'B', \qquad AC \cap A'C', \qquad BC \cap B'C'$$

lie on one line.



The diagram shows the idea, but a coordinate example makes the theorem checkable.

Coordinate Example of Desargues' Theorem

Let the center of perspectivity be

$$O = [0 : 0 : 1].$$

Take

$$A = [1 : 0 : 1], \qquad B = [0 : 1 : 1], \qquad C = [1 : 1 : 1],$$

and

$$A' = [2 : 0 : 1], \qquad B' = [0 : 2 : 1], \qquad C' = [2 : 2 : 1].$$

Then A, A', O are collinear; B, B', O are collinear; and C, C', O are collinear.
Now compute corresponding side intersections.
The line AB has equation

$$X + Y - Z = 0,$$

and the line $A'B'$ has equation

$$X + Y - 2Z = 0.$$

Their intersection satisfies

$$Z = 0, \qquad X + Y = 0.$$

Thus

$$AB \cap A'B' = [1 : -1 : 0].$$

The line AC is

$$X - Z = 0,$$

and $A'C'$ is

$$X - 2Z = 0.$$

Their intersection is

$$[0 : 1 : 0].$$

The line BC is

$$Y - Z = 0,$$

and $B'C'$ is

$$Y - 2Z = 0.$$

Their intersection is

$$[1 : 0 : 0].$$

The three intersection points

$$[1 : -1 : 0], \qquad [0 : 1 : 0], \qquad [1 : 0 : 0]$$

all satisfy

$$Z = 0.$$

Hence they are collinear on the line at infinity.

This example is especially useful because it shows a Euclidean shadow of Desargues' theorem: two homothetic triangles have corresponding sides that are parallel, and their intersections occur at infinity.

6.9 Pappus' Theorem in Coordinates

Pappus' theorem gives another fundamental incidence statement. If A, B, C lie on one line and A', B', C' lie on another line, then the three intersection points

$$AB' \cap A'B, \qquad AC' \cap A'C, \qquad BC' \cap B'C$$

are collinear.

Concrete Coordinate Verification

Take the two lines

$$y = 0 \qquad \text{and} \qquad y = 1.$$

Let

$$A = (0, 0), \quad B = (1, 0), \quad C = (2, 0),$$

and

$$A' = (0, 1), \quad B' = (1, 1), \quad C' = (2, 1).$$

First compute

$$X = AB' \cap A'B.$$

The line AB' is

$$y = x.$$

The line $A'B$ is

$$x + y = 1.$$

Solving gives

$$x = \frac{1}{2}, \qquad y = \frac{1}{2}.$$

Thus

$$X = \left(\frac{1}{2}, \frac{1}{2}\right).$$

Next compute

$$Y = AC' \cap A'C.$$

The line AC' is

$$y = \frac{x}{2},$$

and the line $A'C$ is

$$y = 1 - \frac{x}{2}.$$

Solving gives

$$x = 1, \qquad y = \frac{1}{2}.$$

Thus

$$Y = \left(1, \frac{1}{2}\right).$$

Finally compute

$$Z = BC' \cap B'C.$$

The line BC' is

$$y = x - 1,$$

and the line $B'C$ is

$$y = 2 - x.$$

Solving gives

$$x = \frac{3}{2}, \qquad y = \frac{1}{2}.$$

Thus

$$Z = \left(\frac{3}{2}, \frac{1}{2}\right).$$

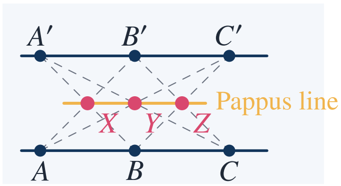
The three points are

$$\left(\frac{1}{2}, \frac{1}{2}\right), \qquad \left(1, \frac{1}{2}\right), \qquad \left(\frac{3}{2}, \frac{1}{2}\right).$$

They all lie on the line

$$y = \frac{1}{2}.$$

This verifies Pappus' theorem in this concrete case.



6.10 Axiomatic Geometry: What Is Being Assumed?

After projective geometry, the next conceptual step is axiomatics. An axiomatic theory begins with undefined objects and relations. For example, one may leave "point," "line," and "lies on" undefined, and specify only axioms governing them.

Typical incidence axioms for projective planes are:

I. Any two distinct points lie on exactly one line.
II. Any two distinct lines meet in exactly one point.
III. There exist four points no three of which are collinear.

The second axiom is false in the Euclidean affine plane because parallel lines do not meet. It becomes true in the projective plane after adding points at infinity.

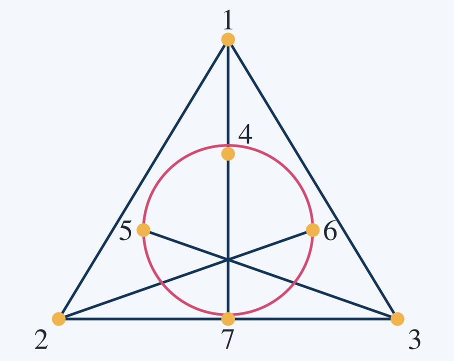
Takeaway

Axioms do not describe what points and lines "really are." They specify the rules that any valid model must satisfy.

6.11 A Finite Model: The Fano Plane

The Fano plane is the smallest projective plane. It has 7 points and 7 lines. Each line contains 3 points, and each point lies on 3 lines.

This finite example is valuable because it destroys the intuition that geometry must be infinite or metric.



Fano plane: a finite projective geometry

One algebraic construction of the Fano plane uses the vector space

$$\mathbb{F}_2^3.$$

The points are the nonzero vectors:

$$100,\ 010,\ 001,\ 110,\ 101,\ 011,\ 111.$$

Since the only nonzero scalar in $\mathbb{F}_2$ is 1, each nonzero vector is its own projective point.
A line consists of three nonzero vectors

$$u, \quad v, \quad u+v.$$

For example,

$$100, \quad 010, \quad 110$$

form a line because

$$100 + 010 = 110.$$

Checking the Projective Plane Axiom in the Fano Plane

Take two distinct points represented by nonzero vectors $u, v \in \mathbb{F}_2^3$. They determine the line

$$\{u, v, u+v\}.$$

This line contains exactly three points.
For instance, the two points

$$101 \qquad \text{and} \qquad 011$$

determine the third point

$$101 + 011 = 110.$$

Hence the line is

$$\{101, 011, 110\}.$$

This verifies concretely that any two distinct points determine a unique line.

6.12 Why Axiomatic Method Matters

The Fano plane shows why the axiomatic method is powerful. A theorem proved from incidence axioms alone is true not only in the usual projective plane over $\mathbb{R}$, but also in finite projective planes such as the Fano plane.

Thus axiomatics separates three questions:

question	meaning
consistency	is there at least one model?
independence	can an axiom fail while the others hold?
categoricity	do the axioms determine one structure up to isomorphism?

The emergence of non-Euclidean geometry is exactly an independence phenomenon: Euclid's parallel postulate is not forced by the other Euclidean axioms.

6.13 Euclid's Parallel Postulate

One form of Euclid's parallel postulate says:

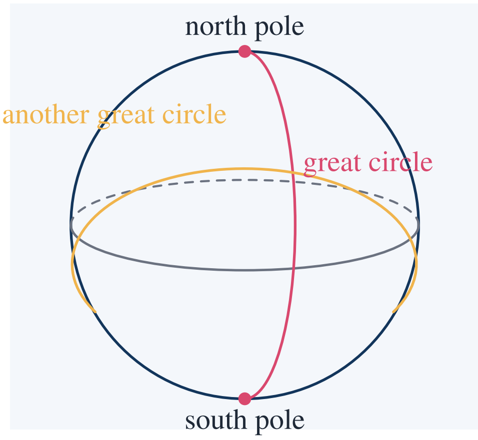
> Given a line L and a point $P \notin L$, there is exactly one line through P parallel to L.

For centuries, mathematicians tried to prove this postulate from the others. The discovery of non-Euclidean geometries showed that this cannot be done. There are geometries where the other basic incidence and order assumptions can be interpreted, but the parallel postulate is replaced by a different rule.

geometry	parallel behavior
Euclidean	exactly one parallel
spherical/elliptic	no parallels
hyperbolic	infinitely many parallels

6.14 Spherical Geometry: No Parallel Lines

On a sphere, the natural analogues of straight lines are great circles. Any two distinct great circles meet in two antipodal points. Therefore spherical geometry has no parallel lines.



Concrete Example: Two Meridians

On Earth, take the meridian of longitude 0° and the meridian of longitude 90°. They are both great circles. They meet at the North Pole and the South Pole.
Thus even though the two meridians look like different "straight lines" on the sphere, they are not parallel. In spherical geometry, no two great circles are parallel.

6.15 A Spherical Triangle with Three Right Angles

Spherical triangles have angle sums greater than π. A standard example is the triangle bounded by the equator and two perpendicular meridians.

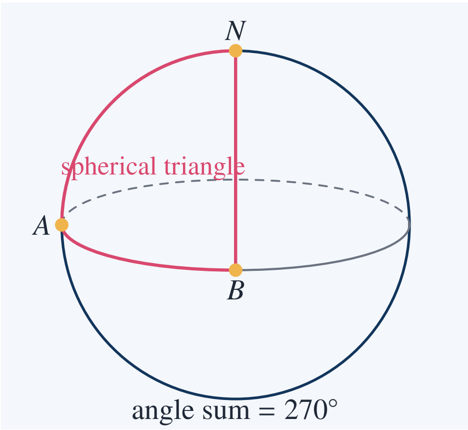
Take the three vertices:

$$\begin{aligned} N &= \text{North Pole}, \\ A &= \text{point on equator at longitude } 0^\circ, \\ B &= \text{point on equator at longitude } 90^\circ. \end{aligned}$$

The sides are arcs of great circles:

$$NA, \quad NB, \quad AB.$$

Each angle is 90°.



Computation of the Angle Sum

At A, the equator and the meridian meet at a right angle:

$$\angle A = 90^\circ.$$

At B, the equator and the other meridian also meet at a right angle:

$$\angle B = 90^\circ.$$

At N, the two meridians differ by 90° in longitude, so

$$\angle N = 90^\circ.$$

Therefore

$$\angle A + \angle B + \angle N = 90^\circ + 90^\circ + 90^\circ = 270^\circ.$$

This is greater than the Euclidean value 180°.

This example shows concretely that the theorem “the angles of a triangle sum to 180°” is not purely logical. It depends on the geometry.

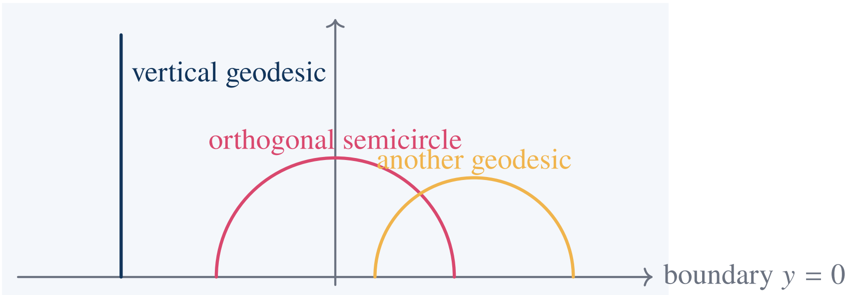
6.16 Hyperbolic Geometry: Infinitely Many Parallels

In the upper half-plane model of hyperbolic geometry, the space is

$$\mathbb{H} = \{z = x + iy : y > 0\}.$$

The boundary line $y = 0$ is not part of the space. Hyperbolic lines are:

$$\text{vertical Euclidean rays,}$$
$$\text{or semicircles orthogonal to the boundary } y = 0.$$



Distance Along a Vertical Hyperbolic Line

The hyperbolic metric in the upper half-plane is

$$ds = \frac{\sqrt{dx^2 + dy^2}}{y}.$$

Along the vertical line $x = 0$, from $i = (0, 1)$ to $2i = (0, 2)$, we have

$$dx = 0,$$

so

$$ds = \frac{dy}{y}.$$

Therefore the hyperbolic distance is

$$d(i, 2i) = \int_1^2 \frac{dy}{y} = \log 2.$$

This shows that even when the picture is drawn in Euclidean coordinates, the distance being measured is not Euclidean distance.

6.17 Many Hyperbolic Parallels Through One Point

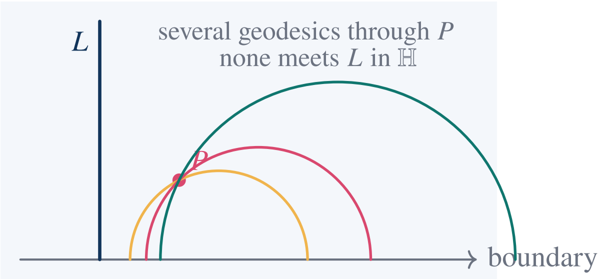
Let L be the vertical geodesic

$$x = 0, \qquad y > 0.$$

Take the point

$$P = (1, 1).$$

There are infinitely many hyperbolic geodesics through P that do not meet L.



Why These Are Hyperbolic Parallels

A hyperbolic geodesic in the upper half-plane must meet the boundary orthogonally. A Euclidean circle centered on the boundary line has this orthogonality property.
For example, the circle centered at 2 with radius $\sqrt{2}$ has equation

$$(x-2)^2 + y^2 = 2.$$

It passes through $P = (1, 1)$, since

$$(1-2)^2 + 1^2 = 2.$$

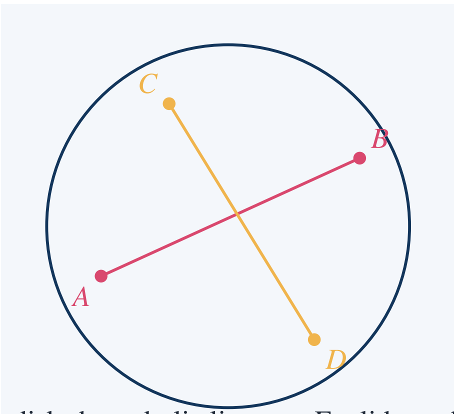
Its boundary endpoints are

$$2-\sqrt{2} \qquad \text{and} \qquad 2+\sqrt{2},$$

both positive. Therefore this geodesic stays to the right of the vertical geodesic $x = 0$ and does not meet L inside the upper half-plane.
Varying the center gives infinitely many such geodesics through P. Thus the Euclidean uniqueness of parallels fails in hyperbolic geometry.

6.18 Beltrami–Klein Model: Projective Geometry Enters Hyperbolic Geometry

One of the most elegant connections in the chapter is that hyperbolic geometry can be modeled projectively. In the Beltrami–Klein disk model, the points are inside a disk and the hyperbolic lines are chords of the disk.



Klein disk: hyperbolic lines are Euclidean chords

The boundary circle is not part of the hyperbolic plane. It is the absolute. Distances can be defined from projective cross ratios. For points A, B on a chord, let P, Q be the two boundary endpoints on the same chord. Then hyperbolic distance is essentially

$$\frac{1}{2}\log(P, Q; A, B).$$

Concrete Distance on a Diameter

Work on the diameter $(-1, 1)$. Let

$$A = 0, \qquad B = r, \qquad 0 < r < 1.$$

The boundary endpoints are

$$P = -1, \qquad Q = 1.$$

The hyperbolic distance in the Klein/Poincare normalization is

$$d(0, r) = \frac{1}{2}\log\frac{1+r}{1-r}.$$

For example, if

$$r = \frac{1}{2},$$

then

$$d(0, \tfrac{1}{2}) = \frac{1}{2}\log\frac{1+\frac{1}{2}}{1-\frac{1}{2}} = \frac{1}{2}\log 3.$$

Thus the boundary is infinitely far away: as $r \to 1^-$,

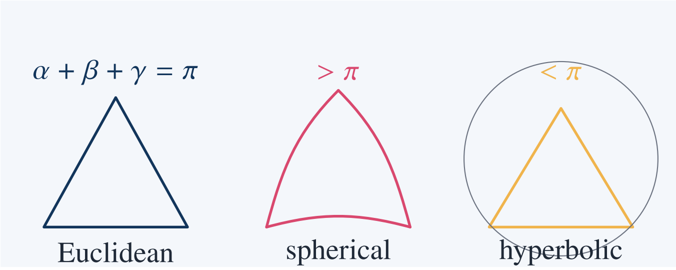
$$\frac{1}{2}\log\frac{1+r}{1-r} \to +\infty.$$

This connects the first half of the chapter to the second. Projective invariants are not merely formal; they can be used to build a non-Euclidean metric.

6.19 Triangle Angle Sums: Three Geometries Compared

The sum of the angles of a triangle separates the three geometries:

geometry	triangle angle sum
Euclidean	$= \pi$
spherical	$> \pi$
hyperbolic	$< \pi$



Concrete Spherical Computation

The spherical triangle formed by the equator and two meridians separated by 90° has three right angles. Hence its angle sum is

$$\frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2} = \frac{3\pi}{2}.$$

The excess over π is

$$\frac{3\pi}{2} - \pi = \frac{\pi}{2}.$$

On the unit sphere, the area of a spherical triangle equals its spherical excess. Therefore this triangle has area

$$\frac{\pi}{2},$$

which is one eighth of the area of the unit sphere:

$$\frac{\pi/2}{4\pi} = \frac{1}{8}.$$

6.20 What Counts as a Model?

A model of an axiomatic system is an interpretation of the primitive terms that makes all the axioms true. This is the central logical idea behind the acceptance of non-Euclidean geometry.

For example:

axiomatic word	spherical model	projective model
point	pair of antipodal points on a sphere	$[X : Y : Z]$
line	great circle, antipodal points identified	$aX + bY + cZ = 0$
incidence	point lies on great circle	$aX + bY + cZ = 0$

In the elliptic interpretation of spherical geometry, antipodal points are identified. Then two lines meet in exactly one point, not two. This gives a geometry with no parallels.

Takeaway

A model converts questions about truth into questions about interpretation. Once a non-Euclidean model satisfies the chosen axioms, the geometry is logically consistent relative to the model's background mathematics.

6.21 Independence of the Parallel Postulate

The existence of hyperbolic models shows that Euclid's parallel postulate cannot be derived from the other axioms. The logic is:

1. Construct a model satisfying the other Euclidean axioms.
2. Show that the parallel postulate fails in that model.
3. Conclude that the parallel postulate is independent of the others.

This is the same style of reasoning that later became standard throughout modern mathematics: independence is proved by building alternative models.

Concrete Failure in the Hyperbolic Upper Half-Plane

Let

$$L = \{(0, y) : y > 0\}$$

be a hyperbolic line, and let

$$P = (1, 1).$$

The circles

$$(x - c)^2 + y^2 = (1 - c)^2 + 1$$

with suitable centers $c > 1$ pass through P and meet the boundary orthogonally. Many of them have both endpoints on the positive real axis, so they do not meet L in the upper half-plane. Thus there are many hyperbolic lines through P that do not meet L. This contradicts Euclid's uniqueness of parallels but is valid in the hyperbolic model.

6.22 A Projective Reading of Non-Euclidean Geometry

The chapter's deepest structural point is that projective geometry and non-Euclidean geometry are not separate topics. Projective geometry teaches which properties survive projection. Non-Euclidean geometry then shows that metric notions themselves can be rebuilt from different invariants.

The Klein disk makes this explicit:

$$\text{points inside a conic} \quad + \quad \text{chords as lines} \quad + \quad \text{cross-ratio distance} \quad = \quad \text{hyperbolic geometry}.$$

This is why the chapter places projective geometry before axiomatics and non-Euclidean geometry. Projective geometry trains the reader to stop treating Euclidean length and angle as sacred. Once that shift is made, non-Euclidean geometry becomes a natural next step.

Key Points

The intellectual progression is:

$$\text{Euclidean metric facts are not projective invariants;}$$

$$\text{incidence can be axiomatized independently of measurement;}$$

$$\text{alternative metrics can be modeled consistently;}$$

$$\text{therefore Euclidean geometry is one model, not the only possible geometry.}$$

6.23 Exercises Solved in This Note

Checklist of Concrete Results

This note explicitly computed or proved:

1. parallel affine lines

$$y = 2x + 1, \qquad y = 2x - 3$$

meet at the projective point

$$[1 : 2 : 0];$$

2. the line through
$$[1:2:1] \quad \text{and} \quad [3:1:1]$$
is
$$X + 2Y - 5Z = 0;$$
3. the intersection of
$$X + Y - Z = 0 \quad \text{and} \quad 2X - Y + Z = 0$$
is
$$[0:1:1];$$
4. the cross ratio
$$(0, 1; 3, \infty) = \frac{3}{2};$$
5. the projective map
$$T(x) = \frac{2x+1}{x+1}$$
preserves the cross ratio in the explicit example above;
6. a coordinate example of Desargues' theorem where the Desargues axis is the line at infinity
$$Z = 0;$$
7. a coordinate verification of Pappus' theorem with Pappus line
$$y = \frac{1}{2};$$
8. the Fano plane construction from
$$\mathbb{F}_2^3;$$
9. the fact that spherical geometry has no parallel great circles;
10. a spherical triangle with angle sum
$$\frac{3\pi}{2};$$
11. the hyperbolic distance
$$d(i, 2i) = \log 2$$
in the upper half-plane;
12. explicit hyperbolic parallels through
$$P = (1, 1)$$
to the geodesic
$$x = 0;$$
13. the Klein disk distance formula on a diameter:
$$d(0, r) = \frac{1}{2} \log \frac{1+r}{1-r}.$$

6.24 Working Summary

The chapter begins with the visual fact that perspective changes the appearance of figures. It ends with the logical fact that different geometries can be valid models of different axiom systems. The bridge between these two ends is projective geometry.

Key Points

Perspective $\Rightarrow$ points at infinity $\Rightarrow$ homogeneous coordinates $\Rightarrow$ incidence theorems $\Rightarrow$ axiomatic models $\Rightarrow$ non-

The practical mathematical lesson is that geometry should be studied through its invariants. Euclidean geometry studies invariants of rigid motion, projective geometry studies invariants of projection, and hyperbolic geometry studies invariants of its own metric. Once this is understood, non-Euclidean geometry is no longer a paradox. It is an alternative structure with its own coherent invariants.

CHAPTER 7

Appendix to Chapter IV: Geometry in More Than Three Dimensions

Source Map

This appendix extends geometry beyond ordinary three-dimensional space. The method is not mystical visualization. Courant and Robbins' path is more concrete: replace visual intuition by coordinates, equations, projections, cross-sections, and combinatorial data. Once geometry is written in the language of $\mathbb{R}^n$, dimensions higher than three become accessible by calculation even when they cannot be directly pictured.

7.1 The Chapter's Route: From Sight to Coordinates

In two and three dimensions, geometry can often be guided by sight. In four or more dimensions, direct visualization fails. The appendix therefore shifts the foundation from vision to algebra.

The progression is:

$$\text{ordinary space} \longrightarrow \text{coordinate space} \longrightarrow n\text{-dimensional distance and angle} \longrightarrow \text{hyperplanes and spheres} \longrightarrow \text{higher-dim}$$

The central move is that a point in n-dimensional space is simply an ordered n-tuple:

$$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n.$$

The dimension is the number of independent coordinates needed to locate a point.

space	point	degrees of freedom
$\mathbb{R}^1$	(x)	1
$\mathbb{R}^2$	(x, y)	2
$\mathbb{R}^3$	(x, y, z)	3
$\mathbb{R}^4$	(x_1, x_2, x_3, x_4)	4
$\mathbb{R}^n$	$(x_1, \dots, x_n)$	n

Takeaway

Higher-dimensional geometry is not inaccessible because we cannot see it. It becomes accessible when geometry is expressed through coordinates, equations, and invariants.

7.2 Distance in $\mathbb{R}^n$

The distance between two points

$$x = (x_1, \dots, x_n), \qquad y = (y_1, \dots, y_n)$$

is defined by the same Pythagorean pattern:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Worked Example: Distance in $\mathbb{R}^4$

Let

$$x = (1, 2, 0, -1), \qquad y = (4, -2, 3, 1).$$

Then

$$x - y = (-3, 4, -3, -2).$$

Therefore

$$\begin{aligned} d(x, y) &= \sqrt{(-3)^2 + 4^2 + (-3)^2 + (-2)^2} \\ &= \sqrt{9 + 16 + 9 + 4} \\ &= \sqrt{38}. \end{aligned}$$

So

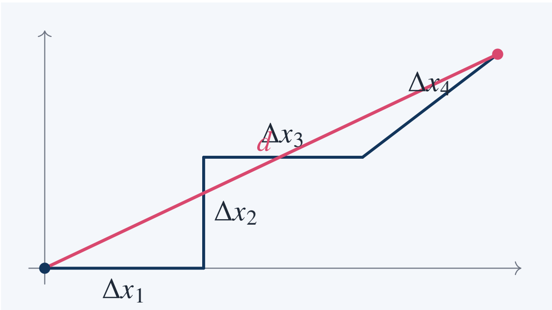
$$d(x, y) = \sqrt{38}.$$

This formula is not an arbitrary definition. It is forced by repeatedly applying the Pythagorean theorem. In $\mathbb{R}^3$,

$$d^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2.$$

In $\mathbb{R}^4$, one adds one more independent square:

$$d^2 = (\Delta x_1)^2 + (\Delta x_2)^2 + (\Delta x_3)^2 + (\Delta x_4)^2.$$



The diagram is only a schematic. It does not literally display four perpendicular axes in the plane. Its role is to show the repeated Pythagorean accumulation of independent coordinate differences.

7.3 Dot Product and Angle

The dot product in $\mathbb{R}^n$ is

$$x \cdot y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

It gives the length formula

$$\|x\| = \sqrt{x \cdot x},$$

and the angle formula

$$\cos\theta = \frac{x \cdot y}{\|x\|\|y\|}.$$

Worked Example: Orthogonality in $\mathbb{R}^4$

Let

$$u = (1, 1, 1, 1), \qquad v = (1, -1, 1, -1).$$

Then

$$u \cdot v = 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot 1 + 1 \cdot (-1) = 1 - 1 + 1 - 1 = 0.$$

Thus

$$u \perp v.$$

Even though u and v are four-dimensional vectors, orthogonality is still detected by one computation:

$$u \cdot v = 0.$$

Worked Example: A Non-Right Angle in $\mathbb{R}^4$

Let

$$a = (1, 0, 1, 0), \qquad b = (1, 1, 0, 0).$$

Then

$$a \cdot b = 1.$$

Also,

$$\|a\| = \sqrt{1^2 + 0^2 + 1^2 + 0^2} = \sqrt{2},$$

and

$$\|b\| = \sqrt{1^2 + 1^2 + 0^2 + 0^2} = \sqrt{2}.$$

Therefore

$$\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}.$$

Hence

$$\theta = 60^\circ.$$

7.4 Hyperplanes

In $\mathbb{R}^2$, a line is given by one linear equation:

$$ax + by = c.$$

In $\mathbb{R}^3$, a plane is given by one linear equation:

$$ax + by + cz = d.$$

In $\mathbb{R}^n$, the corresponding object is a hyperplane:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

A hyperplane has dimension $n - 1$. Thus:

ambient space	one linear equation gives	dimension
$\mathbb{R}^2$	line	1
$\mathbb{R}^3$	plane	2
$\mathbb{R}^4$	hyperplane	3
$\mathbb{R}^n$	hyperplane	$n-1$

Worked Example: A Hyperplane in $\mathbb{R}^4$

Consider
$$H = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : 2x_1 - x_2 + x_3 + 3x_4 = 5\}.$$
This is a 3-dimensional hyperplane inside $\mathbb{R}^4$.
A normal vector is
$$n = (2, -1, 1, 3).$$
Indeed, the equation can be written as
$$n \cdot x = 5.$$
Check whether
$$p = (1, 0, 0, 1)$$
lies on H. Compute
$$2(1) - 0 + 0 + 3(1) = 5.$$
Therefore
$$p \in H.$$
Check
$$q = (0, 0, 0, 0).$$
Then
$$2(0) - 0 + 0 + 3(0) = 0 \neq 5.$$
Thus
$$q \notin H.$$

7.5 Distance from a Point to a Hyperplane

For a hyperplane
$$H = \{x \in \mathbb{R}^n : a \cdot x = b\},$$
the distance from a point p to H is
$$\operatorname{dist}(p, H) = \frac{|a \cdot p - b|}{\|a\|}.$$

Worked Example in $\mathbb{R}^4$

Let
$$H : \ 2x_1 - x_2 + x_3 + 3x_4 = 5.$$

Then

$$a = (2, -1, 1, 3), \qquad b = 5.$$

Let

$$p = (0, 0, 0, 0).$$

Then

$$a \cdot p = 0.$$

Also,

$$\|a\| = \sqrt{2^2 + (-1)^2 + 1^2 + 3^2} = \sqrt{4 + 1 + 1 + 9} = \sqrt{15}.$$

Therefore

$$\operatorname{dist}(p, H) = \frac{|0 - 5|}{\sqrt{15}} = \frac{5}{\sqrt{15}} = \frac{\sqrt{15}}{3}.$$

This computation is important because it shows that even in dimension four, one can still make exact metric statements without drawing the object.

7.6 Spheres in Higher Dimensions

The unit sphere in $\mathbb{R}^n$ is

$$S^{n-1} = \{x \in \mathbb{R}^n : x_1^2 + x_2^2 + \cdots + x_n^2 = 1\}.$$

The notation is slightly surprising at first:

$$S^{n-1} \subset \mathbb{R}^n.$$

The exponent records the dimension of the sphere itself, not the dimension of the surrounding space.

object	equation	dimension
$S^0 \subset \mathbb{R}^1$	$x^2 = 1$	0
$S^1 \subset \mathbb{R}^2$	$x^2 + y^2 = 1$	1
$S^2 \subset \mathbb{R}^3$	$x^2 + y^2 + z^2 = 1$	2
$S^3 \subset \mathbb{R}^4$	$x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1$	3

Cross-Sections of the Three-Sphere

The unit 3-sphere is

$$S^3 = \{(x_1, x_2, x_3, x_4) : x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1\}.$$

Fix

$$x_4 = t.$$

Then the remaining coordinates satisfy

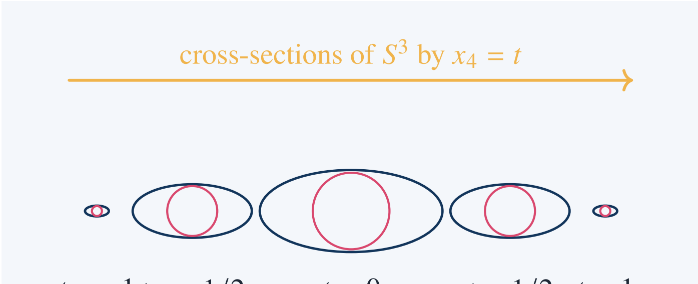
$$x_1^2 + x_2^2 + x_3^2 = 1 - t^2.$$

If $|t| < 1$, this is an ordinary 2-sphere in $\mathbb{R}^3$ with radius

$$\sqrt{1 - t^2}.$$

Thus S^3 may be understood through spherical cross-sections:

$$t = -1 \Rightarrow \text{point}, \quad -1 < t < 1 \Rightarrow \text{sphere}, \quad t = 1 \Rightarrow \text{point}.$$



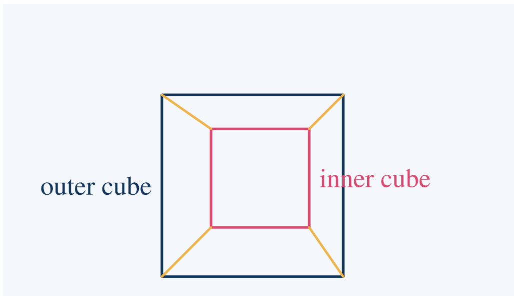
A four-dimensional sphere is studied through ordinary three-dimensional slices.

The picture is not a literal drawing of S^3. It is a sequence of slices. This is one of the main techniques for understanding higher-dimensional objects.

7.7 Projection and Shadow

A higher-dimensional object can be studied through its lower-dimensional projections. This is exactly the same idea used in perspective drawing, but now it is applied to four-dimensional geometry.

For example, a cube in $\mathbb{R}^3$ can be projected to the plane as a hexagon-like or parallelogram-like image. A 4-dimensional cube, or tesseract, can be projected to $\mathbb{R}^3$ and then drawn on paper.



projection of a tesseract

The two cubes in the picture should not be read as one cube inside another cube in ordinary space. They represent a projection of a four-dimensional cube into a lower-dimensional drawing.

Takeaway

A projection is a shadow. It may preserve incidence and adjacency, but it does not preserve all metric properties. The tesseract drawing is a shadow of a four-dimensional object, not the object itself.

7.8 The n-Cube

The unit n-cube is

$$[0, 1]^n = \{(x_1, \dots, x_n) : 0 \le x_i \le 1\}.$$

Its vertices are all points

$$(\varepsilon_1, \dots, \varepsilon_n), \qquad \varepsilon_i \in \{0, 1\}.$$

Thus the number of vertices is

$$2^n.$$

Worked Example: Vertices of the 4-Cube

For $n = 4$, the vertices are

$$(\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4), \qquad \varepsilon_i \in \{0, 1\}.$$

There are

$$2^4 = 16$$

vertices.
Examples are

$$(0,0,0,0), \quad (1,0,0,0), \quad (1,1,0,0), \quad (1,1,1,1).$$

Two vertices are connected by an edge exactly when they differ in one coordinate. For instance,

$$(0,1,1,0)$$

is adjacent to

$$(1,1,1,0), \quad (0,0,1,0), \quad (0,1,0,0), \quad (0,1,1,1).$$

Thus each vertex of the 4-cube has degree 4.

The long diagonal of the unit n-cube runs from

$$(0,0,\ldots,0)$$

to

$$(1,1,\ldots,1).$$

Its length is

$$\sqrt{1^2 + \cdots + 1^2} = \sqrt{n}.$$

Worked Example: Diagonal of the 4-Cube

The long diagonal of the unit 4-cube has length

$$\sqrt{1^2 + 1^2 + 1^2 + 1^2} = \sqrt{4} = 2.$$

Thus a unit tesseract has 4-dimensional diagonal length 2.

7.9 Counting Faces of the n-Cube

A face of an n-cube is determined by choosing some coordinates to vary and fixing the remaining coordinates to 0 or 1.

A k-dimensional face is obtained by:

1. choose k coordinates that are allowed to vary;
2. fix the remaining $n - k$ coordinates to 0 or 1.

Therefore the number of k-faces is

$$\binom{n}{k} 2^{n-k}.$$

Worked Example: Faces of the 4-Cube

For the 4-cube, the number of k-faces is

$$\binom{4}{k}2^{4-k}.$$

Vertices:

$$k=0: \qquad \binom{4}{0}2^4 = 16.$$

Edges:

$$k=1: \qquad \binom{4}{1}2^3 = 4\cdot 8 = 32.$$

Square faces:

$$k=2: \qquad \binom{4}{2}2^2 = 6\cdot 4 = 24.$$

Cubic cells:

$$k=3: \qquad \binom{4}{3}2^1 = 4\cdot 2 = 8.$$

The whole 4-cube:

$$k=4: \qquad \binom{4}{4}2^0 = 1.$$

Thus the boundary of the tesseract has

16 vertices, 32 edges, 24 square faces, 8 cubic cells.

This is a concrete advantage of the coordinate method: the 4-cube cannot be seen directly, but its face numbers can be computed exactly.

7.10 Euler Relation in Four Dimensions

For a convex polyhedron in three dimensions,

$$V-E+F=2.$$

For the boundary of a convex 4-polytope, the corresponding Euler relation is

$$V-E+F-C=0,$$

where C is the number of 3-dimensional cells.

Check for the 4-Cube

For the tesseract,

$$V=16, \qquad E=32, \qquad F=24, \qquad C=8.$$

Then

$$V-E+F-C=16-32+24-8.$$

Compute:

$$16-32=-16, \qquad -16+24=8, \qquad 8-8=0.$$

Thus
$$V - E + F - C = 0.$$

Check for the 4-Simplex

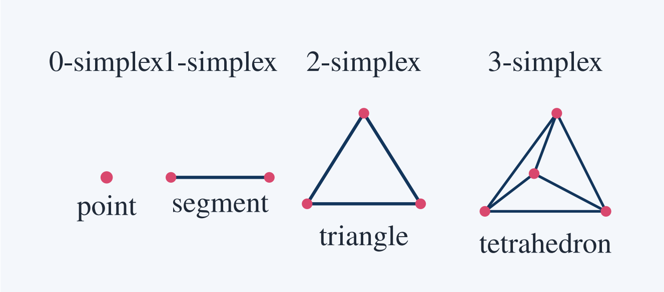
A 4-simplex has
$$5$$
vertices. Its face numbers are:
$$V = \binom{5}{1} = 5,$$
$$E = \binom{5}{2} = 10,$$
$$F = \binom{5}{3} = 10,$$
$$C = \binom{5}{4} = 5.$$
Therefore
$$V - E + F - C = 5 - 10 + 10 - 5 = 0.$$
Again the four-dimensional Euler relation holds.

7.11 The n-Simplex

The simplest n-dimensional polytope is the n-simplex. It is the convex hull of $n+1$ affinely independent points.

The standard n-simplex in $\mathbb{R}^n$ has vertices
$$0,\ e_1,\ e_2, \dots, e_n,$$
where
$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1).$$

dimension	simplex
0	point
1	line segment
2	triangle
3	tetrahedron
4	4-simplex



Volume of the Standard n-Simplex

The standard n-simplex is

$$\Delta_n = \{(x_1, \ldots, x_n) : x_i \geq 0,\ x_1 + \cdots + x_n \leq 1\}.$$

Its volume is

$$\mathrm{Vol}(\Delta_n) = \frac{1}{n!}.$$

Check in low dimensions:
For $n = 1$,

$$\Delta_1 = [0, 1],$$

so its length is

$$1 = \frac{1}{1!}.$$

For $n = 2$,

$$\Delta_2 = \{(x, y) : x, y \geq 0,\ x + y \leq 1\},$$

a right triangle with legs 1 and 1. Its area is

$$\frac{1}{2} = \frac{1}{2!}.$$

For $n = 3$,

$$\Delta_3 = \{(x, y, z) : x, y, z \geq 0,\ x + y + z \leq 1\},$$

a tetrahedron with coordinate intercepts 1, 1, 1. Its volume is

$$\frac{1}{6} = \frac{1}{3!}.$$

7.12 Barycentric Coordinates

A point inside a simplex can be described by barycentric coordinates. For a triangle with vertices A, B, C, a point P has the form

$$P = \lambda_1 A + \lambda_2 B + \lambda_3 C,$$

where

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \qquad \lambda_i \geq 0.$$

For the standard n-simplex with vertices

$$0, e_1, \ldots, e_n,$$

a point

$$x = (x_1, \dots, x_n)$$

inside the simplex satisfies

$$x_i \geq 0, \qquad x_1 + \cdots + x_n \leq 1.$$

Its barycentric coordinates are

$$\lambda_0 = 1 - (x_1 + \cdots + x_n), \qquad \lambda_i = x_i \quad (1 \leq i \leq n).$$

Worked Example in a Triangle

Let the triangle have vertices

$$A = (0,0), \qquad B = (1,0), \qquad C = (0,1).$$

Consider

$$P = \left(\frac{1}{4}, \frac{1}{3}\right).$$

Then

$$\lambda_B = \frac{1}{4}, \qquad \lambda_C = \frac{1}{3},$$

and

$$\lambda_A = 1 - \frac{1}{4} - \frac{1}{3} = 1 - \frac{3}{12} - \frac{4}{12} = \frac{5}{12}.$$

Therefore

$$P = \frac{5}{12}A + \frac{1}{4}B + \frac{1}{3}C.$$

All coefficients are nonnegative and sum to 1, so P lies inside the triangle.

7.13 The Cross-Polytope

The n-dimensional cross-polytope is the convex hull of

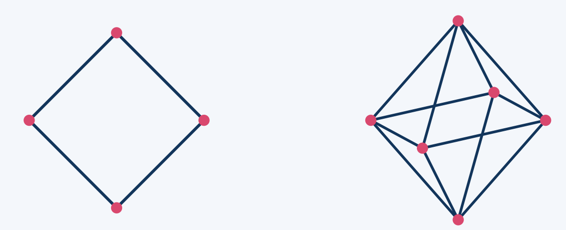
$$\pm e_1, \ \pm e_2, \ \dots, \ \pm e_n.$$

In $\mathbb{R}^2$, it is a diamond. In $\mathbb{R}^3$, it is the octahedron. In $\mathbb{R}^4$, it has 8 vertices:

$$\pm e_1, \quad \pm e_2, \quad \pm e_3, \quad \pm e_4.$$

It can be described by the inequality

$$|x_1| + |x_2| + \cdots + |x_n| \leq 1.$$



2-cross-polytope 3-cross-polytope: octahedron

Worked Example: Vertices of the 4-Cross-Polytope

In $\mathbb{R}^4$, the vertices are

$$\begin{aligned} &(1,0,0,0), \quad (-1,0,0,0), \\ &(0,1,0,0), \quad (0,-1,0,0), \\ &(0,0,1,0), \quad (0,0,-1,0), \\ &(0,0,0,1), \quad (0,0,0,-1). \end{aligned}$$

There are

$$2n = 8$$

vertices when $n = 4$.

7.14 Cube, Simplex, and Cross-Polytope Compared

The three most important regular families in higher dimensions are:

polytope	vertices	defining model	dimension n
simplex	$n+1$	$\mathrm{conv}(0, e_1, \dots, e_n)$	n
cube	2^n	$[0,1]^n$	n
cross-polytope	$2n$	$\mathrm{conv}(\pm e_1, \dots, \pm e_n)$	n

Concrete Comparison in Dimension 4

For $n = 4$:
The 4-simplex has

$$n + 1 = 5$$

vertices.
The 4-cube has

$$2^4 = 16$$

vertices.
The 4-cross-polytope has

$$2n = 8$$

vertices.
Thus these three regular-looking objects are already combinatorially very different in dimension 4.

7.15 Regular Polytopes in Different Dimensions

The classification of regular convex polytopes changes dramatically with dimension.

dimension	regular convex polytopes
2	infinitely many regular polygons
3	five Platonic solids
4	six regular convex 4-polytopes
$n \geq 5$	only three families: simplex, cube, cross-polytope

In dimension 3, the five Platonic solids are:

tetrahedron, cube, octahedron, dodecahedron, icosahedron.

In dimension 4, there are six:

5-cell, 8-cell, 16-cell, 24-cell, 120-cell, 600-cell.

Takeaway

Dimension 4 is exceptional. It has more regular convex polytopes than dimensions $n \geq 5$, where only the simplex, cube, and cross-polytope families survive.

Concrete Identification

The names correspond to numbers of three-dimensional cells:

name	cells	family
5-cell	5 tetrahedra	4-simplex
8-cell	8 cubes	4-cube
16-cell	16 tetrahedra	4-cross-polytope
24-cell	24 octahedra	exceptional
120-cell	120 dodecahedra	exceptional
600-cell	600 tetrahedra	exceptional

7.16 Why Four Dimensions Are Special

The exceptional nature of four dimensions appears repeatedly in geometry and topology. In this appendix, it appears through regular polytopes. The reason is that the combinatorics of possible angle arrangements is more flexible in dimension 4 than in dimension $n \geq 5$.

A rough geometric principle is:

regular polytopes exist only when enough cells fit around lower-dimensional faces

without leaving a gap or overlapping.

In three dimensions, this is the usual vertex-angle condition for Platonic solids. In four dimensions, one must fit polyhedral cells around edges. In dimensions $n \geq 5$, the constraints become so rigid that only the three infinite families remain.

Key Points

The appendix is not merely saying that higher dimensions have analogues of cubes and tetrahedra. It shows that the classification problem itself changes with dimension.

7.17 A Four-Dimensional Object Through Sections

One way to understand a 4-polytope is by slicing it with hyperplanes. The same method already works in three dimensions: slicing a cube by horizontal planes gives squares. In four dimensions, slicing a tesseract by three-dimensional hyperplanes gives ordinary polyhedra.

For the unit 4-cube

$$[0, 1]^4,$$

consider slices by

$$x_4 = t.$$

If

$$0 \le t \le 1,$$

then the slice is

$$[0, 1]^3.$$

So every slice perpendicular to the x_4-axis is an ordinary cube.

Worked Slice of the 4-Cube

Let

$$Q_4 = [0, 1]^4.$$

Fix

$$x_4 = \frac{1}{2}.$$

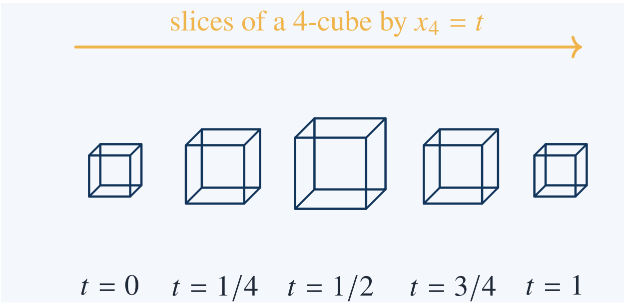
Then the slice is

$$\{(x_1, x_2, x_3, \tfrac{1}{2}) : 0 \le x_1, x_2, x_3 \le 1\}.$$

Ignoring the constant fourth coordinate, this is exactly the ordinary cube

$$[0, 1]^3.$$

Thus a tesseract can be studied as a one-parameter stack of cubes.



This picture is schematic, but the algebraic statement is exact: each constant-x_4 slice of a 4-cube is a 3-cube.

7.18 A Four-Dimensional Object Through Vertices and Edges

Another way to understand a high-dimensional polytope is through its graph: vertices are points, edges connect adjacent vertices.

For the n-cube, vertices are binary strings:

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n), \qquad \varepsilon_i \in \{0, 1\}.$$

Edges connect strings that differ in exactly one coordinate.

Worked Example: A Path in the 4-Cube Graph

Move from

$$0000$$

to

$$1111.$$

One shortest path is

$$0000 \to 1000 \to 1100 \to 1110 \to 1111.$$

Each step changes exactly one coordinate, so each step is an edge of the 4-cube.
The distance in the graph between two vertices is the number of coordinates in which they differ.
This is the Hamming distance. Thus

$$d_{\text{graph}}(0000, 1111) = 4.$$

This gives a discrete way to understand the tesseract without drawing it. Its edge graph is the graph of all binary strings of length 4, connected by single-bit changes.

7.19 High Dimension and Volume Concentration

Higher-dimensional geometry also changes metric intuition. For example, the volume of the unit n-ball behaves very differently from the volume of the unit n-cube.

The unit n-cube

$$[-1, 1]^n$$

has volume

$$2^n.$$

The unit n-ball has volume

$$V_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}.$$

For small n, this gives:

n	V_n
1	2
2	π
3	$\frac{4}{3}\pi$
4	$\frac{1}{2}\pi^2$
5	$\frac{8}{15}\pi^2$

Worked Comparison in Dimension 4

The unit 4-ball has volume

$$V_4 = \frac{\pi^2}{2}.$$

The cube

$$[-1, 1]^4$$

has volume

$$2^4 = 16.$$

Thus the ratio is

$$\frac{V_4}{16} = \frac{\pi^2}{32}.$$

Numerically,

$$\frac{\pi^2}{32} \approx 0.308.$$

So the unit 4-ball occupies about 30.8% of the volume of the surrounding cube.

The key point is not the numerical value alone. It is that high-dimensional volume behaves differently from low-dimensional visual intuition.

7.20 The Diagonal Phenomenon in High Dimensions

In the unit cube

$$[0, 1]^n,$$

the long diagonal has length

$$\sqrt{n}.$$

Therefore the diagonal grows as dimension increases, even though each coordinate remains between 0 and 1.

Worked Example: Distance from Center to a Vertex

The center of the unit n-cube is

$$c = \left(\frac{1}{2}, \ldots, \frac{1}{2}\right).$$

A vertex is

$$v = (0, 0, \ldots, 0).$$

Then

$$d(c, v) = \sqrt{\left(\frac{1}{2}\right)^2 + \cdots + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{n}{4}} = \frac{\sqrt{n}}{2}.$$

Thus in dimension $n = 100$,

$$d(c, v) = \frac{\sqrt{100}}{2} = 5.$$

Even though every coordinate difference is only $1/2$, the accumulated distance is 5.

This is one of the first signs that high-dimensional geometry has its own scale behavior.

7.21 Linear Subspaces and Dimension Counting

A linear subspace of $\mathbb{R}^n$ is a set closed under addition and scalar multiplication. If it has a basis of k independent vectors, then its dimension is k.

Worked Example: A Two-Dimensional Plane in $\mathbb{R}^4$

Let

$$u = (1, 0, 1, 0), \qquad v = (0, 1, 0, 1).$$

The set

$$W = \{su + tv : s, t \in \mathbb{R}\}$$

is a two-dimensional plane inside $\mathbb{R}^4$. Explicitly,

$$su + tv = (s, t, s, t).$$

Thus

$$W = \{(x_1, x_2, x_3, x_4) : x_3 = x_1,\ x_4 = x_2\}.$$

It is described by two independent linear equations:

$$x_3 - x_1 = 0, \qquad x_4 - x_2 = 0.$$

Since it lies in $\mathbb{R}^4$ and has two independent linear constraints, its dimension is

$$4 - 2 = 2.$$

This example shows how dimension can be counted either by parameters or by constraints:

$$\text{dimension} = \text{number of coordinates} - \text{number of independent linear equations}.$$

7.22 Projection onto a Subspace

Projection generalizes the idea of dropping a perpendicular. In $\mathbb{R}^n$, the projection of a vector x onto a unit vector u is

$$\text{proj}_u(x) = (x \cdot u)u.$$

Worked Example in $\mathbb{R}^4$

Let

$$x = (2, 1, 0, 3),$$

and project onto the direction

$$u = \frac{1}{2}(1, 1, 1, 1).$$

First check

$$\|u\|^2 = \frac{1}{4}(1 + 1 + 1 + 1) = 1,$$

so u is a unit vector.
Compute

$$x \cdot u = \frac{1}{2}(2 + 1 + 0 + 3) = 3.$$

Therefore

$$\text{proj}_u(x) = 3u = \frac{3}{2}(1, 1, 1, 1) = \left(\frac{3}{2}, \frac{3}{2}, \frac{3}{2}, \frac{3}{2}\right).$$

Projection is one of the main ways high-dimensional objects are made lower-dimensional without losing all structure.

7.23 A Concrete Optimization Example

High-dimensional geometry is also the language of optimization. The closest point problem is a geometric problem in any dimension.

Closest Point on a Hyperplane

Find the point on

$$H:\ 2x_1 - x_2 + x_3 + 3x_4 = 5$$

closest to the origin.
The normal vector is

$$a = (2, -1, 1, 3).$$

The closest point from the origin to the hyperplane lies in the direction of a, so it has the form

$$p = ta.$$

Impose the hyperplane equation:

$$a \cdot p = 5.$$

Since

$$p = ta,$$

we have

$$a \cdot p = t(a \cdot a).$$

Now

$$a \cdot a = 15.$$

So

$$15t = 5, \qquad t = \frac{1}{3}.$$

Therefore

$$p = \frac{1}{3}(2, -1, 1, 3) = \left(\frac{2}{3}, -\frac{1}{3}, \frac{1}{3}, 1\right).$$

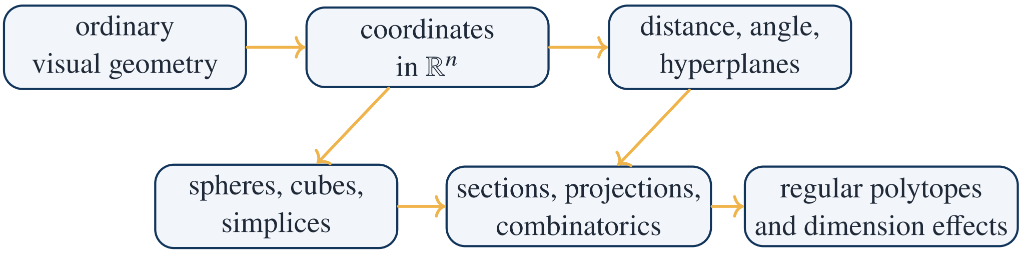
Its distance from the origin is

$$\|p\| = \frac{1}{3}\sqrt{15} = \frac{\sqrt{15}}{3},$$

matching the distance formula computed earlier.

This example shows why the coordinate method is not merely formal. It allows one to solve geometric problems in dimensions that cannot be pictured.

7.24 The Appendix's Method in One Diagram



high-dimensional geometry is controlled by equations, slices, shadows, and counts

7.25 Exercises Solved in This Note

Checklist of Concrete Results

This note explicitly computed or proved:

1. the distance formula in $\mathbb{R}^n$;
2. $$d\big((1,2,0,-1),(4,-2,3,1)\big) = \sqrt{38};$$
3. orthogonality of
$$(1,1,1,1) \quad \text{and} \quad (1,-1,1,-1);$$
4. a $60°$ angle computation in $\mathbb{R}^4$;
5. the structure of the hyperplane
$$2x_1 - x_2 + x_3 + 3x_4 = 5;$$
6. the distance from the origin to this hyperplane:
$$\frac{\sqrt{15}}{3};$$
7. the cross-section of S^3 by $x_4 = t$;
8. the 4-cube vertex count
$$2^4 = 16;$$
9. the 4-cube face counts
$$V = 16, \quad E = 32, \quad F = 24, \quad C = 8;$$
10. the four-dimensional Euler relation
$$V - E + F - C = 0$$
for the 4-cube and 4-simplex;

11. the volume formula

$$\mathrm{Vol}(\Delta_n) = \frac{1}{n!}$$

    for the standard simplex in low dimensions;

12. barycentric coordinates of

$$\left(\frac{1}{4}, \frac{1}{3}\right)$$

    in the standard triangle;

13. the vertices of the 4-cross-polytope;
14. the comparison between simplex, cube, and cross-polytope in dimension 4;
15. the graph-distance interpretation of the 4-cube;
16. the volume comparison between the 4-ball and the 4-cube;
17. the closest point to the origin on a hyperplane in $\mathbb{R}^4$.

7.26 Working Summary

The appendix teaches that higher-dimensional geometry is not an exercise in imagining impossible pictures. It is a disciplined replacement of visual intuition by exact mathematical tools.

Key Points

coordinates $\Rightarrow$ distance and angle $\Rightarrow$ hyperplanes and spheres $\Rightarrow$ sections and projections $\Rightarrow$ p

The most important habit is to ask which representation is appropriate. A four-dimensional object may be studied by coordinates, by its graph of vertices and edges, by its slices, by its projection, or by its face numbers. No single picture captures it completely, but these representations together make the object mathematically precise.

CHAPTER 8

Topology

Source Map

This chapter changes the question from metric geometry to qualitative geometry. Euclidean geometry asks for lengths, angles, congruence, and area. Topology asks which properties survive continuous deformation. Courant and Robbins' route is concrete: begin with elastic deformation, move to graphs and polyhedra, discover Euler characteristic as a numerical invariant, and then use this invariant to distinguish surfaces such as the sphere, torus, and projective plane.

8.1 The Chapter's Route: From Shape to Invariant

The chapter is not about vague deformation. It is about finding quantities that remain unchanged under deformation.

$$\text{continuous deformation} \longrightarrow \text{topological equivalence} \longrightarrow \text{graphs and polyhedra} \longrightarrow \text{Euler characteristic} \longrightarrow \text{classificatio}$$

The central question is:

What can be changed without cutting or gluing, and what cannot?

A circle and an ellipse are topologically the same because one can be deformed continuously into the other. A circle and a figure-eight are not the same, because the figure-eight has a crossing point where two loops meet.

Takeaway

Topology studies properties preserved by continuous deformation. It ignores length and angle, but it does not ignore structure. The structure is encoded by invariants such as connectedness, number of boundary components, orientability, and Euler characteristic.

8.2 Topological Equivalence

Two spaces are topologically equivalent if there is a continuous bijection between them with continuous inverse. Such a map is called a homeomorphism.

Informally, one object may be stretched, bent, or twisted into the other, but not cut, torn, punctured, or glued.

allowed	not allowed
stretching	cutting
bending	gluing distinct points
twisting	creating a hole
smooth deformation	destroying connectedness

circle and ellipse are equivalent; figure-eight has a different local structure

Concrete Test: Removing One Point

A circle S^1 and a line segment $[0, 1]$ are not homeomorphic.
Remove one point from the circle. If the point is removed, the remaining space is connected:

$$S^1 \setminus \{p\}$$

is an open arc.
Remove an interior point from the interval:

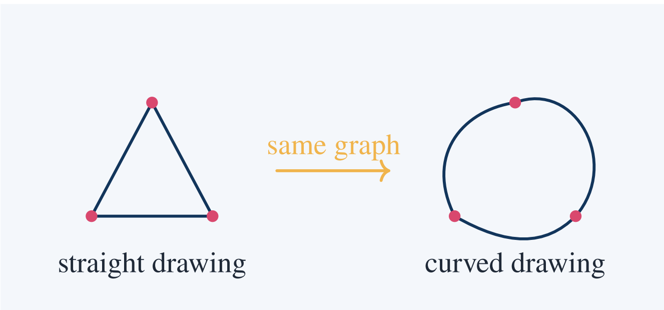
$$[0, 1] \setminus \{1/2\} = [0, 1/2) \cup (1/2, 1].$$

This has two connected components.
Since homeomorphisms preserve the number of connected components after corresponding point removal, the circle and the interval are not homeomorphic.

This is a typical topological argument: do not measure; remove or cut conceptually and count what remains.

8.3 Graphs as Topological Objects

A graph consists of vertices and edges. Topologically, the exact shape of each edge is irrelevant. What matters is which vertices are connected by which edges.



The two drawings above are topologically the same graph. Curving an edge does not change the incidence relation.

Worked Example: Degree Sum Formula

For a finite graph, the degree of a vertex is the number of incident edge ends. The degree sum formula says

$$\sum_{v \in V} \deg(v) = 2E.$$

Proof: every edge has two ends. When all vertex degrees are added, each edge is counted exactly twice. Hence the sum of all degrees is $2E$.

For a triangle graph,

$$V = 3, \qquad E = 3.$$

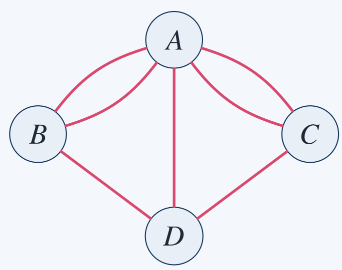
Each vertex has degree 2, so

$$\sum \deg(v) = 2 + 2 + 2 = 6 = 2E.$$

8.4 The Königsberg Bridge Problem

The Königsberg bridge problem asks whether one can walk through the city, cross each bridge exactly once, and return or finish after using all bridges. Euler translated the problem into graph theory.

The land masses become vertices; the bridges become edges.



Königsberg bridge graph

Solved Exercise: Why the Walk Is Impossible

A graph has an Eulerian trail using every edge exactly once only if it has either 0 or 2 vertices of odd degree. If it has 0 odd vertices, the trail can be closed. If it has 2 odd vertices, the trail starts at one odd vertex and ends at the other.

For the Königsberg graph above, the degrees are:

$$\deg(A) = 5, \qquad \deg(B) = 3, \qquad \deg(C) = 3, \qquad \deg(D) = 3.$$

All four vertices have odd degree. Therefore there is no walk crossing each bridge exactly once. This solves the problem without needing the physical lengths or positions of the bridges.

Takeaway

The bridge problem is topological because only connectivity matters. The exact map can be distorted freely; the answer depends only on the graph.

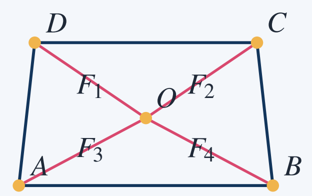
8.5 Euler Characteristic for Planar Graphs

For a connected planar graph drawn without edge crossings, let

$$V = \text{number of vertices}, \qquad E = \text{number of edges}, \qquad F = \text{number of faces}.$$

The outer unbounded region counts as one face. Euler's formula says

$$V - E + F = 2.$$



plus the exterior face

Worked Example: Check Euler's Formula

In the graph above,

$$V = 5$$

because the vertices are A, B, C, D, O.
The edges are the four boundary edges plus four spokes:

$$E = 8.$$

The faces are four interior regions plus the exterior region:

$$F = 5.$$

Therefore

$$V - E + F = 5 - 8 + 5 = 2.$$

Euler's formula holds.

8.6 Why Euler's Formula Is Topological

Euler's formula is stable under subdividing an edge. Suppose one edge is replaced by two edges by inserting a new vertex.

Then

$$V \mapsto V + 1, \qquad E \mapsto E + 1, \qquad F \mapsto F.$$

Thus

$$(V + 1) - (E + 1) + F = V - E + F.$$

The Euler characteristic does not change.

Similarly, if a diagonal is added inside a face, then

$$V \mapsto V, \qquad E \mapsto E + 1, \qquad F \mapsto F + 1.$$

Again,

$$V - (E + 1) + (F + 1) = V - E + F.$$

Concrete Subdivision Check

Start with a triangle:

$$V = 3, \qquad E = 3, \qquad F = 2.$$

Then

$$V - E + F = 3 - 3 + 2 = 2.$$

Insert a new vertex on one edge. The new graph has

$$V = 4, \qquad E = 4, \qquad F = 2.$$

Then

$$V - E + F = 4 - 4 + 2 = 2.$$

The numerical invariant stays unchanged.

This is exactly the kind of quantity topology seeks: one that survives allowable changes.

8.7 Euler Formula for Polyhedra

For a convex polyhedron,

$$V - E + F = 2.$$

This is the same formula as for connected planar graphs, because a polyhedron can be projected onto the plane after removing one face.

Worked Example: Cube

A cube has

$$V = 8, \qquad E = 12, \qquad F = 6.$$

Therefore

$$V - E + F = 8 - 12 + 6 = 2.$$

Worked Example: Tetrahedron

A tetrahedron has

$$V = 4, \qquad E = 6, \qquad F = 4.$$

Therefore

$$V - E + F = 4 - 6 + 4 = 2.$$

Worked Example: Octahedron

An octahedron has

$$V = 6, \qquad E = 12, \qquad F = 8.$$

Therefore
$$V - E + F = 6 - 12 + 8 = 2.$$

8.8 Euler's Formula Restricts Platonic Solids

Euler's formula gives a short route to the classification of the five Platonic solids.

Suppose a regular polyhedron has:

$$p = \text{number of edges of each face},$$

$$q = \text{number of faces meeting at each vertex}.$$

Since each face has p edges and each edge belongs to two faces,

$$pF = 2E.$$

Since q edges meet at each vertex and each edge has two endpoints,

$$qV = 2E.$$

Euler's formula is
$$V - E + F = 2.$$

Substitute
$$V = \frac{2E}{q}, \qquad F = \frac{2E}{p}.$$

Then
$$\frac{2E}{q} - E + \frac{2E}{p} = 2.$$

Divide by $E > 0$:
$$\frac{2}{q} - 1 + \frac{2}{p} = \frac{2}{E} > 0.$$

Therefore
$$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}.$$

Since
$$p \geq 3, \qquad q \geq 3,$$

there are only five possibilities:

$$(p, q) = (3, 3), (4, 3), (3, 4), (5, 3), (3, 5).$$

(p, q)	solid	(V, E, F)
$(3, 3)$	tetrahedron	$(4, 6, 4)$
$(4, 3)$	cube	$(8, 12, 6)$
$(3, 4)$	octahedron	$(6, 12, 8)$
$(5, 3)$	dodecahedron	$(20, 30, 12)$
$(3, 5)$	icosahedron	$(12, 30, 20)$

Takeaway

Euler characteristic is not only descriptive. It has force: it restricts what regular polyhedra can exist.

8.9 Surfaces and Euler Characteristic

For a triangulated surface, define

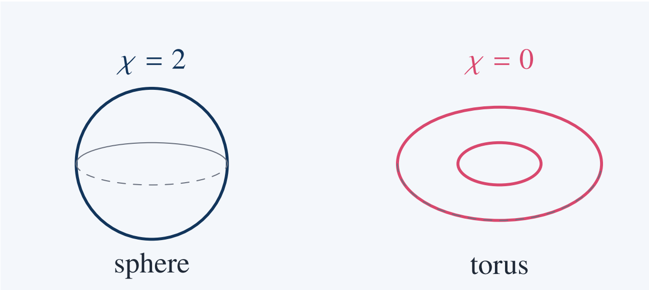
$$\chi = V - E + F.$$

This number is the Euler characteristic of the surface. For the sphere,

$$\chi(S^2) = 2.$$

For the torus,

$$\chi(T^2) = 0.$$



A Cell Decomposition of the Torus

Represent the torus by a square with opposite sides identified:

$$a\text{-side with } a\text{-side}, \qquad b\text{-side with } b\text{-side}.$$

After identification:

$$V = 1,$$

because all four corners become one vertex.
There are

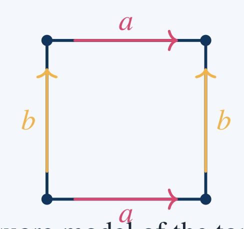
$$E = 2$$

edges, one horizontal edge a and one vertical edge b.
There is

$$F = 1$$

face, the interior of the square.
Therefore

$$\chi(T^2) = V - E + F = 1 - 2 + 1 = 0.$$



square model of the torus

This is an important technique: one studies a surface by cutting it into a polygon and recording side identifications.

8.10 Genus and the Formula $\chi = 2 - 2g$

An orientable closed surface of genus g is a sphere with g handles. Its Euler characteristic is

$$\chi = 2 - 2g.$$

g	surface	χ
0	sphere	2
1	torus	0
2	double torus	−2
3	triple torus	−4

Worked Example: Double Torus

For genus

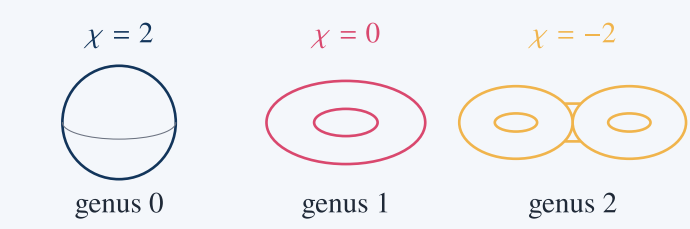
$$g = 2,$$

the formula gives

$$\chi = 2 - 2g = 2 - 4 = -2.$$

Thus a double torus cannot be homeomorphic to a sphere or a torus, because Euler characteristic is a topological invariant:

$$\chi(S^2) = 2, \qquad \chi(T^2) = 0, \qquad \chi(\text{double torus}) = -2.$$



The picture is schematic; the invariant is exact.

8.11 Orientability and the Möbius Strip

A surface is orientable if one can consistently choose a normal direction over the whole surface. The sphere and torus are orientable. The Möbius strip is not.

A Möbius strip is made from a rectangle by gluing two opposite edges after a half-twist.

rectangle model of the Möbius strip

opposite directions indicate a half-twist

Why the Möbius Strip Has One Boundary Component

Start with a rectangle. If the two vertical sides are glued in the same direction, one obtains a cylinder with two boundary circles.
If the two vertical sides are glued in opposite directions, the top edge of the rectangle connects continuously to the bottom edge after the twist. Thus the two horizontal boundary edges become one single boundary curve.
Therefore the Möbius strip has

$$1$$

boundary component, while the cylinder has

$$2$$

boundary components.

Takeaway

Orientability is a topological property. It cannot be detected by measuring lengths, but it can be detected by transporting a direction around the surface and checking whether it returns reversed.

8.12 Cylinder versus Möbius Strip

Both the cylinder and the Möbius strip are made by gluing opposite sides of a rectangle. The difference is the direction of gluing.

surface	edge identification	boundary components
cylinder	same direction	2
Möbius strip	opposite direction	1

Euler Characteristic of the Cylinder and Möbius Strip

Both the cylinder and the Möbius strip can be represented by one rectangle with one pair of opposite sides identified.
Use the cell decomposition:

$$F = 1.$$

After identifying the vertical sides, the two vertical edges become one edge:

$$E_{\text{vertical}} = 1.$$

The top and bottom horizontal edges remain boundary edges:

$$E_{\text{horizontal}} = 2.$$

Thus

$$E = 3.$$

The four corners become two vertices:

$$V = 2.$$

Therefore

$$\chi = V - E + F = 2 - 3 + 1 = 0.$$

So both the cylinder and the Möbius strip have

$$\chi = 0.$$

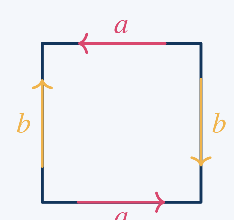
Euler characteristic alone does not distinguish them. Boundary components and orientability are also needed.

This is an important lesson: one invariant may not be enough. Topology often requires a package of invariants.

8.13 The Projective Plane

The real projective plane can be modeled by a disk whose boundary points are identified antipodally. Equivalently, it is obtained from a sphere by identifying opposite points.

A polygon model is a square with both pairs of opposite sides identified with reversed orientation.



polygon model of the projective plane

Euler Characteristic of the Projective Plane

Use the square model with opposite sides identified in reversed directions.
After the identifications, all four vertices become one vertex:

$$V = 1.$$

The horizontal sides form one edge a, and the vertical sides form one edge b:

$$E = 2.$$

The interior of the square is one face:

$$F = 1.$$

Therefore

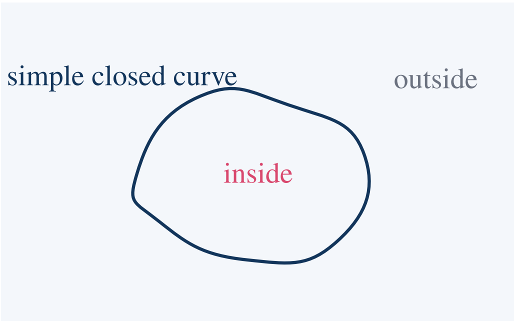
$$\chi(\mathbb{R}P^2) = V - E + F = 1 - 2 + 1 = 0.$$

This cell calculation by itself gives the same Euler characteristic as the torus. Therefore orientability must also be considered: the torus is orientable, while $\mathbb{R}P^2$ is non-orientable.

8.14 The Jordan Curve Theorem

The Jordan curve theorem states that every simple closed curve in the plane separates the plane into exactly two components: an inside and an outside.

A simple closed curve is a continuous loop with no self-intersections.



The theorem looks visually obvious, but it is mathematically subtle. The curve may be extremely irregular. Topology turns the visual intuition into a precise separation statement.

Nonexample: Figure-Eight Curve

A figure-eight is not a simple closed curve because it has a self-intersection.
It does not separate the plane into exactly two components. Instead it creates two bounded lobes and one unbounded exterior region, so the complement has three components.
Thus the hypothesis "simple" in the Jordan curve theorem is essential.

8.15 The Brouwer Fixed Point Theorem in Dimension One

A fixed point of a function f is a point x such that

$$f(x) = x.$$

The one-dimensional version of Brouwer's fixed point theorem says:

If $f : [0, 1] \to [0, 1]$ is continuous, then f has a fixed point.

Proof in One Dimension

Define
$$g(x) = f(x) - x.$$
Since f is continuous, g is continuous.
Because $f(x) \in [0, 1]$, we have
$$f(0) \geq 0,$$
so
$$g(0) = f(0) - 0 \geq 0.$$
Also
$$f(1) \leq 1,$$
so
$$g(1) = f(1) - 1 \leq 0.$$
If
$$g(0) = 0$$
or
$$g(1) = 0,$$
we already have a fixed point. Otherwise,
$$g(0) > 0, \qquad g(1) < 0.$$
By the intermediate value theorem, there exists $c \in (0, 1)$ such that
$$g(c) = 0.$$
Thus
$$f(c) - c = 0,$$
so
$$f(c) = c.$$

Takeaway

The fixed point theorem is topological because it depends on continuity and connectedness, not on a formula for f.

8.16 A Concrete Fixed Point Example

Worked Example

Let

$$f(x) = \frac{x^2 + 1}{3}$$

on

$$[0, 1].$$

First check that f maps $[0, 1]$ into itself:

$$0 \le x^2 \le 1$$

implies

$$\frac{1}{3} \le f(x) \le \frac{2}{3}.$$

Thus

$$f([0, 1]) \subset [0, 1].$$

A fixed point satisfies

$$x = \frac{x^2 + 1}{3}.$$

So

$$3x = x^2 + 1,$$

$$x^2 - 3x + 1 = 0.$$

Hence

$$x = \frac{3 \pm \sqrt{5}}{2}.$$

Only one root lies in $[0, 1]$:

$$x = \frac{3 - \sqrt{5}}{2}.$$

Therefore

$$\frac{3 - \sqrt{5}}{2}$$

is the fixed point.

8.17 Connectedness

A space is connected if it cannot be split into two nonempty separated open pieces. For intervals, the idea is simpler: an interval is connected because one cannot divide it into two parts without creating a gap.

Worked Example: Connected versus Disconnected

The interval

$$[0, 1]$$

is connected.

The set
$$[0,1] \cup [2,3]$$
is disconnected because it is the union of two separated pieces:
$$[0,1] \qquad \text{and} \qquad [2,3].$$
A continuous image of a connected set is connected. Therefore there is no continuous function
$$f : [0,1] \to \{0,1\}$$
with
$$f(0) = 0, \qquad f(1) = 1.$$
If such a function existed, the image would contain both 0 and 1, but $\{0,1\}$ is disconnected.

This is the topological core of the intermediate value theorem.

8.18 Topological Invariants Compared

The chapter's objects can be organized by invariants.

space	χ	connected?	boundary components	orientable?
sphere	2	yes	0	yes
torus	0	yes	0	yes
cylinder	0	yes	2	yes
Möbius strip	0	yes	1	no
projective plane	1	yes	0	no
double torus	−2	yes	0	yes

Remark

No single invariant detects everything. Euler characteristic distinguishes the sphere from the torus, but not the cylinder from the Möbius strip. Orientability and boundary data are needed as well.

8.19 A Short Exact Calculation: Sphere versus Torus

Why a Sphere Is Not a Torus

Assume, for contradiction, that
$$S^2$$
is homeomorphic to
$$T^2.$$
Euler characteristic is invariant under homeomorphism. Therefore one would have
$$\chi(S^2) = \chi(T^2).$$

But

$$\chi(S^2) = 2,$$

while

$$\chi(T^2) = 0.$$

Since

$$2 \neq 0,$$

the two surfaces cannot be homeomorphic.

This is the correct topological style: distinguish spaces by invariants, not by visual appearance.

8.20 Map Coloring and Planar Graphs

A map on the sphere or plane can be converted into a planar graph: regions become vertices, and adjacency becomes edges. The four-color theorem states that four colors always suffice to color any planar map so that adjacent regions have different colors.

A weaker but elementary theorem is the six-color theorem.

Key Inequality for Planar Graphs

For a connected simple planar graph with

$$V \geq 3,$$

one has

$$E \leq 3V - 6.$$

Reason: every face has at least three boundary edges, and each edge borders at most two faces. Therefore

$$3F \leq 2E.$$

Euler's formula gives

$$V - E + F = 2,$$

so

$$F = 2 - V + E.$$

Substitute:

$$3(2 - V + E) \leq 2E.$$

Thus

$$6 - 3V + 3E \leq 2E,$$

so

$$E \leq 3V - 6.$$

Consequence: A Low-Degree Vertex Exists

The average degree of a planar graph is

$$\frac{2E}{V}.$$

Using

$$E \le 3V - 6,$$

we get

$$\frac{2E}{V} \le \frac{6V - 12}{V} = 6 - \frac{12}{V} < 6.$$

Therefore some vertex has degree at most 5. If every vertex had degree at least 6, the average degree would be at least 6, contradiction.

This is the combinatorial engine behind the elementary six-color theorem.

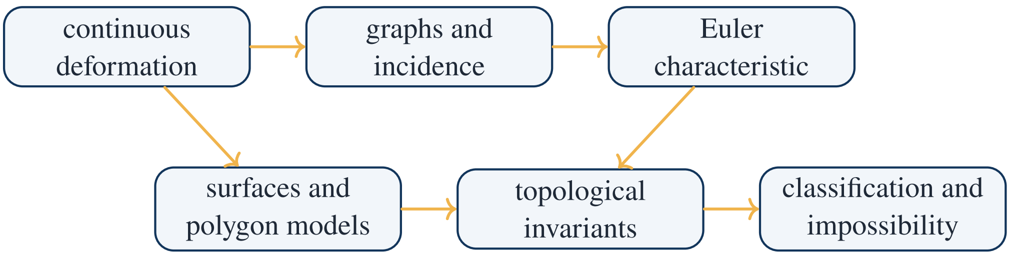
8.21 Six-Color Theorem

Proof Sketch with Induction

Every planar graph can be colored with at most six colors.
Proof by induction on the number of vertices V. For small V, the statement is clear.
Assume it is true for all planar graphs with fewer than V vertices. Let G be a planar graph with V vertices. From the previous result, G has a vertex v of degree at most 5.
Remove v. The remaining graph has $V - 1$ vertices, so by induction it can be colored with six colors. Now put v back. Since v has at most five neighbors, at most five colors are forbidden. One of the six colors remains available. Assign that color to v.
Thus the whole graph is six-colorable.

This proof shows how topology enters graph coloring: Euler's formula forces a low-degree vertex, which makes the coloring induction work.

8.22 The Chapter's Method in One Diagram



topology replaces measurement by invariants

8.23 Exercises Solved in This Note

Checklist of Concrete Results

This note explicitly computed or proved:

1. a point-removal argument showing that a circle and an interval are not homeomorphic;
2. the graph degree sum formula
$$\sum_{v \in V} \deg(v) = 2E;$$
3. impossibility of the Königsberg bridge walk using odd vertex degrees;
4. Euler's formula
$$V - E + F = 2$$
in concrete planar examples;
5. invariance of
$$V - E + F$$
under edge subdivision and diagonal insertion;
6. Euler characteristic checks for the cube, tetrahedron, and octahedron;
7. the inequality
$$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}$$
restricting Platonic solids;
8. the five possible regular polyhedra from the pairs
$$(3, 3), (4, 3), (3, 4), (5, 3), (3, 5);$$
9. the torus cell decomposition with
$$V = 1, \quad E = 2, \quad F = 1, \quad \chi = 0;$$
10. the formula for orientable closed surfaces
$$\chi = 2 - 2g;$$
11. the boundary-component difference between the cylinder and Möbius strip;
12. Euler characteristic computations for the cylinder, Möbius strip, and projective plane;
13. a concrete use of the Jordan curve theorem and a figure-eight nonexample;
14. the one-dimensional Brouwer fixed point theorem;
15. a fixed point calculation for
$$f(x) = \frac{x^2 + 1}{3};$$
16. the planar graph inequality
$$E \leq 3V - 6;$$
17. the existence of a vertex of degree at most 5 in a finite simple planar graph;
18. the six-color theorem by induction.

8.24 Working Summary

The topology chapter teaches a specific mathematical habit: when geometry is too flexible for measurement, look for invariants.

Key Points

$$\text{deformable shape} \quad\Rightarrow\quad \text{graph or cell decomposition} \quad\Rightarrow\quad V-E+F \quad\Rightarrow\quad \text{surface invariant} \quad\Rightarrow\quad \text{classificati}$$

The strongest examples are not philosophical; they are computational. The Königsberg bridge problem is solved by vertex parity. The Platonic solids are restricted by Euler's formula. The sphere and torus are distinguished by Euler characteristic. The cylinder and Möbius strip require boundary and orientability beyond Euler characteristic. These examples show the main power of topology: it turns qualitative shape into exact mathematical obstruction.

Part III

Continuity, Functions, and Analysis

CHAPTER 9

Functions and Limits

Source Map

This chapter develops one of the central transitions in mathematics: from static numbers and figures to variable quantities. Courant and Robbins' path is not to define functions abstractly first and then leave them as formal objects. The chapter proceeds through concrete representations:

$$\text{formula} \longrightarrow \text{table} \longrightarrow \text{graph} \longrightarrow \text{composition and inverse} \longrightarrow \text{limit} \longrightarrow \text{continuity}.$$

The point is that a function is a rule of dependence, while a limit is the tool that allows one to study behavior near a point, at infinity, or through an infinite process.

9.1 The Chapter's Route: From Dependence to Limiting Behavior

A function records how one quantity depends on another. If x is the input and y is the output, then a function specifies a rule

$$y = f(x).$$

This rule may be given by a formula, a graph, a table, a geometric construction, or a physical law.

The chapter's intellectual movement is:

$$\text{variables} \longrightarrow \text{functions} \longrightarrow \text{graphs} \longrightarrow \text{inverse and composition} \longrightarrow \text{limits} \longrightarrow \text{continuous functions}.$$

This progression matters. A formula such as

$$f(x) = x^2$$

is not merely an algebraic expression. It determines a geometric curve, a rate of growth, a symmetry, and a limiting behavior.

Takeaway

A function is a mathematical object that packages dependence. A limit is a mathematical object that packages approach.

9.2 Functions as Rules

A function from a set A to a set B assigns to each element of A exactly one element of B. We write

$$f : A \to B.$$

The set A is the domain, and the set B is the codomain.

Worked Example: Domain and Range

Let

$$f(x) = x^2 - 2x + 3.$$

Complete the square:

$$f(x) = x^2 - 2x + 1 + 2 = (x-1)^2 + 2.$$

If the domain is

$$\mathbb{R},$$

then

$$(x-1)^2 \geq 0,$$

so

$$f(x) \geq 2.$$

The minimum occurs at

$$x = 1,$$

where

$$f(1) = 2.$$

Therefore the range is

$$[2, \infty).$$

The same formula can define different functions if the domain is changed. For example,

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2,$$

is not one-to-one, because

$$f(1) = f(-1) = 1.$$

But

$$g : [0, \infty) \to [0, \infty), \qquad g(x) = x^2,$$

is one-to-one and has inverse

$$g^{-1}(x) = \sqrt{x}.$$

Remark

A function is not only a formula. It is a formula together with a domain and a codomain. Changing the domain can change whether an inverse exists.

9.3 Graphs as Geometry of Functions

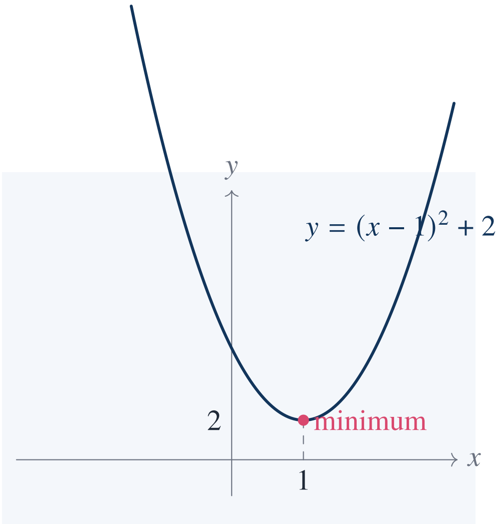
The graph of a function

$$f : A \to \mathbb{R}$$

is the set

$$\{(x, f(x)) : x \in A\}.$$

The graph translates dependence into geometry.



The graph shows instantly what the algebra showed:

$$f(x) = (x-1)^2 + 2$$

has a minimum value 2 at $x = 1$.

Solved Exercise: Find the Intercepts

For

$$f(x) = x^2 - 2x + 3,$$

the y-intercept is

$$f(0) = 3,$$

so the graph meets the y-axis at

$$(0, 3).$$

To find x-intercepts, solve

$$x^2 - 2x + 3 = 0.$$

The discriminant is

$$(-2)^2 - 4(1)(3) = 4 - 12 = -8 < 0.$$

Therefore the graph has no real x-intercepts.

9.4 Linear Functions and Slope

The simplest nonconstant functions are linear:

$$f(x) = mx + b.$$

The number m is the slope:

$$m = \frac{\Delta y}{\Delta x}.$$

Worked Example: Determine a Line from Two Points

Find the linear function passing through

$$(2,5) \qquad \text{and} \qquad (6,13).$$

The slope is

$$m = \frac{13-5}{6-2} = \frac{8}{4} = 2.$$

Thus

$$f(x) = 2x + b.$$

Use the point $(2,5)$:

$$5 = 2(2) + b,$$

so

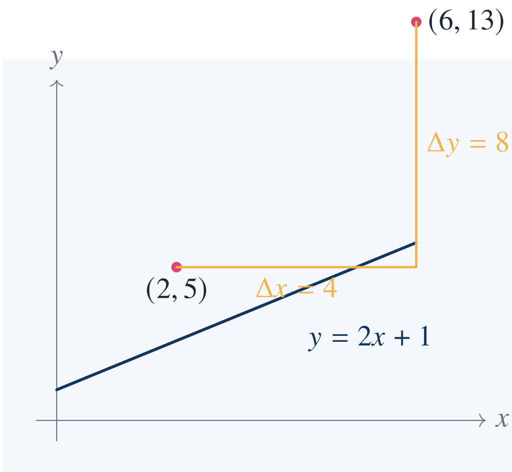
$$b = 1.$$

Therefore

$$f(x) = 2x + 1.$$

Check:

$$f(6) = 2(6) + 1 = 13.$$



The linear function is the first place where graph and algebra perfectly reinforce each other: constant slope means a straight line.

9.5 Quadratic Functions and Completing the Square

Quadratic functions

$$f(x) = ax^2 + bx + c$$

are controlled by the square-completion method:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

Worked Example: Vertex of a Parabola

Let
$$f(x) = 2x^2 - 8x + 7.$$
Factor out 2 from the quadratic part:
$$f(x) = 2(x^2 - 4x) + 7.$$
Complete the square:
$$x^2 - 4x = (x-2)^2 - 4.$$
Thus
$$f(x) = 2((x-2)^2 - 4) + 7 = 2(x-2)^2 - 8 + 7 = 2(x-2)^2 - 1.$$
Therefore the vertex is
$$(2, -1),$$
and the minimum value is
$$-1.$$

This example shows the chapter's method: formulas are not merely manipulated; they are converted into geometric information.

9.6 Composition of Functions

If
$$f : A \to B \qquad \text{and} \qquad g : B \to C,$$
then the composite function is
$$g \circ f : A \to C, \qquad (g \circ f)(x) = g(f(x)).$$

Worked Example: Composition Is Ordered

Let
$$f(x) = x^2 + 1, \qquad g(x) = 3x - 2.$$
Then
$$(g \circ f)(x) = g(x^2 + 1) = 3(x^2 + 1) - 2 = 3x^2 + 1.$$
But
$$(f \circ g)(x) = f(3x - 2) = (3x - 2)^2 + 1.$$
Expand:
$$(3x-2)^2 + 1 = 9x^2 - 12x + 4 + 1 = 9x^2 - 12x + 5.$$
Therefore
$$g \circ f \neq f \circ g.$$

Composition is the algebra of dependence. If x determines $u = f(x)$, and u determines $y = g(u)$, then x determines $y = g(f(x))$.

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[rr, bend right, "g \circ f"'] & u = f(x) \arrow[r, "g"] & y = g(u) \end{tikzcd}$$

9.7 Inverse Functions

A function f has an inverse if each output comes from exactly one input. Geometrically, this means the graph passes the horizontal line test.

Worked Example: Inverting a Function

Let

$$f(x) = \frac{2x-1}{x+3}.$$

Solve

$$y = \frac{2x-1}{x+3}$$

for x. Multiply:

$$y(x+3) = 2x - 1.$$

So

$$yx + 3y = 2x - 1.$$

Move x-terms together:

$$yx - 2x = -1 - 3y.$$

Factor:

$$x(y-2) = -(1+3y).$$

Thus

$$x = \frac{1+3y}{2-y}.$$

Therefore

$$f^{-1}(y) = \frac{1+3y}{2-y}.$$

Replacing y by x,

$$f^{-1}(x) = \frac{1+3x}{2-x}.$$

Check the Inverse

Compute

$$f(f^{-1}(x)) = f\left(\frac{1+3x}{2-x}\right).$$

Then

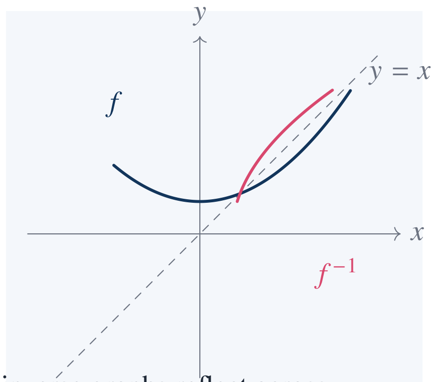
$$2\left(\frac{1+3x}{2-x}\right) - 1 = \frac{2+6x-(2-x)}{2-x} = \frac{7x}{2-x}.$$

Also,

$$\frac{1+3x}{2-x} + 3 = \frac{1+3x+3(2-x)}{2-x} = \frac{7}{2-x}.$$

Therefore

$$f(f^{-1}(x)) = \frac{7x/(2-x)}{7/(2-x)} = x.$$



inverse graphs reflect across $y = x$

The inverse graph is the reflection of the original graph across the line

$$y = x.$$

9.8 Limits as Approaching Behavior

The limit

$$\lim_{x \to a} f(x) = L$$

means that $f(x)$ approaches L as x approaches a, even if $f(a)$ is not defined or is not equal to L.

Worked Example: A Removable Discontinuity

Consider

$$f(x) = \frac{x^2-1}{x-1}.$$

At

$$x = 1,$$

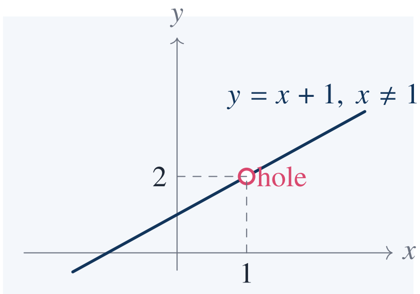
the formula is undefined. But for $x \neq 1$,

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x + 1.$$

Therefore

$$\lim_{x \to 1} \frac{x^2-1}{x-1} = \lim_{x \to 1} (x+1) = 2.$$

The function has a hole at $x = 1$, but the limiting value is 2.



Takeaway

A limit is not necessarily the value of the function at the point. It is the value forced by nearby behavior.

9.9 One-Sided Limits

A two-sided limit exists only when the left-hand and right-hand limits agree:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x).$$

Worked Example: A Jump Discontinuity

Let

$$f(x) = \begin{cases} 1, & x < 0, \\ 3, & x \geq 0. \end{cases}$$

Then

$$\lim_{x \to 0^-} f(x) = 1,$$

but

$$\lim_{x \to 0^+} f(x) = 3.$$

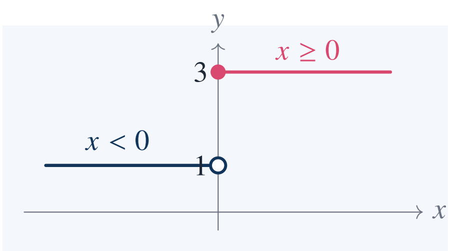
Since

$$1 \neq 3,$$

the two-sided limit

$$\lim_{x \to 0} f(x)$$

does not exist.



9.10 Limit Laws

If

$$\lim_{x\to a} f(x) = L \qquad \text{and} \qquad \lim_{x\to a} g(x) = M,$$

then

$$\lim_{x\to a} (f(x) + g(x)) = L + M,$$

$$\lim_{x\to a} (f(x)g(x)) = LM,$$

and if

$$M \neq 0,$$

then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Worked Example: Rational Limit

Compute

$$\lim_{x\to 2} \frac{x^2 + 3x - 1}{x + 4}.$$

Since the denominator tends to

$$2 + 4 = 6 \neq 0,$$

we may substitute:

$$\lim_{x\to 2} \frac{x^2 + 3x - 1}{x + 4} = \frac{2^2 + 3(2) - 1}{2 + 4} = \frac{4 + 6 - 1}{6} = \frac{9}{6} = \frac{3}{2}.$$

Worked Example: Factor Before Taking Limit

Compute

$$\lim_{x\to 3} \frac{x^2 - 9}{x - 3}.$$

Direct substitution gives

$$\frac{0}{0},$$

which is indeterminate. Factor:

$$x^2 - 9 = (x - 3)(x + 3).$$

For $x \neq 3$,

$$\frac{x^2 - 9}{x - 3} = x + 3.$$

Therefore

$$\lim_{x\to 3} \frac{x^2 - 9}{x - 3} = \lim_{x\to 3} (x + 3) = 6.$$

The expression $0/0$ does not mean the limit is zero. It means the formula needs further analysis.

9.11 The Epsilon–Delta Definition

The precise definition of a limit is:

$$\lim_{x \to a} f(x) = L$$

if for every

$$\epsilon > 0$$

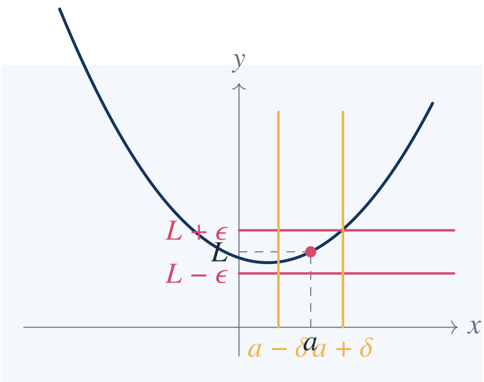
there exists

$$\delta > 0$$

such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \epsilon.$$

The definition says: any desired output tolerance ϵ can be forced by choosing a sufficiently small input tolerance δ.



Epsilon–Delta Proof: $\lim_{x\to 2}(3x + 1) = 7$

We need to show that for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - 2| < \delta \quad \Longrightarrow \quad |(3x + 1) - 7| < \epsilon.$$

Compute:

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|.$$

To make this less than ϵ, it is enough to require

$$|x - 2| < \frac{\epsilon}{3}.$$

Choose

$$\delta = \frac{\epsilon}{3}.$$

Then

$$0 < |x - 2| < \delta$$

implies

$$|(3x + 1) - 7| = 3|x - 2| < 3\delta = \epsilon.$$

Therefore

$$\lim_{x \to 2}(3x + 1) = 7.$$

9.12 A Quadratic Epsilon–Delta Proof

Linear functions are easy because the error is exactly proportional. Quadratic functions require local control.

Epsilon–Delta Proof: $\lim_{x\to 1} x^2 = 1$

We want:

$$|x^2 - 1| < \epsilon$$

when

$$|x - 1| < \delta.$$

Factor:

$$|x^2 - 1| = |x - 1||x + 1|.$$

We need to control $|x + 1|$. Require

$$|x - 1| < 1.$$

Then

$$0 < x < 2,$$

so

$$|x + 1| < 3.$$

Thus

$$|x^2 - 1| < 3|x - 1|.$$

To make this less than ϵ, require

$$|x - 1| < \frac{\epsilon}{3}.$$

Choose

$$\delta = \min\left(1, \frac{\epsilon}{3}\right).$$

Then

$$|x - 1| < \delta$$

implies

$$|x^2 - 1| = |x - 1||x + 1| < 3|x - 1| < 3\delta \leq \epsilon.$$

Therefore

$$\lim_{x\to 1} x^2 = 1.$$

This proof illustrates the real meaning of the definition: choose δ small enough not only to force the final error, but also to keep the function in a locally controlled region.

9.13 Continuity

A function f is continuous at a if

$$\lim_{x\to a} f(x) = f(a).$$

Thus three things must happen:

1. $f(a)$ is defined;
2. $\lim_{x \to a} f(x)$ exists;
3. $\lim_{x \to a} f(x) = f(a)$.

Worked Example: Removable Discontinuity

Define

$$f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

For $x \neq 1$,

$$f(x) = x + 1.$$

Hence

$$\lim_{x \to 1} f(x) = 2.$$

But

$$f(1) = 5.$$

Since

$$\lim_{x \to 1} f(x) \neq f(1),$$

the function is not continuous at $x = 1$.
If instead we define

$$f(1) = 2,$$

then the discontinuity is removed.

Worked Example: Infinite Discontinuity

Let

$$f(x) = \frac{1}{x^2}.$$

As

$$x \to 0,$$

we have

$$f(x) \to +\infty.$$

There is no finite real number L such that

$$\lim_{x \to 0} \frac{1}{x^2} = L.$$

Therefore f is not continuous at 0; in fact f is not even defined at 0.

9.14 Intermediate Value Theorem

The intermediate value theorem says:

If f is continuous on $[a, b]$, and if N lies between

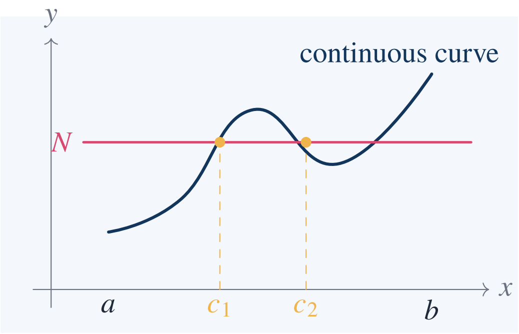
$$f(a) \qquad \text{and} \qquad f(b),$$

then there exists

$$c \in [a, b]$$

such that

$$f(c) = N.$$



Solved Exercise: Existence of a Root

Show that the equation

$$x^3 + x - 1 = 0$$

has a solution in

$$(0, 1).$$

Let

$$f(x) = x^3 + x - 1.$$

This is a polynomial, so it is continuous on $[0, 1]$. Compute:

$$f(0) = 0^3 + 0 - 1 = -1,$$

and

$$f(1) = 1^3 + 1 - 1 = 1.$$

Since

$$-1 < 0 < 1,$$

the intermediate value theorem guarantees that there exists

$$c \in (0, 1)$$

such that

$$f(c) = 0.$$

Therefore the equation

$$x^3 + x - 1 = 0$$

has at least one real solution in $(0, 1)$.

This proof does not find the root explicitly. It proves existence by continuity.

9.15 Bisection Method

The intermediate value theorem also gives a numerical method. If

$$f(a)f(b) < 0,$$

then a root lies between a and b. Repeatedly bisect the interval.

Worked Bisection for $x^3 + x - 1 = 0$

Let

$$f(x) = x^3 + x - 1.$$

We already know

$$f(0) = -1, \qquad f(1) = 1.$$

First midpoint:

$$m_1 = \frac{1}{2}.$$

Compute:

$$f\left(\frac{1}{2}\right) = \frac{1}{8} + \frac{1}{2} - 1 = -\frac{3}{8} < 0.$$

So the root lies in

$$\left(\frac{1}{2}, 1\right).$$

Second midpoint:

$$m_2 = \frac{3}{4}.$$

Compute:

$$f\left(\frac{3}{4}\right) = \frac{27}{64} + \frac{3}{4} - 1 = \frac{27}{64} + \frac{48}{64} - \frac{64}{64} = \frac{11}{64} > 0.$$

So the root lies in

$$\left(\frac{1}{2}, \frac{3}{4}\right).$$

Third midpoint:

$$m_3 = \frac{5}{8}.$$

Compute:

$$f\left(\frac{5}{8}\right) = \frac{125}{512} + \frac{5}{8} - 1 = \frac{125}{512} + \frac{320}{512} - \frac{512}{512} = -\frac{67}{512} < 0.$$

So the root lies in

$$\left(\frac{5}{8}, \frac{3}{4}\right).$$

Thus after three bisections,

$$\frac{5}{8} < c < \frac{3}{4}.$$

The method is slow but rigorous. Each step halves the uncertainty interval.

9.16 Extreme Value Theorem

The extreme value theorem says:

If f is continuous on a closed interval $[a, b]$, then f attains a maximum and a minimum on $[a, b]$.

Worked Example: Minimum and Maximum on a Closed Interval

Let
$$f(x) = x^2 - 4x + 1$$
on
$$[0, 5].$$
Complete the square:
$$f(x) = (x - 2)^2 - 3.$$
The vertex is at
$$x = 2,$$
which lies in $[0, 5]$. Thus the minimum is
$$f(2) = -3.$$
Check endpoints:
$$f(0) = 1,$$
$$f(5) = 25 - 20 + 1 = 6.$$
Therefore the maximum on $[0, 5]$ is
$$6$$
at
$$x = 5.$$

The closed interval matters.

Counterexample on an Open Interval

Consider
$$f(x) = x$$
on
$$(0, 1).$$
The function is continuous, but it has no maximum and no minimum on this open interval.
It has no minimum because values approach 0, but $0 \notin (0, 1)$. It has no maximum because values approach 1, but $1 \notin (0, 1)$.
Thus the conclusion of the extreme value theorem can fail if the interval is not closed.

9.17 Sequences as Functions on the Natural Numbers

A sequence is a function
$$a : \mathbb{N} \to \mathbb{R}.$$
Instead of writing
$$a(n),$$
one usually writes
$$a_n.$$

A sequence converges to L if

$$\lim_{n\to\infty} a_n = L.$$

That means: for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |a_n - L| < \epsilon.$$

Worked Example: $\lim_{n\to\infty} 1/n = 0$

Let

$$a_n = \frac{1}{n}.$$

We prove

$$a_n \to 0.$$

Given

$$\epsilon > 0,$$

choose

$$N > \frac{1}{\epsilon}.$$

Then if

$$n \geq N,$$

we have

$$n > \frac{1}{\epsilon},$$

so

$$\frac{1}{n} < \epsilon.$$

Thus

$$\left|\frac{1}{n} - 0\right| < \epsilon.$$

Therefore

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Worked Example: A Nonconvergent Sequence

Let

$$a_n = (-1)^n.$$

Then

$$a_1 = -1, \quad a_2 = 1, \quad a_3 = -1, \quad a_4 = 1, \ldots$$

The sequence does not approach a single number.
More formally, the subsequence

$$a_{2k} = 1$$

converges to 1, while

$$a_{2k-1} = -1$$

converges to -1. A convergent sequence cannot have two subsequences with different limits. Therefore

$$(-1)^n$$

does not converge.

9.18 Monotone Convergence

A sequence (a_n) is increasing if

$$a_{n+1} \geq a_n$$

for all n. It is bounded above if there exists M such that

$$a_n \leq M$$

for all n.

The monotone convergence theorem says that every increasing sequence bounded above converges.

Worked Example: $a_n = 1 - \frac{1}{n}$

Let

$$a_n = 1 - \frac{1}{n}.$$

Then

$$a_{n+1} - a_n = \left(1 - \frac{1}{n+1}\right) - \left(1 - \frac{1}{n}\right) = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} > 0.$$

Thus (a_n) is increasing.
Also,

$$a_n = 1 - \frac{1}{n} < 1,$$

so it is bounded above by 1.
By monotone convergence, it converges. Directly,

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right) = 1.$$

This theorem is one of the first places where completeness of the real numbers is essential. The rational numbers are not enough for every bounded monotone process to converge inside the same system.

9.19 Nested Intervals

A nested sequence of closed intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

with lengths tending to 0 determines exactly one real number.

Worked Example: Locating $\sqrt{2}$

Use intervals:

$$I_1 = [1, 2],$$

$$I_2 = [1.4, 1.5],$$

$$I_3 = [1.41, 1.42],$$
$$I_4 = [1.414, 1.415].$$

Check:

$$1.414^2 = 1.999396 < 2,$$

and

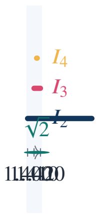
$$1.415^2 = 2.002225 > 2.$$

Therefore

$$\sqrt{2} \in [1.414, 1.415].$$

Continuing this process produces nested intervals whose lengths tend to 0. Their unique common point is

$$\sqrt{2}.$$



Nested intervals connect numerical approximation with the existence of real numbers.

9.20 Limits at Infinity

The notation

$$\lim_{x\to\infty} f(x) = L$$

means that $f(x)$ approaches L as x becomes arbitrarily large.

Worked Example

Compute

$$\lim_{x\to\infty} \frac{3x^2 - 5x + 1}{2x^2 + x - 7}.$$

Divide numerator and denominator by x^2:

$$\frac{3x^2 - 5x + 1}{2x^2 + x - 7} = \frac{3 - \frac{5}{x} + \frac{1}{x^2}}{2 + \frac{1}{x} - \frac{7}{x^2}}.$$

As

$$x \to \infty,$$

we have

$$\frac{1}{x} \to 0, \qquad \frac{1}{x^2} \to 0.$$

Therefore

$$\lim_{x\to\infty} \frac{3x^2 - 5x + 1}{2x^2 + x - 7} = \frac{3}{2}.$$

Worked Example: Horizontal Asymptote

For

$$f(x) = \frac{3x^2 - 5x + 1}{2x^2 + x - 7},$$

the calculation above shows

$$\lim_{x\to\infty} f(x) = \frac{3}{2}.$$

Similarly,

$$\lim_{x\to-\infty} f(x) = \frac{3}{2}.$$

Therefore the graph has horizontal asymptote

$$y = \frac{3}{2}.$$

9.21 The Limit $\lim_{x\to 0} \frac{\sin x}{x} = 1$

One of the most important limits in elementary analysis is

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

It connects geometry with calculus.

For $0 < x < \pi/2$, the unit circle gives the inequalities

$$\sin x < x < \tan x.$$

Dividing by $\sin x$ gives

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}.$$

Taking reciprocals,

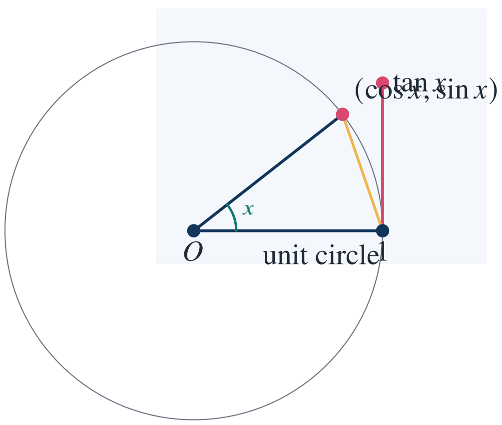
$$\cos x < \frac{\sin x}{x} < 1.$$

As

$$x \to 0,$$
$$\cos x \to 1.$$

By the squeeze theorem,

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$



Takeaway

This limit is not obtained by substituting $x = 0$. It is obtained by comparing three geometric quantities:

$$\sin x, \qquad x, \qquad \tan x.$$

9.22 The Exponential Limit

The number e can be defined by the limit

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

More generally,

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x.$$

Numerical Evidence for *e*

Compute:

$$\left(1+\frac{1}{1}\right)^1 = 2,$$

$$\left(1+\frac{1}{2}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4} = 2.25,$$

$$\left(1+\frac{1}{5}\right)^5 = \left(\frac{6}{5}\right)^5 = \frac{7776}{3125} = 2.48832,$$

$$\left(1+\frac{1}{10}\right)^{10} = 1.1^{10} \approx 2.59374246.$$

The values increase toward

$$e \approx 2.71828.$$

This example shows how a constant can be born from a limiting process, not from a finite algebraic operation.

9.23 Continuity of Polynomial and Rational Functions

Polynomials are continuous everywhere. Rational functions are continuous wherever the denominator is nonzero.

Worked Example: Continuity Domain

Let

$$f(x) = \frac{x^2+1}{x^2-4}.$$

The denominator vanishes when

$$x^2 - 4 = 0,$$

so

$$x = \pm 2.$$

Therefore f is continuous on

$$(-\infty, -2), \quad (-2, 2), \quad (2, \infty),$$

but not continuous at

$$x = -2 \qquad \text{or} \qquad x = 2.$$

At $x = 2$, the denominator is zero and the numerator is

$$2^2 + 1 = 5 \neq 0.$$

Thus there is a vertical asymptote. At $x = -2$, the same is true.

Limit Near the Vertical Asymptote

As

$$x \to 2^+,$$

we have

$$x^2 - 4 = (x - 2)(x + 2) > 0$$

and very small, while

$$x^2 + 1 \to 5.$$

Therefore

$$\frac{x^2 + 1}{x^2 - 4} \to +\infty.$$

As

$$x \to 2^-,$$

we have

$$x - 2 < 0, \qquad x + 2 > 0,$$

so

$$x^+2) > 0$$

and very small, while

$$x^2 + 1 \to 5.$$

Therefore

$$\frac{x^2 + 1}{x^2 - 4} \to +\infty.$$

As

$$x \to 2^-,$$

we have

$$x - 2 < 0, \qquad x + 2 > 0, -4 < 0$$

and very small in magnitude. Therefore

$$\frac{x^2 + 1}{x^2 - 4} \to -\infty.$$

9.24 Uniform versus Pointwise Thinking

Even if the chapter remains elementary, it points toward a deeper distinction: behavior at each point is different from behavior controlled uniformly on a whole interval.

For example, consider

$$f_n(x) = x^n$$

on

$$[0, 1].$$

For each fixed $x \in [0, 1)$,

$$x^n \to 0.$$

At

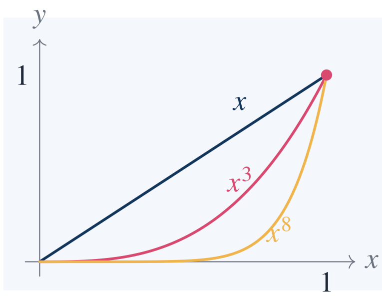
$$x = 1,$$

$$x^n = 1^n = 1.$$

Thus the pointwise limit is

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

Each f_n is continuous, but the limit function is not continuous.



Takeaway

Pointwise limiting behavior can destroy continuity. This example is a first warning that limits of functions require more care than limits of numbers.

9.25 Oscillation and Failure of Limit

Some functions fail to have limits because they oscillate too much.

Worked Example: $\sin(1/x)$ **at** 0

Consider

$$f(x) = \sin \frac{1}{x}.$$

As

$$x \to 0,$$

the quantity

$$\frac{1}{x}$$

becomes unbounded, so the sine oscillates infinitely often.
Take the sequence

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n}.$$

Then

$$x_n \to 0,$$

and

$$\sin \frac{1}{x_n} = \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1.$$

Take another sequence

$$y_n = \frac{1}{\frac{3\pi}{2} + 2\pi n}.$$

Then

$$y_n \to 0,$$

and

$$\sin \frac{1}{y_n} = \sin\left(\frac{3\pi}{2} + 2\pi n\right) = -1.$$

Since two sequences approaching 0 give different limiting values, the limit

$$\lim_{x\to 0} \sin \frac{1}{x}$$

does not exist.

This example is stronger than a jump discontinuity. There is no left or right settling behavior at all.

9.26 The Squeeze Theorem

The squeeze theorem says that if

$$g(x) \le f(x) \le h(x)$$

near a, and

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L,$$

then

$$\lim_{x\to a} f(x) = L.$$

Worked Example: $x^2 \sin(1/x)$

Let

$$f(x) = x^2 \sin \frac{1}{x}.$$

Since

$$-1 \le \sin \frac{1}{x} \le 1,$$

we have

$$-x^2 \le x^2 \sin \frac{1}{x} \le x^2.$$

As

$$x \to 0,$$

$$-x^2 \to 0, \qquad x^2 \to 0.$$

By the squeeze theorem,

$$\lim_{x \to 0} x^2 \sin \frac{1}{x} = 0.$$

This example is important because $\sin(1/x)$ alone has no limit, but the factor x^2 damps the oscillation enough to force a limit.

9.27 A Function Can Be Continuous but Not Monotone

Continuity does not mean monotonicity.

Worked Example

Let

$$f(x) = x^2$$

on

$$[-1, 1].$$

This function is continuous on $[-1, 1]$. But it is not increasing on the whole interval because

$$-1 < 0$$

while

$$f(-1) = 1 > 0 = f(0).$$

It is not decreasing on the whole interval because

$$0 < 1$$

while

$$f(0) = 0 < 1 = f(1).$$

Thus continuity only means no breaks; it does not impose a single direction of movement.

9.28 A Function Can Be Monotone but Not Continuous

Monotonicity does not imply continuity.

Worked Example

Define

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

This function is nondecreasing: if

$$x < y,$$

then

$$f(x) \le f(y).$$

But it is not continuous at 0, since

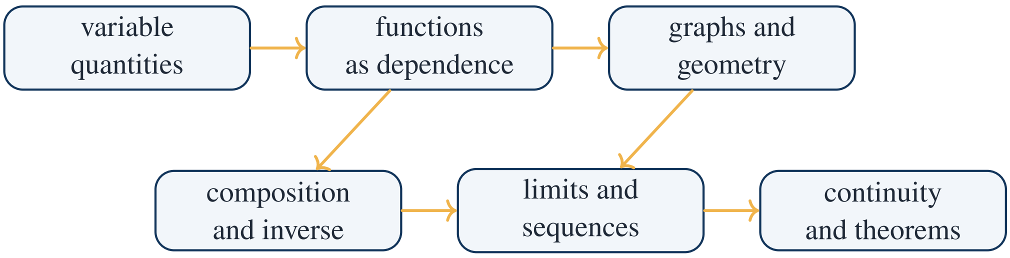
$$\lim_{x \to 0^-} f(x) = 0,$$

while

$$\lim_{x \to 0^+} f(x) = 1.$$

Thus monotonicity permits jumps.

9.29 The Chapter's Method in One Diagram



the chapter turns formulas into behavior and behavior into rigorous limits

9.30 Exercises Solved in This Note

Checklist of Concrete Results

This note explicitly computed or proved:

1. the range of
$$f(x) = x^2 - 2x + 3;$$
2. the intercepts of
$$x^2 - 2x + 3;$$
3. the linear function through
$$(2, 5) \quad \text{and} \quad (6, 13);$$
4. the vertex and minimum of
$$2x^2 - 8x + 7;$$
5. the noncommutativity of composition using
$$f(x) = x^2 + 1, \qquad g(x) = 3x - 2;$$
6. the inverse of
$$f(x) = \frac{2x - 1}{x + 3};$$

7. the removable-discontinuity limit
$$\lim_{x\to 1}\frac{x^2-1}{x-1}=2;$$
8. a jump discontinuity with unequal one-sided limits;
9. $$\lim_{x\to 2}\frac{x^2+3x-1}{x+4}=\frac{3}{2};$$
10. $$\lim_{x\to 3}\frac{x^2-9}{x-3}=6;$$
11. an epsilon–delta proof of
$$\lim_{x\to 2}(3x+1)=7;$$
12. an epsilon–delta proof of
$$\lim_{x\to 1}x^2=1;$$
13. discontinuity analysis of
$$\frac{x^2-1}{x-1}$$
with an altered value at $x=1$;
14. existence of a root of
$$x^3+x-1=0$$
in $(0,1)$;
15. three steps of bisection for
$$x^3+x-1=0;$$
16. maximum and minimum of
$$x^2-4x+1$$
on $[0,5]$;
17. convergence of
$$\frac{1}{n}\to 0;$$
18. nonconvergence of
$$(-1)^n;$$
19. monotone convergence for
$$1-\frac{1}{n};$$
20. nested-interval approximation of
$$\sqrt{2};$$
21. the limit at infinity
$$\lim_{x\to\infty}\frac{3x^2-5x+1}{2x^2+x-7}=\frac{3}{2};$$

22. the geometric squeeze proof of
$$\lim_{x\to 0} \frac{\sin x}{x} = 1;$$
23. the failure of
$$\lim_{x\to 0} \sin \frac{1}{x};$$
24. the squeeze theorem computation
$$\lim_{x\to 0} x^2 \sin \frac{1}{x} = 0.$$

9.31 Working Summary

The chapter teaches that functions must be read dynamically. A formula is not only an expression to simplify; it is a rule whose graph, inverse, composition, growth, discontinuities, and limiting behavior must be analyzed.

Key Points

$$\text{function} \quad\Rightarrow\quad \text{graph} \quad\Rightarrow\quad \text{local behavior} \quad\Rightarrow\quad \text{limit} \quad\Rightarrow\quad \text{continuity} \quad\Rightarrow\quad \text{existence theorems}.$$

The practical mathematical habit is to separate value from behavior:
$$f(a) \quad \text{is a pointwise value, while} \quad \lim_{x\to a} f(x) \quad \text{is nearby behavior.}$$

This distinction explains removable discontinuities, jump discontinuities, asymptotes, oscillatory failure of limits, and the precise meaning of continuity.

CHAPTER 10

Supplement to Chapter VI: More Examples on Limits and Continuity

Source Map

The examples in this supplement are best read slowly. Limits are not a mechanical substitution rule; they are a way of describing what a quantity is forced to approach. The most useful examples are often the borderline ones: a function with a hole, a function with a jump, an oscillating function, or a function whose behavior changes at infinity.

10.1 A Limit May Exist Where the Function Is Not Defined

A simple but important example is

$$f(x) = \frac{x^2 - 4}{x - 2}.$$

At $x = 2$, the expression is not defined. But for $x \neq 2$,

$$\frac{x^2 - 4}{x - 2} = \frac{(x-2)(x+2)}{x - 2} = x + 2.$$

Therefore

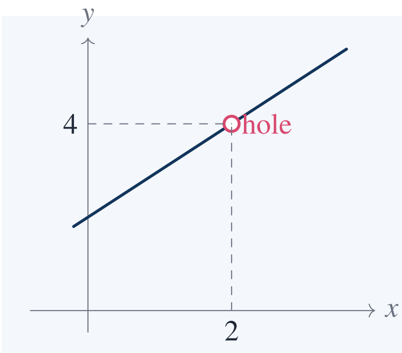
$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4.$$

What actually happens near $x = 2$

Take values near 2:

x	1.9	1.99	2.01	2.1	2.5
$\frac{x^2-4}{x-2}$	3.9	3.99	4.01	4.1	4.5

The value at $x = 2$ is missing, but the nearby values force the limit to be 4.



This example is useful because it separates the question

$$f(2) = ?$$

from the question

$$\lim_{x \to 2} f(x) = ?$$

The first has no answer here; the second has the answer 4.

10.2 Changing One Value Does Not Change the Limit

Define

$$g(x) = \begin{cases} x + 2, & x \neq 2, \\ 100, & x = 2. \end{cases}$$

Then

$$g(2) = 100,$$

but

$$\lim_{x \to 2} g(x) = 4.$$

The reason is that the limit ignores the value at the point itself. It only looks at what happens when x is near the point.

Continuity check

The function g is not continuous at $x = 2$, because

$$\lim_{x \to 2} g(x) = 4$$

while

$$g(2) = 100.$$

For continuity, these two numbers must be equal.

This example is a good warning against drawing a continuous curve merely because most of the graph looks continuous. A single wrong value can break continuity.

10.3 A Limit That Fails Because of a Jump

Consider the function

$$h(x) = \begin{cases} -1, & x < 0, \\ 2, & x \geq 0. \end{cases}$$

Then

$$\lim_{x\to 0^-} h(x) = -1,$$

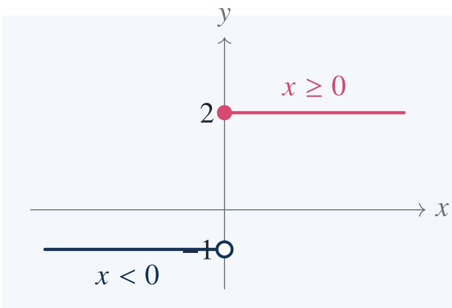
but

$$\lim_{x\to 0^+} h(x) = 2.$$

Since the two one-sided limits are different,

$$\lim_{x\to 0} h(x)$$

does not exist.



This is different from a removable discontinuity. There is no single number that the function approaches from both sides.

10.4 Rationalizing Before Taking a Limit

Some limits are hidden by radicals. For example,

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}$$

cannot be found by direct substitution, since it gives $0/0$. Multiply by the conjugate:

$$\frac{\sqrt{1+x}-1}{x}\cdot\frac{\sqrt{1+x}+1}{\sqrt{1+x}+1} = \frac{(1+x)-1}{x(\sqrt{1+x}+1)}.$$

Thus

$$\frac{\sqrt{1+x}-1}{x} = \frac{1}{\sqrt{1+x}+1} \qquad (x \neq 0).$$

Therefore

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x} = \frac{1}{\sqrt{1+0}+1} = \frac{1}{2}.$$

Same idea with a different point

Compute

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}.$$

Rationalize:

$$\frac{\sqrt{x}-2}{x-4}\cdot\frac{\sqrt{x}+2}{\sqrt{x}+2}=\frac{x-4}{(x-4)(\sqrt{x}+2)}.$$

For $x \neq 4$,

$$\frac{\sqrt{x}-2}{x-4}=\frac{1}{\sqrt{x}+2}.$$

Hence

$$\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}=\frac{1}{2+2}=\frac{1}{4}.$$

The conjugate trick is not a trick in isolation. It reveals the simpler function that was hidden by the square root.

10.5 Limits at Infinity

A rational function at infinity is often controlled by the highest powers of x. For instance,

$$\lim_{x\to\infty}\frac{5x^3-2x+1}{2x^3+x^2-7}$$

is found by dividing numerator and denominator by x^3:

$$\frac{5-\frac{2}{x^2}+\frac{1}{x^3}}{2+\frac{1}{x}-\frac{7}{x^3}}.$$

As $x\to\infty$,

$$\frac{1}{x},\quad \frac{1}{x^2},\quad \frac{1}{x^3}$$

all approach 0. Thus

$$\lim_{x\to\infty}\frac{5x^3-2x+1}{2x^3+x^2-7}=\frac{5}{2}.$$

This is often the simplest way to locate horizontal asymptotes.

10.6 A Function Can Approach Different Infinities on Different Sides

Consider

$$f(x)=\frac{1}{x-1}.$$

As $x\to 1^+$, the denominator is positive and very small, so

$$f(x)\to+\infty.$$

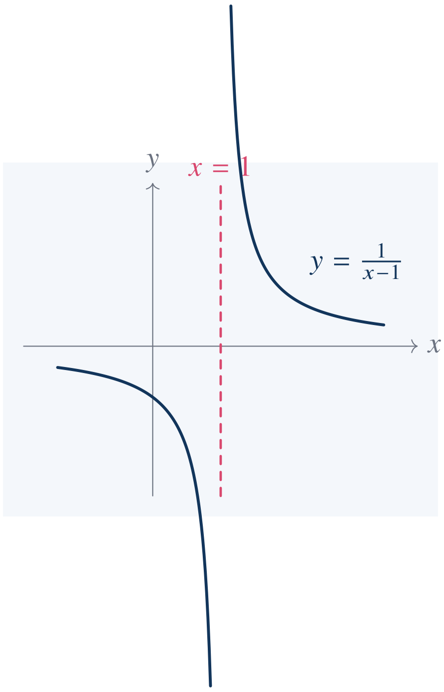
As $x\to 1^-$, the denominator is negative and very small in magnitude, so

$$f(x)\to-\infty.$$

Thus

$$x=1$$

is a vertical asymptote.



This example is a useful reminder that saying "the function goes to infinity" is incomplete unless the side and sign are clear.

10.7 A Precise Epsilon Argument

The limit

$$\lim_{x \to 3} (2x - 1) = 5$$

is simple, but it is a clean place to see the exact definition.

We want

$$|(2x - 1) - 5| < \epsilon.$$

Compute:

$$|(2x - 1) - 5| = |2x - 6| = 2|x - 3|.$$

So it is enough to require

$$|x - 3| < \frac{\epsilon}{2}.$$

Choose

$$\delta = \frac{\epsilon}{2}.$$

Then

$$0 < |x - 3| < \delta$$

implies

$$|(2x - 1) - 5| = 2|x - 3| < 2\delta = \epsilon.$$

Therefore

$$\lim_{x \to 3} (2x - 1) = 5.$$

A slightly less automatic example

Prove

$$\lim_{x \to 2} x^2 = 4.$$

We need

$$|x^2 - 4| < \epsilon.$$

Factor:

$$|x^2 - 4| = |x - 2||x + 2|.$$

If we first require

$$|x - 2| < 1,$$

then

$$1 < x < 3,$$

so

$$3 < x + 2 < 5,$$

and hence

$$|x + 2| < 5.$$

Therefore

$$|x^2 - 4| < 5|x - 2|.$$

Choose

$$\delta = \min\left(1, \frac{\epsilon}{5}\right).$$

Then

$$|x - 2| < \delta$$

implies

$$|x^2 - 4| < 5|x - 2| < 5\delta \leq \epsilon.$$

The only subtlety is the temporary bound $|x - 2| < 1$. It keeps $x + 2$ from wandering too far.

10.8 Continuity from the Definition

A function f is continuous at a if

$$\lim_{x \to a} f(x) = f(a).$$

For a polynomial, this usually works exactly as expected.

Example: A polynomial is continuous at a point

Let

$$f(x) = x^3 - 2x + 4.$$

At

$$x = 1,$$

$$f(1) = 1 - 2 + 4 = 3.$$

Using limit laws,

$$\lim_{x\to 1}(x^3 - 2x + 4) = 1^3 - 2(1) + 4 = 3.$$

Thus

$$\lim_{x\to 1} f(x) = f(1),$$

so f is continuous at 1.

For a rational function, the same substitution works as long as the denominator is not zero.

Example: A rational function

Let

$$r(x) = \frac{x^2 + 1}{x - 3}.$$

At $x = 1$, the denominator is not zero. Hence

$$\lim_{x\to 1} r(x) = \frac{1^2 + 1}{1 - 3} = \frac{2}{-2} = -1.$$

Also

$$r(1) = -1.$$

Therefore r is continuous at 1.
At $x = 3$, the function is not defined, so it cannot be continuous there.

10.9 Intermediate Value Theorem in Use

The intermediate value theorem is often more useful for proving existence than for computing an exact answer.

A root that is guaranteed but not explicitly found

Show that

$$x^5 + x - 3 = 0$$

has a root between 1 and 2.
Let

$$f(x) = x^5 + x - 3.$$

This is a polynomial, hence continuous. Compute:

$$f(1) = 1 + 1 - 3 = -1,$$

and

$$f(2) = 32 + 2 - 3 = 31.$$

Since

$$f(1) < 0 \qquad \text{and} \qquad f(2) > 0,$$

the intermediate value theorem gives some

$$c \in (1, 2)$$

such that
$$f(c) = 0.$$

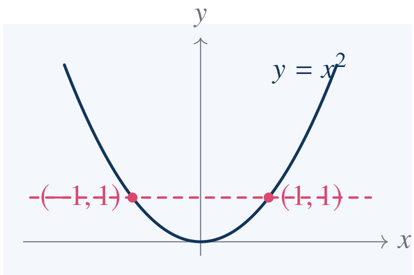
This proof gives existence without a formula for the root. That is already a real result.

A cleaner numerical bracket

For the same function,
$$f(1.1) = 1.1^5 + 1.1 - 3.$$
Since
$$1.1^5 = 1.61051,$$
we get
$$f(1.1) = 1.61051 + 1.1 - 3 = -0.28949.$$
Also,
$$f(1.2) = 1.2^5 + 1.2 - 3.$$
Since
$$1.2^5 = 2.48832,$$
$$f(1.2) = 2.48832 + 1.2 - 3 = 0.68832.$$
So the root actually lies in
$$(1.1, 1.2).$$

10.10 A Continuous Function Need Not Be One-to-One

Consider
$$f(x) = x^2$$
on
$$[-2, 2].$$
The function is continuous, but it is not one-to-one, because
$$f(-1) = 1 = f(1).$$
If we restrict the domain to
$$[0, 2],$$
then the same formula
$$f(x) = x^2$$
becomes one-to-one, and the inverse is
$$f^{-1}(x) = \sqrt{x}.$$



The formula alone does not decide invertibility; the domain matters.

10.11 Oscillation Near a Point

The function

$$f(x) = \sin\frac{1}{x}$$

does not have a limit as

$$x \to 0.$$

The reason is not that it becomes large. It stays between -1 and 1. The problem is that it oscillates infinitely often.

Choose

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n}.$$

Then

$$x_n \to 0$$

and

$$\sin\frac{1}{x_n} = \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1.$$

Choose

$$y_n = \frac{1}{\frac{3\pi}{2} + 2\pi n}.$$

Then

$$y_n \to 0$$

and

$$\sin\frac{1}{y_n} = \sin\left(\frac{3\pi}{2} + 2\pi n\right) = -1.$$

Since two sequences approaching the same point produce different function values, the limit does not exist.

Damping the oscillation

Now consider

$$g(x) = x\sin\frac{1}{x}.$$

Since

$$-1 \le \sin\frac{1}{x} \le 1,$$

we have

$$-|x| \leq x \sin \frac{1}{x} \leq |x|.$$

As

$$x \to 0,$$

both

$$-|x| \quad \text{and} \quad |x|$$

approach 0. By the squeeze theorem,

$$\lim_{x \to 0} x \sin \frac{1}{x} = 0.$$

The same oscillation can either destroy a limit or become harmless, depending on the factor multiplying it.

10.12 A Subtle Continuity Example

Define

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

From the squeeze argument above,

$$\lim_{x \to 0} x \sin \frac{1}{x} = 0.$$

Since

$$f(0) = 0,$$

we have

$$\lim_{x \to 0} f(x) = f(0).$$

Therefore f is continuous at 0.

This example is useful because the graph oscillates infinitely often near 0, but the oscillations shrink in height. Continuity does not mean the graph is visually simple.

10.13 Uniform Control on a Closed Interval

The function

$$f(x) = x^2$$

is continuous on every interval. On the closed interval $[0, 2]$, one can control its change uniformly:

$$|x^2 - y^2| = |x - y||x + y|.$$

If

$$x, y \in [0, 2],$$

then

$$|x + y| \leq 4.$$

So

$$|x^2 - y^2| \leq 4|x - y|.$$

Given $\epsilon > 0$, choosing

$$|x - y| < \frac{\epsilon}{4}$$

guarantees

$$|x^2 - y^2| < \epsilon.$$

On all of $\mathbb{R}$, this same uniform estimate fails because $x + y$ is not bounded.

Why a bounded interval matters

Take

$$x = N, \qquad y = N + \frac{1}{N}.$$

Then

$$|x - y| = \frac{1}{N} \to 0.$$

But

$$|x^2 - y^2| = |x - y||x + y| = \frac{1}{N}\left(2N + \frac{1}{N}\right) = 2 + \frac{1}{N^2}.$$

This does not approach 0. So x^2 is not uniformly continuous on all of $\mathbb{R}$.

This is a more advanced example, but it clarifies why closed bounded intervals are special in analysis.

10.14 A Final Group of Short Calculations

Limit by cancellation

$$\lim_{x \to -1} \frac{x^2 - 1}{x + 1} = \lim_{x \to -1} \frac{(x-1)(x+1)}{x+1} = \lim_{x \to -1} (x - 1) = -2.$$

Limit by highest degree

$$\lim_{x \to \infty} \frac{7x^4 + x}{2x^4 - 3x^2 + 1} = \frac{7}{2}.$$

A non-existent two-sided limit

$$f(x) = \frac{|x|}{x}.$$

For $x > 0$,

$$f(x) = 1.$$

For $x < 0$,

$$f(x) = -1.$$

Thus

$$\lim_{x \to 0^+} \frac{|x|}{x} = 1, \qquad \lim_{x \to 0^-} \frac{|x|}{x} = -1.$$

The two-sided limit does not exist.

A continuous extension

Define

$$f(x) = \frac{\sin x}{x} \qquad (x \neq 0).$$

Since

$$\lim_{x \to 0} \frac{\sin x}{x} = 1,$$

we can make f continuous at 0 by defining

$$f(0) = 1.$$

CHAPTER 11

Maxima and Minima

Source Map

This chapter is about finding the best possible value under given constraints. The useful question is rarely just "differentiate and set equal to zero." One must first identify the variable, the constraint, the quantity to be optimized, and the reason the critical value is actually maximal or minimal.

11.1 Global and Local Extremes

A function f has a global maximum at $x = a$ on a set D if

$$f(x) \le f(a) \qquad \text{for all } x \in D.$$

It has a global minimum at $x = a$ if

$$f(x) \ge f(a) \qquad \text{for all } x \in D.$$

A local maximum only compares values near a. This distinction matters.

Example: local maximum but not global maximum

Let

$$f(x) = x^3 - 3x.$$

Then

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

The critical points are

$$x = -1, \qquad x = 1.$$

Since

$$f''(x) = 6x,$$

we get

$$f''(-1) = -6 < 0,$$

so $x = -1$ is a local maximum. Its value is

$$f(-1) = (-1)^3 - 3(-1) = 2.$$

But $f(x) \to +\infty$ as $x \to +\infty$. Hence $x = -1$ is not a global maximum on $\mathbb{R}$.

This example forces a basic rule: after finding a critical point, one must still ask whether the domain is bounded, whether endpoints exist, and whether the function escapes to infinity.

11.2 Closed Intervals: Critical Points and Endpoints

If f is continuous on a closed interval $[a, b]$, then its maximum and minimum occur either at an endpoint or at a critical point inside the interval.

Worked example

Find the maximum and minimum of

$$f(x) = x^3 - 3x$$

on

$$[-2, 2].$$

First compute

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

The critical points in $[-2, 2]$ are

$$x = -1, \qquad x = 1.$$

Now evaluate f at the two endpoints and the two critical points:

$$f(-2) = -8 + 6 = -2,$$

$$f(-1) = -1 + 3 = 2,$$
$$f(1) = 1 - 3 = -2,$$
$$f(2) = 8 - 6 = 2.$$

Therefore the maximum value is

$$2,$$

attained at

$$x = -1, \quad x = 2.$$

The minimum value is

$$-2,$$

attained at

$$x = -2, \quad x = 1.$$

The endpoints are not optional. In this example, one maximum occurs at an interior critical point and another maximum occurs at an endpoint.

11.3 Quadratic Extremes Without Calculus

A quadratic can be optimized by completing the square. This method is often cleaner than differentiation.

Worked example

Find the minimum of

$$f(x) = 2x^2 - 12x + 17.$$

Complete the square:

$$f(x) = 2(x^2 - 6x) + 17.$$

Since

$$x^2 - 6x = (x-3)^2 - 9,$$

we get

$$f(x) = 2((x-3)^2 - 9) + 17 = 2(x-3)^2 - 18 + 17 = 2(x-3)^2 - 1.$$

Because

$$2(x-3)^2 \geq 0,$$

the minimum value is

$$-1,$$

attained at

$$x = 3.$$

There is no maximum on $\mathbb{R}$, since $f(x) \to +\infty$ as $|x| \to \infty$.

11.4 Fixed Perimeter: Rectangle of Maximum Area

Among rectangles with a fixed perimeter, the square has maximum area.

Let the rectangle have sides x and y, and suppose the perimeter is

$$2x + 2y = P.$$

Then

$$x + y = \frac{P}{2}.$$

The area is

$$A = xy.$$

Using

$$y = \frac{P}{2} - x,$$

we get

$$A(x) = x\left(\frac{P}{2} - x\right) = -\left(x - \frac{P}{4}\right)^2 + \frac{P^2}{16}.$$

Therefore

$$A(x) \leq \frac{P^2}{16},$$

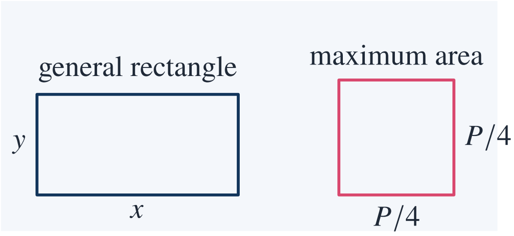
with equality precisely when

$$x = \frac{P}{4}.$$

Then

$$y = \frac{P}{4}.$$

So the maximizing rectangle is a square.



Same result by AM–GM

Since
$$x + y = \frac{P}{2},$$
AM–GM gives
$$\frac{x+y}{2} \geq \sqrt{xy}.$$
Thus
$$\frac{P}{4} \geq \sqrt{A}.$$
Squaring,
$$A \leq \frac{P^2}{16}.$$
Equality holds exactly when
$$x = y.$$

11.5 Fixed Area: Rectangle of Minimum Perimeter

The dual problem is also useful. Among rectangles with fixed area, the square has minimum perimeter.

Suppose
$$xy = A.$$
The perimeter is
$$P = 2x + 2y.$$
Since
$$y = \frac{A}{x},$$
$$P(x) = 2x + \frac{2A}{x}, \qquad x > 0.$$
Differentiate:
$$P'(x) = 2 - \frac{2A}{x^2}.$$
Set
$$P'(x) = 0.$$
Then
$$2 - \frac{2A}{x^2} = 0,$$
so
$$x^2 = A, \qquad x = \sqrt{A}.$$

Thus

$$y = \frac{A}{\sqrt{A}} = \sqrt{A}.$$

The rectangle is again a square.

To verify this is a minimum,

$$P''(x) = \frac{4A}{x^3} > 0$$

for $x > 0$.

11.6 Largest Rectangle Under a Parabola

Find the rectangle of maximum area under

$$y = 1 - x^2$$

above the x-axis, symmetric about the y-axis.

Let the right endpoint be

$$x = a, \qquad 0 < a < 1.$$

Then the left endpoint is $-a$, the width is

$$2a,$$

and the height is

$$1 - a^2.$$

So the area is

$$A(a) = 2a(1 - a^2) = 2a - 2a^3.$$

Differentiate:

$$A'(a) = 2 - 6a^2.$$

Set

$$A'(a) = 0.$$

Then

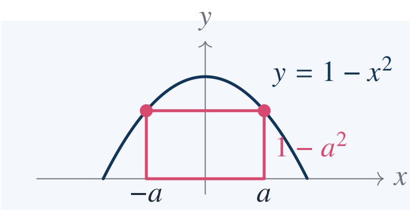
$$2 - 6a^2 = 0,$$

so

$$a^2 = \frac{1}{3}, \qquad a = \frac{1}{\sqrt{3}}.$$

The maximum area is

$$A_{\max} = 2 \cdot \frac{1}{\sqrt{3}}\left(1 - \frac{1}{3}\right) = \frac{2}{\sqrt{3}} \cdot \frac{2}{3} = \frac{4}{3\sqrt{3}}.$$



Endpoint check

At

$$a = 0,$$

the rectangle has width 0, so

$$A = 0.$$

At

$$a = 1,$$

the rectangle has height 0, so

$$A = 0.$$

The interior critical point

$$a = \frac{1}{\sqrt{3}}$$

therefore gives the maximum.

11.7 Shortest Broken Path: Reflection Method

Let A and B be two points on the same side of a line L. Find the point $P \in L$ minimizing

$$AP + PB.$$

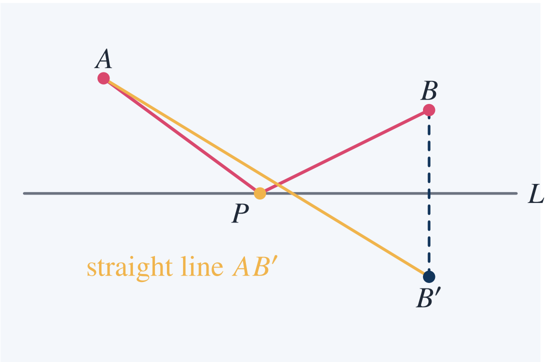
Reflect B across L to a point B'. For any $P \in L$,

$$PB = PB'.$$

Therefore

$$AP + PB = AP + PB'.$$

The shortest broken path from A to B via L becomes the shortest straight path from A to B'. Hence P must be the intersection of the line AB' with L.



Angle condition

The minimizing path satisfies

$$\angle(AP, L) = \angle(L, PB).$$

This is the law of reflection: angle of incidence equals angle of reflection. It is not guessed from physics; it follows from straightening the broken path by reflection.

11.8 Open Box from a Square Sheet

A square sheet of side length s has equal squares of side length x cut from the four corners. The sides are folded up to form an open box.

The base has side length

$$s - 2x,$$

and the height is

$$x.$$

Thus the volume is

$$V(x) = x(s-2x)^2, \qquad 0 < x < \frac{s}{2}.$$

Expand:

$$V(x) = x(s^2 - 4sx + 4x^2) = s^2x - 4sx^2 + 4x^3.$$

Differentiate:

$$V'(x) = s^2 - 8sx + 12x^2.$$

Set equal to zero:

$$12x^2 - 8sx + s^2 = 0.$$

Use the quadratic formula:

$$x = \frac{8s \pm \sqrt{64s^2 - 48s^2}}{24} = \frac{8s \pm 4s}{24}.$$

So

$$x = \frac{s}{2} \qquad \text{or} \qquad x = \frac{s}{6}.$$

The value

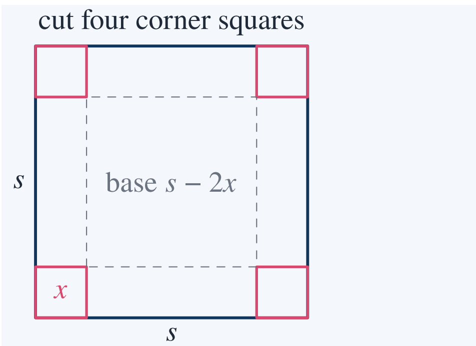
$$x = \frac{s}{2}$$

gives zero base area, so it is not the maximizing box. Therefore

$$x = \frac{s}{6}.$$

The maximum volume is

$$V_{\max} = \frac{s}{6}\left(s - \frac{2s}{6}\right)^2 = \frac{s}{6}\left(\frac{2s}{3}\right)^2 = \frac{s}{6} \cdot \frac{4s^2}{9} = \frac{2s^3}{27}.$$



11.9 Cylinder with Fixed Volume and Minimum Surface Area

A closed cylinder has radius r, height h, volume

$$V = \pi r^2 h,$$

and surface area

$$S = 2\pi r^2 + 2\pi r h.$$

Suppose the volume V is fixed. Then

$$h = \frac{V}{\pi r^2}.$$

Substitute into the surface area:

$$S(r) = 2\pi r^2 + 2\pi r \frac{V}{\pi r^2} = 2\pi r^2 + \frac{2V}{r}.$$

Differentiate:

$$S'(r) = 4\pi r - \frac{2V}{r^2}.$$

Set

$$S'(r) = 0.$$

Then

$$4\pi r = \frac{2V}{r^2},$$

so

$$4\pi r^3 = 2V,$$

$$r^3 = \frac{V}{2\pi}.$$

Using

$$h = \frac{V}{\pi r^2},$$

and

$$V = 2\pi r^3,$$

we get

$$h = \frac{2\pi r^3}{\pi r^2} = 2r.$$

Thus the surface area is minimized when

$$h = 2r.$$

Interpretation

The optimal closed cylinder has height equal to its diameter:

$$h = 2r.$$

This is a genuine optimization result, not a design convention. If the cylinder is too tall, side area is wasted; if it is too flat, top and bottom area are wasted.

11.10 Maximum Product with Fixed Sum

Let

$$x_1, x_2, \ldots, x_n > 0$$

and suppose

$$x_1 + x_2 + \cdots + x_n = S.$$

The product

$$P = x_1 x_2 \cdots x_n$$

is maximized when all variables are equal:

$$x_1 = x_2 = \cdots = x_n = \frac{S}{n}.$$

This follows from AM–GM:

$$\frac{x_1 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \cdots x_n}.$$

So

$$\frac{S}{n} \geq P^{1/n}.$$

Raise both sides to the n-th power:

$$P \leq \left(\frac{S}{n}\right)^n.$$

Equality holds exactly when

$$x_1 = \cdots = x_n.$$

Example

Among positive numbers x, y, z with

$$x + y + z = 12,$$

the product xyz is maximized at

$$x = y = z = 4.$$

The maximum product is

$$4 \cdot 4 \cdot 4 = 64.$$

11.11 Distance from a Point to a Line

Find the point on the line

$$y = 2x + 1$$

closest to

$$P = (3, 0).$$

A point on the line has the form

$$Q = (x, 2x + 1).$$

Minimize the squared distance:

$$D(x) = (x - 3)^2 + (2x + 1)^2.$$

Expand:

$$D(x) = x^2 - 6x + 9 + 4x^2 + 4x + 1 = 5x^2 - 2x + 10.$$

Differentiate:

$$D'(x) = 10x - 2.$$

Set

$$D'(x) = 0.$$

Then

$$10x - 2 = 0, \qquad x = \frac{1}{5}.$$

Thus

$$y = 2 \cdot \frac{1}{5} + 1 = \frac{7}{5}.$$

The closest point is

$$Q = \left(\frac{1}{5}, \frac{7}{5}\right).$$

The minimum squared distance is

$$D\left(\frac{1}{5}\right) = \left(\frac{1}{5} - 3\right)^2 + \left(\frac{7}{5}\right)^2 = \left(-\frac{14}{5}\right)^2 + \frac{49}{25} = \frac{196}{25} + \frac{49}{25} = \frac{245}{25} = \frac{49}{5}.$$

So the minimum distance is

$$\frac{7}{\sqrt{5}}.$$

Why minimizing squared distance is legitimate

Distance is

$$\sqrt{D(x)}.$$

Since the square-root function is increasing on $[0, \infty)$, minimizing

$$\sqrt{D(x)}$$

is equivalent to minimizing

$$D(x).$$

This avoids unnecessary radicals.

11.12 A Constraint Problem with Lagrange Multipliers

Maximize

$$f(x, y) = xy$$

subject to

$$x^2 + y^2 = 1.$$

Use Lagrange multipliers:

$$\nabla f = \lambda \nabla g,$$

where

$$g(x, y) = x^2 + y^2.$$

Then

$$\nabla f = (y, x), \qquad \nabla g = (2x, 2y).$$

So

$$y = 2\lambda x,$$

$$x = 2\lambda y.$$

Multiplying the equations gives

$$xy = 4\lambda^2 xy.$$

If $xy \neq 0$, then

$$4\lambda^2 = 1, \qquad \lambda = \pm\frac{1}{2}.$$

From the equations, this gives

$$y = x \qquad \text{or} \qquad y = -x.$$

If

$$y = x,$$

then the constraint gives

$$2x^2 = 1, \qquad x = \pm\frac{1}{\sqrt{2}}.$$

Then

$$xy = x^2 = \frac{1}{2}.$$

If

$$y = -x,$$

then

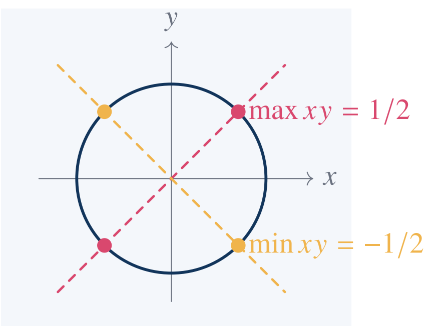
$$xy = -x^2 = -\frac{1}{2}.$$

Thus the maximum is

$$\frac{1}{2}$$

and the minimum is

$$-\frac{1}{2}.$$



11.13 Fermat's Principle and Snell's Law

Suppose light travels from point A in one medium to point B in another medium. Let the interface be the x-axis. Suppose

$$A = (0, a), \qquad B = (d, -b),$$

with

$$a, b, d > 0.$$

Let the crossing point be

$$P = (x, 0).$$

If the speeds in the two media are

$$v_1, \qquad v_2,$$

then the travel time is

$$T(x) = \frac{\sqrt{x^2 + a^2}}{v_1} + \frac{\sqrt{(d-x)^2 + b^2}}{v_2}.$$

Differentiate:

$$T'(x) = \frac{x}{v_1\sqrt{x^2 + a^2}} - \frac{d-x}{v_2\sqrt{(d-x)^2 + b^2}}.$$

At the minimum,

$$T'(x) = 0.$$

Thus

$$\frac{x}{v_1\sqrt{x^2 + a^2}} = \frac{d-x}{v_2\sqrt{(d-x)^2 + b^2}}.$$

But

$$\sin\theta_1 = \frac{x}{\sqrt{x^2 + a^2}},$$

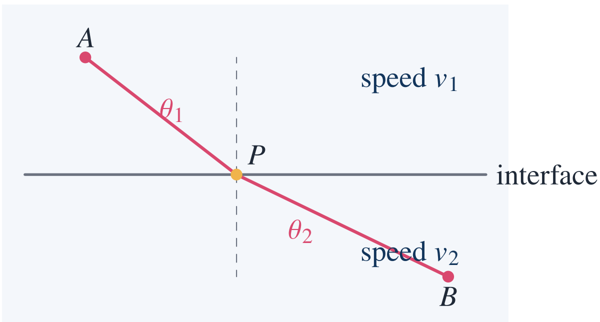
and

$$\sin\theta_2 = \frac{d-x}{\sqrt{(d-x)^2 + b^2}}.$$

Therefore

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}.$$

This is Snell's law in speed form.



11.14 Isoperimetric Thought: Why the Circle Wins

The isoperimetric problem asks: among all plane curves with fixed perimeter, which encloses the greatest area?

The answer is the circle. A full proof is deep, but one can see the governing principle from simple competitors.

For a fixed perimeter P, a square has area

$$A_{\square} = \left(\frac{P}{4}\right)^2 = \frac{P^2}{16}.$$

A circle with circumference P has radius

$$r = \frac{P}{2\pi},$$

so its area is

$$A_{\circ} = \pi r^2 = \pi\left(\frac{P}{2\pi}\right)^2 = \frac{P^2}{4\pi}.$$

Compare:

$$\frac{P^2}{4\pi} > \frac{P^2}{16}$$

because

$$16 > 4\pi \quad \Longleftrightarrow \quad 4 > \pi.$$

Thus, for the same perimeter, the circle beats the square.

Numerical example

Let

$$P = 12.$$

Then the square has side

$$3$$

and area

$$9.$$

The circle has radius

$$r = \frac{12}{2\pi} = \frac{6}{\pi},$$

so area

$$\pi r^2 = \pi \cdot \frac{36}{\pi^2} = \frac{36}{\pi} \approx 11.46.$$

The circle encloses more area with the same boundary length.

11.15 Second Derivative Test and Its Failure

The second derivative test says:

If

$$f'(a) = 0$$

and

$$f''(a) > 0,$$

then a is a local minimum.

If

$$f'(a) = 0$$

and

$$f''(a) < 0,$$

then a is a local maximum.

But if

$$f''(a) = 0,$$

the test gives no conclusion.

Failure example: minimum

Let

$$f(x) = x^4.$$

Then

$$f'(x) = 4x^3, \qquad f''(x) = 12x^2.$$

At

$$x = 0,$$

$$f'(0) = 0, \qquad f''(0) = 0.$$

The second derivative test says nothing. But

$$x^4 \geq 0$$

for all x, and equality occurs at $x = 0$. Thus $x = 0$ is a local and global minimum.

Failure example: neither maximum nor minimum

Let

$$f(x) = x^3.$$

Then

$$f'(x) = 3x^2, \qquad f''(x) = 6x.$$

At

$$x = 0,$$

$$f'(0) = 0, \qquad f''(0) = 0.$$

But $x = 0$ is neither a maximum nor a minimum, since $x^3 < 0$ for $x < 0$ and $x^3 > 0$ for $x > 0$.

The condition $f'(a) = 0$ identifies a candidate, not a verdict.

11.16 Endpoint Maximum Without Critical Point

Let

$$f(x) = x^2$$

on

$$[0,1].$$

Then

$$f'(x) = 2x.$$

The only critical point in the interval is

$$x = 0.$$

But the maximum occurs at

$$x = 1,$$

where

$$f(1) = 1.$$

This point is not an interior critical point.

Lesson from this example

On a closed interval, the candidates for global extrema are:

$$\text{interior critical points} \qquad \text{and} \qquad \text{endpoints.}$$

Ignoring endpoints loses correct answers.

11.17 A Parameter Optimization Problem

Find the point on

$$y = x^2$$

closest to

$$(0,1).$$

A point on the parabola is

$$(x, x^2).$$

The squared distance to $(0,1)$ is

$$D(x) = x^2 + (x^2 - 1)^2.$$

Expand:

$$D(x) = x^2 + x^4 - 2x^2 + 1 = x^4 - x^2 + 1.$$

Differentiate:

$$D'(x) = 4x^3 - 2x = 2x(2x^2 - 1).$$

Thus the critical points are

$$x = 0, \qquad x = \pm\frac{1}{\sqrt{2}}.$$

Evaluate:

$$D(0) = 1.$$

For

$$x^2 = \frac{1}{2},$$

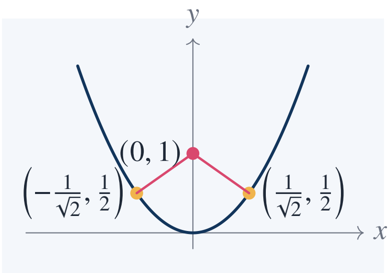
$$D = x^4 - x^2 + 1 = \frac{1}{4} - \frac{1}{2} + 1 = \frac{3}{4}.$$

Thus the closest points are

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{2}\right) \qquad \text{and} \qquad \left(-\frac{1}{\sqrt{2}}, \frac{1}{2}\right).$$

The minimum distance is

$$\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}.$$



CHAPTER 12

The Calculus

Source Map

Calculus begins when one tries to measure quantities that are not fixed: instantaneous velocity instead of average velocity, tangent direction instead of secant direction, area under a curve instead of area of a polygon. The essential operation is a limiting process. Difference quotients lead to derivatives; sums of thin rectangles lead to integrals.

12.1 Average Velocity and Instantaneous Velocity

Suppose a particle moves along a line and its position at time t is

$$s(t) = t^2.$$

The average velocity from $t = 1$ to $t = 1 + h$ is

$$\frac{s(1+h) - s(1)}{h}.$$

Compute:

$$s(1+h) = (1+h)^2 = 1 + 2h + h^2,$$

so

$$\frac{s(1+h) - s(1)}{h} = \frac{1 + 2h + h^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h.$$

As

$$h \to 0,$$

the average velocity tends to

$$2.$$

Thus the instantaneous velocity at $t = 1$ is

$$s'(1) = 2.$$

Same computation at a general time

Let

$$s(t) = t^2.$$

Then

$$\frac{s(t+h)-s(t)}{h} = \frac{(t+h)^2 - t^2}{h}.$$

Expand:

$$(t+h)^2 - t^2 = t^2 + 2th + h^2 - t^2 = 2th + h^2.$$

Therefore

$$\frac{s(t+h)-s(t)}{h} = 2t + h.$$

Taking $h \to 0$,

$$s'(t) = 2t.$$

12.2 The Derivative as a Limit

The derivative of f at x is

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h},$$

provided the limit exists.

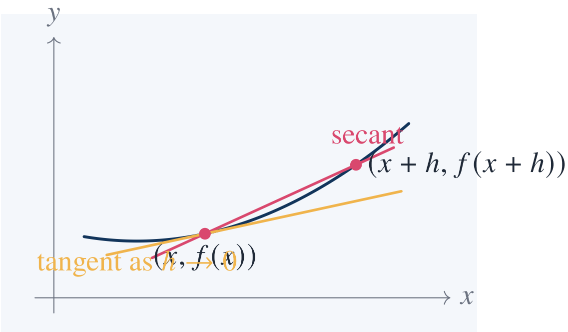
The quotient

$$\frac{f(x+h)-f(x)}{h}$$

is the slope of the secant line through

$$(x, f(x)) \qquad \text{and} \qquad (x+h, f(x+h)).$$

The derivative is the limiting slope of these secants.



12.3 Power Rule from the Definition

For

$$f(x) = x^3,$$

the derivative is computed directly:

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^3 - x^3}{h}.$$

Expand:

$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3.$$

Then

$$(x+h)^3 - x^3 = 3x^2h + 3xh^2 + h^3.$$

Divide by h:

$$\frac{(x+h)^3 - x^3}{h} = 3x^2 + 3xh + h^2.$$

Taking $h \to 0$,

$$f'(x) = 3x^2.$$

Derivative of x^4

Let

$$f(x) = x^4.$$

Then

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^4 - x^4}{h}.$$

Using the binomial expansion,

$$(x+h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4.$$

Thus

$$\frac{(x+h)^4 - x^4}{h} = 4x^3 + 6x^2h + 4xh^2 + h^3.$$

Let $h \to 0$:

$$f'(x) = 4x^3.$$

The pattern is

$$\frac{d}{dx}x^n = nx^{n-1}$$

for positive integers n.

12.4 Tangent Line

The tangent line to $y = f(x)$ at $x = a$ has equation

$$y = f(a) + f'(a)(x-a).$$

Worked example

Find the tangent line to

$$y = x^3 - 2x$$

at

$$x = 1.$$

First,

$$f(1) = 1^3 - 2(1) = -1.$$

Next,

$$f'(x) = 3x^2 - 2,$$

so

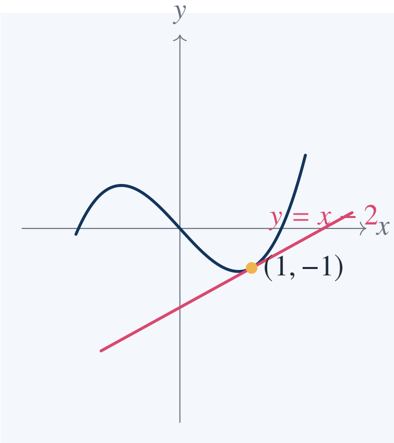
$$f'(1) = 3 - 2 = 1.$$

Therefore the tangent line is

$$y = -1 + 1(x - 1).$$

Hence

$$y = x - 2.$$



12.5 Differentiability Implies Continuity

If f is differentiable at a, then f is continuous at a.

Proof

Assume

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists. Then

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a).$$

Taking $x \to a$, the first factor tends to $f'(a)$, a finite number, and the second factor tends to 0. Hence

$$\lim_{x \to a} (f(x) - f(a)) = 0.$$

Therefore

$$\lim_{x \to a} f(x) = f(a),$$

so f is continuous at a.

The converse is false.

Counterexample

Let

$$f(x) = |x|.$$

This function is continuous at 0, since

$$\lim_{x \to 0} |x| = 0 = f(0).$$

But

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h}.$$

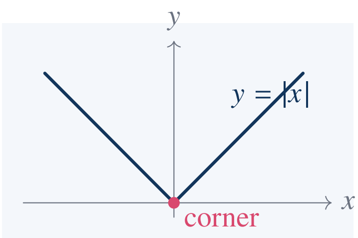
If $h > 0$, this quotient is

$$1.$$

If $h < 0$, this quotient is

$$-1.$$

The two one-sided limits are different. Therefore $f'(0)$ does not exist.



12.6 Product Rule

For differentiable functions f and g,

$$(fg)' = f'g + fg'.$$

Proof from the definition

Start with

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

Add and subtract $f(x+h)g(x)$:

$$\frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}.$$

Group:

$$f(x+h)\frac{g(x+h) - g(x)}{h} + g(x)\frac{f(x+h) - f(x)}{h}.$$

Let $h \to 0$. Since differentiability implies continuity,

$$f(x+h) \to f(x).$$

Thus

$$(fg)'(x) = f(x)g'(x) + g(x)f'(x).$$

Example

Let

$$y = (x^2 + 1)(x^3 - 2x).$$

Then

$$y' = 2x(x^3 - 2x) + (x^2 + 1)(3x^2 - 2).$$

Simplify if desired:

$$y' = 2x^4 - 4x^2 + 3x^4 + x^2 - 2 = 5x^4 - 3x^2 - 2.$$

12.7 Chain Rule

If

$$y = f(u), \qquad u = g(x),$$

then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

In function notation,

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

Worked example

Differentiate

$$y = (3x^2 - 1)^5.$$

Let

$$u = 3x^2 - 1.$$

Then

$$y = u^5.$$

So

$$\frac{dy}{du} = 5u^4, \qquad \frac{du}{dx} = 6x.$$

Therefore

$$\frac{dy}{dx} = 5(3x^2 - 1)^4 \cdot 6x.$$

Thus

$$y' = 30x(3x^2 - 1)^4.$$

Another example

Differentiate

$$y = \sqrt{1 + x^2}.$$

Write

$$y = (1 + x^2)^{1/2}.$$

Then

$$y' = \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x = \frac{x}{\sqrt{1+x^2}}.$$

12.8 Implicit Differentiation

Sometimes y is not given explicitly as a function of x. For the circle

$$x^2 + y^2 = 1,$$

differentiate both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0.$$

Thus

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Tangent line to the unit circle

Find the tangent line to

$$x^2 + y^2 = 1$$

at

$$\left(\frac{3}{5}, \frac{4}{5}\right).$$

The slope is

$$-\frac{x}{y} = -\frac{3/5}{4/5} = -\frac{3}{4}.$$

Therefore the tangent line is

$$y - \frac{4}{5} = -\frac{3}{4}\left(x - \frac{3}{5}\right).$$

Multiply by 20:

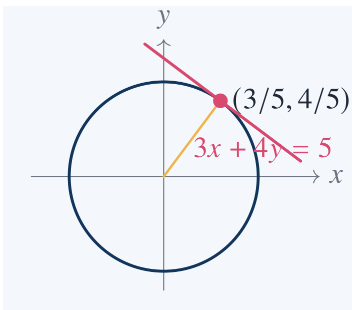
$$20y - 16 = -15x + 9.$$

Thus

$$15x + 20y = 25.$$

Divide by 5:

$$3x + 4y = 5.$$



The tangent line is perpendicular to the radius at the point of contact. The equation

$$3x + 4y = 5$$

makes this visible: its normal vector is $(3, 4)$, parallel to the radius vector

$$\left(\frac{3}{5}, \frac{4}{5}\right).$$

12.9 Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Example

Let

$$f(x) = x^2$$

on

$$[1, 3].$$

The average slope is

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

Since

$$f'(x) = 2x,$$

we need

$$2c = 4,$$

so

$$c = 2.$$

Thus the tangent at $x = 2$ is parallel to the secant line through

$$(1, 1) \qquad \text{and} \qquad (3, 9).$$

12.10 Increasing and Decreasing Functions

If

$$f'(x) > 0$$

on an interval, then f is increasing there. If

$$f'(x) < 0,$$

then f is decreasing there.

Worked example

Analyze
$$f(x) = x^3 - 3x.$$
We have
$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$
The sign chart is:

x	$(-\infty,-1)$	$(-1,1)$	$(1,\infty)$
$f'(x)$	$+$	$-$	$+$

Therefore f is increasing on
$$(-\infty,-1)$$
and
$$(1,\infty),$$
and decreasing on
$$(-1,1).$$
Thus $x = -1$ is a local maximum and $x = 1$ is a local minimum.

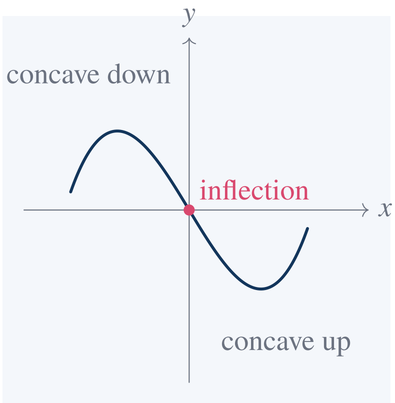
12.11 Concavity and Inflection

The second derivative measures concavity.

If
$$f''(x) > 0,$$
the graph is concave upward. If
$$f''(x) < 0,$$
the graph is concave downward.

Worked example

Let
$$f(x) = x^3 - 3x.$$
Then
$$f''(x) = 6x.$$
So
$$f''(x) < 0 \quad \text{for } x < 0,$$
and
$$f''(x) > 0 \quad \text{for } x > 0.$$
The concavity changes at
$$x = 0.$$
Since
$$f(0) = 0,$$
the inflection point is
$$(0,0).$$



12.12 Area Under a Curve

For a nonnegative function f on $[a, b]$, the area under the curve is approximated by rectangles:

$$\sum_{k=1}^{n} f(x_k)\Delta x.$$

The definite integral is the limit of such sums:

$$\int_a^b f(x)\,dx.$$

Area under $y = x$ from 0 to 1

Geometrically, the region under $y = x$ on $[0, 1]$ is a right triangle with base 1 and height 1. Hence

$$\int_0^1 x\,dx = \frac{1}{2}.$$

Same result from sums

Divide $[0, 1]$ into n equal intervals:

$$\Delta x = \frac{1}{n}.$$

Use right endpoints

$$x_k = \frac{k}{n}.$$

Then the rectangle sum is

$$\sum_{k=1}^{n} \frac{k}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^{n} k.$$

Since

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2},$$

we get

$$\frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}.$$

Let $n \to \infty$:

$$\lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2}.$$

Thus

$$\int_0^1 x\,dx = \frac{1}{2}.$$

12.13 Area Under $y = x^2$

Compute

$$\int_0^1 x^2\,dx$$

from sums.

Divide $[0, 1]$ into n pieces:

$$\Delta x = \frac{1}{n}, \qquad x_k = \frac{k}{n}.$$

The right endpoint sum is

$$\sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^{n} k^2.$$

Use

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Then

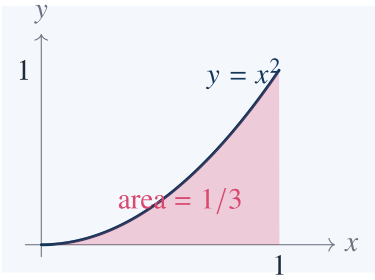
$$\frac{1}{n^3} \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6n^3} = \frac{(n+1)(2n+1)}{6n^2}.$$

Let $n \to \infty$:

$$\lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{2}{6} = \frac{1}{3}.$$

Therefore

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$



12.14 Fundamental Theorem of Calculus

Define

$$A(x) = \int_a^x f(t)\,dt.$$

If f is continuous, then

$$A'(x) = f(x).$$

The integral accumulates area; the derivative recovers the instantaneous rate at which the area is accumulating.

Concrete check

Let

$$f(t) = t^2.$$

Then

$$A(x) = \int_0^x t^2\,dt = \frac{x^3}{3}.$$

Differentiate:

$$A'(x) = x^2.$$

Thus

$$A'(x) = f(x).$$

The second form is:

$$\int_a^b f(x)\,dx = F(b) - F(a),$$

where

$$F'(x) = f(x).$$

Worked example

Compute

$$\int_1^3 (2x+1)\,dx.$$

An antiderivative is

$$F(x) = x^2 + x.$$

Therefore

$$\int_1^3 (2x+1)\,dx = F(3) - F(1).$$

Compute:

$$F(3) = 9 + 3 = 12,$$

$$F(1) = 1 + 1 = 2.$$

Thus

$$\int_1^3 (2x+1)\,dx = 10.$$

12.15 Net Area Versus Geometric Area

The definite integral measures signed area. If the function is below the x-axis, the contribution is negative.

Example

Compute

$$\int_{-1}^{1} x\,dx.$$

An antiderivative is

$$F(x) = \frac{x^2}{2}.$$

Thus

$$\int_{-1}^{1} x\,dx = F(1) - F(-1) = \frac{1}{2} - \frac{1}{2} = 0.$$

This does not mean there is no geometric area. It means the negative area on $[-1, 0]$ cancels the positive area on $[0, 1]$.

The geometric area between $y = x$ and the x-axis from -1 to 1 is

$$\int_{-1}^{1} |x|\,dx = 2\int_{0}^{1} x\,dx = 1.$$

12.16 Substitution

Substitution reverses the chain rule.

Worked example

Compute

$$\int 2x(x^2+1)^5\,dx.$$

Let

$$u = x^2 + 1.$$

Then

$$du = 2x\,dx.$$

Thus

$$\int 2x(x^2+1)^5\,dx = \int u^5\,du.$$

Therefore

$$\int u^5\,du = \frac{u^6}{6} + C.$$

Substitute back:

$$\int 2x(x^2+1)^5\,dx = \frac{(x^2+1)^6}{6} + C.$$

Definite integral with substitution

Compute

$$\int_{0}^{1} 2x(x^2+1)^5\,dx.$$

Use
$$u = x^2 + 1.$$
When
$$x = 0,$$
$$u = 1.$$
When
$$x = 1,$$
$$u = 2.$$
Thus
$$\int_0^1 2x(x^2+1)^5\,dx = \int_1^2 u^5\,du = \left[\frac{u^6}{6}\right]_1^2.$$
Hence
$$\frac{2^6}{6} - \frac{1^6}{6} = \frac{64-1}{6} = \frac{63}{6} = \frac{21}{2}.$$

12.17 Integration by Parts

Integration by parts reverses the product rule:

$$\int u\,dv = uv - \int v\,du.$$

Worked example

Compute
$$\int xe^x\,dx.$$
Choose
$$u = x, \qquad dv = e^x\,dx.$$
Then
$$du = dx, \qquad v = e^x.$$
So
$$\int xe^x\,dx = xe^x - \int e^x\,dx.$$
Therefore
$$\int xe^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Check by differentiating:

$$\frac{d}{dx}\big(e^x(x-1)\big) = e^x(x-1) + e^x = xe^x.$$

12.18 Exponential Growth

The function e^x is characterized by

$$\frac{d}{dx}e^x = e^x.$$

If a quantity grows at a rate proportional to itself, then

$$\frac{dy}{dt} = ky.$$

Separate variables:

$$\frac{1}{y}\,dy = k\,dt.$$

Integrate:

$$\ln|y| = kt + C.$$

Thus

$$y = Ce^{kt}.$$

Worked example

Suppose

$$\frac{dy}{dt} = 3y, \qquad y(0) = 5.$$

Then

$$y = Ce^{3t}.$$

Using

$$y(0) = 5,$$

we get

$$C = 5.$$

Therefore

$$y(t) = 5e^{3t}.$$

12.19 Logarithmic Differentiation

For functions involving powers depending on x, logarithms simplify the derivative.

Derivative of x^x

Let

$$y = x^x, \qquad x > 0.$$

Take logarithms:

$$\ln y = x \ln x.$$

Differentiate:

$$\frac{y'}{y} = \ln x + 1.$$

Therefore

$$y' = y(\ln x + 1).$$

Since

$$y = x^x,$$

$$\frac{d}{dx}x^x = x^x(\ln x + 1).$$

12.20 Newton's Method

Newton's method solves

$$f(x) = 0$$

by replacing the curve with its tangent line.

At x_n, the tangent line is

$$y = f(x_n) + f'(x_n)(x - x_n).$$

Set $y = 0$:

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n).$$

Thus

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Compute $\sqrt{2}$

Solve

$$x^2 - 2 = 0.$$

Here

$$f(x) = x^2 - 2, \qquad f'(x) = 2x.$$

Newton's method gives

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

Start with

$$x_0 = 1.$$

Then

$$x_1 = \frac{1}{2}(1 + 2) = \frac{3}{2}.$$

Next,

$$x_2 = \frac{1}{2}\left(\frac{3}{2} + \frac{2}{3/2}\right) = \frac{1}{2}\left(\frac{3}{2} + \frac{4}{3}\right) = \frac{1}{2} \cdot \frac{17}{6} = \frac{17}{12}.$$

Next,

$$x_3 = \frac{1}{2}\left(\frac{17}{12} + \frac{24}{17}\right) = \frac{1}{2} \cdot \frac{577}{204} = \frac{577}{408}.$$

Numerically,

$$\frac{577}{408} \approx 1.414215686,$$

which is already very close to

$$\sqrt{2} \approx 1.414213562.$$

12.21 Taylor Approximation

A differentiable function can often be approximated near a point by a polynomial built from its derivatives. Around $x = 0$,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

Approximate $e^{0.1}$

Using terms through x^3,

$$e^{0.1} \approx 1 + 0.1 + \frac{0.1^2}{2} + \frac{0.1^3}{6}.$$

Compute:

$$1 + 0.1 = 1.1,$$
$$\frac{0.1^2}{2} = \frac{0.01}{2} = 0.005,$$
$$\frac{0.1^3}{6} = \frac{0.001}{6} \approx 0.0001667.$$

Thus

$$e^{0.1} \approx 1.1051667.$$

The true value is about

$$1.1051702.$$

For sine and cosine,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

Small-angle approximation

For small x,

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}.$$

For

$$x = 0.1,$$
$$\sin(0.1) \approx 0.1 - \frac{0.001}{6} = 0.0998333\ldots.$$

This matches the usual small-angle behavior.

12.22 Arc Length

For a curve

$$y = f(x)$$

on $[a, b]$, the arc length is

$$L = \int_a^b \sqrt{1 + (f'(x))^2}\, dx.$$

Worked example: line segment

Let

$$f(x) = mx + b$$

on $[a, b]$. Then

$$f'(x) = m.$$

So

$$L = \int_a^b \sqrt{1+m^2}\,dx = (b-a)\sqrt{1+m^2}.$$

This agrees with the distance formula. The endpoints are

$$(a, ma+b), \qquad (b, mb+b).$$

The horizontal difference is

$$b - a,$$

and the vertical difference is

$$m(b-a).$$

Thus the length is

$$\sqrt{(b-a)^2 + m^2(b-a)^2} = (b-a)\sqrt{1+m^2}.$$

12.23 A Simple Area Between Curves

Find the area between

$$y = x$$

and

$$y = x^2$$

from $x = 0$ to $x = 1$.

On $[0, 1]$,

$$x \geq x^2.$$

Therefore the area is

$$\int_0^1 (x - x^2)\,dx.$$

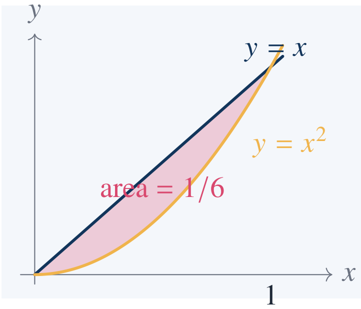
Compute:

$$\int_0^1 x\,dx = \frac{1}{2},$$

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$

Thus

$$\int_0^1 (x - x^2)\,dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$



12.24 Volume by Slicing

If cross-sectional area at position x is $A(x)$, then volume is

$$V = \int_a^b A(x)\,dx.$$

Cone volume

A right circular cone has height h and base radius R. Place its tip at $x = 0$ and base at $x = h$. At distance x from the tip, the radius is

$$r(x) = \frac{R}{h}x.$$

Thus the cross-sectional area is

$$A(x) = \pi r(x)^2 = \pi \frac{R^2}{h^2}x^2.$$

Therefore

$$V = \int_0^h \pi \frac{R^2}{h^2}x^2\,dx.$$

Compute:

$$V = \pi \frac{R^2}{h^2}\left[\frac{x^3}{3}\right]_0^h = \pi \frac{R^2}{h^2} \cdot \frac{h^3}{3} = \frac{1}{3}\pi R^2 h.$$

12.25 Improper Integral

An improper integral may converge even over an infinite interval.

Convergent example

Compute

$$\int_1^\infty \frac{1}{x^2}\,dx.$$

By definition,

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty} \int_1^b x^{-2}\,dx.$$

An antiderivative is

$$-\frac{1}{x}.$$

Thus

$$\int_1^b x^{-2}\,dx = \left[-\frac{1}{x}\right]_1^b = -\frac{1}{b} + 1.$$

Let $b \to \infty$:

$$-\frac{1}{b} + 1 \to 1.$$

Therefore

$$\int_1^\infty \frac{1}{x^2}\,dx = 1.$$

Divergent example

Compute

$$\int_1^\infty \frac{1}{x}\,dx.$$

By definition,

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx.$$

This is

$$\lim_{b\to\infty} [\ln x]_1^b = \lim_{b\to\infty} \ln b.$$

Since

$$\ln b \to \infty,$$

the integral diverges.

The comparison between

$$\frac{1}{x} \quad \text{and} \quad \frac{1}{x^2}$$

is a basic example of how decay rate controls total accumulated area.

12.26 One Calculation Combining Differentiation and Integration

Let

$$F(x) = \int_0^{x^2} \sqrt{1+t^3}\,dt.$$

Find

$$F'(x).$$

Set

$$G(u) = \int_0^u \sqrt{1+t^3}\,dt.$$

Then

$$F(x) = G(x^2).$$

By the fundamental theorem of calculus,

$$G'(u) = \sqrt{1+u^3}.$$

By the chain rule,

$$F'(x) = G'(x^2) \cdot 2x.$$

Thus

$$F'(x) = 2x\sqrt{1 + (x^2)^3} = 2x\sqrt{1 + x^6}.$$

Common mistake

It is wrong to write

$$F'(x) = \sqrt{1 + x^6}$$

and stop. The upper limit is x^2, not x. The extra factor

$$2x$$

comes from the chain rule.

CHAPTER 13

Supplement to Chapter VIII: Mathematical Experiments

Source Map

The word "experiment" here does not mean replacing proof by numerical guessing. It means using computation, approximation, tables, and limiting cases to locate the right mathematical statement before proving it. A good experiment should leave behind a precise conjecture, an error estimate, or a counterexample.

13.1 Difference Quotients as an Experiment

Take

$$f(x) = x^2.$$

At $x = 2$, the difference quotient is

$$\frac{f(2+h) - f(2)}{h} = \frac{(2+h)^2 - 4}{h} = \frac{4 + 4h + h^2 - 4}{h} = 4 + h.$$

Numerically:

h	1	0.5	0.1	0.01	−0.01	−0.1
$\frac{f(2+h) - f(2)}{h}$	5	4.5	4.1	4.01	3.99	3.9

The experiment suggests

$$f'(2) = 4.$$

The exact computation confirms it:

$$\lim_{h \to 0} (4 + h) = 4.$$

A bad experiment if one only samples from one side

Let

$$f(x) = |x|.$$

At $x = 0$,

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h}.$$

If one only tests positive h, the quotient is always 1. That might mislead one into thinking $f'(0) = 1$. But for negative h, the quotient is

$$-1.$$

Thus the two-sided derivative does not exist. Numerical testing must examine the singular side of a problem, not only the convenient side.

13.2 Riemann Sums for $\int_0^1 x^2\,dx$

Divide $[0,1]$ into n equal intervals and use right endpoints:

$$x_k = \frac{k}{n}, \qquad \Delta x = \frac{1}{n}.$$

The sum is

$$S_n = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^{n} k^2.$$

Since

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$

we get

$$S_n = \frac{(n+1)(2n+1)}{6n^2}.$$

For a few values:

n	2	5	10	50	100
S_n	0.625	0.44	0.385	0.3434	0.33835

The values move toward

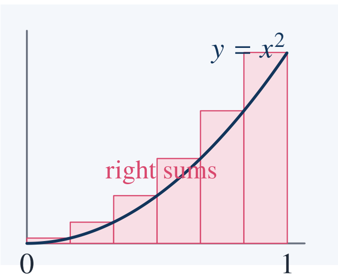
$$\frac{1}{3}.$$

Indeed,

$$\lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}.$$

Therefore

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$



13.3 Trapezoids: Same Integral, Better Approximation

For

$$\int_0^1 x^2\,dx,$$

the trapezoidal rule with n subintervals is

$$T_n = \frac{\Delta x}{2}\left(f(0) + 2f\left(\frac{1}{n}\right) + 2f\left(\frac{2}{n}\right) + \cdots + 2f\left(\frac{n-1}{n}\right) + f(1)\right).$$

Here

$$\Delta x = \frac{1}{n}, \qquad f(x) = x^2.$$

For $n = 4$,

$$T_4 = \frac{1}{8}\left(0 + 2\cdot\frac{1}{16} + 2\cdot\frac{4}{16} + 2\cdot\frac{9}{16} + 1\right).$$

The expression inside parentheses is

$$\frac{2}{16} + \frac{8}{16} + \frac{18}{16} + 1 = \frac{28}{16} + 1 = \frac{44}{16} = \frac{11}{4}.$$

Thus

$$T_4 = \frac{1}{8}\cdot\frac{11}{4} = \frac{11}{32} = 0.34375.$$

The exact value is

$$\frac{1}{3} \approx 0.33333.$$

The trapezoidal rule overestimates here because x^2 is convex.

13.4 A Function That Is Continuous but Violent

Define

$$f(x) = \begin{cases} x\sin\frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Since

$$-1 \leq \sin\frac{1}{x} \leq 1,$$

we have

$$-|x| \leq x\sin\frac{1}{x} \leq |x|.$$

As $x \to 0$,

$$-|x| \to 0, \qquad |x| \to 0.$$

By the squeeze theorem,

$$\lim_{x\to 0} x\sin\frac{1}{x} = 0.$$

Thus

$$\lim_{x\to 0} f(x) = f(0),$$

so f is continuous at 0.

But the graph oscillates infinitely many times near 0. Continuity does not mean visual calmness.

Derivative at 0

Check differentiability at 0:

$$\frac{f(h)-f(0)}{h} = \frac{h\sin(1/h)}{h} = \sin\frac{1}{h}.$$

The limit

$$\lim_{h\to 0}\sin\frac{1}{h}$$

does not exist. Therefore f is continuous at 0, but not differentiable there.

13.5 Work as an Integral

If a variable force $F(x)$ moves an object from $x = a$ to $x = b$, the work is

$$W = \int_a^b F(x)\,dx.$$

For a spring obeying Hooke's law,

$$F(x) = kx.$$

The work required to stretch it from 0 to A is

$$W = \int_0^A kx\,dx = k\left[\frac{x^2}{2}\right]_0^A = \frac{1}{2}kA^2.$$

Numerical example

If

$$k = 12 \qquad \text{and} \qquad A = 0.5,$$

then

$$W = \frac{1}{2}\cdot 12\cdot(0.5)^2 = 6\cdot 0.25 = 1.5.$$

The work is

$$1.5$$

in the corresponding units.

This example is better than a constant-force example because the integral is not decorative. It is exactly what replaces multiplication when the force is changing.

13.6 Arc Length as an Experimental Formula

For a curve

$$y = f(x),$$

divide the interval into small pieces. On one piece,

$$\Delta s \approx \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Since

$$\Delta y \approx f'(x)\Delta x,$$

we get

$$\Delta s \approx \sqrt{1 + (f'(x))^2}\,\Delta x.$$

Thus

$$L = \int_a^b \sqrt{1 + (f'(x))^2}\,dx.$$

Check with a line

Let

$$f(x) = mx + b$$

on $[a, b]$. Then

$$f'(x) = m.$$

Therefore

$$L = \int_a^b \sqrt{1 + m^2}\,dx = (b - a)\sqrt{1 + m^2}.$$

This agrees with the ordinary distance formula. The vertical change is

$$m(b - a),$$

so the length is

$$\sqrt{(b - a)^2 + m^2(b - a)^2} = (b - a)\sqrt{1 + m^2}.$$

13.7 Exponential Growth Beats Powers

Compare

$$e^x \qquad \text{and} \qquad x^5.$$

For small x, x^5 may look large. But at infinity,

$$e^x$$

eventually dominates every power x^n.

One precise statement is

$$\lim_{x\to\infty} \frac{x^n}{e^x} = 0$$

for every fixed positive integer n.

For $n = 5$, apply L'Hospital's rule repeatedly:

$$\lim_{x\to\infty} \frac{x^5}{e^x} = \lim_{x\to\infty} \frac{5x^4}{e^x} = \lim_{x\to\infty} \frac{20x^3}{e^x} = \lim_{x\to\infty} \frac{60x^2}{e^x} = \lim_{x\to\infty} \frac{120x}{e^x} = \lim_{x\to\infty} \frac{120}{e^x} = 0.$$

Numerical check

x	5	10	20
x^5	3125	100000	3200000
e^x	$148.4\ldots$	$22026.4\ldots$	4.85×10^8

At $x = 5$, the power is larger. At $x = 20$, the exponential is already much larger. The limit statement is about eventual behavior, not about small values.

13.8 Logarithms Grow Slower Than Powers

For any $a > 0$,

$$\lim_{x\to\infty} \frac{\log x}{x^a} = 0.$$

For example,

$$\lim_{x\to\infty} \frac{\log x}{\sqrt{x}} = 0.$$

By L'Hospital's rule,

$$\lim_{x\to\infty} \frac{\log x}{x^{1/2}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{2}x^{-1/2}} = \lim_{x\to\infty} \frac{2}{\sqrt{x}} = 0.$$

Consequence for $\log(n!)$

Since

$$\log(n!) = \sum_{k=1}^{n} \log k,$$

one can compare it with an integral:

$$\int_1^n \log x\,dx \le \sum_{k=1}^{n} \log k \le \int_1^{n+1} \log x\,dx.$$

Now

$$\int \log x\,dx = x\log x - x.$$

Therefore

$$\log(n!) \sim n\log n$$

at the rough order-of-magnitude level. More accurately, Stirling's formula gives

$$\log(n!) = n\log n - n + O(\log n).$$

The experiment here is that sums can often be estimated by areas. This is one of the main ways calculus enters discrete mathematics.

13.9 A Geometric Series of Functions

For $|x| < 1$,

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}.$$

The partial sum is

$$S_n(x) = 1 + x + \cdots + x^n = \frac{1-x^{n+1}}{1-x}.$$

If $|x| < 1$, then

$$x^{n+1} \to 0,$$

so

$$S_n(x) \to \frac{1}{1-x}.$$

Numerical experiment at $x = \frac{1}{2}$

n	0	1	2	3	4
$S_n(1/2)$	1	1.5	1.75	1.875	1.9375

The limiting value is

$$\frac{1}{1 - 1/2} = 2.$$

At $x = 1$, the same formal expression becomes

$$1 + 1 + 1 + \cdots,$$

which diverges. At $x = -1$, it becomes

$$1 - 1 + 1 - 1 + \cdots,$$

which does not converge in the ordinary sense. The interval of convergence is part of the object, not an afterthought.

13.10 The Exponential Series

The exponential function has the series expansion

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

Approximate e

At $x = 1$,

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots.$$

Using terms through $1/5!$,

$$1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} = 2 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120}.$$

With denominator 120,

$$2 = \frac{240}{120}, \quad \frac{1}{2} = \frac{60}{120}, \quad \frac{1}{6} = \frac{20}{120}, \quad \frac{1}{24} = \frac{5}{120}.$$

So the sum is

$$\frac{240 + 60 + 20 + 5 + 1}{120} = \frac{326}{120} = 2.716666\ldots.$$

The true value is

$$e = 2.718281\ldots.$$

The terms decrease very quickly because factorials grow faster than powers.

13.11 Euler's Formula from Series

Use the power series

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \cdots.$$

Set

$$z = ix.$$

Then

$$e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \cdots.$$

Since

$$i^2 = -1, \quad i^3 = -i, \quad i^4 = 1,$$

we get

$$e^{ix} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right).$$

But

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots,$$

and

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots.$$

Therefore

$$e^{ix} = \cos x + i \sin x.$$

The case $x = \pi$

Putting $x = \pi$ gives

$$e^{i\pi} = \cos\pi + i\sin\pi.$$

Since

$$\cos\pi = -1, \qquad \sin\pi = 0,$$

we get

$$e^{i\pi} = -1.$$

Thus

$$e^{i\pi} + 1 = 0.$$

This is a genuine mathematical experiment: extending the exponential series to imaginary inputs reveals trigonometry inside exponentiation.

13.12 Harmonic Series and Slow Divergence

The harmonic series is

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots.$$

It diverges, but slowly.

Group terms:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) + \cdots.$$

In each block after the first two, each term is at least the reciprocal of the last denominator in the block:

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

$$\frac{1}{5} + \cdots + \frac{1}{8} > 4 \cdot \frac{1}{8} = \frac{1}{2},$$

$$\frac{1}{9} + \cdots + \frac{1}{16} > 8 \cdot \frac{1}{16} = \frac{1}{2}.$$

So the partial sums eventually exceed

$$1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots,$$

and hence diverge.

Why numerical testing can mislead

Even at $n = 1000$,

$$\sum_{k=1}^{1000} \frac{1}{k}$$

is only about

$$7.485.$$

Slow growth can look like convergence if the experiment is too small. The proof above sees the long-term structure.

13.13 The Zeta Function at 2

The series

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges. Numerically:

N	1	2	5	10	100
$\sum_{n=1}^{N} \frac{1}{n^2}$	1	1.25	1.4636...	1.5497...	1.6350...

The exact value is

$$\zeta(2) = \frac{\pi^2}{6}.$$

Since

$$\frac{\pi^2}{6} \approx 1.644934,$$

the convergence is visible but not extremely fast.

Integral bound for the tail

For $N \geq 1$,

$$\sum_{n=N+1}^{\infty} \frac{1}{n^2} < \int_N^{\infty} \frac{dx}{x^2}.$$

Compute:

$$\int_N^{\infty} x^{-2}\, dx = \left[-\frac{1}{x}\right]_N^{\infty} = \frac{1}{N}.$$

So after N terms, the error is less than

$$\frac{1}{N}.$$

For $N = 100$, the error is less than

$$0.01.$$

13.14 Euler's Product for Sine

The zeros of

$$\sin x$$

occur at

$$x = 0,\ \pm\pi,\ \pm 2\pi,\ \pm 3\pi, \ldots.$$

Euler's product is

$$\frac{\sin x}{x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right).$$

A finite version is already suggestive:

$$\frac{\sin x}{x} \approx \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right).$$

The factors force zeros at

$$x = \pm\pi, \quad \pm 2\pi, \quad \pm 3\pi.$$

Recovering $\zeta(2)$ from the product

Expand the product formally near $x = 0$:

$$\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right) = 1 - \left(\sum_{n=1}^{\infty} \frac{1}{n^2\pi^2}\right) x^2 + \cdots.$$

But

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots = 1 - \frac{x^2}{6} + \cdots.$$

Comparing the coefficient of x^2,

$$\sum_{n=1}^{\infty} \frac{1}{n^2\pi^2} = \frac{1}{6}.$$

Therefore

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

This is not a rigorous proof as written unless convergence of the product is justified, but it is an extremely productive experiment: zeros of a function predict a product expansion, and the product expansion predicts a sum.

13.15 Prime Counts as a Statistical Experiment

Let

$$\pi(x) = \#\{p \le x : p \text{ prime}\}.$$

A simple table gives:

x	10	100	1000	10000
$\pi(x)$	4	25	168	1229

The prime number theorem says

$$\pi(x) \sim \frac{x}{\log x}.$$

Compare:

x	10	100	1000	10000
$x/\log x$	$4.34\ldots$	$21.71\ldots$	$144.76\ldots$	$1085.73\ldots$

The estimate is rough at small values but improves in relative scale.

Why $1/\log x$ appears heuristically

The number

$$n$$

survives divisibility by small primes $p \leq y$ with approximate probability

$$\prod_{p \leq y} \left(1 - \frac{1}{p}\right).$$

Euler-type product heuristics suggest this behaves like

$$\frac{C}{\log y}$$

for a constant C. If y is chosen on the scale needed to test primality, the probability that a large integer near x is prime is of order

$$\frac{1}{\log x}.$$

Therefore the expected number of primes up to x is on the scale

$$\int_2^x \frac{dt}{\log t},$$

and roughly

$$\frac{x}{\log x}.$$

This heuristic is not the proof of the prime number theorem. Its value is that it explains why the density of primes should decay logarithmically rather than like $1/x$, $1/\sqrt{x}$, or a constant.

13.16 A Better Approximation: Logarithmic Integral

The approximation

$$\frac{x}{\log x}$$

is simple, but

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}$$

usually tracks $\pi(x)$ better.

For example,

$$\pi(1000) = 168.$$

Meanwhile

$$\frac{1000}{\log 1000} \approx 144.76.$$

The logarithmic integral gives a larger and more accurate value.

The reason is that prime density changes with t. Replacing the whole interval $[2, x]$ by the final density $1/\log x$ undercounts, because earlier numbers have larger density

$$\frac{1}{\log t}.$$

13.17 Numerical Evidence Is Not Proof

The polynomial

$$f(n) = n^2 + n + 41$$

gives many primes at first:

n	0	1	2	3	4	5
$f(n)$	41	43	47	53	61	71

But at

$$n = 40,$$

$$f(40) = 40^2 + 40 + 41 = 1681 = 41^2.$$

So the prime-generating pattern fails.

The point of the example

A mathematical experiment can reveal a pattern, but a pattern over many cases is still not a theorem. The right conclusion from this experiment is not "the polynomial always gives primes." The right conclusion is:

find the mechanism that makes it fail or prove why it cannot fail.

Here the mechanism is immediate:

$$f(40) = 41^2.$$

13.18 A Small Experiment with Finite Differences

Suppose the values of a sequence are

n	0	1	2	3	4
a_n	1	4	9	16	25

The first differences are

$$3,\ 5,\ 7,\ 9.$$

The second differences are

$$2,\ 2,\ 2.$$

Constant second differences suggest a quadratic formula. Indeed,

$$a_n = (n+1)^2.$$

Another finite-difference experiment

Let

n	0	1	2	3	4
a_n	2	5	10	17	26

First differences:

$$3,\ 5,\ 7,\ 9.$$

Second differences:

$$2,\ 2,\ 2.$$

So again the formula is quadratic:

$$a_n = an^2 + bn + c.$$

Since

$$a_0 = 2,$$

we get

$$c = 2.$$

Since

$$a_1 = 5,$$

$$a + b + 2 = 5,$$

so

$$a + b = 3.$$

Since

$$a_2 = 10,$$

$$4a + 2b + 2 = 10,$$

so

$$2a + b = 4.$$

Subtracting

$$a + b = 3$$

from

$$2a + b = 4$$

gives

$$a = 1.$$

Then

$$b = 2.$$

Therefore

$$a_n = n^2 + 2n + 2.$$

Finite differences are a clean example of experimental mathematics done responsibly: compute, conjecture the degree, solve for coefficients, then verify.

13.19 One Experimental Rule Worth Keeping

When a numerical pattern appears, test it in three ways:

test	what it checks
edge cases	whether the pattern survives small, zero, negative, or boundary inputs
asymptotics	whether the pattern has the right large-scale growth
structural reason	whether there is an identity, invariant, or theorem forcing it

For example, the data

$$1,\ 4,\ 9,\ 16,\ 25$$

suggests squares. The structural reason is the formula

$$a_n = (n+1)^2.$$

The prime polynomial

$$n^2 + n + 41$$

also looks convincing at first, but its structural failure at

$$n = 40$$

is decisive.

Part IV

Modern Developments and Working Logs

CHAPTER 14

Recent Developments

Source Map

This chapter is not a list of fashionable results. Its value is that it shows how old elementary questions can lead into modern mathematics: prime numbers lead to analytic and probabilistic methods; map coloring leads to graph theory and computation; curves and surfaces lead to topology, knots, soap films, and minimal surfaces; infinitesimals reappear through nonstandard analysis.

14.1 A Formula for Primes

A tempting question is whether primes can be produced by a simple formula. Euler's polynomial

$$f(n) = n^2 + n + 41$$

gives primes for many initial values.

n	0	1	2	3	4	5	6	7
$n^2 + n + 41$	41	43	47	53	61	71	83	97

But the pattern cannot continue indefinitely. At

$$n = 40,$$

we get

$$f(40) = 40^2 + 40 + 41 = 1681 = 41^2.$$

So the formula fails.

Why this failure is structural

For

$$f(n) = n^2 + n + 41,$$

take

$$n = 41k.$$

Then

$$f(41k) = (41k)^2 + 41k + 41.$$

Factor out 41:

$$f(41k) = 41(41k^2 + k + 1).$$

Thus for every positive integer k,

$$f(41k)$$

is composite. The failure at $n = 40$ is not an accident; the polynomial has built-in divisibility obstructions.

There are artificial formulas that encode primes, but they usually hide the difficulty rather than solve it. A meaningful formula should explain why the outputs are prime, not merely disguise a primality test.

14.2 Goldbach and Twin Primes

Goldbach's conjecture says that every even integer greater than 2 is the sum of two primes:

$$2m = p + q.$$

Small cases are easy:

$$4 = 2 + 2,$$

$$6 = 3 + 3,$$

$$8 = 3 + 5,$$

$$10 = 5 + 5,$$

$$12 = 5 + 7,$$

$$20 = 3 + 17 = 7 + 13.$$

A counting experiment

For

$$N = 50,$$

some decompositions are

$$50 = 3 + 47 = 7 + 43 = 13 + 37 = 19 + 31.$$

So 50 has several Goldbach representations.
For

$$N = 100,$$

examples include

$$100 = 3 + 97 = 11 + 89 = 17 + 83 = 29 + 71 = 41 + 59 = 47 + 53.$$

The number of representations tends to grow on average, but irregularly.

The twin prime conjecture asks whether infinitely many prime pairs

$$p,\ p + 2$$

exist. Examples:

$$(3,5), \quad (5,7), \quad (11,13), \quad (17,19), \quad (29,31).$$

A naive probabilistic model says that an integer near x has probability about

$$\frac{1}{\log x}$$

of being prime. Then the chance that both n and $n+2$ are prime should look roughly like

$$\frac{C}{(\log n)^2}$$

for a correction constant C. This suggests that the expected number of twin primes up to x should be comparable to

$$C \int_2^x \frac{dt}{(\log t)^2}.$$

This integral tends to infinity, so the heuristic predicts infinitely many twin primes.

Remark

The heuristic is not a proof. It treats primality as if it were nearly random, but divisibility conditions are highly structured. The value of the heuristic is that it predicts the correct scale of the problem.

14.3 Fermat's Last Theorem

Fermat's Last Theorem states that for every integer

$$n > 2,$$

there are no positive integers x, y, z satisfying

$$x^n + y^n = z^n.$$

For

$$n = 2,$$

solutions exist:

$$3^2 + 4^2 = 5^2.$$

The theorem says that the moment the exponent is larger than 2, this phenomenon disappears.

The case $n = 4$: descent idea

A classical part of Fermat's argument proves that

$$x^4 + y^4 = z^4$$

has no positive integer solution. It is enough to rule out

$$x^4 + y^4 = w^2,$$

because if

$$x^4 + y^4 = z^4,$$

then

$$x^4 + y^4 = (z^2)^2.$$

The descent strategy is:

1. Assume a smallest positive solution exists.
2. Use the structure of primitive Pythagorean triples.
3. Construct a smaller positive solution.
4. Contradict minimality.

The method is powerful because it does not merely check examples. It shows that any alleged solution would force an infinite decreasing chain of positive integers, which is impossible.

Why exponent reduction matters

If Fermat's equation has a solution for exponent $n = ab$, then

$$x^{ab} + y^{ab} = z^{ab}$$

can be rewritten as

$$(x^a)^b + (y^a)^b = (z^a)^b.$$

So a solution for exponent ab gives a solution for exponent b.
Therefore it is enough to prove the theorem for exponent 4 and for odd prime exponents. This reduction is elementary but strategically important.

The modern proof connects the equation

$$x^p + y^p = z^p$$

to elliptic curves and modular forms. The striking point is not only that the old problem was solved, but that its solution required a bridge between number theory and geometry.

14.4 The Continuum Hypothesis

The set of natural numbers has cardinality

$$\aleph_0.$$

The set of real numbers has cardinality

$$2^{\aleph_0}.$$

Cantor proved that

$$2^{\aleph_0} > \aleph_0.$$

The continuum hypothesis asks whether there is a set whose cardinality lies strictly between these two:

$$\aleph_0 < |S| < 2^{\aleph_0}.$$

The continuum hypothesis says that no such set exists.

Cantor diagonal argument

Suppose the real numbers in $(0, 1)$ could be listed:

$$r_1 = 0.a_{11}a_{12}a_{13}\cdots,$$
$$r_2 = 0.a_{21}a_{22}a_{23}\cdots,$$
$$r_3 = 0.a_{31}a_{32}a_{33}\cdots.$$

Construct a new decimal

$$s = 0.b_1b_2b_3\cdots$$

by choosing b_n different from a_{nn}, avoiding 9 to prevent decimal ambiguity. Then s differs from r_n in the n-th digit. Therefore s is not on the list.
So the real numbers are uncountable.

The independence result is deeper: within the usual Zermelo–Fraenkel set theory with choice, the continuum hypothesis can neither be proved nor disproved, assuming the axioms are consistent.

Remark

The lesson is not that the question is meaningless. The lesson is that a formal axiom system may leave some natural questions undecided.

14.5 Set-Theoretic Notation

A useful discipline in modern mathematics is to distinguish membership, inclusion, and equality.

notation	meaning
$x \in A$	x is an element of A
$A \subseteq B$	every element of A is an element of B
$A = B$	$A \subseteq B$ and $B \subseteq A$
$A \cup B$	elements in A or B
$A \cap B$	elements in both A and B
$A \setminus B$	elements in A but not in B

Example

Let

$$A = \{1, 2, \{3, 4\}\}.$$

Then

$$1 \in A,$$
$$3 \notin A,$$

but

$$\{3, 4\} \in A.$$

Also,

$$\{1, 2\} \subseteq A,$$

but

$$\{3,4\} \subseteq A$$

is false, because $3 \notin A$ and $4 \notin A$.

This small example prevents a common confusion: being an element is not the same as being a subset.

14.6 The Four-Color Theorem

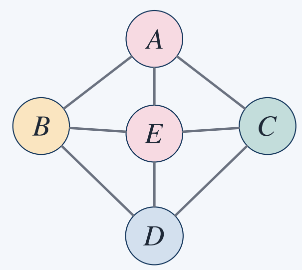
The four-color theorem says that every planar map can be colored with at most four colors so that adjacent regions receive different colors.

A map can be converted into a graph:

$$\text{regions} \longleftrightarrow \text{vertices},$$

$$\text{shared border} \longleftrightarrow \text{edge}.$$

The theorem becomes: every planar graph is four-colorable.



a planar graph with a four-coloring

Euler inequality behind the six-color theorem

For a connected simple planar graph with

$$V \geq 3,$$

Euler's formula gives

$$V - E + F = 2.$$

Every face has at least three edges, and each edge borders at most two faces, so

$$3F \leq 2E.$$

Since

$$F = 2 - V + E,$$

we get

$$3(2 - V + E) \leq 2E.$$

Thus

$$6 - 3V + 3E \leq 2E,$$

so

$$E \le 3V - 6.$$

The average degree is

$$\frac{2E}{V} \le 6 - \frac{12}{V} < 6.$$

Therefore some vertex has degree at most 5.

This proves the key step for the six-color theorem. The four-color theorem is harder because degree 5 is not enough for the same simple induction.

14.7 Hausdorff Dimension and Fractals

Ordinary dimension assigns:

$$\dim(\text{line}) = 1, \qquad \dim(\text{square}) = 2, \qquad \dim(\text{cube}) = 3.$$

Fractals force a more flexible notion of dimension.

For a self-similar set made of N scaled copies of itself, each reduced by a factor r, the similarity dimension d satisfies

$$Nr^d = 1.$$

Equivalently,

$$d = \frac{\log N}{\log(1/r)}.$$

Cantor set dimension

The middle-third Cantor set consists of

$$N = 2$$

copies of itself, each scaled by

$$r = \frac{1}{3}.$$

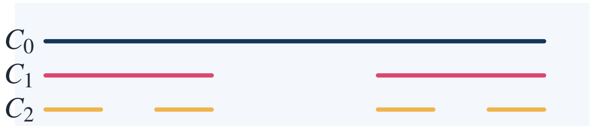
Thus

$$d = \frac{\log 2}{\log 3}.$$

This is approximately

$$0.6309.$$

So the Cantor set has dimension strictly between 0 and 1.



middle-thirds construction

Koch curve dimension

The Koch curve replaces one segment by

$$N = 4$$

segments, each scaled by

$$r = \frac{1}{3}.$$

So its dimension is

$$d = \frac{\log 4}{\log 3}.$$

Since

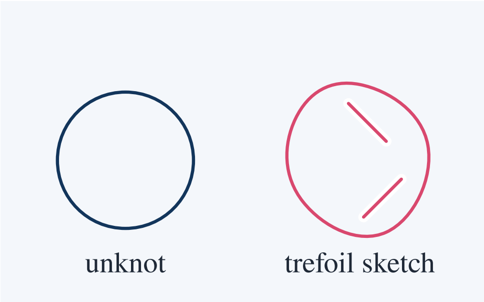
$$1 < \frac{\log 4}{\log 3} < 2,$$

the Koch curve is more than a curve but less than a region.

14.8 Knots

A knot is a closed loop in three-dimensional space. Two knots are equivalent if one can be deformed into the other without cutting the loop or passing it through itself.

The unknot is equivalent to a simple circle. The trefoil is the simplest nontrivial knot.



Tricolorability test for the trefoil

A knot diagram is tricolorable if its arcs can be colored with three colors so that at each crossing either all three incident arcs have the same color or all three have different colors.
The unknot cannot be nontrivially tricolored: any coloring satisfying the rule collapses to one color.
The trefoil has a nontrivial tricoloring. Therefore the trefoil is not equivalent to the unknot.

This is the essential topological move: replace a visual deformation question by an invariant that survives deformation.

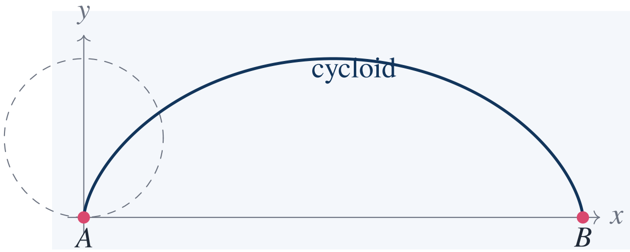
14.9 A Problem in Mechanics

A typical mechanics problem asks for the shape or path forced by minimizing energy or action. The simplest exact example is the brachistochrone problem: find the curve along which a bead slides under gravity from A to B in least time.

The answer is not a straight line and not a circular arc. It is a cycloid.

A cycloid can be parametrized as

$$x = a(\theta - \sin\theta),$$
$$y = a(1 - \cos\theta).$$



Why straight-line intuition fails

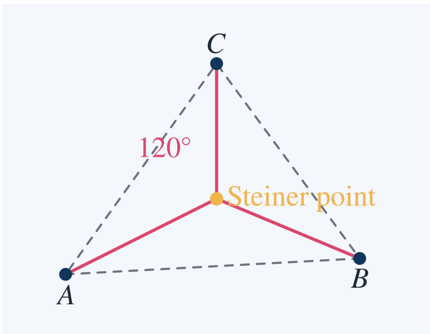
A straight line is shortest in distance, but time is not distance. A curve that initially drops more steeply gives the bead higher speed earlier. The cycloid balances two effects:

$$\text{longer path} \qquad \text{versus} \qquad \text{greater speed gained early.}$$

The minimizing object is determined by a variational problem, not by ordinary Euclidean length.

14.10 Steiner's Problem

Given three points in the plane, find the shortest network connecting them. For a triangle with all angles less than 120°, the minimizing network uses one interior junction where three segments meet at 120°.



Why 120° appears

Suppose three segments meet at an interior minimizing junction. If two of the angles were less than 120°, one can locally shorten the network by moving the junction. At equilibrium, the three unit direction vectors along the edges must sum to zero:

$$u_1 + u_2 + u_3 = 0.$$

Three equal-length unit vectors can sum to zero only when they are separated by angles of

$$120^\circ.$$

If one angle of the original triangle is at least $120°$, the Steiner point disappears; the shortest network is just the two sides meeting at that large angle.

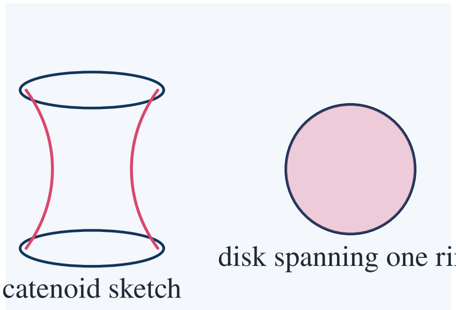
14.11 Soap Films and Minimal Surfaces

Soap films minimize area subject to boundary constraints. The simplest case is a circular wire frame. The minimal surface spanning it is the flat disk.

For two parallel circular rings, the soap film may form a catenoid. A catenoid is generated by rotating a catenary

$$y = a\cosh(x/a)$$

around an axis.



Mean curvature condition

A smooth soap film surface has zero mean curvature:

$$H = 0.$$

For a plane,

$$H = 0.$$

For a sphere of radius R,

$$H = \frac{1}{R}$$

up to convention, so a sphere is not a minimal surface in free space. It is a constant mean curvature surface, relevant when pressure differs across the surface.

The physical film gives an experiment, but the mathematical object is the solution of an area-minimization problem.

14.12 Plateau's Problem

Plateau's problem asks whether every reasonable closed curve in space bounds a surface of least area.

For a planar circle, the answer is easy:

$$\text{the minimal surface is the disk.}$$

For a twisted space curve, the surface may be highly nontrivial.

Why existence is not automatic

A minimizing sequence of surfaces may have areas approaching the infimum:

$$A_1 \geq A_2 \geq A_3 \geq \cdots .$$

But this does not by itself produce a limiting surface. The sequence might develop folds, singularities, or concentrate area.

A proof of existence must show compactness in a suitable class of surfaces and lower semicontinuity of area:

$$A(S) \leq \liminf_{n\to\infty} A(S_n).$$

These are analytical conditions, not just geometric intuition.

This is a typical modern development: an elementary-looking question becomes a problem in analysis, topology, and geometric measure theory.

14.13 Nonstandard Analysis

Nonstandard analysis gives a rigorous framework for infinitesimals. It extends the real numbers to a larger system containing nonzero numbers ε such that

$$|\varepsilon| < \frac{1}{n}$$

for every positive integer n.

Such an ε is infinitesimal.

The derivative can be written in the old Leibniz style:

$$f'(x) = \mathrm{st}\left(\frac{f(x+\varepsilon) - f(x)}{\varepsilon}\right),$$

where st denotes the standard part.

Derivative of x^2 using an infinitesimal

Let

$$f(x) = x^2.$$

Then

$$\frac{f(x+\varepsilon) - f(x)}{\varepsilon} = \frac{(x+\varepsilon)^2 - x^2}{\varepsilon}.$$

Expand:

$$(x+\varepsilon)^2 - x^2 = 2x\varepsilon + \varepsilon^2.$$

So

$$\frac{(x+\varepsilon)^2 - x^2}{\varepsilon} = 2x + \varepsilon.$$

Since ε is infinitesimal,

$$\mathrm{st}(2x + \varepsilon) = 2x.$$

Therefore

$$f'(x) = 2x.$$

Integral as a hyperfinite sum

Divide $[0, 1]$ into N equal pieces where N is an infinite hyperinteger. Then

$$\Delta x = \frac{1}{N}$$

is infinitesimal. For

$$f(x) = x^2,$$

the hyperfinite right sum is

$$\sum_{k=1}^{N} \left(\frac{k}{N}\right)^2 \frac{1}{N} = \frac{1}{N^3} \sum_{k=1}^{N} k^2.$$

Using

$$\sum_{k=1}^{N} k^2 = \frac{N(N+1)(2N+1)}{6},$$

we get

$$\frac{(N+1)(2N+1)}{6N^2}.$$

Rewrite:

$$\frac{(N+1)(2N+1)}{6N^2} = \frac{2N^2+3N+1}{6N^2} = \frac{1}{3} + \frac{1}{2N} + \frac{1}{6N^2}.$$

The last two terms are infinitesimal, so the standard part is

$$\frac{1}{3}.$$

Thus

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$

Nonstandard analysis does not make the old infinitesimal calculations informal. It supplies a formal universe where such calculations can be made exact.

14.14 A Modern Pattern Across the Chapter

Several examples in this chapter share the same mathematical pattern: start with a concrete problem, encode it into a structure, and then find an invariant or limiting principle.

problem	encoding	tool
prime distribution	$\pi(x)$, $\log x$	asymptotic analysis
map coloring	planar graph	graph invariants
fractals	self-similar scaling	dimension
knots	knot diagrams	topological invariants
soap films	spanning surfaces	area minimization
infinitesimals	$^*\mathbb{R}$	standard part map

The strength of these developments is not technical decoration. Each one takes a naive question and turns it into a precise mathematical object.

CHAPTER A

Problems, Exercises, and Solution Notes

Source Map

The exercises below are written as reading notes rather than as a separate problem book. Each problem is meant to test whether the idea of a chapter has become usable: compute something, prove a small statement, find a counterexample, or identify the invariant that controls the situation.

A.1 Number Theory

Problem 1. Euclidean algorithm

Find

$$\gcd(252, 198).$$

Solution

Use Euclidean division:

$$252 = 198 + 54,$$

$$198 = 3 \cdot 54 + 36,$$

$$54 = 1 \cdot 36 + 18,$$

$$36 = 2 \cdot 18 + 0.$$

Hence

$$\gcd(252, 198) = 18.$$

Problem 2. Bezout coefficients

Find integers m, n such that

$$252m + 198n = 18.$$

Solution

Work backward:

$$18 = 54 - 36.$$

Since

$$36 = 198 - 3 \cdot 54,$$

we get

$$18 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198.$$

Since

$$54 = 252 - 198,$$

$$18 = 4(252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198.$$

Thus one choice is

$$m = 4, \qquad n = -5.$$

Problem 3. Irrationality of $\sqrt{2}$

Prove that

$$\sqrt{2} \notin \mathbb{Q}.$$

Solution

Assume

$$\sqrt{2} = \frac{a}{b}$$

in lowest terms, with $a, b \in \mathbb{Z}$, $b \neq 0$. Then

$$2 = \frac{a^2}{b^2},$$

so

$$a^2 = 2b^2.$$

Thus a^2 is even, hence a is even. Write

$$a = 2c.$$

Then

$$4c^2 = 2b^2,$$

so

$$b^2 = 2c^2.$$

Thus b is also even. This contradicts the assumption that a/b was in lowest terms. Therefore

$$\sqrt{2}$$

is irrational.

Problem 4. A modular inverse

Find the inverse of 7 modulo 26.

Solution

We need

$$7x \equiv 1 \pmod{26}.$$

Use Euclidean algorithm:

$$26 = 3 \cdot 7 + 5,$$
$$7 = 1 \cdot 5 + 2,$$
$$5 = 2 \cdot 2 + 1.$$

Back substitute:

$$1 = 5 - 2 \cdot 2.$$

Since

$$2 = 7 - 5,$$
$$1 = 5 - 2(7 - 5) = 3 \cdot 5 - 2 \cdot 7.$$

Since

$$5 = 26 - 3 \cdot 7,$$
$$1 = 3(26 - 3 \cdot 7) - 2 \cdot 7 = 3 \cdot 26 - 11 \cdot 7.$$

Therefore

$$-11 \cdot 7 \equiv 1 \pmod{26}.$$

So

$$7^{-1} \equiv -11 \equiv 15 \pmod{26}.$$

Check:

$$7 \cdot 15 = 105 \equiv 1 \pmod{26}.$$

Problem 5. Chinese remainder theorem

Solve

$$x \equiv 2 \pmod{3}, \qquad x \equiv 3 \pmod{5}, \qquad x \equiv 2 \pmod{7}.$$

Solution

First solve the first two congruences. Write

$$x = 2 + 3k.$$

Then

$$2 + 3k \equiv 3 \pmod{5},$$

so

$$3k \equiv 1 \pmod{5}.$$

Since

$$3^{-1} \equiv 2 \pmod{5},$$

we get

$$k \equiv 2 \pmod{5}.$$

Thus
$$k = 2 + 5t,$$
and
$$x = 2 + 3(2 + 5t) = 8 + 15t.$$
Now impose
$$x \equiv 2 \pmod 7.$$
Then
$$8 + 15t \equiv 2 \pmod 7.$$
Since
$$8 \equiv 1, \qquad 15 \equiv 1 \pmod 7,$$
we get
$$1 + t \equiv 2 \pmod 7,$$
so
$$t \equiv 1 \pmod 7.$$
Therefore
$$x = 8 + 15(1 + 7s) = 23 + 105s.$$
The solution is
$$x \equiv 23 \pmod{105}.$$

A.2 Sets, Logic, and Probability

Problem 6. Inclusion–exclusion

In a class of 60 students, 32 take algebra, 28 take analysis, and 15 take both. How many take neither?

Solution

Let
$$A = \{\text{students taking algebra}\}, \qquad B = \{\text{students taking analysis}\}.$$
Then
$$|A \cup B| = |A| + |B| - |A \cap B|.$$
So
$$|A \cup B| = 32 + 28 - 15 = 45.$$
Therefore the number taking neither is
$$60 - 45 = 15.$$

Problem 7. A set identity

Prove
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

Solution

Use complements:

$$A \setminus (B \cap C) = A \cap (B \cap C)^c.$$

By De Morgan's law,

$$(B \cap C)^c = B^c \cup C^c.$$

Hence

$$A \setminus (B \cap C) = A \cap (B^c \cup C^c).$$

Distribute:

$$A \cap (B^c \cup C^c) = (A \cap B^c) \cup (A \cap C^c).$$

But

$$A \cap B^c = A \setminus B, \qquad A \cap C^c = A \setminus C.$$

Therefore

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

Problem 8. Negating a quantified statement

Negate:

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ |a_n - L| < \epsilon.$$

Solution

Negate one quantifier at a time:

$$\exists \epsilon > 0,\ \forall N \in \mathbb{N},\ \exists n \geq N$$

such that

$$|a_n - L| \geq \epsilon.$$

Thus the negation is

$$\exists \epsilon > 0,\ \forall N \in \mathbb{N},\ \exists n \geq N \text{ such that } |a_n - L| \geq \epsilon.$$

This is the precise statement that the sequence does not converge to L.

Problem 9. Bayes' formula

A disease has prevalence 1%. A test has sensitivity 99% and false positive rate 4%. If a person tests positive, what is the probability that the person has the disease?

Solution

Let

$$D = \{\text{disease}\}, \qquad + = \{\text{positive test}\}.$$

Given:

$$\mathbb{P}(D) = 0.01, \qquad \mathbb{P}(D^c) = 0.99,$$

$$\mathbb{P}(+ \mid D) = 0.99, \qquad \mathbb{P}(+ \mid D^c) = 0.04.$$

First compute

$$\begin{aligned}\mathbb{P}(+) &= \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c) \\ &= 0.99 \cdot 0.01 + 0.04 \cdot 0.99 \\ &= 0.0099 + 0.0396 \\ &= 0.0495.\end{aligned}$$

Bayes' formula gives

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.0099}{0.0495} = \frac{1}{5}.$$

So the probability is

$$20\%.$$

A.3 Geometric Construction and Fields

Problem 10. Constructibility obstruction

Show that

$$\sqrt[3]{5}$$

is not constructible by straightedge and compass.

Solution

The number

$$\sqrt[3]{5}$$

is a root of

$$x^3 - 5.$$

By the rational root test, any rational root must be among

$$\pm 1, \ \pm 5.$$

None of these is a root:

$$\begin{aligned} 1^3 - 5 &= -4, &\qquad (-1)^3 - 5 &= -6, \\ 5^3 - 5 &= 120, &\qquad (-5)^3 - 5 &= -130. \end{aligned}$$

Thus

$$x^3 - 5$$

has no rational root. Since it is cubic, it is irreducible over $\mathbb{Q}$. Therefore

$$[\mathbb{Q}(\sqrt[3]{5}) : \mathbb{Q}] = 3.$$

A constructible algebraic number must have degree dividing a power of 2. Since

$$3 \nmid 2^r$$

for every r, the number

$$\sqrt[3]{5}$$

is not constructible.

Problem 11. Regular pentagon computation

Show that

$$\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}.$$

Solution

Let

$$\zeta = e^{2\pi i/5}.$$

Then

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0.$$

Divide by ζ^2:

$$\zeta^{-2} + \zeta^{-1} + 1 + \zeta + \zeta^2 = 0.$$

Let

$$t = \zeta + \zeta^{-1} = 2\cos \frac{2\pi}{5}.$$

Then

$$\zeta^2 + \zeta^{-2} = t^2 - 2.$$

So

$$(t^2 - 2) + t + 1 = 0.$$

Thus

$$t^2 + t - 1 = 0.$$

The positive root is

$$t = \frac{-1+\sqrt{5}}{2}.$$

Hence

$$\cos \frac{2\pi}{5} = \frac{t}{2} = \frac{\sqrt{5}-1}{4}.$$

Problem 12. Shortest broken path

A point A and a point B lie above a horizontal line L. Find the point $P \in L$ minimizing

$$AP + PB.$$

Solution

Reflect B across L to B'. For every $P \in L$,

$$PB = PB'.$$

Thus

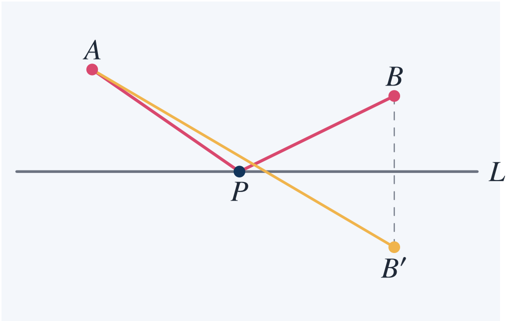
$$AP + PB = AP + PB'.$$

The shortest path from A to B' is the straight segment AB'. Therefore the minimizing point P is

$$P = AB' \cap L.$$

At this point the two angles with the line L are equal:

$$\angle(AP, L) = \angle(L, PB).$$



A.4 Projective Geometry

Problem 13. Parallel lines meet at infinity

Find the projective intersection of the affine lines

$$y = 3x + 1, \qquad y = 3x - 4.$$

Solution

Use homogeneous coordinates:

$$x = \frac{X}{Z}, \qquad y = \frac{Y}{Z}.$$

The first line becomes

$$Y = 3X + Z,$$

or

$$-3X + Y - Z = 0.$$

The second becomes

$$Y = 3X - 4Z,$$

or

$$-3X + Y + 4Z = 0.$$

Subtract:

$$5Z = 0.$$

Thus

$$Z = 0.$$

Substitute into

$$-3X + Y - Z = 0.$$

We get

$$Y = 3X.$$

Therefore the intersection point is

$$[X : Y : Z] = [1 : 3 : 0].$$

This is the point at infinity corresponding to slope 3.

Problem 14. Line through two projective points

Find the line through

$$P = [1 : 0 : 1], \qquad Q = [2 : 3 : 1].$$

Solution

Let the line be

$$aX + bY + cZ = 0.$$

Since P lies on it,

$$a + c = 0.$$

Since Q lies on it,

$$2a + 3b + c = 0.$$

From

$$c = -a,$$

the second equation becomes

$$2a + 3b - a = 0,$$

so

$$a + 3b = 0.$$

Take

$$b = 1.$$

Then

$$a = -3, \qquad c = 3.$$

So the line is

$$-3X + Y + 3Z = 0.$$

Equivalently,

$$3X - Y - 3Z = 0.$$

A.5 Topology

Problem 15. Euler characteristic of a cube

Verify Euler's formula for a cube.

Solution

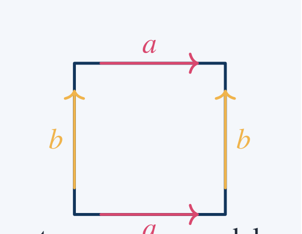
A cube has
$$V = 8$$
vertices,
$$E = 12$$
edges, and
$$F = 6$$
faces. Therefore
$$V - E + F = 8 - 12 + 6 = 2.$$
This agrees with Euler's formula for convex polyhedra.

Problem 16. Euler characteristic of a torus

Use the square model of the torus to compute its Euler characteristic.

Solution

Represent the torus by a square with opposite sides identified in the same direction. After identification, all four corners become one vertex:
$$V = 1.$$
There are two edge classes:
$$E = 2.$$
The interior of the square is one face:
$$F = 1.$$
Thus
$$\chi = V - E + F = 1 - 2 + 1 = 0.$$
So
$$\chi(T^2) = 0.$$



torus square model

Problem 17. Planar graph inequality

Prove that a connected simple planar graph with $V \geq 3$ satisfies
$$E \leq 3V - 6.$$

Solution

Euler's formula gives

$$V - E + F = 2.$$

Since the graph is simple and planar, each face has at least 3 boundary edges. Counting face-edge incidences gives

$$3F \le 2E.$$

From Euler's formula,

$$F = 2 - V + E.$$

Substitute:

$$3(2 - V + E) \le 2E.$$

So

$$6 - 3V + 3E \le 2E.$$

Hence

$$E \le 3V - 6.$$

Problem 18. Nonplanarity of K_5

Use the planar graph inequality to show that K_5 is not planar.

Solution

For K_5,

$$V = 5.$$

Every pair of vertices is connected, so

$$E = \binom{5}{2} = 10.$$

If K_5 were planar, then

$$E \le 3V - 6 = 15 - 6 = 9.$$

But

$$E = 10 > 9.$$

Contradiction. Therefore

$$K_5$$

is not planar.

A.6 Limits and Continuity

Problem 19. Removable discontinuity

Compute

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}.$$

Solution

Factor:

$$x^2 - 4 = (x-2)(x+2).$$

For $x \neq 2$,

$$\frac{x^2-4}{x-2} = x+2.$$

Thus

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4.$$

Problem 20. Epsilon–delta proof

Prove

$$\lim_{x\to 1}(4x-3) = 1.$$

Solution

We need

$$|(4x-3)-1| < \epsilon.$$

Compute:

$$|(4x-3)-1| = |4x-4| = 4|x-1|.$$

Choose

$$\delta = \frac{\epsilon}{4}.$$

Then

$$0 < |x-1| < \delta$$

implies

$$|(4x-3)-1| = 4|x-1| < 4\delta = \epsilon.$$

Therefore

$$\lim_{x\to 1}(4x-3) = 1.$$

Problem 21. A non-existent limit

Show that

$$\lim_{x\to 0} \frac{|x|}{x}$$

does not exist.

Solution

If

$$x > 0,$$

then

$$|x| = x,$$

so

$$\frac{|x|}{x} = 1.$$

Thus

$$\lim_{x \to 0^+} \frac{|x|}{x} = 1.$$

If

$$x < 0,$$

then

$$|x| = -x,$$

so

$$\frac{|x|}{x} = -1.$$

Thus

$$\lim_{x \to 0^-} \frac{|x|}{x} = -1.$$

The one-sided limits are different. Therefore the two-sided limit does not exist.

Problem 22. Squeeze theorem

Compute

$$\lim_{x \to 0} x^2 \cos \frac{1}{x}.$$

Solution

Since

$$-1 \leq \cos \frac{1}{x} \leq 1,$$

we have

$$-x^2 \leq x^2 \cos \frac{1}{x} \leq x^2.$$

As

$$x \to 0,$$

$$-x^2 \to 0, \qquad x^2 \to 0.$$

By the squeeze theorem,

$$\lim_{x \to 0} x^2 \cos \frac{1}{x} = 0.$$

A.7 Differentiation

Problem 23. Derivative from the definition

Find the derivative of

$$f(x) = x^2 + 3x$$

from the definition.

Solution

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Compute:

$$f(x+h) = (x+h)^2 + 3(x+h) = x^2 + 2xh + h^2 + 3x + 3h.$$

Thus

$$f(x+h) - f(x) = 2xh + h^2 + 3h.$$

Divide by h:

$$\frac{f(x+h) - f(x)}{h} = 2x + h + 3.$$

Let $h \to 0$:

$$f'(x) = 2x + 3.$$

Problem 24. Tangent line

Find the tangent line to

$$y = x^3 - x$$

at

$$x = 2.$$

Solution

Let

$$f(x) = x^3 - x.$$

Then

$$f(2) = 8 - 2 = 6.$$

Also

$$f'(x) = 3x^2 - 1.$$

Thus

$$f'(2) = 12 - 1 = 11.$$

The tangent line is

$$y - 6 = 11(x - 2).$$

So

$$y = 11x - 16.$$

Problem 25. Product and chain rule

Differentiate

$$y = (x^2 + 1)(3x - 2)^4.$$

Solution

Use product rule:

$$y' = 2x(3x-2)^4 + (x^2+1) \cdot 4(3x-2)^3 \cdot 3.$$

Thus

$$y' = 2x(3x-2)^4 + 12(x^2+1)(3x-2)^3.$$

Factor:

$$y' = (3x-2)^3 \left(2x(3x-2) + 12(x^2+1)\right).$$

Simplify inside:

$$2x(3x-2) + 12(x^2+1) = 6x^2 - 4x + 12x^2 + 12.$$

So

$$y' = (3x-2)^3(18x^2 - 4x + 12).$$

Problem 26. Implicit differentiation

Find the slope of the tangent line to

$$x^2 + xy + y^2 = 3$$

at

$$(1, 1).$$

Solution

Differentiate:

$$2x + \frac{d}{dx}(xy) + 2y\frac{dy}{dx} = 0.$$

Since

$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y,$$

we get

$$2x + x\frac{dy}{dx} + y + 2y\frac{dy}{dx} = 0.$$

Collect terms:

$$(x+2y)\frac{dy}{dx} = -(2x+y).$$

Thus

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y}.$$

At

$$(1, 1),$$

$$\frac{dy}{dx} = -\frac{2+1}{1+2} = -1.$$

So the slope is

$$-1.$$

A.8 Maxima and Minima

Problem 27. Closed interval extrema

Find the maximum and minimum of

$$f(x) = x^3 - 3x + 1$$

on

$$[-2, 2].$$

Solution

Compute

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

Critical points:

$$x = -1, \qquad x = 1.$$

Evaluate endpoints and critical points:

$$f(-2) = -8 + 6 + 1 = -1,$$

$$f(-1) = -1 + 3 + 1 = 3,$$

$$f(1) = 1 - 3 + 1 = -1,$$

$$f(2) = 8 - 6 + 1 = 3.$$

Thus the maximum value is

$$3,$$

attained at

$$x = -1, \quad x = 2.$$

The minimum value is

$$-1,$$

attained at

$$x = -2, \quad x = 1.$$

Problem 28. Fixed perimeter rectangle

Among rectangles with perimeter 20, find the maximum possible area.

Problem 29. Closest point on a parabola

Find the point on

$$y = x^2$$

closest to

$$(0, 2).$$

Solution

A point on the parabola is

$$(x, x^2).$$

The squared distance to $(0, 2)$ is

$$D(x) = x^2 + (x^2 - 2)^2.$$

Expand:

$$D(x) = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4.$$

Differentiate:

$$D'(x) = 4x^3 - 6x = 2x(2x^2 - 3).$$

Critical points:

$$x = 0, \qquad x = \pm\sqrt{\frac{3}{2}}.$$

Evaluate:

$$D(0) = 4.$$

If

$$x^2 = \frac{3}{2},$$

then

$$D = x^4 - 3x^2 + 4 = \frac{9}{4} - \frac{9}{2} + 4 = \frac{9 - 18 + 16}{4} = \frac{7}{4}.$$

Thus the closest points are

$$\left(\sqrt{\frac{3}{2}}, \frac{3}{2}\right), \qquad \left(-\sqrt{\frac{3}{2}}, \frac{3}{2}\right).$$

The minimum distance is

$$\sqrt{\frac{7}{4}} = \frac{\sqrt{7}}{2}.$$

A.9 Integration

Problem 30. Area under a parabola

Compute

$$\int_0^2 x^2\,dx.$$

Solution

An antiderivative of x^2 is

$$\frac{x^3}{3}.$$

Therefore

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

Problem 31. Area between curves

Find the area between

$$y = x$$

and

$$y = x^2$$

on

$$[0, 1].$$

Solution

On $[0, 1]$,

$$x \geq x^2.$$

Thus the area is

$$\int_0^1 (x - x^2)\,dx.$$

Compute:

$$\int_0^1 x\,dx = \frac{1}{2},$$

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$

Therefore

$$\int_0^1 (x - x^2)\,dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Problem 32. Substitution

Compute

$$\int 2x(x^2 + 5)^7\,dx.$$

Solution

Let

$$u = x^2 + 5.$$

Then

$$du = 2x\,dx.$$

So

$$\int 2x(x^2 + 5)^7\,dx = \int u^7\,du = \frac{u^8}{8} + C.$$

Substitute back:

$$\int 2x(x^2 + 5)^7\,dx = \frac{(x^2 + 5)^8}{8} + C.$$

Problem 33. Integration by parts

Compute

$$\int x \cos x\,dx.$$

Solution

Use

$$\int u\,dv = uv - \int v\,du.$$

Let

$$u = x, \qquad dv = \cos x\,dx.$$

Then

$$du = dx, \qquad v = \sin x.$$

Thus

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx.$$

Since

$$\int \sin x \, dx = -\cos x,$$

we get

$$\int x \cos x \, dx = x \sin x + \cos x + C.$$

Problem 34. Improper integral

Determine whether

$$\int_1^\infty \frac{1}{x^3} \, dx$$

converges, and compute it if it does.

Solution

By definition,

$$\int_1^\infty x^{-3} \, dx = \lim_{b\to\infty} \int_1^b x^{-3} \, dx.$$

An antiderivative is

$$-\frac{1}{2x^2}.$$

Thus

$$\int_1^b x^{-3} \, dx = \left[-\frac{1}{2x^2}\right]_1^b = -\frac{1}{2b^2} + \frac{1}{2}.$$

Let

$$b \to \infty.$$

Then

$$-\frac{1}{2b^2} \to 0.$$

So

$$\int_1^\infty \frac{1}{x^3} \, dx = \frac{1}{2}.$$

The integral converges.

A.10 Sequences and Series

Problem 35. Geometric series

Compute

$$\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n.$$

Solution

For

$$|r| < 1,$$

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Here

$$r = \frac{2}{3}.$$

Thus

$$\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{1-\frac{2}{3}} = \frac{1}{\frac{1}{3}} = 3.$$

Problem 36. Harmonic series divergence

Prove that

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges.

Solution

Group terms:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots .$$

For each block,

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

$$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > 4 \cdot \frac{1}{8} = \frac{1}{2}.$$

In general, the block

$$\frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}}$$

has 2^k terms, each at least

$$\frac{1}{2^{k+1}}.$$

So each block is at least

$$\frac{1}{2}.$$

Therefore the partial sums exceed

$$1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots ,$$

which diverges. Hence the harmonic series diverges.

Problem 37. Ratio test

Determine whether

$$\sum_{n=1}^{\infty} \frac{n!}{10^n}$$

converges.

Solution

Let

$$a_n = \frac{n!}{10^n}.$$

Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{10^{n+1}} \cdot \frac{10^n}{n!} = \frac{n+1}{10}.$$

As

$$n \to \infty,$$

$$\frac{n+1}{10} \to \infty.$$

In particular, the ratio eventually exceeds 1, so the terms do not even tend to 0. Therefore the series diverges.

A.11 Higher-Dimensional Geometry

Problem 38. Distance in $\mathbb{R}^4$

Compute the distance between

$$p = (1, -1, 2, 0)$$

and

$$q = (3, 2, -1, 4).$$

Solution

Subtract:

$$p - q = (-2, -3, 3, -4).$$

Thus

$$d(p, q) = \sqrt{(-2)^2 + (-3)^2 + 3^2 + (-4)^2}.$$

Compute:

$$d(p, q) = \sqrt{4 + 9 + 9 + 16} = \sqrt{38}.$$

Problem 39. A hyperplane distance

Find the distance from the origin to the hyperplane

$$2x_1 - x_2 + 2x_3 - x_4 = 6.$$

Solution

The normal vector is

$$a = (2, -1, 2, -1).$$

The hyperplane is

$$a \cdot x = 6.$$

The distance from 0 to

$$a \cdot x = b$$

is

$$\frac{|b|}{\|a\|}.$$

Here

$$\|a\| = \sqrt{2^2 + (-1)^2 + 2^2 + (-1)^2} = \sqrt{4 + 1 + 4 + 1} = \sqrt{10}.$$

Therefore the distance is

$$\frac{6}{\sqrt{10}} = \frac{3\sqrt{10}}{5}.$$

Problem 40. Faces of the 4-cube

Compute the number of vertices, edges, square faces, and cubic cells of the 4-cube.

Solution

For the n-cube, the number of k-faces is

$$\binom{n}{k} 2^{n-k}.$$

For $n = 4$:
Vertices:

$$k = 0: \qquad \binom{4}{0} 2^4 = 16.$$

Edges:

$$k = 1: \qquad \binom{4}{1} 2^3 = 4 \cdot 8 = 32.$$

Square faces:

$$k = 2: \qquad \binom{4}{2} 2^2 = 6 \cdot 4 = 24.$$

Cubic cells:

$$k = 3: \qquad \binom{4}{3} 2^1 = 4 \cdot 2 = 8.$$

Thus the 4-cube has

$$16 \text{ vertices}, \quad 32 \text{ edges}, \quad 24 \text{ square faces}, \quad 8 \text{ cubic cells}.$$

A.12 Modern Examples

Problem 41. Cantor set dimension

Compute the similarity dimension of the middle-third Cantor set.

Solution

The Cantor set consists of

$$N = 2$$

copies of itself, each scaled by

$$r = \frac{1}{3}.$$

The similarity dimension d satisfies

$$Nr^d = 1.$$

Thus

$$2\left(\frac{1}{3}\right)^d = 1.$$

So

$$2 = 3^d.$$

Taking logarithms:

$$d = \frac{\log 2}{\log 3}.$$

Problem 42. Koch curve dimension

Compute the similarity dimension of the Koch curve.

Solution

Each stage replaces one segment by

$$N = 4$$

segments, each scaled by

$$r = \frac{1}{3}.$$

Thus

$$4\left(\frac{1}{3}\right)^d = 1.$$

So

$$4 = 3^d.$$

Therefore

$$d = \frac{\log 4}{\log 3}.$$

Since

$$3 < 4 < 9,$$

we have

$$1 < \frac{\log 4}{\log 3} < 2.$$

So the Koch curve has dimension strictly between 1 and 2.

Problem 43. Prime polynomial failure

Show that

$$n^2 + n + 41$$

does not always produce primes.

Solution

Take

$$n = 40.$$

Then

$$n^2 + n + 41 = 40^2 + 40 + 41.$$

Compute:

$$40^2 = 1600,$$

so

$$1600 + 40 + 41 = 1681.$$

But

$$1681 = 41^2.$$

Therefore

$$n^2 + n + 41$$

is composite at $n = 40$.

A.13 Mixed Problems

Problem 44. A limit and derivative connection

Let

$$f(x) = x^3.$$

Compute

$$\lim_{h \to 0} \frac{f(2+h) - f(2)}{h}.$$

Solution

$$f(2+h) = (2+h)^3 = 8 + 12h + 6h^2 + h^3.$$

Thus

$$f(2+h) - f(2) = 12h + 6h^2 + h^3.$$

Divide by h:

$$\frac{f(2+h) - f(2)}{h} = 12 + 6h + h^2.$$

Let $h \to 0$:

$$\lim_{h \to 0} \frac{f(2+h) - f(2)}{h} = 12.$$

This equals

$$f'(2),$$

since

$$f'(x) = 3x^2$$

and

$$f'(2) = 12.$$

Problem 45. An optimization problem with area

A rectangle is placed under the curve

$$y = 4 - x^2$$

above the x-axis, symmetric about the y-axis. Find the maximum area.

Solution

Let the right endpoint be

$$x = a, \qquad 0 < a < 2.$$

Then the rectangle has width

$$2a$$

and height

$$4 - a^2.$$

Thus

$$A(a) = 2a(4 - a^2) = 8a - 2a^3.$$

Differentiate:

$$A'(a) = 8 - 6a^2.$$

Set

$$A'(a) = 0.$$

Then

$$8 - 6a^2 = 0,$$

so

$$a^2 = \frac{4}{3}, \qquad a = \frac{2}{\sqrt{3}}.$$

The maximum area is

$$A_{\max} = 2 \cdot \frac{2}{\sqrt{3}} \left(4 - \frac{4}{3}\right).$$

Compute:

$$4 - \frac{4}{3} = \frac{8}{3}.$$

Thus

$$A_{\max} = \frac{4}{\sqrt{3}} \cdot \frac{8}{3} = \frac{32}{3\sqrt{3}} = \frac{32\sqrt{3}}{9}.$$

Problem 46. A topology obstruction

Show that a sphere and a torus are not homeomorphic using Euler characteristic.

Solution

Euler characteristic is a topological invariant. For the sphere,

$$\chi(S^2) = 2.$$

For the torus,

$$\chi(T^2) = 0.$$

If

$$S^2$$

were homeomorphic to

$$T^2,$$

then their Euler characteristics would be equal. But

$$2 \neq 0.$$

Therefore the sphere and torus are not homeomorphic.

Problem 47. A series from calculus

Compute

$$\sum_{n=1}^{\infty} \frac{1}{2^n}.$$

Solution

This is a geometric series:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots .$$

Here

$$a = \frac{1}{2}, \qquad r = \frac{1}{2}.$$

Thus

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1.$$

Problem 48. A final check on rigor

Find the flaw in the following argument:

$$\frac{x^2-1}{x-1} = x + 1.$$

Set $x = 1$. Then

$$\frac{0}{0} = 2.$$

Solution

The identity

$$\frac{x^2 - 1}{x - 1} = x + 1$$

is valid only for

$$x \neq 1,$$

because the cancellation divides by

$$x - 1.$$

At

$$x = 1,$$

the original expression is undefined. One may say

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2,$$

but one may not say

$$\frac{0}{0} = 2.$$

The error is confusing equality of functions on a punctured domain with a value at the missing point.

CHAPTER B

Suggestions for Further Reading

Source Map

A useful reading list is not a display of book titles. It should explain what each book is good for, what mathematical habit it trains, and what one should try to extract from it. The books below are organized by purpose: foundations, analysis, algebra, geometry, topology, number theory, probability, and modern directions.

B.1 How to Read Beyond This Book

A serious reading plan should not be linear in the weak sense of finishing one book and only then opening the next. A better method is to read by mathematical function.

For each book, I use four questions:

1. What problem does this book make precise?
2. What new language does it introduce?
3. Which examples does it make unavoidable?
4. Which theorem changes the way I think?

For instance, reading about topology is not mainly about memorizing definitions of open and closed sets. The point is to learn how invariants distinguish spaces when geometry no longer has a metric. Reading algebra is not mainly about symbol manipulation. The point is to understand structure-preserving maps, quotient objects, and classification by invariants.

B.2 General Mathematical Maturity

Courant and Robbins, *What Is Mathematics?*

This remains the central text for these notes. Its strength is the way it moves from elementary questions to deep structures without pretending that the elementary questions are shallow.

The important habit to extract is this:

start from a concrete problem, then uncover the hidden structure.

Examples:

concrete problem	hidden structure
construct a regular polygon	field extensions
draw a map with few colors	planar graphs
compare shapes under deformation	topological invariants
find maximum area	optimization and variational thinking

Polya, *How to Solve It*

Polya is useful not because it gives tricks, but because it studies the psychology of problem solving. The best part of the book is the insistence on changing representation.

A problem may become easier after one replaces:

numbers by geometry, geometry by algebra, algebra by induction, a direct proof by contradiction.

Reading exercise

Take a problem already solved in these notes, such as the maximum-area rectangle with fixed perimeter. Solve it in three ways:

completing the square, AM–GM, calculus.

Then compare what each method sees. Completing the square sees the parabola; AM–GM sees symmetry; calculus sees the critical point.

Hardy, *A Mathematician's Apology*

This book is not a technical textbook. It is useful for understanding the aesthetic and cultural side of pure mathematics. I would not use it as a guide to research, but it is valuable for seeing why mathematicians care about depth, permanence, and elegance.

The danger is to imitate Hardy's style without Hardy's substance. The correct lesson is not that applied mathematics is inferior. The better lesson is that a mathematical result should have internal necessity.

B.3 Analysis and Calculus

Spivak, *Calculus*

Spivak is the natural next step after an elementary calculus chapter. It turns computational calculus into proof-based analysis.

The key transition is:

differentiate this function becomes why does the derivative exist?

Topics to read carefully:

topic	what to extract
ϵ–δ limits	quantifier control
continuity	local behavior and global consequences
mean value theorem	derivatives control functions
Riemann integration	area as a limiting process
Taylor theorem	local polynomial approximation

A useful Spivak-style problem

Prove directly from the definition that

$$\lim_{x \to a} x^2 = a^2.$$

Start from

$$|x^2 - a^2| = |x - a||x + a|.$$

The point is not the result itself. The point is learning how to control

$$|x + a|$$

by first requiring x to stay close to a.

Rudin, *Principles of Mathematical Analysis*

Rudin is terse and unforgiving. It is not the best first analysis book, but it is an excellent book for testing whether one has real command of definitions.

The central concepts are:

$$\text{metric spaces}, \quad \text{compactness}, \quad \text{connectedness}, \quad \text{uniform convergence}, \quad \text{measure-like thinking}.$$

The most important theorem to internalize is Heine–Borel in Euclidean space:

$$K \subset \mathbb{R}^n \text{ is compact} \iff K \text{ is closed and bounded}.$$

This theorem explains why optimization on closed bounded sets behaves so much better than optimization on open or unbounded domains.

Abbott, *Understanding Analysis*

Abbott is more explanatory than Rudin. It is especially good for understanding why the real numbers require completeness.

A good reading project is:

$$\text{nested intervals} \implies \text{supremum property} \implies \text{Bolzano–Weierstrass} \implies \text{compactness}.$$

Concrete point

The sequence

$$a_n^2 < 2$$

inside rational approximations can approach

$$\sqrt{2},$$

but

$$\sqrt{2} \notin \mathbb{Q}.$$

Thus the rational numbers have holes. Completeness is exactly what fills those holes.

B.4 Linear Algebra

Axler, *Linear Algebra Done Right*

Axler is useful because it treats linear algebra structurally rather than as matrix arithmetic. The main object is a linear map

$$T : V \to W,$$

not merely a rectangular array of numbers.

Important ideas:

concept	meaning
$\ker T$	directions killed by T
$\operatorname{im} T$	directions reached by T
eigenvalue	one-dimensional invariant behavior
diagonalization	decomposition into simple actions
inner product	geometry inside vector spaces

Reading exercise

For

$$T(x, y, z) = (x + y, y + z),$$

compute:

$$\ker T, \qquad \operatorname{im} T.$$

Solving

$$x + y = 0, \qquad y + z = 0$$

gives

$$x = -y, \qquad z = -y.$$

Thus

$$\ker T = \{(-t, t, -t) : t \in \mathbb{R}\}.$$

Since the two output coordinates can be chosen freely,

$$\operatorname{im} T = \mathbb{R}^2.$$

This small example already shows rank-nullity:

$$\dim \ker T + \dim \operatorname{im} T = 1 + 2 = 3.$$

Strang, *Introduction to Linear Algebra*

Strang is better for computation, geometry, and applications. It is especially strong for understanding the four fundamental subspaces of a matrix.

Read Strang when the goal is to see linear algebra as:

solving equations, least squares, orthogonal projection, singular value decomposition.

Read Axler when the goal is to understand:

linear maps, eigenstructure, invariant subspaces, operator theory.

B.5 Abstract Algebra

Dummit and Foote, *Abstract Algebra*

This is a large reference-style book. It is useful because it gives many examples and covers groups, rings, fields, modules, and Galois theory.

The most important route for the themes of these notes is:

$$\text{groups} \quad \text{rings} \quad \text{fields} \quad \text{field extensions} \quad \text{Galois theory}.$$

The connection with straightedge and compass construction is direct:

$$\text{constructible numbers} \quad \Longleftrightarrow \quad \text{quadratic field towers}.$$

Artin, *Algebra*

Artin is more conceptual and geometric. It is good for seeing algebra as the study of symmetry and structure.

A useful reading goal is to understand the same object in three languages:

language	example
group action	symmetries of a polygon
linear representation	matrices acting on vector spaces
field extension	numbers obtained by adjoining roots

Reading exercise

The symmetries of an equilateral triangle form the dihedral group

$$D_3.$$

It has

$$6$$

elements: three rotations and three reflections. Check that the rotations form a subgroup

$$C_3 \subset D_3.$$

Then verify that a reflection s and a rotation r satisfy

$$srs = r^{-1}.$$

This relation captures the noncommutativity of the triangle's symmetry group.

Stewart, *Galois Theory*

For the specific bridge between classical construction problems and modern algebra, a focused book on Galois theory is useful.

The essential ideas to extract are:

$$\text{field extension degree}, \qquad \text{minimal polynomial}, \qquad \text{automorphism group}, \qquad \text{solvability by radicals}.$$

In the context of these notes, the decisive fact is:

$$\sqrt[3]{2}$$

has degree 3 over $\mathbb{Q}$, while constructible numbers have degrees dividing powers of 2. Therefore duplicating the cube is impossible by straightedge and compass.

B.6 Geometry and Topology

Coxeter, *Introduction to Geometry*

Coxeter is excellent for synthetic geometry, regular polytopes, projective geometry, and higher-dimensional figures. It is especially valuable after reading the chapters on projective geometry and higher-dimensional geometry.

Useful themes:

theme	why it matters
regular polytopes	classification by symmetry
projective geometry	incidence over metric
inversion	circles and lines treated together
higher dimensions	geometry beyond visualization

Stillwell, *The Four Pillars of Geometry*

Stillwell is good for organizing geometry into four viewpoints:

$$\text{Euclidean}, \quad \text{projective}, \quad \text{non-Euclidean}, \quad \text{transformational}.$$

This matches the intellectual movement in the geometry chapters: geometry is not one subject with one method, but a family of structures distinguished by their invariants.

Munkres, *Topology*

Munkres is the standard next step for point-set topology. The early chapters should be read slowly.

Core definitions:

$$\text{topological space}, \quad \text{basis}, \quad \text{continuity}, \quad \text{compactness}, \quad \text{connectedness}.$$

The first serious test is to understand why continuity can be defined without a formula:

$$f : X \to Y \text{ is continuous}$$

if for every open set $U \subseteq Y$,

$$f^{-1}(U)$$

is open in X.

Reading exercise

Let

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

For the open interval

$$U = (1, 4),$$

compute

$$f^{-1}(U).$$

We need

$$1 < x^2 < 4.$$

Thus

$$1 < |x| < 2.$$

So

$$f^{-1}(U) = (-2, -1) \cup (1, 2),$$

which is open. This is the inverse-image definition of continuity at work.

Hatcher, *Algebraic Topology*

Hatcher should be read after one has some point-set topology and algebra. It is not an easy book, but it explains how algebraic invariants distinguish spaces.

Main objects:

$$\pi_1(X), \qquad H_n(X), \qquad \text{covering spaces}, \qquad \text{cell complexes}.$$

For these notes, the natural bridge is Euler characteristic:

$$\chi = V - E + F.$$

In algebraic topology this becomes

$$\chi(X) = \sum_{n \geq 0} (-1)^n \operatorname{rank} H_n(X).$$

Thus a combinatorial invariant becomes a homological invariant.

B.7 Number Theory

Hardy and Wright, *An Introduction to the Theory of Numbers*

This is a classical text. It is broad and demanding. It connects elementary number theory with deeper analytic questions.

Topics to read first:

$$\text{divisibility}, \qquad \text{congruences}, \qquad \text{quadratic residues}, \qquad \text{continued fractions}, \qquad \text{distribution of primes}.$$

Reading exercise

Show that if

$$p$$

is an odd prime and

$$a^2 \equiv 1 \pmod{p},$$

then

$$a \equiv 1 \pmod{p} \qquad \text{or} \qquad a \equiv -1 \pmod{p}.$$

Indeed,

$$a^2 - 1 \equiv 0 \pmod{p},$$

so

$$(a-1)(a+1) \equiv 0 \pmod{p}.$$

Since p is prime, it divides one factor:

$$p \mid a-1 \qquad \text{or} \qquad p \mid a+1.$$

Ireland and Rosen, *A Classical Introduction to Modern Number Theory*

This book is a strong bridge toward algebraic number theory. It is especially good for reciprocity laws, finite fields, and arithmetic structure.

Read it after becoming comfortable with:

$$\mathbb{Z}/n\mathbb{Z}, \qquad \text{cyclic groups}, \qquad \text{polynomials over finite fields}.$$

Apostol, *Introduction to Analytic Number Theory*

Apostol is useful for analytic methods: arithmetic functions, Dirichlet convolution, Mobius inversion, and prime number estimates.

The key algebraic operation is Dirichlet convolution:

$$(f * g)(n) = \sum_{d \mid n} f(d) g\left(\frac{n}{d}\right).$$

This turns divisibility into an algebra of functions.

Reading exercise

Let

$$1(n) = 1$$

for every n, and let μ be the Mobius function. The identity

$$\sum_{d \mid n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1 \end{cases}$$

means

$$\mu * 1 = \varepsilon,$$

where $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$. This is the multiplicative inverse relation behind Mobius inversion.

B.8 Probability, Statistics, and Data

Feller, *An Introduction to Probability Theory and Its Applications*

Feller is rich in examples and combinatorial reasoning. It is excellent for training probabilistic intuition.

Important themes:

$$\text{random walks}, \qquad \text{generating functions}, \qquad \text{limit theorems}, \qquad \text{rare events}.$$

Grimmett and Stirzaker, *Probability and Random Processes*

This is more systematic. It is good for Markov chains, conditioning, expectation, and convergence.

A useful minimum goal:

$$\text{master conditional expectation}$$

rather than treating it as a formula.

Reading exercise

Let X be the result of a fair die roll. Compute

$$\mathbb{E}[X \mid X \text{ is even}].$$

The possible even values are

$$2, 4, 6,$$

each equally likely under the condition. Therefore

$$\mathbb{E}[X \mid X \text{ is even}] = \frac{2+4+6}{3} = 4.$$

Wasserman, *All of Statistics*

This is useful for a fast but mathematically serious overview of statistics: estimation, confidence intervals, hypothesis testing, regression, and nonparametric ideas.

The best way to read it is not to memorize tests. Instead, track the core statistical objects:

$$\text{model}, \qquad \text{estimator}, \qquad \text{risk}, \qquad \text{bias}, \qquad \text{variance}, \qquad \text{likelihood}.$$

B.9 Logic, Foundations, and Computation

Enderton, *A Mathematical Introduction to Logic*

Enderton is a good first serious logic book. It introduces syntax, semantics, completeness, compactness, and incompleteness.

The main distinction to learn:

$$\Gamma \vdash \varphi \qquad \text{versus} \qquad \Gamma \models \varphi.$$

The first means provability from rules. The second means truth in every model.

Boolos, Burgess, and Jeffrey, *Computability and Logic*

This is strong for computability, incompleteness, and formal systems. It is especially relevant if one wants to understand why some mathematical questions are undecidable inside a given formal framework.

Important concepts:

$$\text{recursive function}, \qquad \text{Turing machine}, \qquad \text{arithmetization}, \qquad \text{incompleteness}.$$

Sipser, *Introduction to the Theory of Computation*

Sipser is clear for automata, computability, and complexity. It is useful if one wants the computational side of modern mathematical thinking.

A compact chain of ideas:

$$\text{finite automata} \subset \text{pushdown automata} \subset \text{Turing machines.}$$

Each enlargement changes what can be recognized or computed.

B.10 Differential Equations and Dynamical Systems

Arnold, *Ordinary Differential Equations*

Arnold is geometrically minded. It treats differential equations as flows, not merely as formulas to solve.

The important conceptual shift:

$$\frac{dx}{dt} = f(x)$$

defines a vector field, and solutions are trajectories of that field.

Reading exercise

Solve

$$\frac{dx}{dt} = kx.$$

Separate variables:

$$\frac{1}{x}\,dx = k\,dt.$$

Integrate:

$$\ln|x| = kt + C.$$

Thus

$$x(t) = Ce^{kt}.$$

The sign of k determines whether the equilibrium $x = 0$ is attracting or repelling.

Strogatz, *Nonlinear Dynamics and Chaos*

Strogatz is excellent for intuition and examples. It is especially good for bifurcations, phase portraits, and simple nonlinear systems.

Good first topics:

fixed points, stability, bifurcations, limit cycles, chaos.

B.11 Differential Geometry and Manifolds

do Carmo, *Differential Geometry of Curves and Surfaces*

This is a natural continuation of the geometry chapters. It explains curvature for curves and surfaces concretely.

Basic objects:

curvature of a curve, first fundamental form, second fundamental form, Gaussian curvature.

A key theorem:

$$\text{Gaussian curvature is intrinsic.}$$

This means it can be measured from distances on the surface itself, not from how the surface sits in three-dimensional space.

Lee, *Introduction to Smooth Manifolds*

Lee is systematic and precise. It is a serious next step after multivariable calculus and linear algebra.

Core language:

$$\text{manifold}, \qquad \text{chart}, \qquad \text{atlas}, \qquad \text{tangent space}, \qquad \text{differential form}.$$

The first real conceptual step is that a manifold is locally Euclidean but may be globally non-Euclidean:

$$\text{near each point it looks like } \mathbb{R}^n, \quad \text{but globally it may be a sphere, torus, or projective space.}$$

B.12 Combinatorics and Graph Theory

Diestel, *Graph Theory*

Diestel is a strong modern graph theory text. It connects naturally with the Königsberg bridges, planar graphs, coloring, and Euler characteristic.

Topics to read:

$$\text{trees}, \qquad \text{connectivity}, \qquad \text{matchings}, \qquad \text{planarity}, \qquad \text{coloring}.$$

Reading exercise

A tree with V vertices has

$$E = V - 1$$

edges. Prove by induction. Removing a leaf from a tree with V vertices leaves a tree with $V - 1$ vertices and one fewer edge. Thus if the smaller tree has

$$V - 2$$

edges, the original has

$$V - 1$$

edges.

Stanley, *Enumerative Combinatorics*

Stanley is deeper and more advanced. It is useful for generating functions, partially ordered sets, and algebraic combinatorics.

A first useful object is the generating function:

$$A(x) = \sum_{n\geq 0} a_n x^n.$$

It turns a sequence into an algebraic object.

For example, if

$$a_n = 1$$

for all $n \geq 0$, then

$$A(x) = 1 + x + x^2 + \cdots = \frac{1}{1-x}.$$

B.13 A Minimal Reading Path

If I had to reduce the list to a focused path, I would choose:

stage	books
mathematical maturity	Polya, Courant–Robbins
proof-based calculus	Spivak, Abbott
linear structure	Axler, Strang
algebraic structure	Artin, Dummit–Foote
geometry and topology	Coxeter, Stillwell, Munkres
number theory	Hardy–Wright, Ireland–Rosen
probability and statistics	Feller, Wasserman
modern directions	Hatcher, Lee, Strogatz

This path is not meant to be finished quickly. It is meant to create a coherent mathematical spine: proof, structure, space, number, randomness, and dynamics.

B.14 What to Record While Reading

For each substantial chapter, I would record four things.

item	what to write
one definition	the definition that changes the language of the subject
one theorem	the theorem that converts the definition into power
one example	a concrete object where the theorem can be computed
one counterexample	the example that shows why a hypothesis is necessary

Example using compactness

Definition:

$$K \text{ is compact if every open cover has a finite subcover.}$$

Theorem:

$$K \subset \mathbb{R}^n \text{ is compact} \iff K \text{ is closed and bounded.}$$

Example:

$$[0,1]$$

is compact.
Counterexample:

$$(0,1)$$

is bounded but not compact. The open cover

$$\left\{ \left(\frac{1}{n}, 1 \right) : n \geq 2 \right\}$$

covers $(0, 1)$, but no finite subcollection covers all of $(0, 1)$.

This method prevents passive reading. A mathematical text has been understood only when one can produce definitions, theorems, examples, and counterexamples without copying the author.

CHAPTER C

Glossary and Notation

Source Map

This glossary records the notation used throughout these reading notes. I use each entry not only as a definition, but also as a warning against common misreadings. Many mathematical mistakes come from confusing similar-looking symbols: $\in$ and $\subseteq$, local and global extrema, equality and isomorphism, convergence and boundedness.

C.1 Logical Symbols

symbol	meaning
$P \Rightarrow Q$	if P is true, then Q is true
$P \iff Q$	P and Q are equivalent; each implies the other
$\forall x$	for every x
$\exists x$	there exists x
$\exists! x$	there exists exactly one x
$\neg P$	not P
$P \wedge Q$	P and Q
$P \vee Q$	P or Q, usually inclusive
$\therefore$	therefore
$\because$	because

Negating quantifiers

The negation of

$$\forall x \in A,\ \exists y \in B,\ P(x,y)$$

is

$$\exists x \in A,\ \forall y \in B,\ \neg P(x,y).$$

Example:

$$\forall \epsilon > 0,\ \exists N,\ \forall n \geq N,\ |a_n - L| < \epsilon$$

means $a_n \to L$. Its negation is

$$\exists \epsilon > 0,\ \forall N,\ \exists n \geq N \text{ such that } |a_n - L| \geq \epsilon.$$

C.2 Sets

symbol	meaning
$x \in A$	x is an element of A
$x \notin A$	x is not an element of A
$A \subseteq B$	every element of A is an element of B
$A \subset B$	A$\subseteq$ B and usually $A \neq B$
$A = B$	A$\subseteq$ B and $B \subseteq A$
$A \cup B$	union: elements in A or B
$A \cap B$	intersection: elements in both A and B
$A \setminus B$	elements in A but not in B
A^c	complement of A, relative to the ambient universe
$\varnothing$	empty set
$\mathcal{P}(A)$	power set of A, the set of all subsets of A
$\lvert A \rvert$	cardinality of A

Element versus subset

Let

$$A = \{1, 2, \{3, 4\}\}.$$

Then

$$\{3, 4\} \in A,$$

but

$$3 \notin A, \qquad 4 \notin A.$$

Also

$$\{1, 2\} \subseteq A.$$

The statement

$$\{3, 4\} \subseteq A$$

is false, because $3 \notin A$ and $4 \notin A$.

De Morgan laws

$$(A \cup B)^c = A^c \cap B^c,$$
$$(A \cap B)^c = A^c \cup B^c.$$

For set difference,

$$A \setminus B = A \cap B^c.$$

Therefore

$$A \setminus (B \cup C) = A \cap (B \cup C)^c = A \cap B^c \cap C^c.$$

C.3 Number Systems

symbol	meaning
$\mathbb{N}$	natural numbers; convention may be $\{1, 2, 3, \ldots\}$ or $\{0, 1, 2, \ldots\}$
$\mathbb{Z}$	integers
$\mathbb{Q}$	rational numbers
$\mathbb{R}$	real numbers
$\mathbb{C}$	complex numbers
$\mathbb{F}_p$	finite field with p elements, where p is prime
$\mathbb{Z}/n\mathbb{Z}$	integers modulo n
$\mathbb{R}^n$	n-dimensional real coordinate space
$\mathbb{C}^n$	n-dimensional complex coordinate space

Containment chain

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

The inclusion

$$\mathbb{Q} \subseteq \mathbb{R}$$

is strict because

$$\sqrt{2} \in \mathbb{R} \quad \text{but} \quad \sqrt{2} \notin \mathbb{Q}.$$

The inclusion

$$\mathbb{R} \subseteq \mathbb{C}$$

is strict because

$$i \in \mathbb{C} \quad \text{but} \quad i \notin \mathbb{R}.$$

C.4 Divisibility and Congruences

symbol	meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$\gcd(a, b)$	greatest common divisor
$\operatorname{lcm}(a, b)$	least common multiple
$a \equiv b \pmod{n}$	n$\mid (a - b)$
$a^{-1} \pmod{n}$	multiplicative inverse of a modulo n, if it exists
p	often denotes a prime
$\pi(x)$	number of primes less than or equal to x

Modular inverse condition

The inverse

$$a^{-1} \pmod{n}$$

exists if and only if

$$\gcd(a, n) = 1.$$

Example:

$$7^{-1} \pmod{26} = 15,$$

because

$$7 \cdot 15 = 105 \equiv 1 \pmod{26}.$$

C.5 Functions

symbol	meaning
$f : A \to B$	f is a function with domain A and codomain B
$f(x)$	value of f at input x
$\mathrm{dom}(f)$	domain of f
$\mathrm{im}(f)$	image or range of f
f^{-1}	inverse function, when it exists
$f^{-1}(S)$	preimage of a set S; does not require f to be invertible
$g \circ f$	composition: $(g \circ f)(x) = g(f(x))$
id_A	identity map on A

Inverse function versus preimage

Let

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

The function f has no inverse on all of $\mathbb{R}$, because

$$f(1) = f(-1) = 1.$$

But the preimage notation still makes sense:

$$f^{-1}(\{4\}) = \{-2, 2\}.$$

So $f^{-1}(S)$ as a preimage is different from f^{-1} as an inverse function.

C.6 Limits and Continuity

symbol	meaning
$\lim_{x\to a} f(x) = L$	f(x) approaches L as x approaches a
$\lim_{x\to a^-} f(x)$	left-hand limit
$\lim_{x\to a^+} f(x)$	right-hand limit
$\lim_{x\to\infty} f(x)$	limit as x grows without bound
f continuous at a	$\lim_{x\to a} f(x) = f(a)$
$a_n \to L$	sequence a_n converges to L
$O(g(x))$	bounded in magnitude by a constant multiple of $g(x)$
$o(g(x))$	negligible compared with $g(x)$

Limit versus value

For

$$f(x) = \frac{x^2 - 1}{x - 1},$$

the expression is undefined at

$$x = 1.$$

But for

$$x \neq 1,$$

$$f(x) = x + 1.$$

Therefore

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

The limit exists even though the value at the point does not.

Big-O and little-o examples

As

$$x \to 0,$$

$$\sin x = x + O(x^3).$$

This means the error

$$\sin x - x$$

is bounded by a constant multiple of

$$|x|^3$$

near 0.
Also,

$$x^2 = o(x) \qquad (x \to 0),$$

because

$$\frac{x^2}{x} = x \to 0.$$

C.7 Differentiation

symbol	meaning
$f'(x)$	derivative of f at x
$\frac{df}{dx}$	Leibniz notation for derivative
$\frac{dy}{dx}$	derivative of y with respect to x
$f''(x)$	second derivative
$f^{(n)}(x)$	n-th derivative
∇f	gradient of f
Df	derivative or differential of f, depending on context
$\partial_i f$	partial derivative with respect to the i-th variable
$\frac{\partial f}{\partial x}$	partial derivative with respect to x

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}.$$

Tangent line

The tangent line to

$$y = f(x)$$

at

$$x = a$$

is

$$y = f(a) + f'(a)(x - a).$$

Example:

$$f(x) = x^2, \qquad a = 3.$$

Then

$$f(3) = 9, \qquad f'(x) = 2x, \qquad f'(3) = 6.$$

So the tangent line is

$$y = 9 + 6(x - 3) = 6x - 9.$$

C.8 Integration

symbol	meaning
$\int_a^b f(x)\,dx$	definite integral of f from a to b
$\int f(x)\,dx$	indefinite integral; family of antiderivatives
$F'(x) = f(x)$	F is an antiderivative of f
dx	variable of integration
$\sum_{k=1}^n$	finite summation
$\prod_{k=1}^n$	finite product

Fundamental theorem of calculus

If

$$F'(x) = f(x),$$

then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Example:

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

Signed area warning

$$\int_{-1}^1 x\,dx = 0,$$

but the geometric area between $y = x$ and the x-axis from -1 to 1 is

$$\int_{-1}^1 |x|\,dx = 1.$$

A definite integral records signed area unless absolute value is used.

C.9 Linear Algebra

symbol	meaning
V, W	vector spaces
v, w	vectors
0	zero vector, depending on context
$\mathrm{span}(v_1, \ldots, v_k)$	all linear combinations of $v_1, \ldots, v_k$
$\dim V$	dimension of V
$T : V \to W$	linear map
$\ker T$	kernel of T
$\mathrm{im}\, T$	image of T
$\mathrm{rank}\, T$	$\dim \mathrm{im}\, T$
A^T	transpose of matrix A
A^{-1}	inverse matrix, if it exists
$\det A$	determinant of A
$\mathrm{tr}\, A$	trace of A
λ	often an eigenvalue
$Av = \lambda v$	v is an eigenvector with eigenvalue λ

Rank-nullity

For a linear map
$$T : V \to W$$
with V finite-dimensional,
$$\dim V = \dim \ker T + \dim \mathrm{im}\, T.$$
Example:
$$T : \mathbb{R}^3 \to \mathbb{R}^2, \qquad T(x, y, z) = (x + y, y + z).$$
Solving
$$x + y = 0, \qquad y + z = 0$$
gives
$$(x, y, z) = (-t, t, -t).$$
Thus
$$\dim \ker T = 1.$$
Since the image is all of $\mathbb{R}^2$,
$$\dim \mathrm{im}\, T = 2.$$
So
$$3 = 1 + 2.$$

C.10 Algebra

symbol	meaning	
G	group	
e	identity element of a group	
$H \leq G$	H is a subgroup of G	
$H \trianglelefteq G$	H is a normal subgroup of G	
G/H	quotient group	
$\lvert G \rvert$	order of a finite group	
$\langle g \rangle$	subgroup generated by g	
R	ring	
$I \trianglelefteq R$	I is an ideal of R	
R/I	quotient ring	
K, F	fields	
K/F	field extensionK:F>	K:F
	degree of field extension	
$F(\alpha)$	smallest field containing F and α	
$\mathrm{Gal}(K/F)$	Galois group of K/F	

Field extension degree and constructibility

If a real algebraic number α is constructible by straightedge and compass, then

$$[\mathbb{Q}(\alpha) : \mathbb{Q}]$$

divides a power of 2.
For

$$\alpha = \sqrt[3]{2},$$

the minimal polynomial is

$$x^3 - 2.$$

It is irreducible over $\mathbb{Q}$, so

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3.$$

Since 3 does not divide any power of 2, $\sqrt[3]{2}$ is not constructible.

C.11 Geometry

symbol	meaning
AB	line segment joining A and B, or its length depending on context
$\overline{AB}$	segment from A to B
$\angle ABC$	angle with vertex B
$\triangle ABC$	triangle with vertices A, B, C
$A \parallel B$	parallel objects, usually lines
$A \perp B$	perpendicular objects
$d(P, Q)$	distance between points P, Q
$\lVert v \rVert$	length or norm of vector v
$u \cdot v$	dot product
S^n	n-sphere
$\mathbb{P}^n$	n-dimensional projective space

Projective coordinates

A point in the projective plane is written

$$[X : Y : Z],$$

where

$$[X : Y : Z] = [\lambda X : \lambda Y : \lambda Z] \qquad \lambda \neq 0.$$

Affine points are embedded by

$$(x, y) \mapsto [x : y : 1].$$

Points with

$$Z = 0$$

are points at infinity.

C.12 Topology

symbol	meaning
X, Y	topological spaces
U, V	often open sets
$\overline{A}$	closure of A
A°	interior of A
∂A	boundary of A
$X \simeq Y$	X is homotopy equivalent to Y, depending on context
$X \cong Y$	X is isomorphic or homeomorphic to Y, depending on context
$\chi(X)$	Euler characteristic of X
$\pi_1(X)$	fundamental group of X
$H_n(X)$	n-th homology group of X

Euler characteristic

For a triangulated surface,

$$\chi = V - E + F.$$

For a sphere,

$$\chi(S^2) = 2.$$

For a torus,

$$\chi(T^2) = 0.$$

Since Euler characteristic is a topological invariant,

$$S^2$$

and

$$T^2$$

are not homeomorphic.

C.13 Probability and Statistics

symbol	meaning
Ω	sample space
$\mathcal{F}$	sigma-algebra of events
$\mathbb{P}(A)$	probability of event A
A^c	complement of event A
$A \perp B$	A and B are independent, if used probabilistically
X, Y	random variables
$\mathbb{E}[X]$	expectation of X
$\mathrm{Var}(X)$	variance of X
$\mathrm{Cov}(X, Y)$	covariance
$X \sim \mu$	X has distribution μ
$X \sim N(\mu, \sigma^2)$	normal distribution with mean μ, variance σ^2
$\mathbf{1}_A$	indicator of event A

Conditional probability

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) > 0.$$

Bayes' formula:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Variance identity

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

For a Bernoulli random variable

$$X \sim \text{Bernoulli}(p),$$

$$\mathbb{E}[X] = p,$$

and

$$\mathbb{E}[X^2] = p,$$

because $X^2 = X$. Hence

$$\text{Var}(X) = p - p^2 = p(1-p).$$

C.14 Asymptotic Notation

symbol	meaning
$f(x) = O(g(x))$	$\lvert f(x)\rvert \le C\lvert g(x)\rvert$ eventually or near a point
$f(x) = o(g(x))$	$f(x)/g(x) \to 0$
$f(x) \sim g(x)$	$f(x)/g(x) \to 1$
$f(x) = \Theta(g(x))$	$f = O(g)$ and $g = O(f)$
$f(x) \ll g(x)$	number-theoretic notation for $f = O(g)$
$f(x) \gg g(x)$	number-theoretic notation for $g = O(f)$

Examples

As

$$x \to \infty,$$

$$x^2 + 3x + 1 \sim x^2.$$

Indeed,

$$\frac{x^2 + 3x + 1}{x^2} = 1 + \frac{3}{x} + \frac{1}{x^2} \to 1.$$

Also,

$$\log x = o(x^\alpha)$$

for every

$$\alpha > 0.$$

For example,

$$\frac{\log x}{\sqrt{x}} \to 0.$$

C.15 Common Ambiguities

notation	possible ambiguity
$\lvert x\rvert$	absolute value if x is a number; cardinality if x is a set; order if x is a group element
AB	segment, line, or length, depending on context
f^{-1}	inverse function or preimage notation
$\cong$	isomorphism, congruence, or homeomorphism, depending on field
0	zero number, zero vector, zero map, or zero object
1	number one, identity element, or identity map depending on context
dim	dimension of vector space, manifold, or fractal dimension
∂	partial derivative or boundary operator

A useful discipline

When a symbol is overloaded, identify its type.
For example:

$$|G|$$

means the number of elements of a finite group G, while

$$|x|$$

for

$$x \in \mathbb{R}$$

means absolute value.
The notation is the same, but the objects have different types:

$$G \text{ is a group,} \qquad x \text{ is a real number.}$$

C.16 Frequently Used Identities

$$(a+b)^2 = a^2 + 2ab + b^2.$$

$$a^2 - b^2 = (a-b)(a+b).$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2).$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2).$$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}, \qquad r \neq 1.$$

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}, \qquad |r| < 1.$$

$$e^{ix} = \cos x + i \sin x.$$

$$\sin^2 x + \cos^2 x = 1.$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x.$$

$$\sin(2x) = 2 \sin x \cos x.$$

C.17 Standard Derivatives

$f(x)$	$f'(x)$
c	0
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\arctan x$	$\frac{1}{1+x^2}$

C.18 Standard Integrals

$f(x)$	$\int f(x)\,dx$
$x^n,\ n \neq -1$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln\|x\| + C$
e^x	$e^x + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\frac{1}{1+x^2}$	$\arctan x + C$

C.19 Named Principles

name	usable form
Pigeonhole principle	if $n+1$ objects are placed into n boxes, one box contains at least two objects
Euclidean algorithm	successive division computes $\gcd(a, b)$
AM–GM	$\frac{x_1 + \cdots + x_n}{n} \geq (x_1 \cdots x_n)^{1/n}$ for $x_i > 0$
Intermediate value theorem	continuous functions on intervals take all intermediate values
Extreme value theorem	continuous functions on closed bounded intervals attain maximum and minimum
Mean value theorem	some tangent slope equals the average slope
Fundamental theorem of calculus	differentiation and integration are inverse operations
Euler formula	$V - E + F = 2$ for convex polyhedra
Bayes' theorem	$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$

C.20 Short Index of Symbols

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	standard number systems
$\in, \subseteq, \cup, \cap, \setminus$	set notation
$\forall, \exists, \Rightarrow, \Longleftrightarrow$	logic
$\lim, \sum, \prod, \int, \frac{d}{dx}$	analysis
ker, im, det, tr	linear algebra
$\gcd, \equiv \pmod n, \pi(x)$	number theory
χ, π_1, H_n	topology
$\mathbb{P}, \mathbb{E}$, Var	probability
$O, o, \sim, \Theta$	asymptotics

CHAPTER D

Personal Research Index

Source Map

This chapter is a personal index of research directions suggested by the preceding reading notes. I treat each topic through three pieces: a concrete model problem, the invariant or method that controls it, and a possible extension. The purpose is not to claim finished research results, but to mark where elementary questions begin to touch serious mathematics.

D.1 Algebraic Obstructions in Classical Geometry

The straightedge-and-compass problems are a good model of mathematical maturity: the visible problem is geometric, but the obstruction is algebraic.

problem	required number	obstruction
duplicate cube	$\sqrt[3]{2}$	$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$
trisect $60°$	$2\cos 20°$	$x^3 - 3x - 1$
regular heptagon	$2\cos(2\pi/7)$	$t^3 + t^2 - 2t - 1$
regular pentagon	$\cos(2\pi/5)$	quadratic, constructible

Index calculation: heptagon obstruction

Let

$$\zeta = e^{2\pi i/7}, \qquad t = \zeta + \zeta^{-1} = 2\cos\frac{2\pi}{7}.$$

Since

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 + \zeta^5 + \zeta^6 = 0,$$

divide by ζ^3:

$$\zeta^{-3} + \zeta^{-2} + \zeta^{-1} + 1 + \zeta + \zeta^2 + \zeta^3 = 0.$$

Using

$$\zeta^2 + \zeta^{-2} = t^2 - 2,$$

and

$$\zeta^3 + \zeta^{-3} = t^3 - 3t,$$

we get

$$(t^3 - 3t) + (t^2 - 2) + t + 1 = 0.$$

Hence

$$t^3 + t^2 - 2t - 1 = 0.$$

This cubic has no rational root:

$$f(1) = -1, \qquad f(-1) = 1.$$

Therefore it is irreducible over $\mathbb{Q}$, and the degree is 3. Since a constructible number must lie in a tower of quadratic extensions, the regular heptagon is not constructible.

Research direction

The general problem is to understand how a geometric operation changes the field of coordinates. The elementary construction theory suggests a broader question:

Given a class of allowed geometric operations, what field extensions can it generate?

For straightedge and compass the answer is quadratic towers. For origami, marked ruler constructions, or conic intersections, the permitted algebra changes. The research value is that geometry becomes a controlled algebraic machine.

D.2 Projective Invariants and Models of Geometry

Projective geometry is useful because it separates incidence from metric measurement. The main invariant that survives projection is not distance but cross ratio.

For four points on the affine line,

$$(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}.$$

Index calculation: cross ratio

Compute

$$(0, 1; 3, \infty).$$

Use the limit form:

$$(0, 1; 3, \infty) = \lim_{d\to\infty} \frac{(3-0)(d-1)}{(3-1)(d-0)}.$$

Thus

$$(0, 1; 3, \infty) = \frac{3}{2} \lim_{d\to\infty} \frac{d-1}{d} = \frac{3}{2}.$$

The same invariant appears in hyperbolic geometry. In the Klein disk model, distances along a chord can be expressed by a logarithm of a cross ratio.

Hyperbolic distance on a diameter

On the interval $(-1, 1)$, take

$$A = 0, \qquad B = r, \qquad 0 < r < 1.$$

The boundary points are

$$P = -1, \qquad Q = 1.$$

The distance is

$$d(0, r) = \frac{1}{2} \log \frac{1+r}{1-r}.$$

For

$$r = \frac{1}{2},$$

$$d(0, \tfrac{1}{2}) = \frac{1}{2} \log \frac{1+\frac{1}{2}}{1-\frac{1}{2}} = \frac{1}{2} \log 3.$$

As $r \to 1^-$,

$$\frac{1}{2} \log \frac{1+r}{1-r} \to +\infty.$$

So the boundary is infinitely far away in the hyperbolic metric.

Research direction

A natural direction is to study how much geometry can be rebuilt from projective data. The Klein model suggests that metric information may be encoded indirectly through boundary structure and cross ratios.

A compact question to keep:

When does an incidence geometry admit a natural metric?

D.3 Topology as Invariant Hunting

Topology becomes serious when one stops relying on pictures and starts using invariants.

space	basic invariant
S^2	$\chi = 2$
T^2	$\chi = 0$
double torus	$\chi = -2$
Möbius strip	non-orientable, one boundary component
$\mathbb{R}P^2$	non-orientable, closed

Index calculation: torus from a square

Represent the torus by a square with opposite sides identified in the same direction.
After gluing, all four vertices become one vertex:

$$V = 1.$$

The two horizontal sides become one edge, and the two vertical sides become one edge:

$$E = 2.$$

The interior is one face:

$$F = 1.$$

Thus

$$\chi(T^2) = V - E + F = 1 - 2 + 1 = 0.$$

Distinguishing sphere and torus

If

$$S^2$$

were homeomorphic to

$$T^2,$$

then their Euler characteristics would be equal. But

$$\chi(S^2) = 2, \qquad \chi(T^2) = 0.$$

Therefore

$$S^2 \not\cong T^2.$$

Research direction

Euler characteristic is only the first invariant. It distinguishes the sphere from the torus, but it does not distinguish every pair of spaces. For example, the torus and the Klein bottle both have Euler characteristic 0, but one is orientable and the other is not.

The research habit here is:

when one invariant fails, refine the invariant.

Possible next invariants:

$$\pi_1(X), \qquad H_n(X), \qquad \text{cohomology ring}, \qquad \text{characteristic classes}.$$

D.4 Knot Diagrams and Algebraic Tests

A knot is visually simple but mathematically subtle. The unknot and trefoil cannot be distinguished by length, curvature, or ordinary drawing. They must be distinguished by an invariant.

Tricolorability as a first knot invariant

A knot diagram is tricolorable if its arcs can be colored using three colors such that at every crossing either:

all three incident arcs have the same color,

or

all three incident arcs have different colors.

The unknot has no nontrivial tricoloring. The trefoil has one. Hence the trefoil is not equivalent to the unknot.

Research direction

The deeper direction is to replace diagrammatic tests with algebraic objects:

knot group, Alexander polynomial, Jones polynomial, knot homologies.

The theme matches earlier chapters: a geometric equivalence problem becomes an algebraic invariant problem.

D.5 Optimization and Variational Thinking

Elementary maximum-minimum problems are the entry point to variational mathematics. The same pattern appears in shortest paths, light refraction, soap films, geodesics, and minimal surfaces.

Steiner point angle condition

Suppose three segments meet at an interior minimizing junction. Let their unit direction vectors be

$$u_1, u_2, u_3.$$

At equilibrium, a small movement of the junction should not decrease total length. This gives

$$u_1 + u_2 + u_3 = 0.$$

Three unit vectors sum to zero exactly when the angle between each pair is

$$120^\circ.$$

Thus the Steiner network for three points, when the Steiner point is interior, has three edges meeting at

$$120^\circ.$$

Fermat reflection principle

For points A, B on one side of a line L, the shortest broken path

$$A \to P \to B, \qquad P \in L,$$

is found by reflecting B across L to B'. Then

$$AP + PB = AP + PB'.$$

The shortest path is the straight segment from A to B'. Therefore

$$P = AB' \cap L.$$

The minimizing condition is

$$\angle(AP, L) = \angle(L, PB).$$

Research direction

The natural extension is to replace finite-dimensional optimization by function-space optimization. Instead of choosing a number x, one chooses a curve, surface, or map.

Examples:

problem	quantity minimized
geodesic	length
soap film	area
brachistochrone	time
harmonic map	energy
minimal surface	area functional

A useful first equation is the Euler–Lagrange equation:

$$\frac{\partial L}{\partial y} - \frac{d}{dx}\frac{\partial L}{\partial y'} = 0.$$

D.6 Analysis of Pathological Examples

The examples

$$\sin\frac{1}{x}, \qquad x\sin\frac{1}{x}, \qquad x^2\sin\frac{1}{x}$$

are small but important. They separate boundedness, convergence, continuity, and differentiability.

Continuity without differentiability

Define

$$f(x) = \begin{cases} x\sin\frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Since

$$-|x| \leq x\sin\frac{1}{x} \leq |x|,$$

the squeeze theorem gives

$$\lim_{x\to 0} x\sin\frac{1}{x} = 0.$$

Thus f is continuous at 0.
But

$$\frac{f(h) - f(0)}{h} = \sin\frac{1}{h}.$$

The limit does not exist as $h \to 0$. Therefore f is not differentiable at 0.

Differentiability restored by stronger damping

Define

$$g(x) = \begin{cases} x^2\sin\frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then

$$\frac{g(h) - g(0)}{h} = h\sin\frac{1}{h}.$$

Since

$$-|h| \leq h\sin\frac{1}{h} \leq |h|,$$

we get

$$\lim_{h \to 0} h \sin \frac{1}{h} = 0.$$

Thus

$$g'(0) = 0.$$

Research direction

The broader question is how local regularity is measured.

Important scales:

$$C^0, \quad C^1, \quad C^k, \quad C^\infty, \quad \text{analytic}, \quad \text{Sobolev}, \quad \text{Hölder}.$$

A simple research habit is to ask:

Which theorem fails if this regularity assumption is weakened?

D.7 Uniform Convergence and Loss of Continuity

Pointwise convergence can destroy continuity.

Standard example

Let

$$f_n(x) = x^n$$

on

$$[0, 1].$$

For $0 \le x < 1$,

$$x^n \to 0.$$

At $x = 1$,

$$x^n = 1.$$

Thus the pointwise limit is

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

Each f_n is continuous, but the limit is not continuous.

This example is a compact warning: convergence of values at each point is not enough to preserve structure.

Research direction

Uniform convergence is the correct repair: if

$$f_n \to f$$

uniformly and each f_n is continuous, then f is continuous.

The broader pattern is that strong convergence preserves more structure than weak convergence. This distinction reappears in functional analysis, PDE, and probability.

D.8 Prime Distribution as a Statistical Object

The primes are deterministic, but their large-scale distribution often looks statistical.

$$\pi(x) = \#\{p \le x : p \text{ prime}\}.$$

The prime number theorem says

$$\pi(x) \sim \frac{x}{\log x}.$$

Small numerical comparison

x	10	100	1000	10000
$\pi(x)$	4	25	168	1229
$x/\log x$	$4.34\ldots$	$21.71\ldots$	$144.76\ldots$	$1085.73\ldots$

The approximation is not exact, but the scale is already visible.

Heuristic density

A large integer near x behaves as if its probability of being prime were approximately

$$\frac{1}{\log x}.$$

Then the expected number of primes in a short interval

$$[x, x + H]$$

should be about

$$\frac{H}{\log x},$$

provided H is not too small.

Research direction

The natural research questions concern fluctuations around the main term:

$$\pi(x) - \mathrm{Li}(x),$$

gaps between consecutive primes,

$$p_{n+1} - p_n,$$

and correlations such as twin primes:

$$p, \ p + 2.$$

The important habit is to separate:

$$\text{main term}, \qquad \text{error term}, \qquad \text{local irregularity}.$$

D.9 Fractal Dimension and Scaling

Fractals make dimension measurable by scaling rather than by ordinary visual classification.

For a self-similar set made of N copies, each scaled by r, the similarity dimension d satisfies

$$Nr^d = 1.$$

Cantor set

For the middle-third Cantor set,

$$N = 2, \qquad r = \frac{1}{3}.$$

Thus

$$2\left(\frac{1}{3}\right)^d = 1.$$

So

$$2 = 3^d,$$

and

$$d = \frac{\log 2}{\log 3}.$$

Koch curve

For the Koch curve,

$$N = 4, \qquad r = \frac{1}{3}.$$

Thus

$$4\left(\frac{1}{3}\right)^d = 1,$$

so

$$d = \frac{\log 4}{\log 3}.$$

Since

$$3 < 4 < 9,$$
$$1 < \frac{\log 4}{\log 3} < 2.$$

The Koch curve has dimension between a curve and a region.

Research direction

Fractal dimension gives a quantitative invariant for irregular sets. The next step is to compare different notions of dimension:

Hausdorff dimension, box-counting dimension, packing dimension.

A useful test question:

When do these dimensions agree, and when do they differ?

D.10 Higher-Dimensional Geometry and Concentration

In high dimensions, ordinary geometric intuition breaks down. The unit cube

$$[0, 1]^n$$

has long diagonal

$$\sqrt{n}.$$

The distance from the center

$$c = \left(\frac{1}{2}, \dots, \frac{1}{2}\right)$$

to a vertex is

$$\frac{\sqrt{n}}{2}.$$

Distance from center to vertex

Let

$$v = (0, \dots, 0).$$

Then

$$d(c, v) = \sqrt{\left(\frac{1}{2}\right)^2 + \cdots + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{n}}{2}.$$

For

$$n = 100,$$

this distance is

$$5.$$

So even though each coordinate difference is only $1/2$, the accumulated distance is large.

Research direction

The basic phenomenon is concentration of measure: in high dimension, volume often accumulates in regions that are not visually intuitive.

Natural examples:

$$\text{sphere}, \qquad \text{cube}, \qquad \text{Gaussian space}, \qquad \text{simplex}.$$

A concrete question:

Where is most of the volume of a high-dimensional ball or cube?

This question connects geometry, probability, statistics, and data science.

D.11 Finite Fields and Discrete Geometry

The Fano plane gives a small but serious model of projective geometry. It has

$$7$$

points and

$$7$$

lines. It can be built from

$$\mathbb{F}_2^3.$$

The points are the nonzero vectors:

$$100,\ 010,\ 001,\ 110,\ 101,\ 011,\ 111.$$

A line consists of

$$u, \quad v, \quad u+v.$$

A line in the Fano plane

Take

$$u = 101, \qquad v = 011.$$

Then

$$u + v = 110$$

over $\mathbb{F}_2$. Therefore

$$\{101, 011, 110\}$$

is a line.

Research direction

Finite geometry suggests a discrete analogue of continuous geometry. It leads naturally to:

finite projective spaces, error-correcting codes, design theory, incidence combinatorics.

A compact research question:

How much of projective geometry survives over finite fields?

D.12 Proof Auditing

A personal research index should include not only topics, but also standards for proof. A proof is weak if it hides its main difficulty inside phrases like "clearly," "obviously," or "by intuition."

audit question	purpose
What is the exact statement?	prevents proving a weaker claim
Where is compactness used?	detects hidden existence assumptions
Where is continuity used?	prevents invalid intermediate-value arguments
Where is irreducibility used?	separates algebraic degree from a chosen polynomial
Does a local argument imply a global result?	prevents false optimization conclusions
Does the invariant fully classify the objects?	prevents overclaiming

Example of an audit failure

Claim:

$$f'(a) = 0 \implies a \text{ is a maximum or minimum.}$$

Counterexample:

$$f(x) = x^3.$$

Then

$$f'(x) = 3x^2,$$

so

$$f'(0) = 0.$$

But 0 is neither a local maximum nor a local minimum. The derivative condition gives a candidate, not a conclusion.

D.13 Problems I Would Keep Returning To

Problem A: classify constructibility under altered tools

Straightedge and compass give quadratic extensions. What happens if one allows one additional operation, such as angle trisection or conic intersection?

A starting calculation:

$$x^3 - 2$$

is impossible for straightedge and compass because it has degree 3. But a tool capable of extracting cube roots would change the field tower.

The research issue is to classify the algebra generated by the tool, not only to solve one construction problem.

Problem B: compare invariants for surfaces

Euler characteristic distinguishes:

$$S^2 \quad \text{and} \quad T^2.$$

But it does not distinguish every surface. Add orientability and boundary components.

surface	χ	orientable?	boundary components
torus	0	yes	0
Klein bottle	0	no	0
cylinder	0	yes	2
Möbius strip	0	no	1

The research question is how far a finite list of invariants goes toward classification.

Problem C: quantify prime randomness

The prime number theorem gives the first-order density

$$\frac{1}{\log x}.$$

The twin prime heuristic asks for a second-order correlation:

$$n \text{ prime and } n+2 \text{ prime.}$$

The key issue is that primality is not independent. Divisibility by small primes creates local congruence obstructions.

A useful finite experiment is to count twin primes up to increasing cutoffs and compare with

$$\int_2^x \frac{dt}{(\log t)^2}.$$

Problem D: understand minimizers through first variation

For length networks, equilibrium gives vector balance. For surfaces, area minimization gives mean curvature zero:

$$H = 0.$$

For energy functionals, the Euler–Lagrange equation gives the governing differential equation.

The research habit is to derive the equation from the variation, not to guess the equation from the final answer.

D.14 A Working Index of Methods

method	where it appears
Euclidean algorithm	number theory, modular inverses
field degree	constructibility, algebraic impossibility
cross ratio	projective geometry, hyperbolic metric
Euler characteristic	polyhedra, surfaces, topology
compactness	existence of extrema, convergence subsequences
squeeze theorem	oscillatory limits
AM–GM	optimization, inequalities
Lagrange multipliers	constrained extrema
finite differences	detecting polynomial structure
asymptotic comparison	prime counts, growth rates
scaling law	fractal dimension, dimensional analysis
invariant construction	topology, knots, geometry

D.15 Research Note Template

For future reading, I would record each serious idea in the following format.

field	entry
Problem	State the question in one sentence.
Toy model	Give the smallest example where the phenomenon appears.
Invariant	Name the quantity or structure that survives transformations.
Obstruction	State what prevents the desired object from existing.
Computation	Do one explicit calculation.
Failure mode	Give a counterexample to an overstrong claim.
Next theorem	Name the theorem that would push the problem further.

Example entry: constructible regular heptagon

Problem:

Can a regular heptagon be constructed by straightedge and compass?

Toy model:

$$t = 2\cos\frac{2\pi}{7}.$$

Invariant:

$$[\mathbb{Q}(t) : \mathbb{Q}].$$

Computation:

$$t^3 + t^2 - 2t - 1 = 0.$$

Obstruction:

$$[\mathbb{Q}(t) : \mathbb{Q}] = 3,$$

but constructible numbers have degree dividing a power of 2.
Conclusion:

The regular heptagon is not constructible.